

LightSpeed™ VCT

Reference Protocol Guide

5761709-1EN

Revision 3

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This manual supports the following configurations:

LightSpeed™ VCT

LightSpeed™ VCT XT

LightSpeed™ VCT XTe

Not all configurations are available in all regions.

This product is certified as a LightSpeed™ Multislice CT System.

LightSpeed™ VCT

Reference Protocol Guide, English

5761709-1EN

Revision: 3

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Revision History

Revision History

REV	DATE	REASON FOR CHANGE
1	Nov. 2016	Initial Release
2	Dec. 2016	Typo
3	Mar. 2023	Typo

Purpose

Purpose

This document provides a listing of the parameters provided in the reference protocols contained under the GE tab selector. Each table lists the Protocol Number, Protocol Name, Post Processing software associated with, Scan Type, SFOV, Pitch/Table Speed/Row, Gantry Rotation Time, Slice Thickness, Beam Collimation, kV, mA/Min-Max/Noise Index/Average mA, Recon Algorithm Type used, CTDIvol, DLP (mGy-cm), Scan Length, and Phantom Type used and Description.

If the ASiR option is installed on the system, there will be a % Dose Reduction column and % ASiR column listed in the table indicating the % Dose Reduction and % ASiR value in the reference protocol. For protocols with manual mA, the mA listed is the actual mA used for the scan. The Reference mA is calculated as:

$$\frac{1.00}{(1.00 - \% \text{ Dose Reduction})} \times \text{Actual mA}$$

Manual mA mode allows you to scan without enabling Auto mA mode. When building protocols, make sure the mA value field contains a reasonable mA value in the event that AutomA is turned off. In the GE reference protocols, the Average (Avg) mA provided in the tables within this document are what is used in calculating the CTDI and DLP for protocols using AutomA or SmartmA. The Avg mA value provided in the table is an estimate based on an average patient size (32- 36 cm) . The actual average mA for each image is calculated by using all of the mA values for each scan rotation, as determined by Auto mA for the patient. The Auto mA Theory section in the User Manual, Scan Chapter and in the Technical Reference Manual, General Information Chapter details how Auto mA does this calculation.

Reference Adult Protocols - Installed

Options : ASiR, Sub 0.4sec, VCT85kW and VCT Hi Power

Head

Table 2-1

[Interval]* - For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (s)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy-cm) ASiR	Scan Length (mm)	Phantom (cm)	Description
21.1	Routine Head 1s Group1		Axial	Head	4i	1	5	20	140	120	40	Std	40	32.00	192.01	55	16	Routine head protocol for evaluation of the brain for abnormalities.
	Group2		Axial	Head	4i	1	5	20	120	190	40	Std	40	36.47	291.78	75	16	
21.2	Routine Head 2s Group1		Axial	Head	4i	2	5	20	140	60	40	Std	40	32.00	192.01	55	16	Routine head protocol for evaluation of the brain for abnormalities.
	Group2		Axial	Head	4i	2	5	20	120	95	40	Std	40	36.47	291.78	75	16	
21.3	Trauma Head 2.5mm		Axial	Head	8i	0.5	2.5	20	140	335	40	Std	40	44.67	625.37	137.5	16	Emergency head protocol for evaluation of the brain and cranium for abnormalities.
21.4	Routine Head 0.5s Group1		Axial	Head	4i	0.5	5	20	140	240	40	Std	40	32.00	192.01	55	16	Routine head protocol for evaluation of the brain for abnormalities.
	Group2		Axial	Head	4i	0.5	5	20	120	360	40	Std	40	35.52	284.17	75	16	
21.5	Circle of Willis 0.4s.	MIROI	Axial	Head	1i	1	5 [0]	5	120	50	0	Std	0	125.58	62.79	0	16	Test bolus for CT angiography of the head for evaluation of carotid and cerebral vasculature.
			Helical	Head	0.531:1 10.625	0.4	0.625	20	120	180	40	Std Plus IQE	50	26.02	252.82	80	16	CT angiography of the head for evaluation of carotid and cerebral vasculature.
21.6	CoW/ Carotid 0.625mm 0.4s.Smart mA	MIROI	Axial	Head	1i	1	5 [0]	5	120	80	0	Std	0	180.84	90.42	0	16	Test bolus for CT angiography of the head for evaluation of carotid and cerebral vasculature.
			Helical	Head	0.531:1 10.625	0.4	0.625	20	120	SmartmA 100-650mA NI=8 Avg mA 415	30	Std Plus IQE	40	61.66	1155.86	170	16	CT angiography of the head for evaluation of carotid and cerebral vasculature.
21.8	Helical Head	DMPR	Helical	Head	0.531:1 10.625	0.5	0.625	20	120	180	40	Std Plus IQE	40	32.52	527.31	145	16	Helical scan mode for evaluation of the brain for cerebral abnormality.
21.10	VHS Head 4D CTA	MIROI	Axial	Head	1i	1	5 [0]	5	120	50	0	Std	0	125.58	62.79	0	16	Test bolus for CT angiography of the head for evaluation of carotid and cerebral vasculature.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (s)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy-cm) ASiR	Scan Length (mm)	Phantom (cm)	Description
			Helical-S	Head	0.984:1 39.375 18 pass	0.4	5 [10]	40	120	150	50	Std Plus	50	248.10	3101.19	110	16	CT angiography of the head using Volume Helical scan mode for evaluation of cerebral vasculature.
21.11	VHS Perfusion 350-370 Strength Contrast		Axial	Head	4i	1	5	20	120	200	40	Std	40	38.39	537.48	135	16	Non-Enhance Brain, evaluation for hemorrhage or infarction.
		Perfusion	Helical-S/P	Head	0.984:1 39.375 28 pass	0.4	5	40	80	200	0	Std	50	186.02	2325.25	110	16	CT Perfusion using Volume Helical scan mode for evaluation of cerebral perfusion over time.
			Axial	Head	4i	1	5	20	120	200	40	Std	40	38.39	537.48	135	16	Post Contrast Brain, Evaluation for residual tissue enhancement.
21.12	CT Perfusion 300-320 Strength Contrast		Axial	Head	4i	1	5	20	120	200	40	Std	40	38.39	537.48	135	16	Non-Enhance Brain, evaluation for hemorrhage or infarction.
		Perfusion	Cine	Head	8i	1	5 [0]	40	80	200	0	Std	0	652.82	2611.27	35	16	CT Perfusion using Cine scan mode for evaluation of cerebral perfusion over time.
			Axial	Head	4i	1	5	20	120	200	40	Std	40	38.39	537.48	135	16	Non-Enhance Brain, evaluation for hemorrhage or infarction.
21.13	CT Perfusion 350-370 Strength Contrast		Axial	Head	4i	1	5	20	120	200	40	Std	40	38.39	537.48	135	16	Non-Enhance Brain, evaluation for hemorrhage or infarction.
		Perfusion	Cine	Head	8i	1	5 [0]	40	80	200	0	Std	0	587.55	2350.19	35	16	CT Perfusion using Cine scan mode for evaluation of cerebral perfusion over time.
			Axial	Head	4i	1	5	20	120	200	40	Std	40	38.39	537.48	135	16	Post Contrast Brain, Evaluation for residual tissue enhancement.
21.14	CT Perfusion Brain Tumor - Dual Phase		Axial	Head	4i	1	5	20	120	200	40	Std	40	38.39	537.48	135	16	Non-Enhance Brain, evaluation for hemorrhage or infarction.
	Group 1	Perfusion	Cine	Head	8i	1	5 [0]	40	80	200	0	Std	0	587.55	2350.19	35	16	CT Perfusion using Cine scan mode for evaluation of cerebral perfusion over time.
	Group 2	Perfusion	Axial	Head	8i	1	5 [0]	40	80	200	0	Std	0	130.54	522.15	35	16	Delayed phase CT Perfusion using Axial scan mode for evaluation of Brain tumor permeability.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (s)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy-cm) ASiR	Scan Length (mm)	Phantom (cm)	Description
			Axial	Head	4i	1	5	20	120	200	40	Std	40	38.39	537.48	135	16	Post Contrast Brain, Evaluation for residual tissue enhancement.
21.15	CT Perfusion 350-370 Strength Contrast (Axial)		Axial	Head	4i	1	5	20	120	200	40	Std	40	38.39	537.48	135	16	Non-Enhance Brain, evaluation for hemorrhage or infarction.
		Perfusion	Axial	Head	8i	1	5 [0]	40	80	200	0	Std	0	287.18	1148.73	35	16	CT Perfusion using Axial scan mode for evaluation of cerebral perfusion over time.
			Axial	Head	4i	1	5	20	120	200	40	Std	40	38.39	537.48	135	16	Post Contrast Brain, Evaluation for residual tissue enhancement.
21.16	CT Perfusion 350-370 Strength Contrast (Axial Shuttle Mode)		Axial	Head	4i	1	5	20	120	200	40	Std	40	38.39	537.48	135	16	Non-Enhance Brain, evaluation for hemorrhage or infarction.
		Perfusion	Axial-S	Head	8i 17 pass	0.4	5	40	80	500	0	Std	0	221.91	1775.31	75	16	CT Perfusion using Volume Shuttle (Axial) scan mode for evaluation of cerebral perfusion over time.
			Axial	Head	4i	1	5	20	120	200	40	Std	40	38.39	537.48	135	16	Post Contrast Brain, Evaluation for residual tissue enhancement.

Orbit

Table 2-2

[Interval]* - For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy-cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
22.1	Sinus Supine Helical + DMPR	DMPR	Helical	Head	0.531:1 10.625	0.4	0.625	20	120	85	30	Bone Plus IQE	50	12.29	112.48	74.375	16	Routine Helical Sinus protocol, Evaluation for any abnormality
22.3	Orbits Helical + DMPR	DMPR	Helical	Head	0.531:1 10.625	0.5	0.625	20	120	155	30	Std Plus IQE	40	28.00	221.29	61.875	16	Routine Helical Orbit protocol, Evaluation for abnormality soft tissues and bone structures of the face.
22.6	Helical IAC 0.625mm Axial mode - Use Reformat for Coronal		Helical	Head	0.531:1 10.625	0.6	0.625	20	140	190	30	Std Plus IQE	40	57.23	319.79	38.75	16	Axial scan mode, Evaluation for inner ear abnormalities in Axial plane
22.8	Axial and Coronal IAC		Axial	Head	16i	1	1.25	20	140	195	30	Std	40	52.00	208.01	38.75	16	Helical scan mode, Evaluation for inner ear abnormalities in Axial plane
			Axial	Head	16i	1	1.25	20	140	235	30	Std	40	62.67	250.68	38.75	16	Helical scan mode, Evaluation for inner ear abnormalities in Coronal plane

Neck

Table 2-3

[Interval]* - For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
23.1	C-Spine C5-C7 Axial		Axial	SmallBody	8i	0.5	2.5	20	140	520	20	Std	40	28.28	395.86	137.5	32	C-Spine C5-C7 area Axial mode, Evaluation for soft tissue, disc and bone abnormalities.
23.2	C-Spine C1-C4 Axial		Axial	SmallBody	8i	1	2.5	20	120	190	20	Std	40	14.17	113.36	77.5	32	C-Spine C1-C4 area Axial mode, Evaluation for soft tissue, disc and bone abnormalities.
23.3	C-Spine Helical + DMPR	DMPR	Helical	SmallBody	0.516:1 20.625	0.8	0.625	40	120	270	20	Bone Plus IQE	40	29.20	421.46	110	32	C-Spine Helical mode with DMPR, Evaluation for soft tissue, disc and bone abnormalities.
23.4	C-Spine Helical + DMPR/ SmartmA	DMPR	Helical	SmallBody	0.516:1 20.625	0.8	0.625	40	120	SmartmA 100-650mA NI=11.34 Avg mA 250	20	Bone Plus IQE	40	27.04	390.24	110	32	C-Spine Helical mode with DMPR and SmartmA, Evaluation for soft tissue, disc and bone abnormalities.
23.8	Carotid 0.625mm 0.4s. 0.5:1 SmartmA/ SmartPrep		Helical	SmallBody	0.516:1 20.625	0.4	0.625	40	140	SmartmA 100-650mA NI=11 Avg mA 345	40	Std Plus IQE	50	26.46	541.51	170	32	Carotid CTA (pitch 0.5:1) with SmartmA and SmartPrep, Evaluation of carotid vasculature.
23.9	Carotid 0.625mm 0.4s. 0.9:1 SmartmA/ SmartPrep		Helical	SmallBody	0.984:1 39.375	0.4	0.625	40	120	SmartmA 80-700mA NI=11 Avg mA 375	40	Std Plus IQE	50	10.62	215.89	170	32	Carotid CTA (pitch 0.9:1) with SmartmA and SmartPrep, Evaluation of carotid vasculature.
23.10	Carotid/ CoW 0.625mm 0.4s. AutomA	MIROI	Axial	SmallBody	1i	1	5 [0]	5	120	80	0	Std	0	99.00	49.50	0	32	Test bolus for Carotid CTA (pitch 0.5:1) of the head for evaluation of carotid and cerebral vasculature.
			Helical	SmallBody	0.516:1 20.625	0.4	0.625	40	140	AutomA 80-650mA NI=11 Avg mA 360	40	Std Plus IQE	50	27.61	565.06	170	32	Carotid and Brain CTA (pitch 0.5:1) with AutomA, Evaluation of carotid and cerebral vasculature.
23.11	Carotid/ CoW 0.625mm 0.4s. Smart mA/ SmartPrep		Helical	SmallBody	0.969:1 19.375	0.4	0.625	20	140	SmartmA 100-650mA NI=11 Avg mA 390	40	Std Plus IQE	50	17.51	330.81	170	32	Carotid CTA (pitch 0.9:1) with SmartmA and SmartPrep, Evaluation of carotid and cerebral vasculature.
23.12	Soft Tissue Neck 2.5mm AutomA		Helical	SmallBody	0.984:1 39.375	0.5	2.5	40	120	AutomA 100-335mA NI=9.1 Avg mA 175	30	Std Plus	30	6.20	113.26	150	32	Neck with AutomA, Evaluation of soft tissue structures for abnormality

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
23.13	Soft Tissue Neck + DMPR/ SmartmA	DMPR	Helical	SmallBody	0.969:1 19.375	0.5	0.625	20	120	SmartmA 100-600mA NI=12.6 Avg mA 255	40	Std Plus IQE	50	9.82	165.46	150	32	Neck with DMPR and SmartmA, Evaluation of soft tissue structures for abnormality

Upper Extremity

Table 2-4

[Interval]* - For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
24.1	Shoulder 1.25mm + DMPR	DMPR	Helical	LargeBody	0.516:1 20.625	0.6	1.25 [0.625]	40	140	360	40	Std Plus IQE	40	47.03	632.79	100	32	Shoulder with DMPR, Evaluation of soft tissue and bone of shoulder area
24.2	Shoulder 1.25mm + DMPR/ SmartmA	DMPR	Helical	LargeBody	0.516:1 20.625	0.6	1.25 [0.625]	40	140	SmartmA 100-650mA NI=20 Avg mA 355	40	Std Plus IQE	40	46.37	624.00	100	32	Shoulder with DMPR and SmartmA, Evaluation of soft tissue and bone of shoulder area

Chest

Table 2-5

[Interval]* - For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
25.1	Routine Chest 0.4s. 5mm SmartmA		Helical	LargeBody	1.375:1 55.0	0.4	5	40	120	SmartmA 100-650mA NI=13 Avg mA 250	50	Std Plus	50	5.76	142.87	200	32	Chest with SmartmA, Evaluation of mediastinum and lungs fields
25.2	Routine Chest 0.4s.+ DMPR/ SmartmA	DMPR	Helical	LargeBody	0.984:1 39.375	0.4	0.625	40	120	SmartmA 100-750mA NI=22 Avg mA 310	50	Std Plus	50	9.98	232.46	200	32	Chest with SmartmA and DMPR, Evaluation of mediastinum and lungs fields
25.3	Chest with HiRes 0.4s. SmartmA		Helical	LargeBody	0.969:1 19.375	0.4	5	20	120	SmartmA 80-650mA NI=12 Avg mA 300	40	Chest Plus	40	10.50	229.59	200	32	Chest with SmartmA, Evaluation of mediastinum and lung fields with higher resolution images of the lungs
25.4	Hi Res Chest 0.5s. Axial		Axial	LargeBody	1i	0.5	1.25 [10]	1.25	120	190	50	Bone+	50	2.22	53.35	230	32	Axial scan mode, Evaluation of mediastinum and lung fields in high resolution mode
25.5	Chest Abd Pelvis 0.4/ 0.5s. SmartmA 5mm (2 groups) Group1		Helical	LargeBody	0.984:1 39.375	0.4	5	40	120	SmartmA 100-650mA NI=13 Avg mA 185	50	Std Plus	50	5.96	108.95	150	32	Multi group scan for Chest, Abdomen, Pelvis (0.4/0.5sec) with SmartmA, Evaluation of chest and abdominal anatomy for abnormalities
	Group2		Helical	LargeBody	1.375:1 55.0	0.5	5	40	120	SmartmA 100-650mA NI=13 Avg mA 185	50	Std Plus	50	5.33	182.72	295	32	
25.6	Chest Abd Pelvis 0.5s + DMPR SmartPrep	DMPR	Helical	LargeBody	1.375:1 55.0	0.5	0.625	40	120	SmartmA 100-750mA NI=22 Avg mA 270	50	Std Plus	50	7.78	387.24	450	32	Helical scan mode (0.5sec) with SmartmA, Evaluation of chest and abdominal anatomy for abnormalities
25.7	Pulmonary Embolus 0.5s.	MIROI	Axial	LargeBody	1i	1	5 [0]	5	120	80	0	Std	0	93.89	46.95	0	32	Timing Bolus for Helical scan mode 1.25mm thickness and helical pitch of 0.9:1, evaluation of Chest for pulmonary embolism
			Helical	LargeBody	0.984:1 39.375	0.5	1.25	40	120	350	40	Std	40	14.09	300.25	180	32	Helical scan mode 1.25mm thickness and helical pitch of 0.9:1, Evaluation Chest for pulmonary embolism

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy-cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
25.8	Pulmonary Embolism 0.5s. SmartmA	MIROI	Axial	LargeBody	1i	1	5 [0]	5	120	80	0	Std	0	93.89	46.95	0	32	Timing Bolus for Helical scan mode 1.25mm thickness and helical pitch of 0.9:1 with SmartmA, evaluation of Chest for pulmonary embolism
			Helical	LargeBody	0.984:1 39.375	0.5	1.25	40	120	SmartmA 100-750mA NI=20 Avg mA 300	40	Std	40	12.07	256.93	180	32	Helical scan mode 1.25mm thickness and helical pitch of 0.9:1 with SmartmA, Evaluation Chest for pulmonary embolism
25.9	Aortic Dissection 0.625mm + DMPR/ SmartmA/ SmartPrep	DMPR	Helical	LargeBody	1.375:1 55.0	0.5	0.625	40	120	SmartmA 100-750mA NI=22 Avg mA 360	40	Std Plus	40	10.37	361.20	300	32	Helical scan mode 0.625mm thickness and helical pitch of 1.375:1 with SmartmA, SmartPrep and DMPR, Evaluation for aortic dissection
25.10	Aortic Dissection 1.25mm + DMPR/ SmartmA/ SmartPrep	DMPR	Helical	LargeBody	1.375:1 55.0	0.5	1.25 [0.8]	40	120	SmartmA 100-750mA NI=20 Avg mA 330	40	Std Plus	40	9.51	330.68	300	32	Helical scan mode 1.25mm thickness and helical pitch of 1.375:1 with SmartmA, SmartPrep and DMPR, Evaluation for aortic dissection
25.11	Lung Assessment	Advanced Lung Analysis	Helical	LargeBody	0.969:1 19.375	0.5	1.25	20	120	120	0	Bone	0	5.25	114.77	200	32	Helical scan mode for acquisition of data for Advanced Lung Analysis (ALA) software for lung nodule assessment
25.12	Gated Chest Cardiac Helical		Helical	LargeBody	1.375:1 55.0	0.4	5	40	120	60	0	Std Plus	0	1.38	13.55	50	32	Localizer scan for Helical Gated scan for evaluation of thoracic vasculature
		MIROI	Axial	LargeBody	1i	1	5 [0]	5	120	60	0	Std	0	98.59	49.29	0	32	Timing bolus scan for Helical Gated scan for evaluation of thoracic vasculature
			Cardiac SEG	CardiacLarge	0.160:1 6.4	0.35	0.625	40	120	ECG Mod 110-270mA Phase 70-80 Avg mA 270	40	Std SS Seg	40	30.98	1037.93	300	32	Helical Gated scan for evaluation of thoracic vasculature
25.13	Gated Chest SnapShot Pulse		Helical	LargeBody	1.375:1 55.0	0.4	5	40	120	60	0	Std Plus	0	1.38	13.55	50	32	Localizer scan for Cine Gated scan for evaluation of thoracic vasculature
		MIROI	Axial	LargeBody	1i	1	5 [0]	5	120	60	0	Std	0	98.59	49.29	0	32	Timing bolus scan for Cine Gated scan for evaluation of thoracic vasculature
			Cardiac Pulse Cine	CardiacLarge	56i	0.35	0.625 [35]	40	120	270	40	Std SS Seg	40	14.42	454.39	314.375	32	Cine Gated scan for evaluation of thoracic vasculature
25.14	SmartScore - Gated 0.35s	Smart Score	Cine	LargeBody	8i	0.35	2.5	20	120	430	0	Std Segment	0	8.99	89.95	97.5	32	Cine gated scan mode for acquisition of data for coronary artery calcification analysis

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy-cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
25.15	SnapShot Pulse 30-65 BPM		Helical	LargeBody	1.375:1 55.0	0.4	5	40	120	60	0	Std Plus	0	1.38	13.55	50	32	Localizer scan for Prospectively gated Cardiac Cine scan for evaluation of coronary arteries, intended for stable heart rates 30-65 bpm
		MIROI	Axial	LargeBody	1i	1	5 [0]	5	120	60	0	Std	0	98.59	49.29	0	32	Timing bolus scan for Prospectively gated Cardiac Cine scan for evaluation of coronary arteries, intended for stable heart rates 30-65 bpm
			Cardiac Pulse Cine	CardiacSmall	56i	0.35	0.625 [35]	40	120	270	40	Std SS Seg	40	12.69	133.24	104.375	32	Prospectively gated Cardiac Cine scan for evaluation of coronary arteries, intended for stable heart rates 30-65 bpm
25.16	SnapShot Segment 30-74 BPM		Helical	LargeBody	1.375:1 55.0	0.4	5	40	120	60	0	Std Plus	0	1.38	13.55	50	32	Localizer scan for Retrospectively gated Cardiac Helical scan for evaluation of coronary arteries, intended for heart rates 30-74 bpm
		MIROI	Axial	LargeBody	1i	1	5 [0]	5	120	60	0	Std	0	98.59	49.29	0	32	Timing bolus scan for Retrospectively gated Cardiac Helical scan for evaluation of coronary arteries, intended for heart rates 30-74 bpm
			Cardiac SEG	CardiacLarge	0.160:1 6.4	0.35	0.625	40	120	ECG Mod 120-270mA Phase 70-80 Avg mA 270	40	Std SS Seg	40	33.25	448.83	100	32	Retrospectively gated Cardiac Helical scan for evaluation of coronary arteries, intended for heart rates 30-74 bpm
25.17	SnapShot Burst 75-113 BPM		Helical	LargeBody	1.375:1 55.0	0.4	5	40	120	60	0	Std Plus	0	1.38	13.55	50	32	Localizer scan for Retrospectively gated Cardiac Helical scan for evaluation of coronary arteries, intended for heart rates 75-113 bpm
		MIROI	Axial	LargeBody	1i	1	5 [0]	5	120	60	0	Std	0	98.59	49.29	0	32	Timing bolus scan for Retrospectively gated Cardiac Helical scan for evaluation of coronary arteries, intended for heart rates 75-113 bpm
			Cardiac SSB	CardiacLarge	0.160:1 6.4	0.35	0.625	40	120	ECG Mod 120-270mA Phase 70-80 Avg mA 270	40	Std SSB	40	33.25	448.83	100	32	Retrospectively gated Cardiac Helical scan for evaluation of coronary arteries, intended for heart rates 75-113 bpm
25.18	SnapShot Burst Plus 114+ BPM		Helical	LargeBody	1.375:1 55.0	0.4	5	40	120	60	0	Std Plus	0	1.38	13.55	50	32	Localizer scan for Retrospectively gated Cardiac Helical scan for evaluation of coronary arteries, intended for heart rates over 113 bpm

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy-cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
		MIROI	Axial	LargeBody	1i	1	5 [0]	5	120	60	0	Std	0	98.59	49.29	0	32	Timing bolus scan for Retrospectively gated Cardiac Helical scan for evaluation of coronary arteries, intended for heart rates over 113 bpm
			Cardiac SSBP	CardiacLarge	0.160:1 6.4	0.35	0.625	40	120	ECG Mod 120-270mA Phase 70-80 Avg mA 270	40	Std SSB	40	33.25	448.83	100	32	Retrospectively gated Cardiac Helical scan for evaluation of coronary arteries, intended for heart rates over 113 bpm
25.19	Advantage 4D		Helical	LargeBody	1.375:1 55.0	0.6	5	40	120	170	40	Std Plus	40	5.88	145.59	200	32	Localizer scan for Advantage 4D series
		Advantage 4D	Cine	LargeBody	8i	1	2.5	20	120	70	30	Std	30	35.67	570.67	157.5	32	Cine scan mode for acquisition data for Advantage 4D for retrospective gating
25.20	VHS Chest 4D CTA SmartPrep		Helical-S	LargeBody	1.375:1 55.0 11 pass	0.4	5 [10]	40	120	SmartmA 150-500mA NI=14 Avg mA 210	40	Std Plus	40	67.29	2119.57	300	32	Volume Helical scan mode with SmartPrep for time resolved CT angiography of chest vasculature

Abdomen

Table 2-6

[Interval]* - For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
26.1	Abdomen Pelvis 0.5s. 5mm SmartmA		Helical	LargeBody	1.375:1 55.0	0.5	5	40	120	SmartmA 50-650mA NI=11.57 Avg mA 250	50	Std Plus	50	7.2	322.54	400	32	Basic Abdomen Pelvis (0.5sec) with Smart mA, Evaluation for abnormalities of abdominal and pelvic structures
26.2	Abdomen Pelvis 0.6s. 5mm + DMPR SmartmA/ SmartPrep	DMPR	Helical	LargeBody	0.984:1 39.375	0.6	0.625	40	120	SmartmA 80-750mA NI=18 Avg mA 260	50	Std Plus	50	12.56	543.37	400	32	Basic Abdomen Pelvis (0.6sec) with Smart mA, SmartPrep and DMPR, Evaluation for abnormalities of abdominal and pelvic structures
26.4	Renal Stone 0.5s. 5mm		Helical	LargeBody	1.375:1 55.0	0.5	5	40	120	190	50	Std Plus	50	5.47	245.13	400	32	Helical scan, Evaluation of kidney, ureters and bladder for renal stones
26.5	Renal Stone 0.5s. 5mm SmartmA		Helical	LargeBody	1.375:1 55.0	0.5	5	40	120	SmartmA 50-650mA NI=12 Avg mA 190	50	Std Plus	50	5.47	245.13	400	32	Helical scan with SmartmA, Evaluation of kidney, ureters and bladder for renal stones
26.7	AAA 0.5s. + DMPR 0.625mm	MIROI	Axial	LargeBody	1i	1	5 [0]	5	120	80	0	Std	0	112.67	56.34	0	32	Timing bolus for Helical scan with DMPR, Evaluation for abdominal aortic aneurysm
		DMPR	Helical	LargeBody	1.375:1 55.0	0.5	0.625	40	140	340	50	Std Plus	50	13.88	496.35	309.375	32	Helical scan with DMPR, Evaluation for abdominal aortic aneurysm
26.8	AAA 0.6s. + DMPR 0.625mm SmartmA/ SmartPrep	DMPR	Helical	LargeBody	1.375:1 55.0	0.6	0.625	40	120	SmartmA 100-700mA NI=19 Avg mA 275	50	Std Plus	50	9.51	339.51	309.375	32	Helical scan with DMPR, Smart mA and SmartPrep, Evaluation for abdominal aortic aneurysm
26.10	Dual Phase Liver		Helical	LargeBody	1.375:1 55.0	0.6	5	40	120	150	50	Std Plus	50	5.19	92.16	130	32	Non contrast evaluation of upper abdominal structures including liver
		MIROI	Axial	LargeBody	1i	1	5 [0]	5	120	80	0	Std	0	112.67	56.34	0	32	Timing bolus for Multi-Group acquisition for early arterial and portal venous phase of the liver
	Group1		Helical	LargeBody	1.375:1 55.0	0.5	2.5	40	120	275	50	Std Plus	50	7.92	140.87	130	32	Acquisition for early arterial phase of the liver
	Group2		Helical	LargeBody	1.375:1 55.0	0.5	5	40	120	210	50	Std Plus	50	6.05	270.94	400	32	Acquisition for portal venous phase of the liver

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy-cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
26.11	Tri Phase Liver		Helical	LargeBody	1.375:1 55.0	0.6	5	40	120	150	50	Std Plus	50	5.19	92.16	130	32	Non contrast evaluation of upper abdominal structures including liver
		MIROI	Axial	LargeBody	1i	1	5 [0]	5	120	80	0	Std	0	112.67	56.34	0	32	Timing bolus for Multi-Group acquisition for early arterial, late arterial and portal venous phase of the liver
	Group1		Helical	LargeBody	1.375:1 55.0	0.5	2.5	40	120	275	50	Std Plus	50	7.92	140.87	130	32	Acquisition for early arterial phase of the liver
	Group2		Helical	LargeBody	1.375:1 55.0	0.5	5	40	120	225	50	Std Plus	50	6.48	290.29	400	32	Acquisition for late arterial phase of the liver
	Group3		Helical	LargeBody	1.375:1 55.0	0.5	5	40	120	225	50	Std Plus	50	6.48	290.29	400	32	Acquisition for portal venous phase of the liver
26.12	Dual Phase Pancreas SmartmA/ SmartPrep		Helical	LargeBody	0.984:1 39.375	0.5	5	40	120	SmartmA 80-650mA NI=12 Avg mA 200	50	Std Plus	50	8.05	82.74	70	32	Non contrast evaluation of upper abdominal structures including pancreas
	Group1		Helical	LargeBody	0.984:1 39.375	0.5	2.5	40	120	SmartmA 50-700mA NI=15 Avg mA 260	50	Std Plus	50	10.46	191.28	150	32	Acquisition with SmartmA and SmartPrep for arterial phase of the pancreas
	Group2		Helical	LargeBody	0.984:1 39.375	0.5	5	40	120	SmartmA 50-700mA NI=15 Avg mA 225	50	Std Plus	50	9.06	319.47	320	32	Acquisition for venous phase of the pancreas
26.13	Dual Phase Liver SmartmA/ SmartPrep		Helical	LargeBody	1.375:1 55.0	0.5	5	40	120	SmartmA 80-500mA NI=12 Avg mA 160	50	Std Plus	50	4.61	81.96	130	32	Non contrast evaluation of upper abdominal structures including liver
	Group1		Helical	LargeBody	1.375:1 55.0	0.5	2.5	40	120	SmartmA 50-700mA NI=15 Avg mA 275	50	Std Plus	50	7.92	140.87	130	32	Acquisition with SmartmA and SmartPrep for early arterial phase of the liver
	Group2		Helical	LargeBody	0.984:1 39.375	0.5	5	40	120	SmartmA 50-700mA NI=11 Avg mA 225	50	Std Plus	50	9.06	391.91	400	32	Acquisition with SmartmA and SmartPrep for portal venous phase of the liver
26.14	Tri Phase Liver SmartmA/ SmartPrep		Helical	LargeBody	0.984:1 39.375	0.5	5	40	120	SmartmA 80-500mA NI=12 Avg mA 200	50	Std Plus	50	8.05	131.04	130	32	Non contrast evaluation of upper abdominal structures including liver
	Group1		Helical	LargeBody	1.375:1 55.0	0.5	2.5	40	120	SmartmA 50-700mA NI=15 Avg mA 275	50	Std Plus	50	7.92	140.87	130	32	Acquisition with SmartmA and SmartPrep for early arterial phase of the liver
	Group2		Helical	LargeBody	1.375:1 55.0	0.5	5	40	120	SmartmA 50-700mA NI=11 Avg mA 225	50	Std Plus	50	6.48	290.29	400	32	Acquisition with SmartmA and SmartPrep for late arterial phase of the liver
	Group3		Helical	LargeBody	1.375:1 55.0	0.5	5	40	120	SmartmA 50-700mA NI=11 Avg mA 225	50	Std Plus	50	6.48	290.29	400	32	Acquisition with SmartmA and SmartPrep for portal venous phase of the liver
26.15	VHS Liver Perfusion		Helical	LargeBody	0.984:1 39.375	0.6	5	40	120	85	50	Std	50	4.11	66.80	130	32	Non contrast evaluation of upper abdominal structures including liver

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
		Perfusion	Helical-S/P	LargeBody	1.375:1 55.0 21 pass	0.4	5	40	120	150	50	Std Plus	50	109.99	1704.89	140	32	Volume Helical scan mode for evaluation of liver tumor perfusion over time
26.16	VHS 4D CTA		Helical	LargeBody	0.984:1 39.375	0.6	5	40	120	170	50	Std	50	8.21	133.61	130	32	Localizer scan for Volume Helical Shuttle scan mode for evaluation of abdominal vasculature
		MIROI	Axial	LargeBody	1i	1	5 [0]	5	120	80	0	Std	0	112.67	56.34	0	32	Timing bolus scan for Volume Helical Shuttle scan mode for evaluation of abdominal vasculature
			Helical-S	LargeBody	1.375:1 55.0 18 pass	0.5	5 [10]	40	140	SmartmA 150-650mA NI=12 Avg mA 175	50	Std Plus	50	162.84	3175.35	180	32	Volume Helical Shuttle scan mode for time resolved CT angiography of abdominal vasculature
26.17	Runoff 0.625mm	MIROI	Axial	LargeBody	1i	1	5 [0]	5	120	80	0	Std	0	112.67	56.34	0	32	Timing bolus scan for evaluation of vascular structure of abdomen, femurs and lower extremities (Runoff)
			Helical	LargeBody	1.375:1 55.0	0.5	0.625	40	140	410	40	Std Plus	40	16.74	1648.79	936.875	32	Helical scanmode 0.625mm, Evaluation of vascular structure of abdomen, femurs and lower extremities (Runoff)
26.18	Runoff 1.25mm	MIROI	Axial	LargeBody	1i	1	5 [0]	5	120	80	0	Std	0	112.67	56.34	0	32	Timing bolus scan for evaluation of vascular structure of abdomen, femurs and lower extremities (Runoff)
			Helical	LargeBody	1.375:1 55.0	0.5	1.25 [0.7]	40	140	350	40	Std Plus	40	14.29	1496.13	998.9	32	Helical scanmode 1.25mm, Evaluation of vascular structure of abdomen, femurs and lower extremities (Runoff)
26.19	Runoff 0.625mm SmartmA/ SmartPrep		Helical	LargeBody	1.375:1 55.0	0.5	0.625	40	140	SmartmA 100-700mA NI=19 Avg mA 330	40	Std Plus	40	13.47	1326.48	936.875	32	Helical scanmode 0.625mm with SmartmA and SmartPrep, Evaluation of vascular structure of abdomen, femurs and lower extremities (Runoff)
26.20	Runoff 1.25mm SmartmA/ SmartPrep		Helical	LargeBody	1.375:1 55.0	0.5	1.25 [0.7]	40	140	SmartmA 100-700mA NI=20 Avg mA 360	40	Std Plus	40	14.70	1538.87	998.9	32	Helical scanmode 1.25mm with SmartmA and SmartPrep, Evaluation of vascular structure of abdomen, femurs and lower extremities (Runoff)
26.22	CT Colonography 1.25mm	Advantage CTC Pro	Helical	LargeBody	1.375:1 55.0	0.5	1.25	40	120	100	50	Std Plus	50	2.88	129.02	400	32	Helical scan mode supine acquisition for using Advantage CTC Pro for evaluation of the colon
		Advantage CTC Pro	Helical	LargeBody	1.375:1 55.0	0.5	1.25	40	120	100	50	Std Plus	50	2.88	129.02	400	32	Helical scan mode prone acquisition for using Advantage CTC Pro for evaluation of the colon

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy-cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
26.41	Body Interventional 5mm/3i		Helical	LargeBody	1.375:1 27.50	0.5	5	20	120	SmartmA 50-650mA NI=11.57 Avg mA 360	0	Std	0	11.41	251.58	195	32	Localization scan for CT interventional scan
			SmartView SEG	LargeBody	3i	0.4	5	10	120	80	0	Std Segment	0	684.58	684.58	0	32	CT Fluoro scan for interventional scan
26.42	Body Interventional 5mm/1i		Helical	LargeBody	1.375:1 27.50	0.5	5	20	120	SmartmA 50-650mA NI=11.57 Avg mA 360	0	Std	0	11.41	251.58	195	32	Localization scan for CT interventional scan
			SmartView SEG	LargeBody	1i	0.4	5	5	120	80	0	Std Segment	0	845.04	422.52	0	32	CT Fluoro scan for interventional scan

L-Spine

Table 2-7

[Interval]* - For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy-cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
27.1	L-Spine 3 Level Axial Group1		Axial	LargeBody	8i	1	2.5	20	120	225	30	Std	40	19.08	38.15	17.5	32	Axial scan mode, Evaluation of soft tissue and boney structures lumbar spine
	Group2		Axial	LargeBody	8i	1	2.5	20	120	225	30	Std	40	19.08	38.15	17.5	32	Axial scan mode, Evaluation of soft tissue and boney structures lumbar spine
	Group3		Axial	LargeBody	8i	1	2.5	20	140	250	30	Std	40	30.03	60.06	17.5	32	Axial scan mode, Evaluation of soft tissue and boney structures lumbar spine
27.3	L-Spine Survey Helical 2.5mm 0.6s. SmartmA		Helical	LargeBody	0.516:1 20.625	0.6	2.5	40	140	SmartmA 100-700mA NI=14 Avg mA 380	30	Std Plus	40	49.64	1040.25	175	32	Helical scan mode 2.5mm (0.6sec), Limited evaluation of lumbar spine
27.4	L-Spine Survey Helical 2.5mm 0.8s. SmartmA		Helical	LargeBody	0.516:1 20.625	0.8	2.5	40	140	SmartmA 100-700mA NI=14 Avg mA 365	30	Std Plus	40	63.57	1332.02	175	32	Helical scan mode 2.5mm (0.8sec), Limited evaluation of lumbar spine
27.6	L-Spine Helical 0.625mm + DMPR/ SmartmA	DMPR	Helical	LargeBody	0.516:1 20.625	0.6	0.625	40	140	SmartmA 100-715mA NI=20 Avg mA 485	30	Std Plus IQE	40	63.36	1327.68	175	32	Helical scan mode 0.625mm with DMPR and SmartmA, Evaluation of soft tissue and boney structures lumbar spine
27.7	L-Spine Helical 1.25mm + DMPR/ SmartmA - Large Patient	DMPR	Helical	LargeBody	0.516:1 20.625	1	1.25	40	140	SmartmA 100-715mA NI=18 Avg mA 490	30	Std Plus	40	106.68	2234.87	175	32	Helical scan mode 1.25mm with DMPR and SmartmA, Evaluation of soft tissue and boney structures lumbar spine

Pelvis

Table 2-8
[Interval]* - For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.
For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
28.1	Pelvis for Fracture + DMPR/ SmartmA	DMPR	Helical	LargeBody	0.984:1 39.375	0.8	1.25 [0.625]	40	120	SmartmA 100-700mA NI=20 Avg mA 300	40	Bone Plus IQE	40	19.32	258.70	101.25	32	Helical scan with DMPR and SmartmA, Evaluation of pelvis bony structures for fracture

Lower Extremity

Table 2-9

[Interval]* - For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
29.1	Lower Extremity Survey 2.5mm		Helical	LargeBody	0.516:1 20.625	0.5	2.5	40	120	175	30	Std Plus	30	13.45	126.94	60	32	Helical scan mode 2.5mm, Limited evaluation of lower extremity for abnormalities
29.2	Ankle 0.625 mm Single Joint		Helical	MediumBody	0.531:1 10.625	0.5	0.625	20	120	175	30	Std Plus IQE	40	14.50	75.64	35	32	Helical scan mode 0.625mm, Evaluation of soft tissue and bony anatomy of the ankle area
29.3	Ankle 0.625 mm Both Joints		Helical	LargeBody	0.531:1 10.625	0.5	0.625	20	120	210	30	Std Plus IQE	30	16.76	87.38	35	32	Helical scan mode 0.625mm, Evaluation of soft tissue and bony anatomy of the ankle area
29.4	Knee 0.625 mm Both Joints		Helical	LargeBody	0.531:1 10.625	0.6	0.625	20	120	AutomA 100-335mA NI=20 Avg mA 190	30	Std Plus IQE	30	18.19	94.84	35	32	Helical scan mode 0.625mm, Evaluation of soft tissue and bony anatomy of both knees

**Reference Pediatric Protocols -
Installed Options : ASiR, Sub 0.4
sec, VCT85kW and VCT Hi Power**

Head

Table 3-1

[Interval]* - For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy-cm) ASiR	Scan Length (mm)	Phantom (cm)	Description
31.1	PED HEAD TO 18 Months		Axial	Ped Head	4i	0.5	5	20	120	170	30	Std	30	14.47	173.68	115	16	Routine head (0.5s) for infant (up to 18 months) 5mm
31.2	PED HEAD 18 MOS TO 5YRS		Axial	SmallHead	4i	0.5	5	20	120	235	30	Std	30	20.01	240.08	115	16	Routine head (0.5s) for children (18 months to 5 years) 5mm
31.3	PED HEAD TO 5YRS TO 18YRS Group1		Axial	SmallHead	8i	1	2.5	20	120	170	30	Std	30	28.95	115.79	37.5	16	Routine head (1.0s) for children (5 years to 18 years) 2.5mm
	Group2		Axial	SmallHead	4i	1	5	20	120	125	30	Std	30	21.28	170.27	75	16	Routine head (1.0s) for children (5 years to 18 years) 5mm

Orbit

Table 3-2

[Interval]* - For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy-cm) ASiR	Scan Length (mm)	Phantom (cm)	Description
32.1	Sinus Supine Helical + DMPR	DMPR	Helical	SmallHead	0.531:1 10.625	0.4	0.625	20	120	65	20	Bone Plus IQE	30	8.33	76.29	74.375	16	Routine Helical scan mode with DMPR, Evaluation of sinus area for abnormality in pediatric patients
32.3	Orbits Helical	DMPR	Helical	SmallHead	0.531:1 10.625	0.4	0.625	20	120	145	20	Std Plus IQE	30	18.59	146.96	61.875	16	Helical scan mode, Evaluation of orbital and face structures for abnormality in pediatric patients
32.6	Helical IAC 0.625mm Axial mode - Use Reformat for Coronal		Helical	SmallHead	0.531:1 10.625	0.4	0.625	20	120	130	20	Std Plus IQE	30	16.67	93.21	38.75	16	Helical scan mode, Evaluation of inner ear structures for abnormality in pediatric patients

Chest

Table 3-3

[Interval]* - For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy-cm) ASiR	Scan Length (mm)	Phantom (cm)	Description
35.1.1	RC 6.0-7.5 kg (13.2-16.5 lbs)		Helical	PedBody	1.375:1 27.5	0.4	3.75	20	100	55	30	Std Plus	30	0.76	7.62	75	32	Weight and height based routine chest protocol
35.1.3	SnapShot Segment 30-74 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	85	30	Std Plus	30	0.59	5.80	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	Std	0	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
			Cardiac SEG	CardiacSmall	0.160:1 6.4	0.35	0.625	40	80	ECG Mod 55-195mA Phase 40-80 AvgmA195	30	Std SS Seg	40	8.51	93.56	75	32	Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
35.1.4	SnapShot Burst 75-104 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	85	30	Std Plus	30	0.59	5.80	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	Std	0	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
			Cardiac SSB	CardiacSmall	0.160:1 6.4	0.35	0.625	40	80	ECG Mod 55-195mA Phase 40-80 AvgmA195	30	Std SSB	40	8.51	93.56	75	32	Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
35.1.5	SnapShot Burst+ >105 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	85	30	Std Plus	30	0.59	5.80	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	Std	0	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
			Cardiac SSBP	CardiacSmall	0.160:1 6.4	0.35	0.625	40	80	ECG Mod 55-195mA Phase 35-85 AvgmA195	30	Std SSB	40	9.13	100.42	75	32	Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
35.1.7	Non ECG Gated		Helical	PedBody	0.984:1 39.375	0.4	0.625	40	80	SmartmA 50-200mA NI=18 AvgmA140	30	Std Plus	30	1.36	14.70	75	32	Weight and height based non ECG gated protocol for cardiac with SmartmA
35.2.1	RC 7.5-9.5 kg (16.5-20.9lbs)		Helical	PedBody	1.375:1 27.5	0.4	3.75	20	100	85	30	Std Plus	30	1.18	11.77	75	32	Weight and height based routine chest protocol
35.2.3	SnapShot Segment 30-74 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	85	30	Std Plus	30	0.59	5.80	50	32	Weight and height based protocol for localization

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy-cm) ASiR	Scan Length (mm)	Phantom (cm)	Description
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	Std	0	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
			Cardiac SEG	CardiacSmall	0.160:1 6.4	0.35	0.625	40	80	ECG Mod 55-210mA Phase 40-80 AvgmA210	30	Std SS Seg	40	9.13	100.39	75	32	Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
35.2.4	SnapShot Burst 75-104 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	85	30	Std Plus	30	0.59	5.80	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	Std	0	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
			Cardiac SSB	CardiacSmall	0.160:1 6.4	0.35	0.625	40	80	ECG Mod 55-210mA Phase 40-80 AvgmA210	30	Std SSB	40	9.13	100.39	75	32	Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
35.2.5	SnapShot Burst+ >105 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	85	30	Std Plus	30	0.59	5.80	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	Std	0	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
			Cardiac SSBP	CardiacSmall	0.160:1 6.4	0.35	0.625	40	80	ECG Mod 55-210mA Phase 35-85 AvgmA210	30	Std SSB	40	9.82	107.99	75	32	Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
35.2.7	Non ECG Gated		Helical	PedBody	0.984:1 39.375	0.4	0.625	40	80	SmartmA 50-220mA NI=18 AvgmA155	30	Std Plus	30	1.51	16.28	75	32	Weight and height based non ECG gated protocol for cardiac with SmartmA
35.3.1	RC 9.5-11.5 kg (20.9-25.4lbs)		Helical	SmallBody	1.375:1 27.5	0.4	3.75	20	120	70	30	Std Plus	30	1.52	15.77	78.75	32	Weight and height based routine chest protocol
35.3.3	SnapShot Segment 30-74 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	85	30	Std Plus	30	0.59	5.80	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	Std	0	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
			Cardiac SEG	CardiacSmall	0.160:1 6.4	0.35	0.625	40	80	ECG Mod 70-225mA Phase 40-80 AvgmA225	30	Std SS Seg	40	9.88	108.64	75	32	Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
35.3.4	SnapShot Burst 75-104 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	85	30	Std Plus	30	0.59	5.80	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	Std	0	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy-cm) ASiR	DLP (mGy-cm) ASiR	Scan Length (mm)	Phantom (cm)	Description
			Cardiac SSB	CardiacSmall	0.160:1 6.4	0.35	0.625	40	80	ECG Mod 70-225mA Phase 40-80 AvgmA225	30	Std SSB	40	9.88	108.64	75	32	Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
35.3.5	SnapShot Burst+ >105 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	85	30	Std Plus	30	0.59	5.80	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	Std	0	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
			Cardiac SSBP	CardiacSmall	0.160:1 6.4	0.35	0.625	40	80	ECG Mod 70-225mA Phase 35-85 AvgmA225	30	Std SSB	40	10.57	116.23	75	32	Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
35.3.7	Non ECG Gated		Helical	PedBody	0.984:1 39.375	0.4	0.625	40	80	SmartmA 50-240mA NI=18 AvgmA170	30	Std Plus	30	1.65	17.85	75	32	Weight and height based non ECG gated protocol for cardiac with SmartmA
35.4.1	RC 11.5-14.5kg (25.4-32.0lbs)		Helical	SmallBody	1.375:1 27.5	0.4	5	20	120	65	30	Std Plus	30	1.41	16.23	90	32	Weight and height based routine chest protocol
35.4.3	SnapShot Segment 30-74 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	85	30	Std Plus	30	0.59	5.80	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	Std	0	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
			Cardiac SEG	CardiacSmall	0.160:1 6.4	0.35	0.625	40	80	ECG Mod 70-240mA Phase 40-80 AvgmA240	30	Std SS Seg	40	10.50	115.46	75	32	Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
35.4.4	SnapShot Burst 75-104 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	85	30	Std Plus	30	0.59	5.80	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	Std	0	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
			Cardiac SSB	CardiacSmall	0.160:1 6.4	0.35	0.625	40	80	ECG Mod 70-240mA Phase 40-80 AvgmA240	30	Std SSB	40	10.50	115.46	75	32	Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
35.4.5	SnapShot Burst+ >105 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	85	30	Std Plus	30	0.59	5.80	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	Std	0	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
			Cardiac SSBP	CardiacSmall	0.160:1 6.4	0.35	0.625	40	80	ECG Mod 70-240mA Phase 35-85 AvgmA240	30	Std SSB	40	11.25	123.79	75	32	Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy-cm) ASiR	Scan Length (mm)	Phantom (cm)	Description
35.4.7	Non ECG Gated		Helical	PedBody	0.984:1 39.375	0.4	0.625	40	80	SmartmA 50-280mA NI=18 AvgmA195	30	Std Plus	30	1.90	20.48	75	32	Weight and height based non ECG gated protocol for cardiac with SmartmA
35.5.1	RC 14.5-18.5kg (32.0-40.8lbs)		Helical	SmallBody	1.375:1 27.5	0.4	5	20	120	70	30	Std Plus	30	1.52	17.48	90	32	Weight and height based routine chest protocol
35.5.3	SnapShot Segment 30-74 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	85	30	Std Plus	30	0.59	5.80	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	Std	0	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
			Cardiac SEG	CardiacSmall	0.160:1 6.4	0.35	0.625	40	80	ECG Mod 70-265mA Phase 40-80 AvgmA265	30	Std SS Seg	40	11.53	126.82	75	32	Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
35.5.4	SnapShot Burst 75-104 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	85	30	Std Plus	30	0.59	5.80	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	Std	0	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
			Cardiac SSB	CardiacSmall	0.160:1 6.4	0.35	0.625	40	80	ECG Mod 70-265mA Phase 40-80 AvgmA265	30	Std SSB	40	11.53	126.82	75	32	Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
35.5.5	SnapShot Burst+ >105 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	85	30	Std Plus	30	0.59	5.80	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	Std	0	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
			Cardiac SSBP	CardiacSmall	0.160:1 6.4	0.35	0.625	40	80	ECG Mod 70-265mA Phase 35-85 AvgmA265	30	Std SSB	40	12.40	136.37	75	32	Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
35.5.7	Non ECG Gated		Helical	PedBody	0.984:1 39.375	0.4	0.625	40	80	SmartmA 50-300mA NI=18 AvgmA210	30	Std Plus	30	2.04	22.05	75	32	Weight and height based non ECG gated protocol for cardiac with SmartmA
35.6.1	RC 18.5-22.5kg (40.8-49.6lbs)		Helical	SmallBody	1.375:1 27.5	0.4	5	20	120	75	30	Std Plus	30	1.63	18.73	90	32	Weight and height based routine chest protocol
35.6.3	SnapShot Segment 30-74 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	85	30	Std Plus	30	0.59	5.80	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	Std	0	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy-cm) ASiR	Scan Length (mm)	Phantom (cm)	Description
			Cardiac SEG	CardiacSmall	0.160:1 6.4	0.35	0.625	40	80	ECG Mod 70-280mA Phase 40-80 AvgmA280	30	Std SS Seg	40	12.16	133.72	75	32	Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
35.6.4	SnapShot Burst 75-104 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	85	30	Std Plus	30	0.59	5.80	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	Std	0	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
			Cardiac SSB	CardiacSmall	0.160:1 6.4	0.35	0.625	40	80	ECG Mod 70-280mA Phase 40-80 AvgmA280	30	Std SSB	40	12.16	133.72	75	32	Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
35.6.5	SnapShot Burst+ >105 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	85	30	Std Plus	30	0.59	5.80	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	Std	0	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
			Cardiac SSBP	CardiacSmall	0.160:1 6.4	0.35	0.625	40	80	ECG Mod 70-280mA Phase 35-85 AvgmA280	30	Std SSB	40	13.09	144.01	75	32	Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
35.6.7	Non ECG Gated		Helical	PedBody	0.984:1 39.375	0.4	0.625	40	80	SmartmA 100-320mA NI=18 AvgmA225	30	Std Plus	30	2.19	23.63	75	32	Weight and height based non ECG gated protocol for cardiac with SmartmA
35.7.1	RC 22.5-31.5kg (49.6-69.5lbs)		Helical	LargeBody	1.375:1 55.0	0.4	5	40	120	75	30	Std Plus	30	1.73	22.98	85	32	Weight and height based routine chest protocol
35.7.3	SnapShot Segment 30-74 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	85	30	Std Plus	30	0.59	5.80	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	Std	0	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
			Cardiac SEG	CardiacSmall	0.160:1 6.4	0.35	0.625	40	80	ECG Mod 75-295mA Phase 40-80 AvgmA295	30	Std SS Seg	40	12.81	140.96	75	32	Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
35.7.4	SnapShot Burst 75-104 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	85	30	Std Plus	30	0.59	5.80	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	Std	0	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
			Cardiac SSB	CardiacSmall	0.160:1 6.4	0.35	0.625	40	80	ECG Mod 75-295mA Phase 40-80 AvgmA295	30	Std SSB	40	12.81	140.96	75	32	Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy-cm) ASiR	Scan Length (mm)	Phantom (cm)	Description
35.7.5	SnapShot Burst+ >105 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	85	30	Std Plus	30	0.59	5.80	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	Std	0	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
			Cardiac SSBP	CardiacSmall	0.160:1 6.4	0.35	0.625	40	80	ECG Mod 75-295mA Phase 35-85 AvgmA295	30	Std SSB	40	13.79	151.74	75	32	Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
35.7.7	Non ECG Gated		Helical	PedBody	0.984:1 39.375	0.4	0.625	40	80	SmartmA 100-360mA NI=18 AvgmA250	30	Std Plus	30	2.43	26.25	75	32	Weight and height based non ECG gated protocol for cardiac with SmartmA
35.8.1	RC 31.5-40.5kg (69.5-89.3lbs)		Helical	LargeBody	1.375:1 55.0	0.4	5	40	120	85	30	Std Plus	30	1.96	32.90	120	32	Weight and height based routine chest protocol
35.8.3	SnapShot Segment 30-74 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	100	85	30	Std Plus	30	1.10	10.76	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	100	50	0	Std	0	39.50	19.75	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
			Cardiac SEG	CardiacSmall	0.160:1 6.4	0.35	0.625	40	100	ECG Mod 70-265mA Phase 40-80 AvgmA265	30	Std SS Seg	30	21.40	235.37	75	32	Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
35.8.4	SnapShot Burst 75-104 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	100	85	30	Std Plus	30	1.10	10.76	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	100	50	0	Std	0	39.50	19.75	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
			Cardiac SSB	CardiacSmall	0.160:1 6.4	0.35	0.625	40	100	ECG Mod 70-265mA Phase 40-80 AvgmA265	30	Std SSB	30	21.40	235.37	75	32	Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
35.8.5	SnapShot Burst+ >105 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	100	85	30	Std Plus	30	1.10	10.76	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	100	50	0	Std	0	39.50	19.75	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
			Cardiac SSBP	CardiacSmall	0.160:1 6.4	0.35	0.625	40	100	ECG Mod 70-265mA Phase 35-85 AvgmA265	30	Std SSB	40	23.01	253.11	75	32	Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
35.8.7	Non ECG Gated		Helical	SmallBody	0.984:1 39.375	0.4	0.625	40	100	SmartmA 100-400mA NI=16 AvgmA280	30	Std Plus	30	5.06	54.57	75	32	Weight and height based non ECG gated protocol for cardiac with SmartmA

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy-cm) ASiR	Scan Length (mm)	Phantom (cm)	Description
35.9.1	RC 40.5-55.0kg(89.3-121.3lbs)		Helical	LargeBody	1.375:1 55.0	0.4	5	40	120	90	30	Std Plus	30	2.07	41.06	150	32	Weight and height based routine chest protocol
35.9.3	SnapShot Segment 30-74 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	100	85	30	Std Plus	30	1.10	10.76	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	100	50	0	Std	0	39.50	19.75	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
			Cardiac SEG	CardiacSmall	0.160:1 6.4	0.35	0.625	40	100	ECG Mod 70-295mA Phase 40-80 AvgmA295	30	Std SS Seg	30	23.71	260.86	75	32	Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
35.9.4	SnapShot Burst 75-104 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	100	85	30	Std Plus	30	1.10	10.76	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	100	50	0	Std	0	39.50	19.75	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
			Cardiac SSB	CardiacSmall	0.160:1 6.4	0.35	0.625	40	100	ECG Mod 70-295mA Phase 40-80 AvgmA295	30	Std SSB	30	23.71	260.86	75	32	Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
35.9.5	SnapShot Burst+ >105 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	100	85	30	Std Plus	30	1.10	10.76	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	100	50	0	Std	0	39.50	19.75	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
			Cardiac SSBP	CardiacSmall	0.160:1 6.4	0.35	0.625	40	100	ECG Mod 70-295mA Phase 35-85 AvgmA295	30	Std SSB	40	25.57	281.32	75	32	Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
35.9.7	Non ECG Gated		Helical	SmallBody	0.984:1 39.375	0.4	0.625	40	100	SmartmA 100-440mA NI=16 AvgmA310	30	Std Plus	30	5.60	60.42	75	32	Weight and height based non ECG gated protocol for cardiac with SmartmA

Abdomen

Table 3-4

[Interval]* - For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy-cm) ASiR	Scan Length (mm)	Phantom (cm)	Description
36.1.1	Abd 6.0-7.5kg (13.2-16.5lbs)		Helical	PedBody	1.375:1 27.5	0.4	3.75	20	120	70	30	Std Plus	30	1.52	15.77	78.75	32	Weight and height based routine abdomen protocol
36.2.1	Abd 7.5-9.5kg (16.5-20.9lbs)		Helical	PedBody	1.375:1 27.5	0.4	3.75	20	120	110	30	Std Plus	30	2.39	27.47	90	32	Weight and height based routine abdomen protocol
36.3.1	Abd 9.5-11.5kg (20.9-25.4lbs)		Helical	PedBody	1.375:1 27.5	0.4	3.75	20	120	125	30	Std Plus	30	2.71	34.26	101.25	32	Weight and height based routine abdomen protocol
36.4.1	Abd 11.5-14.5kg (25.4-32.0lbs)		Helical	SmallBody	1.375:1 27.5	0.4	5	20	120	85	30	Std Plus	30	1.84	24.91	110	32	Weight and height based routine abdomen protocol
36.5.1	Abd 14.5-18.5kg (32.0-40.8lbs)		Helical	SmallBody	1.375:1 27.5	0.4	5	20	120	95	30	Std Plus	30	2.06	27.84	110	32	Weight and height based routine abdomen protocol
36.6.1	Abd 18.5-22.5kg (40.8-49.6lbs)		Helical	Medium Body	1.375:1 27.5	0.4	5	20	120	85	30	Std Plus	30	2.18	33.77	130	32	Weight and height based routine abdomen protocol
36.7.1	Abd 22.5-31.5kg (49.6-69.5lbs)		Helical	LargeBody	1.375:1 55.0	0.4	5	40	120	105	30	Std Plus	30	2.42	47.91	150	32	Weight and height based routine abdomen protocol
36.8.1	Abd 31.5-40.5kg (69.5-89.3lbs)		Helical	LargeBody	1.375:1 55.0	0.4	5	40	120	120	30	Std Plus	30	2.77	54.75	150	32	Weight and height based routine abdomen protocol
36.9.1	Abd 40.5-55.0kg (89.3-121.3lbs)		Helical	LargeBody	1.375:1 55.0	0.4	5	40	120	140	30	Std Plus	30	3.23	80.01	200	32	Weight and height based routine abdomen protocol

Miscellaneous

Table 3-5

[Interval]* - For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy-cm) ASiR	Scan Length (mm)	Phantom (cm)	Description
40.1	Pink Zone Routine 0-20 lbs/0-9 kg		Helical	PedBody	1.375:1 55.0	0.5	5	40	80	SmartmA 65-130mA NI=5 Avg mA 70	0	Std	0	0.61	8.70	95	32	Pediatric and Small patient Routine procedure based protocols
40.2	Pink Zone Routine 21-60 lbs/9.1-27.2 kg		Helical	PedBody	1.375:1 55.0	0.5	5	40	100	SmartmA 80-160mA NI=7 Avg mA 70	0	Std	0	1.13	27.47	195	32	Pediatric and Small patient Routine procedure based protocols
40.3	Pink Zone Routine 61-100 lbs/27.3-45.4 kg		Helical	Medium Body	1.375:1 55.0	0.5	5	40	120	SmartmA 95-190mA NI=10 Avg mA 70	0	Std	0	2.09	71.81	295	32	Pediatric and Small patient Routine procedure based protocols
40.4	Pink Zone Routine 101-200 lbs/45.4-90.7 kg		Helical	LargeBody	1.375:1 55.0	0.5	5	40	120	SmartmA 110-220mA NI=12 Avg mA 70	0	Std	0	2.02	89.30	395	32	Pediatric and Small patient Routine procedure based protocols
40.5	Pink Zone Routine >200 lbs/>90.8 kg		Helical	LargeBody	1.375:1 55.0	0.5	5	40	120	SmartmA 125-300mA NI=15 Avg mA 70	0	Std	0	2.02	89.30	395	32	Pediatric and Small patient Routine procedure based protocols
40.7	Green Zone Lower Dose 0-20 lbs/0-9 kg		Helical	PedBody	1.375:1 55.0	0.5	5	40	80	SmartmA 50-100mA NI=7 Avg mA 70	0	Std	0	0.61	8.70	95	32	Pediatric and Small patient Lower Dose procedure based protocols
40.8	Green Zone Lower Dose 21-60 lbs/9.1-27.2 kg		Helical	PedBody	1.375:1 55.0	0.5	5	40	100	SmartmA 60-120mA NI=9 Avg mA 70	0	Std	0	1.13	27.47	195	32	Pediatric and Small patient Lower Dose procedure based protocols
40.9	Green Zone Lower Dose 61-100 lbs/27.3-45.4 kg		Helical	Medium Body	1.375:1 55.0	0.5	5	40	120	SmartmA 70-140mA NI=11 Avg mA 70	0	Std	0	2.09	71.81	295	32	Pediatric and Small patient Lower Dose procedure based protocols
40.10	Green Zone Lower Dose 101-200 lbs/45.4-90.7 kg		Helical	LargeBody	1.375:1 55.0	0.5	5	40	120	SmartmA 80-160mA NI=13 Avg mA 70	0	Std	0	2.02	89.30	395	32	Pediatric and Small patient Lower Dose procedure based protocols
40.11	Green Zone Lower Dose >200 lbs/>90.8 kg		Helical	LargeBody	1.375:1 55.0	0.5	5	40	120	SmartmA 90-240mA NI=16 Avg mA 70	0	Std	0	2.02	89.30	395	32	Pediatric and Small patient Lower Dose procedure based protocols
40.13	Grey Zone CT Angio 0-20 lbs/0-9 kg		Helical	PedBody	1.375:1 55.0	0.5	2.5	40	80	SmartmA 100-200mA NI=5 Avg mA 140	0	Std	0	1.22	17.71	97.5	32	Pediatric and Small patient Angio procedure based protocols

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (mm)	Phantom (cm)	Description
40.14	Grey Zone CT Angio 21-60 lbs/9.1-27.2 kg		Helical	SmallBody	1.375:1 55.0	0.5	2.5	40	100	SmartmA 120-240mA NI=9.9 Avg mA 140	0	Std	0	2.26	55.50	197.5	32	Pediatric and Small patient Angio procedure based protocols
40.15	Grey Zone CT Angio 61-100 lbs/ 27.3-45.4 kg		Helical	LargeBody	1.375:1 55.0	0.5	2.5	40	120	SmartmA 120-240mA NI=10 Avg mA 140	0	Std	0	4.03	139.28	297.5	32	Pediatric and Small patient Angio procedure based protocols
40.16	Grey Zone CT Angio 101-200 lbs/ 45.4-90.7 kg		Helical	LargeBody	1.375:1 55.0	0.5	2.5	40	120	SmartmA 120-240mA NI=12 Avg mA 140	0	Std	0	4.03	179.62	397.5	32	Pediatric and Small patient Angio procedure based protocols
40.17	Grey Zone CT Angio >200 lbs/ >90.8 kg		Helical	LargeBody	1.375:1 55.0	0.5	2.5	40	120	SmartmA 120-300mA NI=15 Avg mA 140	0	Std	0	4.03	179.62	397.5	32	Pediatric and Small patient Angio procedure based protocols

Reference Adult Protocols - Installed

Options : ASiR, Sub 0.4sec and

VCT85kW

Head

Table 4-1

[Interval]* - For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (s)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (mm)	Phantom (cm)	Description
21.1	Routine Head 1s Group1		Axial	Head	4i	1	5	20	140	120	40	Std	40	32.00	192.01	55	16	Routine head protocol for evaluation of the brain for abnormalities.
	Group2		Axial	Head	4i	1	5	20	120	190	40	Std	40	36.47	291.78	75	16	
21.2	Routine Head 2s Group1		Axial	Head	4i	2	5	20	140	60	40	Std	40	32.00	192.01	55	16	Routine head protocol for evaluation of the brain for abnormalities.
	Group2		Axial	Head	4i	2	5	20	120	95	40	Std	40	36.47	291.78	75	16	
21.3	Trauma Head 2.5mm		Axial	Head	8i	0.5	2.5	20	140	335	40	Std	40	44.67	625.37	137.5	16	Emergency head protocol for evaluation of the brain and cranium for abnormalities.
21.4	Routine Head 0.5s Group1		Axial	Head	4i	0.5	5	20	140	240	40	Std	40	32.00	192.01	55	16	Routine head protocol for evaluation of the brain for abnormalities.
	Group2		Axial	Head	4i	0.5	5	20	120	360	40	Std	40	35.52	284.17	75	16	
21.5	Circle of Willis 0.4s	MIROI	Axial	Head	1i	1	5 [0]	5	120	50	0	Std	0	125.58	62.79	0	16	Test bolus for CT angiography of the head for evaluation of carotid and cerebral vasculature.
			Helical	Head	0.531:1 10.625	0.4	0.625	20	120	180	40	Std Plus IQE	50	26.02	252.82	80	16	CT angiography of the head for evaluation of carotid and cerebral vasculature.
21.6	CoW/ Carotid 0.625mm 0.4s.Smart mA	MIROI	Axial	Head	1i	1	5 [0]	5	120	80	0	Std	0	180.84	90.42	0	16	Test bolus for CT angiography of the head for evaluation of carotid and cerebral vasculature.
			Helical	Head	0.531:1 10.625	0.4	0.625	20	120	SmartmA 100-650mA NI=8 Avg mA 415	30	Std Plus IQE	40	61.66	1155.86	170	16	CT angiography of the head for evaluation of carotid and cerebral vasculature.
21.8	Helical Head	DMPR	Helical	Head	0.531:1 10.625	0.5	0.625	20	120	180	40	Std Plus IQE	40	32.52	527.31	145	16	Helical scan mode for evaluation of the brain for cerebral abnormality.
21.10	VHS Head 4D CTA	MIROI	Axial	Head	1i	1	5 [0]	5	120	50	0	Std	0	125.58	62.79	0	16	Test bolus for CT angiography of the head for evaluation of carotid and cerebral vasculature.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (s)	Slice Thickness (mm) [Interval] (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (mm)	Phantom (cm)	Description
			Helical-S	Head	0.984:1 39.375 18 pass	0.4	5 [10]	40	120	150	50	Std Plus	50	248.10	3101.19	110	16	CT angiography of the head using Volume Helical scan mode for evaluation of cerebral vasculature.
21.11	VHS Perfusion 350-370 Strength Contrast		Axial	Head	4i	1	5	20	120	200	40	Std	40	38.39	537.48	135	16	Non-Enhance Brain, evaluation for hemorrhage or infarction.
		Perfusion	Helical-S/P	Head	0.984:1 39.375 28 pass	0.4	5	40	80	200	0	Std	50	186.02	2325.25	110	16	CT Perfusion using Volume Helical scan mode for evaluation of cerebral perfusion over time.
			Axial	Head	4i	1	5	20	120	200	40	Std	40	38.39	537.48	135	16	Post Contrast Brain, Evaluation for residual tissue enhancement.
21.12	CT Perfusion 300-320 Strength Contrast		Axial	Head	4i	1	5	20	120	200	40	Std	40	38.39	537.48	135	16	Non-Enhance Brain, evaluation for hemorrhage or infarction.
		Perfusion	Cine	Head	8i	1	5 [0]	40	80	200	0	Std	0	652.82	2611.27	35	16	CT Perfusion using Cine scan mode for evaluation of cerebral perfusion over time.
			Axial	Head	4i	1	5	20	120	200	40	Std	40	38.39	537.48	135	16	Non-Enhance Brain, evaluation for hemorrhage or infarction.
21.13	CT Perfusion 350-370 Strength Contrast		Axial	Head	4i	1	5	20	120	200	40	Std	40	38.39	537.48	135	16	Non-Enhance Brain, evaluation for hemorrhage or infarction.
		Perfusion	Cine	Head	8i	1	5 [0]	40	80	200	0	Std	0	587.55	2350.19	35	16	CT Perfusion using Cine scan mode for evaluation of cerebral perfusion over time.
			Axial	Head	4i	1	5	20	120	200	40	Std	40	38.39	537.48	135	16	Post Contrast Brain, Evaluation for residual tissue enhancement.
21.14	CT Perfusion Brain Tumor - Dual Phase		Axial	Head	4i	1	5	20	120	200	40	Std	40	38.39	537.48	135	16	Non-Enhance Brain, evaluation for hemorrhage or infarction.
	Group 1	Perfusion	Cine	Head	8i	1	5 [0]	40	80	200	0	Std	0	587.55	2350.19	35	16	CT Perfusion using Cine scan mode for evaluation of cerebral perfusion over time.
	Group 2	Perfusion	Axial	Head	8i	1	5 [0]	40	80	200	0	Std	0	130.54	522.15	35	16	Delayed phase CT Perfusion using Axial scan mode for evaluation of Brain tumor permeability.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (s)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (mm)	Phantom (cm)	Description
			Axial	Head	4i	1	5	20	120	200	40	Std	40	38.39	537.48	135	16	Post Contrast Brain, Evaluation for residual tissue enhancement.
21.15	CT Perfusion 350-370 Strength Contrast (Axial)		Axial	Head	4i	1	5	20	120	200	40	Std	40	38.39	537.48	135	16	Non-Enhance Brain, evaluation for hemorrhage or infarction.
		Perfusion	Axial	Head	8i	1	5 [0]	40	80	200	0	Std	0	287.18	1148.73	35	16	CT Perfusion using Axial scan mode for evaluation of cerebral perfusion over time.
			Axial	Head	4i	1	5	20	120	200	40	Std	40	38.39	537.48	135	16	Post Contrast Brain, Evaluation for residual tissue enhancement.
21.16	CT Perfusion 350-370 Strength Contrast (Axial Shuttle Mode)		Axial	Head	4i	1	5	20	120	200	40	Std	40	38.39	537.48	135	16	Non-Enhance Brain, evaluation for hemorrhage or infarction.
		Perfusion	Axial-S	Head	8i 17 pass	0.4	5	40	80	500	0	Std	0	221.91	1775.31	75	16	CT Perfusion using Volume Shuttle (Axial) scan mode for evaluation of cerebral perfusion over time.
			Axial	Head	4i	1	5	20	120	200	40	Std	40	38.39	537.48	135	16	Post Contrast Brain, Evaluation for residual tissue enhancement.

Orbit

Table 4-2

[Interval]* - For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
22.1	Sinus Supine Helical + DMPR	DMPR	Helical	Head	0.531:1 10.625	0.4	0.625	20	120	85	30	Bone Plus IQE	50	12.29	112.48	74.375	16	Routine Helical Sinus protocol, Evaluation for any abnormality
22.3	Orbits Helical + DMPR	DMPR	Helical	Head	0.531:1 10.625	0.5	0.625	20	120	155	30	Std Plus IQE	40	28.00	221.29	61.875	16	Routine Helical Orbit protocol, Evaluation for abnormality soft tissues and bone structures of the face.
22.6	Helical IAC 0.625mm Axial mode - Use Reformat for Coronal		Helical	Head	0.531:1 10.625	0.6	0.625	20	140	190	30	Std Plus IQE	40	57.23	319.79	38.75	16	Axial scan mode, Evaluation for inner ear abnormalities in Axial plane
22.8	Axial and Coronal IAC		Axial	Head	16i	1	1.25	20	140	195	30	Std	40	52.00	208.01	38.75	16	Helical scan mode, Evaluation for inner ear abnormalities in Axial plane
			Axial	Head	16i	1	1.25	20	140	235	30	Std	40	62.67	250.68	38.75	16	Helical scan mode, Evaluation for inner ear abnormalities in Coronal plane

Neck

Table 4-3

[Interval]* - For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
23.1	C-Spine C5-C7 Axial		Axial	SmallBody	8i	0.5	2.5	20	140	490	20	Std	40	26.64	373.03	137.5	32	C-Spine C5-C7 area Axial mode, Evaluation for soft tissue, disc and bone abnormalities.
23.2	C-Spine C1-C4 Axial		Axial	SmallBody	8i	1	2.5	20	120	190	20	Std	40	14.17	113.36	77.5	32	C-Spine C1-C4 area Axial mode, Evaluation for soft tissue, disc and bone abnormalities.
23.3	C-Spine Helical + DMPR	DMPR	Helical	SmallBody	0.516:1 20.625	0.8	0.625	40	120	270	20	Bone Plus IQE	40	29.20	421.46	110	32	C-Spine Helical mode with DMPR, Evaluation for soft tissue, disc and bone abnormalities.
23.4	C-Spine Helical + DMPR/ SmartmA	DMPR	Helical	SmallBody	0.516:1 20.625	0.8	0.625	40	120	SmartmA 100-650mA NI=11.34 Avg mA 250	20	Bone Plus IQE	40	27.04	390.24	110	32	C-Spine Helical mode with DMPR and SmartmA, Evaluation for soft tissue, disc and bone abnormalities.
23.8	Carotid 0.625mm 0.4s. 0.5:1 SmartmA/ SmartPrep		Helical	SmallBody	0.516:1 20.625	0.4	0.625	40	140	SmartmA 100-610mA NI=11 Avg mA 345	40	Std Plus IQE	50	26.46	541.51	170	32	Carotid CTA (pitch 0.5:1) with SmartmA and SmartPrep, Evaluation of carotid vasculature.
23.9	Carotid 0.625mm 0.4s. 0.9:1 SmartmA/ SmartPrep		Helical	SmallBody	0.984:1 39.375	0.4	0.625	40	120	SmartmA 80-700mA NI=11 Avg mA 375	40	Std Plus IQE	50	10.62	215.89	170	32	Carotid CTA (pitch 0.9:1) with SmartmA and SmartPrep, Evaluation of carotid vasculature.
23.10	Carotid/ CoW 0.625mm 0.4s. AutomA	MIROI	Axial	SmallBody	1i	1	5 [0]	5	120	80	0	Std	0	99.00	49.50	0	32	Test bolus for Carotid CTA (pitch 0.5:1) of the head for evaluation of carotid and cerebral vasculature.
			Helical	SmallBody	0.516:1 20.625	0.4	0.625	40	140	AutomA 80-610mA NI=11 Avg mA 360	40	Std Plus IQE	50	27.61	565.06	170	32	Carotid and Brain CTA (pitch 0.5:1) with AutomA, Evaluation of carotid and cerebral vasculature.
23.11	Carotid/ CoW 0.625mm 0.4s. Smart mA/ SmartPrep		Helical	SmallBody	0.969:1 19.375	0.4	0.625	20	140	SmartmA 100-610mA NI=11 Avg mA 365	40	Std Plus IQE	50	16.39	309.60	170	32	Carotid CTA (pitch 0.9:1) with SmartmA and SmartPrep, Evaluation of carotid and cerebral vasculature.
23.12	Soft Tissue Neck 2.5mm AutomA		Helical	SmallBody	0.984:1 39.375	0.5	2.5	40	120	AutomA 100-335mA NI=9.1 Avg mA 175	30	Std Plus	30	6.20	113.26	150	32	Neck with AutomA, Evaluation of soft tissue structures for abnormality

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
23.13	Soft Tissue Neck + DMPR/ SmartmA	DMPR	Helical	SmallBody	0.969:1 19.375	0.5	0.625	20	120	SmartmA 100-600mA NI=12.6 Avg mA 255	40	Std Plus IQE	50	9.82	165.46	150	32	Neck with DMPR and SmartmA, Evaluation of soft tissue structures for abnormality

Upper Extremity

Table 4-4

[Interval]* - For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
24.1	Shoulder 1.25mm + DMPR	DMPR	Helical	LargeBody	0.516:1 20.625	0.6	1.25 [0.625]	40	140	360	40	Std Plus IQE	40	47.03	632.79	100	32	Shoulder with DMPR, Evaluation of soft tissue and bone of shoulder area
24.2	Shoulder 1.25mm + DMPR/ SmartmA	DMPR	Helical	LargeBody	0.516:1 20.625	0.6	1.25 [0.625]	40	140	SmartmA 100-610mA NI=20 Avg mA 355	40	Std Plus IQE	40	46.37	624.00	100	32	Shoulder with DMPR and SmartmA, Evaluation of soft tissue and bone of shoulder area

Chest

Table 4-5

[Interval]* - For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
25.1	Routine Chest 0.4s. 5mm SmartmA		Helical	LargeBody	1.375:1 55.0	0.4	5	40	120	SmartmA 100-650mA NI=13 Avg mA 250	50	Std Plus	50	5.76	142.87	200	32	Chest with SmartmA, Evaluation of mediastinum and lungs fields
25.2	Routine Chest 0.4s.+ DMPR/ SmartmA	DMPR	Helical	LargeBody	0.984:1 39.375	0.4	0.625	40	120	SmartmA 100-700mA NI=22 Avg mA 310	50	Std Plus	50	9.98	232.46	200	32	Chest with SmartmA and DMPR, Evaluation of mediastinum and lungs fields
25.3	Chest with HiRes 0.4s. SmartmA		Helical	LargeBody	0.969:1 19.375	0.4	5	20	120	SmartmA 80-650mA NI=12 Avg mA 300	40	Chest Plus	40	10.50	229.59	200	32	Chest with SmartmA, Evaluation of mediastinum and lung fields with higher resolution images of the lungs
25.4	Hi Res Chest 0.5s. Axial		Axial	LargeBody	1i	0.5	1.25 [10]	1.25	120	190	50	Bone+	50	2.22	53.35	230	32	Axial scan mode, Evaluation of mediastinum and lung fields in high resolution mode
25.5	Chest Abd Pelvis 0.4/ 0.5s. SmartmA 5mm (2 groups) Group1		Helical	LargeBody	0.984:1 39.375	0.4	5	40	120	SmartmA 100-650mA NI=13 Avg mA 185	50	Std Plus	50	5.96	108.95	150	32	Multi group scan for Chest, Abdomen, Pelvis (0.4/0.5 sec) with SmartmA, Evaluation of chest and abdominal anatomy for abnormalities
	Group2		Helical	LargeBody	1.375:1 55.0	0.5	5	40	120	SmartmA 100-650mA NI=13 Avg mA 185	50	Std Plus	50	5.33	182.72	295	32	
25.6	Chest Abd Pelvis 0.5s + DMPR SmartPrep	DMPR	Helical	LargeBody	1.375:1 55.0	0.5	0.625	40	120	SmartmA 100-700mA NI=22 Avg mA 270	50	Std Plus	50	7.78	387.24	450	32	Helical scan mode (0.5sec) with SmartmA, Evaluation of chest and abdominal anatomy for abnormalities
25.7	Pulmonary Embolus 0.5s.	MIROI	Axial	LargeBody	1i	1	5 [0]	5	120	80	0	Std	0	93.89	46.95	0	32	Timing Bolus for Helical scan mode 1.25mm thickness and helical pitch of 0.9:1, evaluation of Chest for pulmonary embolism
			Helical	LargeBody	0.984:1 39.375	0.5	1.25	40	120	350	40	Std	40	14.09	300.25	180	32	Helical scan mode 1.25mm thickness and helical pitch of 0.9:1, Evaluation Chest for pulmonary embolism

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy-cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
25.8	Pulmonary Embolism 0.5s. SmartmA	MIROI	Axial	LargeBody	1i	1	5 [0]	5	120	80	0	Std	0	93.89	46.95	0	32	Timing Bolus for Helical scan mode 1.25mm thickness and helical pitch of 0.9:1 with SmartmA, evaluation of Chest for pulmonary embolism
			Helical	LargeBody	0.984:1 39.375	0.5	1.25	40	120	SmartmA 100-700mA NI=20 Avg mA 300	40	Std	40	12.07	256.93	180	32	Helical scan mode 1.25mm thickness and helical pitch of 0.9:1 with SmartmA, Evaluation Chest for pulmonary embolism
25.9	Aortic Dissection 0.625mm + DMPR/ SmartmA/ SmartPrep	DMPR	Helical	LargeBody	1.375:1 55.0	0.5	0.625	40	120	SmartmA 100-700mA NI=22 Avg mA 360	40	Std Plus	40	10.37	361.20	300	32	Helical scan mode 0.625mm thickness and helical pitch of 1.375:1 with SmartmA, SmartPrep and DMPR, Evaluation for aortic dissection
25.10	Aortic Dissection 1.25mm + DMPR/ SmartmA/ SmartPrep	DMPR	Helical	LargeBody	1.375:1 55.0	0.5	1.25 [0.8]	40	120	SmartmA 100-700mA NI=20 Avg mA 330	40	Std Plus	40	9.51	330.68	300	32	Helical scan mode 1.25mm thickness and helical pitch of 1.375:1 with SmartmA, SmartPrep and DMPR, Evaluation for aortic dissection
25.11	Lung Assessment	Advanced Lung Analysis	Helical	LargeBody	0.969:1 19.375	0.5	1.25	20	120	120	0	Bone	0	5.25	114.77	200	32	Helical scan mode for acquisition of data for Advanced Lung Analysis (ALA) software for lung nodule assessment
25.12	Gated Chest Cardiac Helical		Helical	LargeBody	1.375:1 55.0	0.4	5	40	120	60	0	Std Plus	0	1.38	13.55	50	32	Localizer scan for Helical Gated scan for evaluation of thoracic vasculature
		MIROI	Axial	LargeBody	1i	1	5 [0]	5	120	60	0	Std	0	98.59	49.29	0	32	Timing bolus scan for Helical Gated scan for evaluation of thoracic vasculature
			Cardiac SEG	CardiacLarge	0.160:1 6.4	0.35	0.625	40	120	ECG Mod 110-270mA Phase 70-80 Avg mA 270	40	Std SS Seg	40	30.98	1037.93	300	32	Helical Gated scan for evaluation of thoracic vasculature
25.13	Gated Chest SnapShot Pulse		Helical	LargeBody	1.375:1 55.0	0.4	5	40	120	60	0	Std Plus	0	1.38	13.55	50	32	Localizer scan for Cine Gated scan for evaluation of thoracic vasculature
		MIROI	Axial	LargeBody	1i	1	5 [0]	5	120	60	0	Std	0	98.59	49.29	0	32	Timing bolus scan for Cine Gated scan for evaluation of thoracic vasculature
			Cardiac Pulse Cine	CardiacLarge	56i	0.35	0.625 [35]	40	120	270	40	Std SS Seg	40	14.42	454.39	314.375	32	Cine Gated scan for evaluation of thoracic vasculature
25.14	SmartScore - Gated 0.35s	Smart Score	Cine	LargeBody	8i	0.35	2.5	20	120	430	0	Std Segment	0	8.99	89.95	97.5	32	Cine gated scan mode for acquisition of data for coronary artery calcification analysis

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy-cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
25.15	SnapShot Pulse 30-65 BPM		Helical	LargeBody	1.375:1 55.0	0.4	5	40	120	60	0	Std Plus	0	1.38	13.55	50	32	Localizer scan for Prospectively gated Cardiac Cine scan for evaluation of coronary arteries, intended for stable heart rates 30-65 bpm
		MIROI	Axial	LargeBody	1i	1	5 [0]	5	120	60	0	Std	0	98.59	49.29	0	32	Timing bolus scan for Prospectively gated Cardiac Cine scan for evaluation of coronary arteries, intended for stable heart rates 30-65 bpm
			Cardiac Pulse Cine	CardiacSmall	56i	0.35	0.625 [35]	40	120	270	40	Std SS Seg	40	12.69	133.24	104.375	32	Prospectively gated Cardiac Cine scan for evaluation of coronary arteries, intended for stable heart rates 30-65 bpm
25.16	SnapShot Segment 30-74 BPM		Helical	LargeBody	1.375:1 55.0	0.4	5	40	120	60	0	Std Plus	0	1.38	13.55	50	32	Localizer scan for Retrospectively gated Cardiac Helical scan for evaluation of coronary arteries, intended for heart rates 30-74 bpm
		MIROI	Axial	LargeBody	1i	1	5 [0]	5	120	60	0	Std	0	98.59	49.29	0	32	Timing bolus scan for Retrospectively gated Cardiac Helical scan for evaluation of coronary arteries, intended for heart rates 30-74 bpm
			Cardiac SEG	CardiacLarge	0.160:1 6.4	0.35	0.625	40	120	ECG Mod 120-270mA Phase 70-80 Avg mA 270	40	Std SS Seg	40	33.25	448.83	100	32	Retrospectively gated Cardiac Helical scan for evaluation of coronary arteries, intended for heart rates 30-74 bpm
25.17	SnapShot Burst 75-113 BPM		Helical	LargeBody	1.375:1 55.0	0.4	5	40	120	60	0	Std Plus	0	1.38	13.55	50	32	Localizer scan for Retrospectively gated Cardiac Helical scan for evaluation of coronary arteries, intended for heart rates 75-113 bpm
		MIROI	Axial	LargeBody	1i	1	5 [0]	5	120	60	0	Std	0	98.59	49.29	0	32	Timing bolus scan for Retrospectively gated Cardiac Helical scan for evaluation of coronary arteries, intended for heart rates 75-113 bpm
			Cardiac SSB	CardiacLarge	0.160:1 6.4	0.35	0.625	40	120	ECG Mod 120-270mA Phase 70-80 Avg mA 270	40	Std SSB	40	33.25	448.83	100	32	Retrospectively gated Cardiac Helical scan for evaluation of coronary arteries, intended for heart rates 75-113 bpm
25.18	SnapShot Burst Plus 114+ BPM		Helical	LargeBody	1.375:1 55.0	0.4	5	40	120	60	0	Std Plus	0	1.38	13.55	50	32	Localizer scan for Retrospectively gated Cardiac Helical scan for evaluation of coronary arteries, intended for heart rates over 113 bpm

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy-cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
		MIROI	Axial	LargeBody	1i	1	5 [0]	5	120	60	0	Std	0	98.59	49.29	0	32	Timing bolus scan for Retrospectively gated Cardiac Helical scan for evaluation of coronary arteries, intended for heart rates over 113 bpm
			Cardiac SSBP	CardiacLarge	0.160:1 6.4	0.35	0.625	40	120	ECG Mod 120-270mA Phase 70-80 Avg mA 270	40	Std SSB	40	33.25	448.83	100	32	Retrospectively gated Cardiac Helical scan for evaluation of coronary arteries, intended for heart rates over 113 bpm
25.19	Advantage 4D		Helical	LargeBody	1.375:1 55.0	0.6	5	40	120	170	40	Std Plus	40	5.88	145.59	200	32	Localizer scan for Advantage 4D series
		Advantage 4D	Cine	LargeBody	8i	1	2.5	20	120	70	30	Std	30	35.67	570.67	157.5	32	Cine scan mode for acquisition data for Advantage 4D for retrospective gating
25.20	VHS Chest 4D CTA SmartPrep		Helical-S	LargeBody	1.375:1 55.0 11 pass	0.4	5 [10]	40	120	SmartmA 150-500mA NI=14 Avg mA 210	40	Std Plus	40	67.29	2119.57	300	32	Volume Helical scan mode with SmartPrep for time resolved CT angiography of chest vasculature

Abdomen

Table 4-6

[Interval]* - For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
26.1	Abdomen Pelvis 0.5s. 5mm SmartmA		Helical	LargeBody	1.375:1 55.0	0.5	5	40	120	SmartmA 50-650mA NI=11.57 Avg mA 250	50	Std Plus	50	7.20	322.54	400	32	Basic Abdomen Pelvis (0.5sec) with Smart mA, Evaluation for abnormalities of abdominal and pelvic structures
26.2	Abdomen Pelvis 0.6s. 5mm + DMPR SmartmA/ SmartPrep	DMPR	Helical	LargeBody	0.984:1 39.375	0.6	0.625	40	120	SmartmA 80-700mA NI=18 Avg mA 260	50	Std Plus	50	12.56	543.37	400	32	Basic Abdomen Pelvis (0.6sec) with Smart mA, SmartPrep and DMPR, Evaluation for abnormalities of abdominal and pelvic structures
26.4	Renal Stone 0.5s. 5mm		Helical	LargeBody	1.375:1 55.0	0.5	5	40	120	190	50	Std Plus	50	5.47	245.13	400	32	Helical scan, Evaluation of kidney, ureters and bladder for renal stones
26.5	Renal Stone 0.5s. 5mm SmartmA		Helical	LargeBody	1.375:1 55.0	0.5	5	40	120	SmartmA 50-650mA NI=12 Avg mA 190	50	Std Plus	50	5.47	245.13	400	32	Helical scan with SmartmA, Evaluation of kidney, ureters and bladder for renal stones
26.7	AAA 0.5s. + DMPR 0.625mm	MIROI	Axial	LargeBody	1i	1	5 [0]	5	120	80	0	Std	0	112.67	56.34	0	32	Timing bolus for Helical scan with DMPR, Evaluation for abdominal aortic aneurysm
		DMPR	Helical	LargeBody	1.375:1 55.0	0.5	0.625	40	140	305	50	Std Plus	50	12.45	444.70	309.375	32	Helical scan with DMPR, Evaluation for abdominal aortic aneurysm
26.8	AAA 0.5s. + DMPR 0.625mm SmartmA/ SmartPrep	DMPR	Helical	LargeBody	1.375:1 55.0	0.6	0.625	40	120	SmartmA 100-700mA NI=19 Avg mA 275	50	Std Plus	50	9.51	339.51	309.375	32	Helical scan with DMPR, Smart mA and SmartPrep, Evaluation for abdominal aortic aneurysm
26.10	Dual Phase Liver		Helical	LargeBody	1.375:1 55.0	0.6	5	40	120	150	50	Std Plus	50	5.19	92.16	130	32	Non contrast evaluation of upper abdominal structures including liver
		MIROI	Axial	LargeBody	1i	1	5 [0]	5	120	80	0	Std	0	112.67	56.34	0	32	Timing bolus for Multi-Group acquisition for early arterial and portal venous phase of the liver
	Group1		Helical	LargeBody	1.375:1 55.0	0.5	2.5	40	120	275	50	Std Plus	50	7.92	140.87	130	32	Acquisition for early arterial phase of the liver
	Group2		Helical	LargeBody	1.375:1 55.0	0.5	5	40	120	210	50	Std Plus	50	6.05	270.94	400	32	Acquisition for portal venous phase of the liver

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy-cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
26.11	Tri Phase Liver		Helical	LargeBody	1.375:1 55.0	0.6	5	40	120	150	50	Std Plus	50	5.19	92.16	130	32	Non contrast evaluation of upper abdominal structures including liver
		MIROI	Axial	LargeBody	1i	1	5 [0]	5	120	80	0	Std	0	112.67	56.34	0	32	Timing bolus for Multi-Group acquisition for early arterial, late arterial and portal venous phase of the liver
	Group1		Helical	LargeBody	1.375:1 55.0	0.5	2.5	40	120	275	50	Std Plus	50	7.92	140.87	130	32	Acquisition for early arterial phase of the liver
	Group2		Helical	LargeBody	1.375:1 55.0	0.5	5	40	120	225	50	Std Plus	50	6.48	290.29	400	32	Acquisition for late arterial phase of the liver
	Group3		Helical	LargeBody	1.375:1 55.0	0.5	5	40	120	225	50	Std Plus	50	6.48	290.29	400	32	Acquisition for portal venous phase of the liver
26.12	Dual Phase Pancreas SmartmA/ SmartPrep		Helical	LargeBody	0.984:1 39.375	0.5	5	40	120	SmartmA 80-650mA NI=12 Avg mA 200	50	Std Plus	50	8.05	82.74	70	32	Non contrast evaluation of upper abdominal structures including pancreas
	Group1		Helical	LargeBody	0.984:1 39.375	0.5	2.5	40	120	SmartmA 50-700mA NI=15 Avg mA 260	50	Std Plus	50	10.46	191.28	150	32	Acquisition with SmartmA and SmartPrep for arterial phase of the pancreas
	Group2		Helical	LargeBody	0.984:1 39.375	0.5	5	40	120	SmartmA 50-700mA NI=15 Avg mA 225	50	Std Plus	50	9.06	319.47	320	32	Acquisition for venous phase of the pancreas
26.13	Dual Phase Liver SmartmA/ SmartPrep		Helical	LargeBody	1.375:1 55.0	0.5	5	40	120	SmartmA 80-500mA NI=12 Avg mA 160	50	Std Plus	50	4.61	81.96	130	32	Non contrast evaluation of upper abdominal structures including liver
	Group1		Helical	LargeBody	1.375:1 55.0	0.5	2.5	40	120	SmartmA 50-700mA NI=15 Avg mA 275	50	Std Plus	50	7.92	140.87	130	32	Acquisition with SmartmA and SmartPrep for early arterial phase of the liver
	Group2		Helical	LargeBody	0.984:1 39.375	0.5	5	40	120	SmartmA 50-700mA NI=11 Avg mA 225	50	Std Plus	50	9.06	391.91	400	32	Acquisition with SmartmA and SmartPrep for portal venous phase of the liver
26.14	Tri Phase Liver SmartmA/ SmartPrep		Helical	LargeBody	0.984:1 39.375	0.5	5	40	120	SmartmA 80-500mA NI=12 Avg mA 200	50	Std Plus	50	8.05	131.04	130	32	Non contrast evaluation of upper abdominal structures including liver
	Group1		Helical	LargeBody	1.375:1 55.0	0.5	2.5	40	120	SmartmA 50-700mA NI=15 Avg mA 275	50	Std Plus	50	7.92	140.87	130	32	Acquisition with SmartmA and SmartPrep for early arterial phase of the liver
	Group2		Helical	LargeBody	1.375:1 55.0	0.5	5	40	120	SmartmA 50-700mA NI=11 Avg mA 225	50	Std Plus	50	6.48	290.29	400	32	Acquisition with SmartmA and SmartPrep for late arterial phase of the liver
	Group3		Helical	LargeBody	1.375:1 55.0	0.5	5	40	120	SmartmA 50-700mA NI=11 Avg mA 225	50	Std Plus	50	6.48	290.29	400	32	Acquisition with SmartmA and SmartPrep for portal venous phase of the liver
26.15	VHS Liver Perfusion		Helical	LargeBody	0.984:1 39.375	0.6	5	40	120	85	50	Std	50	4.11	66.80	130	32	Non contrast evaluation of upper abdominal structures including liver

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
		Perfusion	Helical-S/P	LargeBody	1.375:1 55.0 21 pass	0.4	5	40	120	150	50	Std Plus	50	109.99	1704.89	140	32	Volume Helical scan mode for evaluation of liver tumor perfusion over time
26.16	VHS 4D CTA		Helical	LargeBody	0.984:1 39.375	0.6	5	40	120	170	50	Std	50	8.21	133.61	130	32	Localizer scan for Volume Helical Shuttle scan mode for evaluation of abdominal vasculature
		MIROI	Axial	LargeBody	1i	1	5 [0]	5	120	80	0	Std	0	112.67	56.34	0	32	Timing bolus scan for Volume Helical Shuttle scan mode for evaluation of abdominal vasculature
			Helical-S	LargeBody	1.375:1 55.0 18 pass	0.5	5 [10]	40	140	SmartmA 150-610mA NI=12 Avg mA 175	50	Std Plus	50	162.84	3175.35	180	32	Volume Helical Shuttle scan mode for time resolved CT angiography of abdominal vasculature
26.17	Runoff 0.625mm	MIROI	Axial	LargeBody	1i	1	5 [0]	5	120	80	0	Std	0	112.67	56.34	0	32	Timing bolus scan for evaluation of vascular structure of abdomen, femurs and lower extremities (Runoff)
			Helical	LargeBody	1.375:1 55.0	0.5	0.625	40	140	365	40	Std Plus	40	14.90	1467.83	936.875	32	Helical scanmode 0.625mm, Evaluation of vascular structure of abdomen, femurs and lower extremities (Runoff)
26.18	Runoff 1.25mm	MIROI	Axial	LargeBody	1i	1	5 [0]	5	120	80	0	Std	0	112.67	56.34	0	32	Timing bolus scan for evaluation of vascular structure of abdomen, femurs and lower extremities (Runoff)
			Helical	LargeBody	1.375:1 55.0	0.5	1.25 [0.7]	40	140	350	40	Std Plus	40	14.29	1496.13	998.9	32	Helical scanmode 1.25mm, Evaluation of vascular structure of abdomen, femurs and lower extremities (Runoff)
26.19	Runoff 0.625mm SmartmA/ SmartPrep		Helical	LargeBody	1.375:1 55.0	0.5	0.625	40	140	SmartmA 100-610mA NI=19 Avg mA 330	40	Std Plus	40	13.47	1326.48	936.875	32	Helical scanmode 0.625mm with SmartmA and SmartPrep, Evaluation of vascular structure of abdomen, femurs and lower extremities (Runoff)
26.20	Runoff 1.25mm SmartmA/ SmartPrep		Helical	LargeBody	1.375:1 55.0	0.5	1.25 [0.7]	40	140	SmartmA 100-610mA NI=20 Avg mA 360	40	Std Plus	40	14.70	1538.87	998.9	32	Helical scanmode 1.25mm with SmartmA and SmartPrep, Evaluation of vascular structure of abdomen, femurs and lower extremities (Runoff)
26.22	CT Colonography 1.25mm	Advantage CTC Pro	Helical	LargeBody	1.375:1 55.0	0.5	1.25	40	120	100	50	Std Plus	50	2.88	129.02	400	32	Helical scan mode supine acquisition for using Advantage CTC Pro for evaluation of the colon
		Advantage CTC Pro	Helical	LargeBody	1.375:1 55.0	0.5	1.25	40	120	100	50	Std Plus	50	2.88	129.02	400	32	Helical scan mode prone acquisition for using Advantage CTC Pro for evaluation of the colon

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy-cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
26.41	Body Interventional 5mm/3i		Helical	LargeBody	1.375:1 27.50	0.5	5	20	120	SmartmA 50-650mA NI=11.57 Avg mA 360	0	Std	0	11.41	251.58	195	32	Localization scan for CT interventional scan
			SmartView SEG	LargeBody	3i	0.4	5	10	120	80	0	Std Segment	0	684.58	684.58	0	32	CT Fluoro scan for interventional scan
26.42	Body Interventional 5mm/1i		Helical	LargeBody	1.375:1 27.50	0.5	5	20	120	SmartmA 50-650mA NI=11.57 Avg mA 360	0	Std	0	11.41	251.58	195	32	Localization scan for CT interventional scan
			SmartView SEG	LargeBody	1i	0.4	5	5	120	80	0	Std Segment	0	845.04	422.52	0	32	CT Fluoro scan for interventional scan

L-Spine

Table 4-7

[Interval]* - For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
27.1	L-Spine 3 Level Axial Group1		Axial	LargeBody	8i	1	2.5	20	120	225	30	Std	40	19.08	38.15	17.5	32	Axial scan mode, Evaluation of soft tissue and boney structures lumbar spine
	Group2		Axial	LargeBody	8i	1	2.5	20	120	225	30	Std	40	19.08	38.15	17.5	32	Axial scan mode, Evaluation of soft tissue and boney structures lumbar spine
	Group3		Axial	LargeBody	8i	1	2.5	20	140	250	30	Std	40	30.03	60.06	17.5	32	Axial scan mode, Evaluation of soft tissue and boney structures lumbar spine
27.3	L-Spine Survey Helical 2.5mm 0.6s. SmartmA		Helical	LargeBody	0.516:1 20.625	0.6	2.5	40	140	SmartmA 100-610mA NI=14 Avg mA 380	30	Std Plus	40	49.64	1040.25	75	32	Helical scan mode 2.5mm (0.6sec), Limited evaluation of lumbar spine
27.4	L-Spine Survey Helical 2.5mm 0.8s. SmartmA		Helical	LargeBody	0.516:1 20.625	0.8	2.5	40	140	SmartmA 100-610mA NI=14 Avg mA 365	30	Std Plus	40	63.57	1332.02	75	32	Helical scan mode 2.5mm (0.8sec), Limited evaluation of lumbar spine
27.6	L-Spine Helical 0.625mm + DMPR/ SmartmA	DMPR	Helical	LargeBody	0.516:1 20.625	0.6	0.625	40	140	SmartmA 100-610mA NI=20 Avg mA 425	30	Std Plus IQE	40	55.52	1163.43	75	32	Helical scan mode 0.625mm with DMPR and SmartmA, Evaluation of soft tissue and boney structures lumbar spine
27.7	L-Spine Helical 1.25mm + DMPR/ SmartmA - Large Patient	DMPR	Helical	LargeBody	0.516:1 20.625	1	1.25	40	140	SmartmA 100-610mA NI=18 Avg mA 425	30	Std Plus	40	92.53	1938.41	75	32	Helical scan mode 1.25mm with DMPR and SmartmA, Evaluation of soft tissue and boney structures lumbar spine

Pelvis

Table 4-8

[Interval]* - For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.
For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
28.1	Pelvis for Fracture + DMPR/ SmartmA	DMPR	Helical	LargeBody	0.984:1 39.375	0.8	1.25 [0.625]	40	120	SmartmA 100-700mA NI=20 Avg mA 300	40	Bone Plus IQE	40	19.32	258.70	101.25	32	Helical scan with DMPR and SmartmA, Evaluation of pelvis bony structures for fracture

Lower Extremity

Table 4-9

[Interval]* - For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
29.1	Lower Extremity Survey 2.5mm		Helical	LargeBody	0.516:1 20.625	0.5	2.5	40	120	175	30	Std Plus	30	13.45	126.94	60	32	Helical scan mode 2.5mm, Limited evaluation of lower extremity for abnormalities
29.2	Ankle 0.625 mm Single Joint		Helical	MediumBody	0.531:1 10.625	0.5	0.625	20	120	175	30	Std Plus IQE	40	14.50	75.64	35	32	Helical scan mode 0.625mm, Evaluation of soft tissue and bony anatomy of the ankle area
29.3	Ankle 0.625 mm Both Joints		Helical	LargeBody	0.531:1 10.625	0.5	0.625	20	120	210	30	Std Plus IQE	30	16.76	87.38	35	32	Helical scan mode 0.625mm, Evaluation of soft tissue and bony anatomy of the ankle area
29.4	Knee 0.625 mm Both Joints		Helical	LargeBody	0.531:1 10.625	0.6	0.625	20	120	AutomA 100-335mA NI=20 Avg mA 190	30	Std Plus IQE	30	18.19	94.84	35	32	Helical scan mode 0.625mm, Evaluation of soft tissue and bony anatomy of both knees

**Reference Pediatric Protocols -
Installed Options : ASiR, Sub 0.4 sec
and VCT85kW**

Head

Table 5-1

[Interval]* - For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy-cm) ASiR	Scan Length (mm)	Phantom (cm)	Description
31.1	PED HEAD TO 18 Months		Axial	Ped Head	4i	0.5	5	20	120	170	30	Std	30	14.47	173.68	115	16	Routine head (0.5s) for infant (up to 18 months) 5mm
31.2	PED HEAD 18 MOS TO 5YRS		Axial	SmallHead	4i	0.5	5	20	120	235	30	Std	30	20.01	240.08	115	16	Routine head (0.5s) for children (18 months to 5 years) 5mm
31.3	PED HEAD TO 5YRS TO 18YRS Group1		Axial	SmallHead	8i	1	2.5	20	120	170	30	Std	30	28.95	115.79	37.5	16	Routine head (1.0s) for children (5 years to 18 years) 2.5mm
	Group2		Axial	SmallHead	4i	1	5	20	120	125	30	Std	30	21.28	170.27	75	16	Routine head (1.0s) for children (5 years to 18 years) 5mm

Orbit

Table 5-2

[Interval]* - For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy-cm) ASiR	Scan Length (mm)	Phantom (cm)	Description
32.1	Sinus Supine Helical + DMPR	DMPR	Helical	SmallHead	0.531:1 10.625	0.4	0.625	20	120	65	20	Bone Plus IQE	30	8.33	76.29	74.375	16	Routine Helical scan mode with DMPR, Evaluation of sinus area for abnormality in pediatric patients
32.3	Orbits Helical	DMPR	Helical	SmallHead	0.531:1 10.625	0.4	0.625	20	120	145	20	Std Plus IQE	30	18.59	146.96	61.875	16	Helical scan mode, Evaluation of orbital and face structures for abnormality in pediatric patients
32.6	Helical IAC 0.625mm Axial mode - Use Reformat for Coronal		Helical	SmallHead	0.531:1 10.625	0.4	0.625	20	120	130	20	Std Plus IQE	30	16.67	93.21	38.75	16	Helical scan mode, Evaluation of inner ear structures for abnormality in pediatric patients

Chest

Table 5-3

[Interval]* - For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy-cm) ASiR	Scan Length (mm)	Phantom (cm)	Description
35.1.1	RC 6.0-7.5 kg (13.2-16.5 lbs)		Helical	PedBody	1.375:1 27.5	0.4	3.75	20	100	55	30	Std Plus	30	0.76	7.62	75	32	Weight and height based routine chest protocol
35.1.3	SnapShot Segment 30-74 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	85	30	Std Plus	30	0.59	5.80	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	Std	0	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
			Cardiac SEG	CardiacSmall	0.160:1 6.4	0.35	0.625	40	80	ECG Mod 55-195mA Phase 40-80 AvgmA195	30	Std SS Seg	40	8.51	93.56	75	32	Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
35.1.4	SnapShot Burst 75-104 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	85	30	Std Plus	30	0.59	5.80	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	Std	0	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
			Cardiac SSB	CardiacSmall	0.160:1 6.4	0.35	0.625	40	80	ECG Mod 55-195mA Phase 40-80 AvgmA195	30	Std SSB	40	8.51	93.56	75	32	Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
35.1.5	SnapShot Burst+ >105 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	85	30	Std Plus	30	0.59	5.80	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	Std	0	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
			Cardiac SSBP	CardiacSmall	0.160:1 6.4	0.35	0.625	40	80	ECG Mod 55-195mA Phase 35-85 AvgmA195	30	Std SSB	40	9.13	100.42	75	32	Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
35.1.7	Non ECG Gated		Helical	PedBody	0.984:1 39.375	0.4	0.625	40	80	SmartmA 50-200mA NI=18 AvgmA140	30	Std Plus	30	1.36	14.70	75	32	Weight and height based non ECG gated protocol for cardiac with SmartmA
35.2.1	RC 7.5-9.5 kg (16.5-20.9lbs)		Helical	PedBody	1.375:1 27.5	0.4	3.75	20	100	85	30	Std Plus	30	1.18	11.77	75	32	Weight and height based routine chest protocol
35.2.3	SnapShot Segment 30-74 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	85	30	Std Plus	30	0.59	5.80	50	32	Weight and height based protocol for localization

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy-cm) ASiR	Scan Length (mm)	Phantom (cm)	Description
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	Std	0	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
			Cardiac SEG	CardiacSmall	0.160:1 6.4	0.35	0.625	40	80	ECG Mod 55-210mA Phase 40-80 AvgmA210	30	Std SS Seg	40	9.13	100.39	75	32	Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
35.2.4	SnapShot Burst 75-104 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	85	30	Std Plus	30	0.59	5.80	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	Std	0	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
			Cardiac SSB	CardiacSmall	0.160:1 6.4	0.35	0.625	40	80	ECG Mod 55-210mA Phase 40-80 AvgmA210	30	Std SSB	40	9.13	100.39	75	32	Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
35.2.5	SnapShot Burst+ >105 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	85	30	Std Plus	30	0.59	5.80	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	Std	0	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
			Cardiac SSBP	CardiacSmall	0.160:1 6.4	0.35	0.625	40	80	ECG Mod 55-210mA Phase 35-85 AvgmA210	30	Std SSB	40	9.82	107.99	75	32	Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
35.2.7	Non ECG Gated		Helical	PedBody	0.984:1 39.375	0.4	0.625	40	80	SmartmA 50-220mA NI=18 AvgmA155	30	Std Plus	30	1.51	16.28	75	32	Weight and height based non ECG gated protocol for cardiac with SmartmA
35.3.1	RC 9.5-11.5 kg (20.9-25.4lbs)		Helical	SmallBody	1.375:1 27.5	0.4	3.75	20	120	70	30	Std Plus	30	1.52	15.77	78.75	32	Weight and height based routine chest protocol
35.3.3	SnapShot Segment 30-74 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	85	30	Std Plus	30	0.59	5.80	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	Std	0	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
			Cardiac SEG	CardiacSmall	0.160:1 6.4	0.35	0.625	40	80	ECG Mod 70-225mA Phase 40-80 AvgmA225	30	Std SS Seg	40	9.88	108.64	75	32	Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
35.3.4	SnapShot Burst 75-104 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	85	30	Std Plus	30	0.59	5.80	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	Std	0	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy-cm) ASiR	DLP (mGy-cm) ASiR	Scan Length (mm)	Phantom (cm)	Description
			Cardiac SSB	CardiacSmall	0.160:1 6.4	0.35	0.625	40	80	ECG Mod 70-225mA Phase 40-80 AvgmA225	30	Std SSB	40	9.88	108.64	75	32	Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
35.3.5	SnapShot Burst+ >105 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	85	30	Std Plus	30	0.59	5.80	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	Std	0	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
			Cardiac SSBP	CardiacSmall	0.160:1 6.4	0.35	0.625	40	80	ECG Mod 70-225mA Phase 35-85 AvgmA225	30	Std SSB	40	10.57	116.23	75	32	Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
35.3.7	Non ECG Gated		Helical	PedBody	0.984:1 39.375	0.4	0.625	40	80	SmartmA 50-240mA NI=18 AvgmA170	30	Std Plus	30	1.65	17.85	75	32	Weight and height based non ECG gated protocol for cardiac with SmartmA
35.4.1	RC 11.5-14.5kg (25.4-32.0lbs)		Helical	SmallBody	1.375:1 27.5	0.4	5	20	120	65	30	Std Plus	30	1.41	16.23	90	32	Weight and height based routine chest protocol
35.4.3	SnapShot Segment 30-74 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	85	30	Std Plus	30	0.59	5.80	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	Std	0	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
			Cardiac SEG	CardiacSmall	0.160:1 6.4	0.35	0.625	40	80	ECG Mod 70-240mA Phase 40-80 AvgmA240	30	Std SS Seg	40	10.50	115.46	75	32	Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
35.4.4	SnapShot Burst 75-104 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	85	30	Std Plus	30	0.59	5.80	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	Std	0	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
			Cardiac SSB	CardiacSmall	0.160:1 6.4	0.35	0.625	40	80	ECG Mod 70-240mA Phase 40-80 AvgmA240	30	Std SSB	40	10.50	115.46	75	32	Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
35.4.5	SnapShot Burst+ >105 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	85	30	Std Plus	30	0.59	5.80	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	Std	0	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
			Cardiac SSBP	CardiacSmall	0.160:1 6.4	0.35	0.625	40	80	ECG Mod 70-240mA Phase 35-85 AvgmA240	30	Std SSB	40	11.25	123.79	75	32	Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy-cm) ASiR	Scan Length (mm)	Phantom (cm)	Description
35.4.7	Non ECG Gated		Helical	PedBody	0.984:1 39.375	0.4	0.625	40	80	SmartmA 50-280mA NI=18 AvgmA195	30	Std Plus	30	1.90	20.48	75	32	Weight and height based non ECG gated protocol for cardiac with SmartmA
35.5.1	RC 14.5-18.5kg (32.0-40.8lbs)		Helical	SmallBody	1.375:1 27.5	0.4	5	20	120	70	30	Std Plus	30	1.52	17.48	90	32	Weight and height based routine chest protocol
35.5.3	SnapShot Segment 30-74 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	85	30	Std Plus	30	0.59	5.80	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	Std	0	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
			Cardiac SEG	CardiacSmall	0.160:1 6.4	0.35	0.625	40	80	ECG Mod 70-265mA Phase 40-80 AvgmA265	30	Std SS Seg	40	11.53	126.82	75	32	Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
35.5.4	SnapShot Burst 75-104 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	85	30	Std Plus	30	0.59	5.80	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	Std	0	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
			Cardiac SSB	CardiacSmall	0.160:1 6.4	0.35	0.625	40	80	ECG Mod 70-265mA Phase 40-80 AvgmA265	30	Std SSB	40	11.53	126.82	75	32	Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
35.5.5	SnapShot Burst+ >105 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	85	30	Std Plus	30	0.59	5.80	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	Std	0	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
			Cardiac SSBP	CardiacSmall	0.160:1 6.4	0.35	0.625	40	80	ECG Mod 70-265mA Phase 35-85 AvgmA265	30	Std SSB	40	12.40	136.37	75	32	Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
35.5.7	Non ECG Gated		Helical	PedBody	0.984:1 39.375	0.4	0.625	40	80	SmartmA 50-300mA NI=18 AvgmA210	30	Std Plus	30	2.04	22.05	75	32	Weight and height based non ECG gated protocol for cardiac with SmartmA
35.6.1	RC 18.5-22.5kg (40.8-49.6lbs)		Helical	SmallBody	1.375:1 27.5	0.4	5	20	120	75	30	Std Plus	30	1.63	18.73	90	32	Weight and height based routine chest protocol
35.6.3	SnapShot Segment 30-74 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	85	30	Std Plus	30	0.59	5.80	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	Std	0	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy-cm) ASiR	Scan Length (mm)	Phantom (cm)	Description
			Cardiac SEG	CardiacSmall	0.160:1 6.4	0.35	0.625	40	80	ECG Mod 70-280mA Phase 40-80 AvgmA280	30	Std SS Seg	40	12.16	133.72	75	32	Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
35.6.4	SnapShot Burst 75-104 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	85	30	Std Plus	30	0.59	5.80	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	Std	0	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
			Cardiac SSB	CardiacSmall	0.160:1 6.4	0.35	0.625	40	80	ECG Mod 70-280mA Phase 40-80 AvgmA280	30	Std SSB	40	12.16	133.72	75	32	Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
35.6.5	SnapShot Burst+ >105 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	85	30	Std Plus	30	0.59	5.80	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	Std	0	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
			Cardiac SSBP	CardiacSmall	0.160:1 6.4	0.35	0.625	40	80	ECG Mod 70-280mA Phase 35-85 AvgmA280	30	Std SSB	40	13.09	144.01	75	32	Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
35.6.7	Non ECG Gated		Helical	PedBody	0.984:1 39.375	0.4	0.625	40	80	SmartmA 100-320mA NI=18 AvgmA225	30	Std Plus	30	2.19	23.63	75	32	Weight and height based non ECG gated protocol for cardiac with SmartmA
35.7.1	RC 22.5-31.5kg (49.6-69.5lbs)		Helical	LargeBody	1.375:1 55.0	0.4	5	40	120	75	30	Std Plus	30	1.73	22.98	85	32	Weight and height based routine chest protocol
35.7.3	SnapShot Segment 30-74 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	85	30	Std Plus	30	0.59	5.80	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	Std	0	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
			Cardiac SEG	CardiacSmall	0.160:1 6.4	0.35	0.625	40	80	ECG Mod 75-295mA Phase 40-80 AvgmA295	30	Std SS Seg	40	12.81	140.96	75	32	Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
35.7.4	SnapShot Burst 75-104 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	85	30	Std Plus	30	0.59	5.80	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	Std	0	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
			Cardiac SSB	CardiacSmall	0.160:1 6.4	0.35	0.625	40	80	ECG Mod 75-295mA Phase 40-80 AvgmA295	30	Std SSB	40	12.81	140.96	75	32	Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy-cm) ASiR	Scan Length (mm)	Phantom (cm)	Description
35.7.5	SnapShot Burst+ >105 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	85	30	Std Plus	30	0.59	5.80	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	Std	0	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
			Cardiac SSBP	CardiacSmall	0.160:1 6.4	0.35	0.625	40	80	ECG Mod 75-295mA Phase 35-85 AvgmA295	30	Std SSB	40	13.79	151.74	75	32	Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
35.7.7	Non ECG Gated		Helical	PedBody	0.984:1 39.375	0.4	0.625	40	80	SmartmA 100-360mA NI=18 AvgmA250	30	Std Plus	30	2.43	26.25	75	32	Weight and height based non ECG gated protocol for cardiac with SmartmA
35.8.1	RC 31.5-40.5kg (69.5-89.3lbs)		Helical	LargeBody	1.375:1 55.0	0.4	5	40	120	85	30	Std Plus	30	1.96	32.90	120	32	Weight and height based routine chest protocol
35.8.3	SnapShot Segment 30-74 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	100	85	30	Std Plus	30	1.10	10.76	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	100	50	0	Std	0	39.50	19.75	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
			Cardiac SEG	CardiacSmall	0.160:1 6.4	0.35	0.625	40	100	ECG Mod 70-265mA Phase 40-80 AvgmA265	30	Std SS Seg	30	21.40	235.37	75	32	Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
35.8.4	SnapShot Burst 75-104 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	100	85	30	Std Plus	30	1.10	10.76	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	100	50	0	Std	0	39.50	19.75	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
			Cardiac SSB	CardiacSmall	0.160:1 6.4	0.35	0.625	40	100	ECG Mod 70-265mA Phase 40-80 AvgmA265	30	Std SSB	30	21.40	235.37	75	32	Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
35.8.5	SnapShot Burst+ >105 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	100	85	30	Std Plus	30	1.10	10.76	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	100	50	0	Std	0	39.50	19.75	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
			Cardiac SSBP	CardiacSmall	0.160:1 6.4	0.35	0.625	40	100	ECG Mod 70-265mA Phase 35-85 AvgmA265	30	Std SSB	40	23.01	253.11	75	32	Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
35.8.7	Non ECG Gated		Helical	SmallBody	0.984:1 39.375	0.4	0.625	40	100	SmartmA 100-400mA NI=16 AvgmA280	30	Std Plus	30	5.06	54.57	75	32	Weight and height based non ECG gated protocol for cardiac with SmartmA

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy-cm) ASiR	Scan Length (mm)	Phantom (cm)	Description
35.9.1	RC 40.5-55.0kg(89.3-121.3lbs)		Helical	LargeBody	1.375:1 55.0	0.4	5	40	120	90	30	Std Plus	30	2.07	41.06	150	32	Weight and height based routine chest protocol
35.9.3	SnapShot Segment 30-74 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	100	85	30	Std Plus	30	1.10	10.76	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	100	50	0	Std	0	39.50	19.75	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
			Cardiac SEG	CardiacSmall	0.160:1 6.4	0.35	0.625	40	100	ECG Mod 70-295mA Phase 40-80 AvgmA295	30	Std SS Seg	30	23.71	260.86	75	32	Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
35.9.4	SnapShot Burst 75-104 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	100	85	30	Std Plus	30	1.10	10.76	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	100	50	0	Std	0	39.50	19.75	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
			Cardiac SSB	CardiacSmall	0.160:1 6.4	0.35	0.625	40	100	ECG Mod 70-295mA Phase 40-80 AvgmA295	30	Std SSB	30	23.71	260.86	75	32	Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
35.9.5	SnapShot Burst+ >105 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	100	85	30	Std Plus	30	1.10	10.76	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	100	50	0	Std	0	39.50	19.75	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
			Cardiac SSBP	CardiacSmall	0.160:1 6.4	0.35	0.625	40	100	ECG Mod 70-295mA Phase 35-85 AvgmA295	30	Std SSB	40	25.57	281.32	75	32	Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
35.9.7	Non ECG Gated		Helical	SmallBody	0.984:1 39.375	0.4	0.625	40	100	SmartmA 100-440mA NI=16 AvgmA310	30	Std Plus	30	5.60	60.42	75	32	Weight and height based non ECG gated protocol for cardiac with SmartmA

Abdomen

Table 5-4

[Interval]* - For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy-cm) ASiR	Scan Length (mm)	Phantom (cm)	Description
36.1.1	Abd 6.0-7.5kg (13.2-16.5lbs)		Helical	PedBody	1.375:1 27.5	0.4	3.75	20	120	70	30	Std Plus	30	1.52	15.77	78.75	32	Weight and height based routine abdomen protocol
36.2.1	Abd 7.5-9.5kg (16.5-20.9lbs)		Helical	PedBody	1.375:1 27.5	0.4	3.75	20	120	110	30	Std Plus	30	2.39	27.47	90	32	Weight and height based routine abdomen protocol
36.3.1	Abd 9.5-11.5kg (20.9-25.4lbs)		Helical	PedBody	1.375:1 27.5	0.4	3.75	20	120	125	30	Std Plus	30	2.71	34.26	101.25	32	Weight and height based routine abdomen protocol
36.4.1	Abd 11.5-14.5kg (25.4-32.0lbs)		Helical	SmallBody	1.375:1 27.5	0.4	5	20	120	85	30	Std Plus	30	1.84	24.91	110	32	Weight and height based routine abdomen protocol
36.5.1	Abd 14.5-18.5kg (32.0-40.8lbs)		Helical	SmallBody	1.375:1 27.5	0.4	5	20	120	95	30	Std Plus	30	2.06	27.84	110	32	Weight and height based routine abdomen protocol
36.6.1	Abd 18.5-22.5kg (40.8-49.6lbs)		Helical	Medium Body	1.375:1 27.5	0.4	5	20	120	85	30	Std Plus	30	2.18	33.77	130	32	Weight and height based routine abdomen protocol
36.7.1	Abd 22.5-31.5kg (49.6-69.5lbs)		Helical	LargeBody	1.375:1 55.0	0.4	5	40	120	105	30	Std Plus	30	2.42	47.91	150	32	Weight and height based routine abdomen protocol
36.8.1	Abd 31.5-40.5kg (69.5-89.3lbs)		Helical	LargeBody	1.375:1 55.0	0.4	5	40	120	120	30	Std Plus	30	2.77	54.75	150	32	Weight and height based routine abdomen protocol
36.9.1	Abd 40.5-55.0kg (89.3-121.3lbs)		Helical	LargeBody	1.375:1 55.0	0.4	5	40	120	140	30	Std Plus	30	3.23	80.01	200	32	Weight and height based routine abdomen protocol

Miscellaneous

Table 5-5

[Interval]* - For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy-cm) ASiR	Scan Length (mm)	Phantom (cm)	Description
40.1	Pink Zone Routine 0-20 lbs/0-9 kg		Helical	PedBody	1.375:1 55.0	0.5	5	40	80	SmartmA 65-130mA NI=5 Avg mA 70	0	Std	0	0.61	8.70	95	32	Pediatric and Small patient Routine procedure based protocols
40.2	Pink Zone Routine 21-60 lbs/9.1-27.2 kg		Helical	PedBody	1.375:1 55.0	0.5	5	40	100	SmartmA 80-160mA NI=7 Avg mA 70	0	Std	0	1.13	27.47	195	32	Pediatric and Small patient Routine procedure based protocols
40.3	Pink Zone Routine 61-100 lbs/27.3-45.4 kg		Helical	Medium Body	1.375:1 55.0	0.5	5	40	120	SmartmA 95-190mA NI=10 Avg mA 70	0	Std	0	2.09	71.81	295	32	Pediatric and Small patient Routine procedure based protocols
40.4	Pink Zone Routine 101-200 lbs/45.4-90.7 kg		Helical	LargeBody	1.375:1 55.0	0.5	5	40	120	SmartmA 110-220mA NI=12 Avg mA 70	0	Std	0	2.02	89.30	395	32	Pediatric and Small patient Routine procedure based protocols
40.5	Pink Zone Routine >200 lbs/>90.8 kg		Helical	LargeBody	1.375:1 55.0	0.5	5	40	120	SmartmA 125-300mA NI=15 Avg mA 70	0	Std	0	2.02	89.30	395	32	Pediatric and Small patient Routine procedure based protocols
40.7	Green Zone Lower Dose 0-20 lbs/0-9 kg		Helical	PedBody	1.375:1 55.0	0.5	5	40	80	SmartmA 50-100mA NI=7 Avg mA 70	0	Std	0	0.61	8.70	95	32	Pediatric and Small patient Lower Dose procedure based protocols
40.8	Green Zone Lower Dose 21-60 lbs/9.1-27.2 kg		Helical	PedBody	1.375:1 55.0	0.5	5	40	100	SmartmA 60-120mA NI=9 Avg mA 70	0	Std	0	1.13	27.47	195	32	Pediatric and Small patient Lower Dose procedure based protocols
40.9	Green Zone Lower Dose 61-100 lbs/ 27.3-45.4 kg		Helical	Medium Body	1.375:1 55.0	0.5	5	40	120	SmartmA 70-140mA NI=11 Avg mA 70	0	Std	0	2.09	71.81	295	32	Pediatric and Small patient Lower Dose procedure based protocols
40.10	Green Zone Lower Dose 101-200 lbs/ 45.4-90.7 kg		Helical	LargeBody	1.375:1 55.0	0.5	5	40	120	SmartmA 80-160mA NI=13 Avg mA 70	0	Std	0	2.02	89.30	395	32	Pediatric and Small patient Lower Dose procedure based protocols
40.11	Green Zone Lower Dose >200 lbs/ >90.8 kg		Helical	LargeBody	1.375:1 55.0	0.5	5	40	120	SmartmA 90-240mA NI=16 Avg mA 70	0	Std	0	2.02	89.30	395	32	Pediatric and Small patient Lower Dose procedure based protocols
40.13	Grey Zone CT Angio 0-20 lbs/0-9 kg		Helical	PedBody	1.375:1 55.0	0.5	2.5	40	80	SmartmA 100-200mA NI=5 Avg mA 140	0	Std	0	1.22	17.71	97.5	32	Pediatric and Small patient Angio procedure based protocols

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (mm)	Phantom (cm)	Description
40.14	Grey Zone CT Angio 21-60 lbs/9.1-27.2 kg		Helical	SmallBody	1.375:1 55.0	0.5	2.5	40	100	SmartmA 120-240mA NI=9.9 Avg mA 140	0	Std	0	2.26	55.50	197.5	32	Pediatric and Small patient Angio procedure based protocols
40.15	Grey Zone CT Angio 61-100 lbs/ 27.3-45.4 kg		Helical	LargeBody	1.375:1 55.0	0.5	2.5	40	120	SmartmA 120-240mA NI=10 Avg mA 140	0	Std	0	4.03	139.28	297.5	32	Pediatric and Small patient Angio procedure based protocols
40.16	Grey Zone CT Angio 101-200 lbs/ 45.4-90.7 kg		Helical	LargeBody	1.375:1 55.0	0.5	2.5	40	120	SmartmA 120-240mA NI=12 Avg mA 140	0	Std	0	4.03	179.62	397.5	32	Pediatric and Small patient Angio procedure based protocols
40.17	Grey Zone CT Angio >200 lbs/ >90.8 kg		Helical	LargeBody	1.375:1 55.0	0.5	2.5	40	120	SmartmA 120-300mA NI=15 Avg mA 140	0	Std	0	4.03	179.62	397.5	32	Pediatric and Small patient Angio procedure based protocols

Reference Adult Protocols - Installed

Options : ASiR and Sub 0.4sec

Head

Table 6-1

[Interval]* - For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (s)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (mm)	Phantom (cm)	Description
21.1	Routine Head 1s Group1		Axial	Head	4i	1	5	20	140	120	40	Std	40	32.00	192.01	55	16	Routine head protocol for evaluation of the brain for abnormalities.
	Group2		Axial	Head	4i	1	5	20	120	190	40	Std	40	36.47	291.78	75	16	
21.2	Routine Head 2s Group1		Axial	Head	4i	2	5	20	140	60	40	Std	40	32.00	192.01	55	16	Routine head protocol for evaluation of the brain for abnormalities.
	Group2		Axial	Head	4i	2	5	20	120	95	40	Std	40	36.47	291.78	75	16	
21.3	Trauma Head 2.5mm		Axial	Head	8i	0.5	2.5	20	140	310	40	Std	40	41.34	578.70	137.5	16	Emergency head protocol for evaluation of the brain and cranium for abnormalities.
21.4	Routine Head 0.5s Group1		Axial	Head	4i	0.5	5	20	140	240	40	Std	40	32.00	192.01	55	16	Routine head protocol for evaluation of the brain for abnormalities.
	Group2		Axial	Head	4i	0.5	5	20	120	360	40	Std	40	35.52	284.17	75	16	
21.5	Circle of Willis 0.4s	MIROI	Axial	Head	1i	1	5 [0]	5	120	50	0	Std	0	125.28	62.79	0	16	Test bolus for CT angiography of the head for evaluation of carotid and cerebral vasculature.
			Helical	Head	0.531:1 10.625	0.4	0.625	20	120	180	40	Std Plus IQE	50	26.02	252.82	80	16	CT angiography of the head for evaluation of carotid and cerebral vasculature.
21.6	CoW/ Carotid 0.625mm 0.4s.Smart mA	MIROI	Axial	Head	1i	1	5 [0]	5	120	80	0	Std	0	180.84	90.42	0	16	Test bolus for CT angiography of the head for evaluation of carotid and cerebral vasculature.
			Helical	Head	0.531:1 10.625	0.4	0.625	20	120	SmartmA 100-600mA NI=8 Avg mA 415	30	Std Plus IQE	40	61.66	1155.86	170	16	CT angiography of the head for evaluation of carotid and cerebral vasculature.
21.8	Helical Head	DMPR	Helical	Head	0.531:1 10.625	0.5	0.625	20	120	180	40	Std Plus IQE	40	32.52	527.31	145	16	Helical scan mode for evaluation of the brain for cerebral abnormality.
21.10	VHS Head 4D CTA	MIROI	Axial	Head	1i	1	5 [0]	5	120	50	0	Std	0	125.58	62.79	0	16	Test bolus for CT angiography of the head for evaluation of carotid and cerebral vasculature.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (s)	Slice Thickness (mm) [Interval] J* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (mm)	Phantom (cm)	Description
			Helical-S	Head	0.984:1 39.375 18 pass	0.4	5 [10]	40	120	150	50	Std Plus	50	248.10	3101.19	110	16	CT angiography of the head using Volume Helical scan mode for evaluation of cerebral vasculature.
21.11	VHS Perfusion 350-370 Strength Contrast		Axial	Head	4i	1	5	20	120	200	40	Std	40	38.39	537.48	135	16	Non-Enhance Brain, evaluation for hemorrhage or infarction.
		Perfusion	Helical-S/P	Head	0.984:1 39.375 28 pass	0.4	5	40	80	200	0	Std	50	186.02	2325.25	110	16	CT Perfusion using Volume Helical scan mode for evaluation of cerebral perfusion over time.
			Axial	Head	4i	1	5	20	120	200	40	Std	40	38.39	537.48	135	16	Post Contrast Brain, Evaluation for residual tissue enhancement.
21.12	CT Perfusion 300-320 Strength Contrast		Axial	Head	4i	1	5	20	120	200	40	Std	40	38.39	537.48	135	16	Non-Enhance Brain, evaluation for hemorrhage or infarction.
		Perfusion	Cine	Head	8i	1	5 [0]	40	80	200	0	Std	0	652.82	2611.27	35	16	CT Perfusion using Cine scan mode for evaluation of cerebral perfusion over time.
			Axial	Head	4i	1	5	20	120	200	40	Std	40	38.39	537.48	135	16	Non-Enhance Brain, evaluation for hemorrhage or infarction.
21.13	CT Perfusion 350-370 Strength Contrast		Axial	Head	4i	1	5	20	120	200	40	Std	40	38.39	537.48	135	16	Non-Enhance Brain, evaluation for hemorrhage or infarction.
		Perfusion	Cine	Head	8i	1	5 [0]	40	80	200	0	Std	0	587.55	2350.19	35	16	CT Perfusion using Cine scan mode for evaluation of cerebral perfusion over time.
			Axial	Head	4i	1	5	20	120	200	40	Std	40	38.39	537.48	135	16	Post Contrast Brain, Evaluation for residual tissue enhancement.
21.14	CT Perfusion Brain Tumor - Dual Phase		Axial	Head	4i	1	5	20	120	200	40	Std	40	38.39	537.48	135	16	Non-Enhance Brain, evaluation for hemorrhage or infarction.
	Group 1	Perfusion	Cine	Head	8i	1	5 [0]	40	80	200	0	Std	0	587.55	2350.19	35	16	CT Perfusion using Cine scan mode for evaluation of cerebral perfusion over time.
	Group 2	Perfusion	Axial	Head	8i	1	5 [0]	40	80	200	0	Std	0	130.54	522.15	35	16	Delayed phase CT Perfusion using Axial scan mode for evaluation of Brain tumor permeability.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (s)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (mm)	Phantom (cm)	Description
			Axial	Head	4i	1	5	20	120	200	40	Std	40	38.39	537.48	135	16	Post Contrast Brain, Evaluation for residual tissue enhancement.
21.15	CT Perfusion 350-370 Strength Contrast (Axial)		Axial	Head	4i	1	5	20	120	200	40	Std	40	38.39	537.48	135	16	Non-Enhance Brain, evaluation for hemorrhage or infarction.
		Perfusion	Axial	Head	8i	1	5 [0]	40	80	200	0	Std	0	287.18	1148.73	35	16	CT Perfusion using Axial scan mode for evaluation of cerebral perfusion over time.
			Axial	Head	4i	1	5	20	120	200	40	Std	40	38.39	537.48	135	16	Post Contrast Brain, Evaluation for residual tissue enhancement.
21.16	CT Perfusion 350-370 Strength Contrast (Axial Shuttle Mode)		Axial	Head	4i	1	5	20	120	200	40	Std	40	38.39	537.48	135	16	Non-Enhance Brain, evaluation for hemorrhage or infarction.
		Perfusion	Axial-S	Head	8i 17 pass	0.4	5	40	80	500	0	Std	0	221.91	1775.31	75	16	CT Perfusion using Volume Shuttle (Axial) scan mode for evaluation of cerebral perfusion over time.
			Axial	Head	4i	1	5	20	120	200	40	Std	40	38.39	537.48	135	16	Post Contrast Brain, Evaluation for residual tissue enhancement.

Orbit

Table 6-2

[Interval]* - For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy-cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
22.1	Sinus Supine Helical + DMPR	DMPR	Helical	Head	0.531:1 10.625	0.4	0.625	20	120	85	30	Bone Plus IQE	50	12.29	112.48	74.375	16	Routine Helical Sinus protocol, Evaluation for any abnormality
22.3	Orbits Helical + DMPR	DMPR	Helical	Head	0.531:1 10.625	0.5	0.625	20	120	155	30	Std Plus IQE	40	28.00	221.29	61.875	16	Routine Helical Orbit protocol, Evaluation for abnormality soft tissues and bone structures of the face.
22.6	Helical IAC 0.625mm Axial mode - Use Reformat for Coronal		Helical	Head	0.531:1 10.625	0.6	0.625	20	140	190	30	Std Plus IQE	40	57.23	319.79	38.75	16	Axial scan mode, Evaluation for inner ear abnormalities in Axial plane
22.8	Axial and Coronal IAC		Axial	Head	16i	1	1.25	20	140	195	30	Std	40	52.00	208.01	38.75	16	Helical scan mode, Evaluation for inner ear abnormalities in Axial plane
			Axial	Head	16i	1	1.25	20	140	235	30	Std	40	62.67	250.68	38.75	16	Helical scan mode, Evaluation for inner ear abnormalities in Coronal plane

Neck

Table 6-3

[Interval]* - For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval] (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
23.1	C-Spine C5-C7 Axial		Axial	SmallBody	8i	0.5	2.5	20	140	410	20	Std	40	22.29	312.12	137.5	32	C-Spine C5-C7 area Axial mode, Evaluation for soft tissue, disc and bone abnormalities.
23.2	C-Spine C1-C4 Axial		Axial	SmallBody	8i	1	2.5	20	120	190	20	Std	40	14.17	113.36	77.5	32	C-Spine C1-C4 area Axial mode, Evaluation for soft tissue, disc and bone abnormalities.
23.3	C-Spine Helical + DMPR	DMPR	Helical	SmallBody	0.516:1 20.625	0.8	0.625	40	120	270	20	Bone Plus IQE	40	29.20	421.46	110	32	C-Spine Helical mode with DMPR, Evaluation for soft tissue, disc and bone abnormalities.
23.4	C-Spine Helical + DMPR/ SmartmA	DMPR	Helical	SmallBody	0.516:1 20.625	0.8	0.625	40	120	SmartmA 100-600mA NI=11.34 Avg mA 250	20	Bone Plus IQE	40	27.04	390.24	110	32	C-Spine Helical mode with DMPR and SmartmA, Evaluation for soft tissue, disc and bone abnormalities.
23.8	Carotid 0.625mm 0.4s. 0.5:1 SmartmA/ SmartPrep		Helical	SmallBody	0.516:1 20.625	0.4	0.625	40	140	SmartmA 100-515mA NI=11 Avg mA 310	40	Std Plus IQE	50	23.78	486.15	170	32	Carotid CTA (pitch 0.5:1) with SmartmA and SmartPrep, Evaluation of carotid vasculature.
23.9	Carotid 0.625mm 0.4s. 0.9:1 SmartmA/ SmartPrep		Helical	SmallBody	0.984:1 39.375	0.4	0.625	40	120	SmartmA 80-600mA NI=11 Avg mA 360	40	Std Plus IQE	50	10.20	207.25	170	32	Carotid CTA (pitch 0.9:1) with SmartmA and SmartPrep, Evaluation of carotid vasculature.
23.10	Carotid/ CoW 0.625mm 0.4s. AutomA	MIROI	Axial	SmallBody	1i	1	5 [0]	5	120	80	0	Std	0	99.00	49.50	0	32	Test bolus for Carotid CTA (pitch 0.5:1) of the head for evaluation of carotid and cerebral vasculature.
			Helical	SmallBody	0.516:1 20.625	0.4	0.625	40	140	AutomA 80-515mA NI=11 Avg mA 310	40	Std Plus IQE	50	23.78	486.15	170	32	Carotid and Brain CTA (pitch 0.5:1) with AutomA, Evaluation of carotid and cerebral vasculature.
23.11	Carotid/ CoW 0.625mm 0.4s. Smart mA/ SmartPrep		Helical	SmallBody	0.969:1 19.375	0.4	0.625	20	140	SmartmA 100-515mA NI=11 Avg mA 310	40	Std Plus IQE	50	13.54	255.40	170	32	Carotid CTA (pitch 0.9:1) with SmartmA and SmartPrep, Evaluation of carotid and cerebral vasculature.
23.12	Soft Tissue Neck 2.5mm AutomA		Helical	SmallBody	0.984:1 39.375	0.5	2.5	40	120	AutomA 100-335mA NI=9.1 Avg mA 175	30	Std Plus	30	6.20	113.26	150	32	Neck with AutomA, Evaluation of soft tissue structures for abnormality

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
23.13	Soft Tissue Neck + DMPR/ SmartmA	DMPR	Helical	SmallBody	0.969:1 19.375	0.5	0.625	20	120	SmartmA 100-600mA NI=12.6 Avg mA 255	40	Std Plus IQE	50	9.82	165.46	150	32	Neck with DMPR and SmartmA, Evaluation of soft tissue structures for abnormality

Upper Extremity

Table 6-4

[Interval]* - For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
24.1	Shoulder 1.25mm + DMPR	DMPR	Helical	LargeBody	0.516:1 20.625	0.6	1.25 [0.625]	40	140	310	40	Std Plus IQE	40	40.50	544.18	100	32	Shoulder with DMPR, Evaluation of soft tissue and bone of shoulder area
24.2	Shoulder 1.25mm + DMPR/ SmartmA	DMPR	Helical	LargeBody	0.516:1 20.625	0.6	1.25 [0.625]	40	140	SmartmA 100-515mA NI=20 Avg mA 310	40	Std Plus IQE	40	40.50	544.18	100	32	Shoulder with DMPR and SmartmA, Evaluation of soft tissue and bone of shoulder area

Chest

Table 6-5

[Interval]* - For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
25.1	Routine Chest 0.4s. 5mm SmartmA		Helical	LargeBody	1.375:1 55.0	0.4	5	40	120	SmartmA 100-600mA NI=13 Avg mA 250	50	Std Plus	50	5.76	142.87	200	32	Chest with SmartmA, Evaluation of mediastinum and lungs fields
25.2	Routine Chest 0.4s.+ DMPR/ SmartmA	DMPR	Helical	LargeBody	0.984:1 39.375	0.4	0.625	40	120	SmartmA 100-600mA NI=22 Avg mA 300	50	Std Plus	50	9.66	224.97	200	32	Chest with SmartmA and DMPR, Evaluation of mediastinum and lungs fields
25.3	Chest with HiRes 0.4s. SmartmA		Helical	LargeBody	0.969:1 19.375	0.4	5	20	120	SmartmA 80-600mA NI=12 Avg mA 300	40	Chest Plus	40	10.50	229.59	200	32	Chest with SmartmA, Evaluation of mediastinum and lung fields with higher resolution images of the lungs
25.4	Hi Res Chest 0.5s. Axial		Axial	LargeBody	1i	0.5	1.25 [10]	1.25	120	190	50	Bone+	50	2.22	53.35	230	32	Axial scan mode, Evaluation of mediastinum and lung fields in high resolution mode
25.5	Chest Abd Pelvis 0.4/ 0.5s. SmartmA 5mm (2 groups) Group1		Helical	LargeBody	0.984:1 39.375	0.4	5	40	120	SmartmA 100-600mA NI=13 Avg mA 185	50	Std Plus	50	5.96	108.95	150	32	Multi group scan for Chest, Abdomen, Pelvis (0.4/0.5 sec) with SmartmA, Evaluation of chest and abdominal anatomy for abnormalities
	Group2		Helical	LargeBody	1.375:1 55.0	0.5	5	40	120	SmartmA 100-600mA NI=13 Avg mA 185	50	Std Plus	50	5.33	182.72	295	32	
25.6	Chest Abd Pelvis 0.5s + DMPR SmartPrep	DMPR	Helical	LargeBody	1.375:1 55.0	0.5	0.625	40	120	SmartmA 100-600mA NI=22 Avg mA 270	50	Std Plus	50	7.78	387.24	450	32	Helical scan mode (0.5sec) with SmartmA, Evaluation of chest and abdominal anatomy for abnormalities
25.7	Pulmonary Embolis 0.5s.	MIROI	Axial	LargeBody	1i	1	5 [0]	5	120	80	0	Std	0	93.89	46.95	0	32	Timing Bolus for Helical scan mode 1.25mm thickness and helical pitch of 0.9:1, evaluation of Chest for pulmonary embolism
			Helical	LargeBody	0.984:1 39.375	0.5	1.25	40	120	350	40	Std	40	14.09	300.25	180	32	Helical scan mode 1.25mm thickness and helical pitch of 0.9:1, Evaluation Chest for pulmonary embolism

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy-cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
25.8	Pulmonary Embolism 0.5s. SmartmA	MIROI	Axial	LargeBody	1i	1	5 [0]	5	120	80	0	Std	0	93.89	46.95	0	32	Timing Bolus for Helical scan mode 1.25mm thickness and helical pitch of 0.9:1 with SmartmA, evaluation of Chest for pulmonary embolism
			Helical	LargeBody	0.984:1 39.375	0.5	1.25	40	120	SmartmA 100-600mA NI=20 Avg mA 300	40	Std	40	12.07	256.93	180	32	Helical scan mode 1.25mm thickness and helical pitch of 0.9:1 with SmartmA, Evaluation Chest for pulmonary embolism
25.9	Aortic Dissection 0.625mm + DMPR/ SmartmA/ SmartPrep	DMPR	Helical	LargeBody	1.375:1 55.0	0.5	0.625	40	120	SmartmA 100-600mA NI=22 Avg mA 360	40	Std Plus	40	10.37	361.20	300	32	Helical scan mode 0.625mm thickness and helical pitch of 1.375:1 with SmartmA, SmartPrep and DMPR, Evaluation for aortic dissection
25.10	Aortic Dissection 1.25mm + DMPR/ SmartmA/ SmartPrep	DMPR	Helical	LargeBody	1.375:1 55.0	0.5	1.25 [0.8]	40	120	SmartmA 100-600mA NI=20 Avg mA 330	40	Std Plus	40	9.51	330.68	300	32	Helical scan mode 1.25mm thickness and helical pitch of 1.375:1 with SmartmA, SmartPrep and DMPR, Evaluation for aortic dissection
25.11	Lung Assessment	Advanced Lung Analysis	Helical	LargeBody	0.969:1 19.375	0.5	1.25	20	120	120	0	Bone	0	5.25	114.77	200	32	Helical scan mode for acquisition of data for Advanced Lung Analysis (ALA) software for lung nodule assessment
25.12	Gated Chest Cardiac Helical		Helical	LargeBody	1.375:1 55.0	0.4	5	40	120	60	0	Std Plus	0	1.38	13.55	50	32	Localizer scan for Helical Gated scan for evaluation of thoracic vasculature
		MIROI	Axial	LargeBody	1i	1	5 [0]	5	120	60	0	Std	0	98.59	49.29	0	32	Timing bolus scan for Helical Gated scan for evaluation of thoracic vasculature
			Cardiac SEG	CardiacLarge	0.160:1 6.4	0.35	0.625	40	120	ECG Mod 110-270mA Phase 70-80 Avg mA 270	40	Std SS Seg	40	30.98	1037.93	300	32	Helical Gated scan for evaluation of thoracic vasculature
25.13	Gated Chest SnapShot Pulse		Helical	LargeBody	1.375:1 55.0	0.4	5	40	120	60	0	Std Plus	0	1.38	13.55	50	32	Localizer scan for Cine Gated scan for evaluation of thoracic vasculature
		MIROI	Axial	LargeBody	1i	1	5 [0]	5	120	60	0	Std	0	98.59	49.29	0	32	Timing bolus scan for Cine Gated scan for evaluation of thoracic vasculature
			Cardiac Pulse Cine	CardiacLarge	56i	0.35	0.625 [35]	40	120	270	40	Std SS Seg	40	14.42	454.39	314.375	32	Cine Gated scan for evaluation of thoracic vasculature
25.14	SmartScore - Gated 0.35s	Smart Score	Cine	LargeBody	8i	0.35	2.5	20	120	430	0	Std Segment	0	8.99	89.95	97.5	32	Cine gated scan mode for acquisition of data for coronary artery calcification analysis

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy-cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
25.15	SnapShot Pulse 30-65 BPM		Helical	LargeBody	1.375:1 55.0	0.4	5	40	120	60	0	Std Plus	0	1.38	13.55	50	32	Localizer scan for Prospectively gated Cardiac Cine scan for evaluation of coronary arteries, intended for stable heart rates 30-65 bpm
		MIROI	Axial	LargeBody	1i	1	5 [0]	5	120	60	0	Std	0	98.59	49.29	0	32	Timing bolus scan for Prospectively gated Cardiac Cine scan for evaluation of coronary arteries, intended for stable heart rates 30-65 bpm
			Cardiac Pulse Cine	CardiacSmall	56i	0.35	0.625 [35]	40	120	270	40	Std SS Seg	40	12.69	133.24	104.375	32	Prospectively gated Cardiac Cine scan for evaluation of coronary arteries, intended for stable heart rates 30-65 bpm
25.16	SnapShot Segment 30-74 BPM		Helical	LargeBody	1.375:1 55.0	0.4	5	40	120	60	0	Std Plus	0	1.38	13.55	50	32	Localizer scan for Retrospectively gated Cardiac Helical scan for evaluation of coronary arteries, intended for heart rates 30-74 bpm
		MIROI	Axial	LargeBody	1i	1	5 [0]	5	120	60	0	Std	0	98.59	49.29	0	32	Timing bolus scan for Retrospectively gated Cardiac Helical scan for evaluation of coronary arteries, intended for heart rates 30-74 bpm
			Cardiac SEG	CardiacLarge	0.160:1 6.4	0.35	0.625	40	120	ECG Mod 120-270mA Phase 70-80 Avg mA 270	40	Std SS Seg	40	33.25	448.83	100	32	Retrospectively gated Cardiac Helical scan for evaluation of coronary arteries, intended for heart rates 30-74 bpm
25.17	SnapShot Burst 75-113 BPM		Helical	LargeBody	1.375:1 55.0	0.4	5	40	120	60	0	Std Plus	0	1.38	13.55	50	32	Localizer scan for Retrospectively gated Cardiac Helical scan for evaluation of coronary arteries, intended for heart rates 75-113 bpm
		MIROI	Axial	LargeBody	1i	1	5 [0]	5	120	60	0	Std	0	98.59	49.29	0	32	Timing bolus scan for Retrospectively gated Cardiac Helical scan for evaluation of coronary arteries, intended for heart rates 75-113 bpm
			Cardiac SSB	CardiacLarge	0.160:1 6.4	0.35	0.625	40	120	ECG Mod 120-270mA Phase 70-80 Avg mA 270	40	Std SSB	40	33.25	448.83	100	32	Retrospectively gated Cardiac Helical scan for evaluation of coronary arteries, intended for heart rates 75-113 bpm
25.18	SnapShot Burst Plus 114+ BPM		Helical	LargeBody	1.375:1 55.0	0.4	5	40	120	60	0	Std Plus	0	1.38	13.55	50	32	Localizer scan for Retrospectively gated Cardiac Helical scan for evaluation of coronary arteries, intended for heart rates over 113 bpm

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy-cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
		MIROI	Axial	LargeBody	1i	1	5 [0]	5	120	60	0	Std	0	98.59	49.29	0	32	Timing bolus scan for Retrospectively gated Cardiac Helical scan for evaluation of coronary arteries, intended for heart rates over 113 bpm
			Cardiac SSBP	CardiacLarge	0.160:1 6.4	0.35	0.625	40	120	ECG Mod 120-270mA Phase 70-80 Avg mA 270	40	Std SSB	40	33.25	448.83	100	32	Retrospectively gated Cardiac Helical scan for evaluation of coronary arteries, intended for heart rates over 113 bpm
25.19	Advantage 4D		Helical	LargeBody	1.375:1 55.0	0.6	5	40	120	170	40	Std Plus	40	5.88	145.59	200	32	Localizer scan for Advantage 4D series
		Advantage 4D	Cine	LargeBody	8i	1	2.5	20	120	70	30	Std	30	35.67	570.67	157.5	32	Cine scan mode for acquisition data for Advantage 4D for retrospective gating
25.20	VHS Chest 4D CTA SmartPrep		Helical-S	LargeBody	1.375:1 55.0 11 pass	0.4	5 [10]	40	120	SmartmA 150-500mA NI=14 Avg mA 210	40	Std Plus	40	67.29	2119.57	300	32	Volume Helical scan mode with SmartPrep for time resolved CT angiography of chest vasculature

Abdomen

Table 6-6

[Interval]* - For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
26.1	Abdomen Pelvis 0.5s. 5mm SmartmA		Helical	LargeBody	1.375:1 55.0	0.5	5	40	120	SmartmA 50-600mA NI=11.57 Avg mA 250	50	Std Plus	50	7.2	322.54	400	32	Basic Abdomen Pelvis (0.5sec) with Smart mA, Evaluation for abnormalities of abdominal and pelvic structures
26.2	Abdomen Pelvis 0.6s. 5mm + DMPR SmartmA/ SmartPrep	DMPR	Helical	LargeBody	0.984:1 39.375	0.6	0.625	40	120	SmartmA 80-600mA NI=18 Avg mA 260	50	Std Plus	50	12.56	543.37	400	32	Basic Abdomen Pelvis (0.6sec) with Smart mA, SmartPrep and DMPR, Evaluation for abnormalities of abdominal and pelvic structures
26.4	Renal Stone 0.5s. 5mm		Helical	LargeBody	1.375:1 55.0	0.5	5	40	120	190	50	Std Plus	50	5.47	245.13	400	32	Helical scan, Evaluation of kidney, ureters and bladder for renal stones
26.5	Renal Stone 0.5s. 5mm SmartmA		Helical	LargeBody	1.375:1 55.0	0.5	5	40	120	SmartmA 50-600mA NI=12 Avg mA 190	50	Std Plus	50	5.47	245.13	400	32	Helical scan with SmartmA, Evaluation of kidney, ureters and bladder for renal stones
26.7	AAA 0.5s. + DMPR 0.625mm	MIROI	Axial	LargeBody	1i	1	5 [0]	5	120	80	0	Std	0	112.67	56.34	0	32	Timing bolus for Helical scan with DMPR, Evaluation for abdominal aortic aneurysm
		DMPR	Helical	LargeBody	1.375:1 55.0	0.5	0.625	40	140	255	50	Std Plus	50	10.41	371.80	309.375	32	Helical scan with DMPR, Evaluation for abdominal aortic aneurysm
26.8	AAA 0.5s. + DMPR 0.625mm SmartmA/ SmartPrep	DMPR	Helical	LargeBody	1.375:1 55.0	0.6	0.625	40	120	SmartmA 100-600mA NI=19 Avg mA 275	50	Std Plus	50	9.51	339.51	309.375	32	Helical scan with DMPR, Smart mA and SmartPrep, Evaluation for abdominal aortic aneurysm
26.10	Dual Phase Liver		Helical	LargeBody	1.375:1 55.0	0.6	5	40	120	150	50	Std Plus	50	5.19	92.16	130	32	Non contrast evaluation of upper abdominal structures including liver
		MIROI	Axial	LargeBody	1i	1	5 [0]	5	120	80	0	Std	0	112.67	56.34	0	32	Timing bolus for Multi-Group acquisition for early arterial and portal venous phase of the liver
	Group1		Helical	LargeBody	1.375:1 55.0	0.5	2.5	40	120	275	50	Std Plus	50	7.92	140.87	130	32	Acquisition for early arterial phase of the liver
	Group2		Helical	LargeBody	1.375:1 55.0	0.5	5	40	120	210	50	Std Plus	50	6.05	270.94	400	32	Acquisition for portal venous phase of the liver

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
26.11	Tri Phase Liver		Helical	LargeBody	1.375:1 55.0	0.6	5	40	120	150	50	Std Plus	50	5.19	92.16	130	32	Non contrast evaluation of upper abdominal structures including liver
		MIROI	Axial	LargeBody	1i	1	5 [0]	5	120	80	0	Std	0	112.67	56.34	0	32	Timing bolus for Multi-Group acquisition for early arterial, late arterial and portal venous phase of the liver
	Group1		Helical	LargeBody	1.375:1 55.0	0.5	2.5	40	120	275	50	Std Plus	50	7.92	140.87	130	32	Acquisition for early arterial phase of the liver
	Group2		Helical	LargeBody	1.375:1 55.0	0.5	5	40	120	225	50	Std Plus	50	6.48	290.29	400	32	Acquisition for late arterial phase of the liver
	Group3		Helical	LargeBody	1.375:1 55.0	0.5	5	40	120	225	50	Std Plus	50	6.48	290.29	400	32	Acquisition for portal venous phase of the liver
26.12	Dual Phase Pancreas SmartmA/ SmartPrep		Helical	LargeBody	0.984:1 39.375	0.5	5	40	120	SmartmA 80-600mA NI=12 Avg mA 200	50	Std Plus	50	8.05	82.74	70	32	Non contrast evaluation of upper abdominal structures including pancreas
	Group1		Helical	LargeBody	0.984:1 39.375	0.5	2.5	40	120	SmartmA 50-600mA NI=15 Avg mA 260	50	Std Plus	50	10.46	191.28	150	32	Acquisition with SmartmA and SmartPrep for arterial phase of the pancreas
	Group2		Helical	LargeBody	0.984:1 39.375	0.5	5	40	120	SmartmA 50-600mA NI=15 Avg mA 225	50	Std Plus	50	9.06	319.47	320	32	Acquisition for venous phase of the pancreas
26.13	Dual Phase Liver SmartmA/ SmartPrep		Helical	LargeBody	1.375:1 55.0	0.5	5	40	120	SmartmA 80-500mA NI=12 Avg mA 160	50	Std Plus	50	4.61	81.96	130	32	Non contrast evaluation of upper abdominal structures including liver
	Group1		Helical	LargeBody	1.375:1 55.0	0.5	2.5	40	120	SmartmA 50-600mA NI=15 Avg mA 275	50	Std Plus	50	7.92	140.87	130	32	Acquisition with SmartmA and SmartPrep for early arterial phase of the liver
	Group2		Helical	LargeBody	0.984:1 39.375	0.5	5	40	120	SmartmA 50-600mA NI=11 Avg mA 225	50	Std Plus	50	9.06	391.91	400	32	Acquisition with SmartmA and SmartPrep for portal venous phase of the liver
26.14	Tri Phase Liver SmartmA/ SmartPrep		Helical	LargeBody	0.984:1 39.375	0.5	5	40	120	SmartmA 80-500mA NI=12 Avg mA 200	50	Std Plus	50	8.05	131.04	130	32	Non contrast evaluation of upper abdominal structures including liver
	Group1		Helical	LargeBody	1.375:1 55.0	0.5	2.5	40	120	SmartmA 50-600mA NI=15 Avg mA 275	50	Std Plus	50	7.92	140.87	130	32	Acquisition with SmartmA and SmartPrep for early arterial phase of the liver
	Group2		Helical	LargeBody	1.375:1 55.0	0.5	5	40	120	SmartmA 50-600mA NI=11 Avg mA 225	50	Std Plus	50	6.48	290.29	400	32	Acquisition with SmartmA and SmartPrep for late arterial phase of the liver
	Group3		Helical	LargeBody	1.375:1 55.0	0.5	5	40	120	SmartmA 50-600mA NI=11 Avg mA 225	50	Std Plus	50	6.48	290.29	400	32	Acquisition with SmartmA and SmartPrep for portal venous phase of the liver
26.15	VHS Liver Perfusion		Helical	LargeBody	0.984:1 39.375	0.6	5	40	120	85	50	Std	50	4.11	66.80	130	32	Non contrast evaluation of upper abdominal structures including liver

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy-cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
		Perfusion	Helical-S/P	LargeBody	1.375:1 55.0 21 pass	0.4	5	40	120	150	50	Std Plus	50	109.99	1704.89	140	32	Volume Helical scan mode for evaluation of liver tumor perfusion over time
26.16	VHS 4D CTA		Helical	LargeBody	0.984:1 39.375	0.6	5	40	120	170	50	Std	50	8.21	133.61	130	32	Localizer scan for Volume Helical Shuttle scan mode for evaluation of abdominal vasculature
		MIROI	Axial	LargeBody	1i	1	5 [0]	5	120	80	0	Std	0	112.67	56.34	0	32	Timing bolus scan for Volume Helical Shuttle scan mode for evaluation of abdominal vasculature
			Helical-S	LargeBody	1.375:1 55.0 18 pass	0.5	5 [10]	40	140	SmartmA 150-515mA NI=12 Avg mA 175	50	Std Plus	50	162.84	3175.35	180	32	Volume Helical Shuttle scan mode for time resolved CT angiography of abdominal vasculature
26.17	Runoff 0.625mm	MIROI	Axial	LargeBody	1i	1	5 [0]	5	120	80	0	Std	0	112.67	56.34	0	32	Timing bolus scan for evaluation of vascular structure of abdomen, femurs and lower extremities (Runoff)
			Helical	LargeBody	1.375:1 55.0	0.5	0.625	40	140	310	40	Std Plus	40	12.65	1246.09	936.875	32	Helical scanmode 0.625mm, Evaluation of vascular structure of abdomen, femurs and lower extremities (Runoff)
26.18	Runoff 1.25mm	MIROI	Axial	LargeBody	1i	1	5 [0]	5	120	80	0	Std	0	112.67	56.34	0	32	Timing bolus scan for evaluation of vascular structure of abdomen, femurs and lower extremities (Runoff)
			Helical	LargeBody	1.375:1 55.0	0.5	1.25 [0.7]	40	140	310	40	Std Plus	40	12.65	1324.58	998.9	32	Helical scanmode 1.25mm, Evaluation of vascular structure of abdomen, femurs and lower extremities (Runoff)
26.19	Runoff 0.625mm SmartmA/ SmartPrep		Helical	LargeBody	1.375:1 55.0	0.5	0.625	40	140	SmartmA 100-515mA NI=19 Avg mA 310	40	Std Plus	40	12.65	1246.09	936.875	32	Helical scanmode 0.625mm with SmartmA and SmartPrep, Evaluation of vascular structure of abdomen, femurs and lower extremities (Runoff)
26.20	Runoff 1.25mm SmartmA/ SmartPrep		Helical	LargeBody	1.375:1 55.0	0.5	1.25 [0.7]	40	140	SmartmA 100-515mA NI=20 Avg mA 310	40	Std Plus	40	12.65	1324.58	998.9	32	Helical scanmode 1.25mm with SmartmA and SmartPrep, Evaluation of vascular structure of abdomen, femurs and lower extremities (Runoff)
26.22	CT Colonography 1.25mm	Advantage CTC Pro	Helical	LargeBody	1.375:1 55.0	0.5	1.25	40	120	100	50	Std Plus	50	2.88	129.02	400	32	Helical scan mode supine acquisition for using Advantage CTC Pro for evaluation of the colon
		Advantage CTC Pro	Helical	LargeBody	1.375:1 55.0	0.5	1.25	40	120	100	50	Std Plus	50	2.88	129.02	400	32	Helical scan mode prone acquisition for using Advantage CTC Pro for evaluation of the colon

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy-cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
26.41	Body Interventional 5mm/3i		Helical	LargeBody	1.375:1 27.50	0.5	5	20	120	SmartmA 50-600mA NI=11.57 Avg mA 360	0	Std	0	11.41	251.58	195	32	Localization scan for CT interventional scan
			SmartView SEG	LargeBody	3i	0.4	5	10	120	80	0	Std Segment	0	684.58	684.58	0	32	CT Fluoro scan for interventional scan
26.42	Body Interventional 5mm/1i		Helical	LargeBody	1.375:1 27.50	0.5	5	20	120	SmartmA 50-600mA NI=11.57 Avg mA 360	0	Std	0	11.41	251.58	195	32	Localization scan for CT interventional scan
			SmartView SEG	LargeBody	1i	0.4	5	5	120	80	0	Std Segment	0	845.04	422.52	0	32	CT Fluoro scan for interventional scan

L-Spine

Table 6-7

[Interval]* - For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
27.1	L-Spine 3 Level Axial Group1		Axial	LargeBody	8i	1	2.5	20	120	225	30	Std	40	19.08	38.15	17.5	32	Axial scan mode, Evaluation of soft tissue and boney structures lumbar spine
	Group2		Axial	LargeBody	8i	1	2.5	20	120	225	30	Std	40	19.08	38.15	17.5	32	Axial scan mode, Evaluation of soft tissue and boney structures lumbar spine
	Group3		Axial	LargeBody	8i	1	2.5	20	140	250	30	Std	40	30.03	60.06	17.5	32	Axial scan mode, Evaluation of soft tissue and boney structures lumbar spine
27.3	L-Spine Survey Helical 2.5mm 0.6s. SmartmA		Helical	LargeBody	0.516:1 20.625	0.6	2.5	40	140	SmartmA 100-515mA NI=14 Avg mA 360	30	Std Plus	40	47.03	985.50	75	32	Helical scan mode 2.5mm (0.6sec), Limited evaluation of lumbar spine
27.4	L-Spine Survey Helical 2.5mm 0.8s. SmartmA		Helical	LargeBody	0.516:1 20.625	0.8	2.5	40	140	SmartmA 100-515mA NI=14 Avg mA 360	30	Std Plus	40	62.70	1313.78	75	32	Helical scan mode 2.5mm (0.8sec), Limited evaluation of lumbar spine
27.6	L-Spine Helical 0.625mm + DMPR/ SmartmA	DMPR	Helical	LargeBody	0.516:1 20.625	0.6	0.625	40	140	SmartmA 100-515mA NI=20 Avg mA 360	30	Std Plus IQE	40	47.03	985.50	75	32	Helical scan mode 0.625mm with DMPR and SmartmA, Evaluation of soft tissue and boney structures lumbar spine
27.7	L-Spine Helical 1.25mm + DMPR/ SmartmA - Large Patient	DMPR	Helical	LargeBody	0.516:1 20.625	1	1.25	40	140	SmartmA 100-515mA NI=18 Avg mA 360	30	Std Plus	40	78.38	1641.95	75	32	Helical scan mode 1.25mm with DMPR and SmartmA, Evaluation of soft tissue and boney structures lumbar spine

Pelvis

Table 6-8
[Interval]* - For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.
For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
28.1	Pelvis for Fracture + DMPR/ SmartmA	DMPR	Helical	LargeBody	0.984:1 39.375	0.8	1.25 [0.625]	40	120	SmartmA 100-600mA NI=20 Avg mA 300	40	Bone Plus IQE	40	19.32	258.70	101.25	32	Helical scan with DMPR and SmartmA, Evaluation of pelvis bony structures for fracture

Lower Extremity

Table 6-9

[Interval]* - For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
29.1	Lower Extremity Survey 2.5mm		Helical	LargeBody	0.516:1 20.625	0.5	2.5	40	120	175	30	Std Plus	30	13.45	126.94	60	32	Helical scan mode 2.5mm, Limited evaluation of lower extremity for abnormalities
29.2	Ankle 0.625 mm Single Joint		Helical	MediumBody	0.531:1 10.625	0.5	0.625	20	120	175	30	Std Plus IQE	40	14.50	75.64	35	32	Helical scan mode 0.625mm, Evaluation of soft tissue and bony anatomy of the ankle area
29.3	Ankle 0.625 mm Both Joints		Helical	LargeBody	0.531:1 10.625	0.5	0.625	20	120	210	30	Std Plus IQE	30	16.76	87.38	35	32	Helical scan mode 0.625mm, Evaluation of soft tissue and bony anatomy of the ankle area
29.4	Knee 0.625 mm Both Joints		Helical	LargeBody	0.531:1 10.625	0.6	0.625	20	120	AutomA 100-335mA NI=20 Avg mA 190	30	Std Plus IQE	30	18.19	94.84	35	32	Helical scan mode 0.625mm, Evaluation of soft tissue and bony anatomy of both knees

Reference Pediatric Protocols - Installed Options : ASiR and Sub 0.4 sec

Head

Table 7-1

[Interval]* - For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy-cm) ASiR	Scan Length (mm)	Phantom (cm)	Description
31.1	PED HEAD TO 18 Months		Axial	Ped Head	4i	0.5	5	20	120	170	30	Std	30	14.47	173.68	115	16	Routine head (0.5s) for infant (up to 18 months) 5mm
31.2	PED HEAD 18 MOS TO 5YRS		Axial	SmallHead	4i	0.5	5	20	120	235	30	Std	30	20.01	240.08	115	16	Routine head (0.5s) for children (18 months to 5 years) 5mm
31.3	PED HEAD TO 5YRS TO 18YRS Group1		Axial	SmallHead	8i	1	2.5	20	120	170	30	Std	30	28.95	115.79	37.5	16	Routine head (1.0s) for children (5 years to 18 years) 2.5mm
	Group2		Axial	SmallHead	4i	1	5	20	120	125	30	Std	30	21.28	170.27	75	16	Routine head (1.0s) for children (5 years to 18 years) 5mm

Orbit

Table 7-2

[Interval]* - For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy-cm) ASiR	Scan Length (mm)	Phantom (cm)	Description
32.1	Sinus Supine Helical + DMPR	DMPR	Helical	SmallHead	0.531:1 10.625	0.4	0.625	20	120	65	20	Bone Plus IQE	30	8.33	76.29	74.375	16	Routine Helical scan mode with DMPR, Evaluation of sinus area for abnormality in pediatric patients
32.3	Orbits Helical	DMPR	Helical	SmallHead	0.531:1 10.625	0.4	0.625	20	120	145	20	Std Plus IQE	30	18.59	146.96	61.875	16	Helical scan mode, Evaluation of orbital and face structures for abnormality in pediatric patients
32.6	Helical IAC 0.625mm Axial mode - Use Reformat for Coronal		Helical	SmallHead	0.531:1 10.625	0.4	0.625	20	120	130	20	Std Plus IQE	30	16.67	93.21	38.75	16	Helical scan mode, Evaluation of inner ear structures for abnormality in pediatric patients

Chest

Table 7-3

[Interval]* - For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy-cm) ASiR	Scan Length (mm)	Phantom (cm)	Description
35.1.1	RC 6.0-7.5 kg (13.2-16.5 lbs)		Helical	PedBody	1.375:1 27.5	0.4	3.75	20	100	55	30	Std Plus	30	0.76	7.62	75	32	Weight and height based routine chest protocol
35.1.3	SnapShot Segment 30-74 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	85	30	Std Plus	30	0.59	5.80	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	Std	0	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
			Cardiac SEG	CardiacSmall	0.160:1 6.4	0.35	0.625	40	80	ECG Mod 55-195mA Phase 40-80 AvgmA195	30	Std SS Seg	40	8.51	93.56	75	32	Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
35.1.4	SnapShot Burst 75-104 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	85	30	Std Plus	30	0.59	5.80	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	Std	0	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
			Cardiac SSB	CardiacSmall	0.160:1 6.4	0.35	0.625	40	80	ECG Mod 55-195mA Phase 40-80 AvgmA195	30	Std SSB	40	8.51	93.56	75	32	Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
35.1.5	SnapShot Burst+ >105 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	85	30	Std Plus	30	0.59	5.80	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	Std	0	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
			Cardiac SSBP	CardiacSmall	0.160:1 6.4	0.35	0.625	40	80	ECG Mod 55-195mA Phase 35-85 AvgmA195	30	Std SSB	40	9.13	100.42	75	32	Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
35.1.7	Non ECG Gated		Helical	PedBody	0.984:1 39.375	0.4	0.625	40	80	SmartmA 50-200mA NI=18 AvgmA140	30	Std Plus	30	1.36	14.70	75	32	Weight and height based non ECG gated protocol for cardiac with SmartmA
35.2.1	RC 7.5-9.5 kg (16.5-20.9lbs)		Helical	PedBody	1.375:1 27.5	0.4	3.75	20	100	85	30	Std Plus	30	1.18	11.77	75	32	Weight and height based routine chest protocol
35.2.3	SnapShot Segment 30-74 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	85	30	Std Plus	30	0.59	5.80	50	32	Weight and height based protocol for localization

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy-cm) ASiR	Scan Length (mm)	Phantom (cm)	Description
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	Std	0	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
			Cardiac SEG	CardiacSmall	0.160:1 6.4	0.35	0.625	40	80	ECG Mod 55-210mA Phase 40-80 AvgmA210	30	Std SS Seg	40	9.13	100.39	75	32	Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
35.2.4	SnapShot Burst 75-104 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	85	30	Std Plus	30	0.59	5.80	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	Std	0	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
			Cardiac SSB	CardiacSmall	0.160:1 6.4	0.35	0.625	40	80	ECG Mod 55-210mA Phase 40-80 AvgmA210	30	Std SSB	40	9.13	100.39	75	32	Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
35.2.5	SnapShot Burst+ >105 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	85	30	Std Plus	30	0.59	5.80	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	Std	0	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
			Cardiac SSBP	CardiacSmall	0.160:1 6.4	0.35	0.625	40	80	ECG Mod 55-210mA Phase 35-85 AvgmA210	30	Std SSB	40	9.82	107.99	75	32	Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
35.2.7	Non ECG Gated		Helical	PedBody	0.984:1 39.375	0.4	0.625	40	80	SmartmA 50-220mA NI=18 AvgmA155	30	Std Plus	30	1.51	16.28	75	32	Weight and height based non ECG gated protocol for cardiac with SmartmA
35.3.1	RC 9.5-11.5 kg (20.9-25.4lbs)		Helical	SmallBody	1.375:1 27.5	0.4	3.75	20	120	70	30	Std Plus	30	1.52	15.77	78.75	32	Weight and height based routine chest protocol
35.3.3	SnapShot Segment 30-74 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	85	30	Std Plus	30	0.59	5.80	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	Std	0	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
			Cardiac SEG	CardiacSmall	0.160:1 6.4	0.35	0.625	40	80	ECG Mod 70-225mA Phase 40-80 AvgmA225	30	Std SS Seg	40	9.88	108.64	75	32	Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
35.3.4	SnapShot Burst 75-104 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	85	30	Std Plus	30	0.59	5.80	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	Std	0	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy-cm) ASiR	DLP (mGy-cm) ASiR	Scan Length (mm)	Phantom (cm)	Description
			Cardiac SSB	CardiacSmall	0.160:1 6.4	0.35	0.625	40	80	ECG Mod 70-225mA Phase 40-80 AvgmA225	30	Std SSB	40	9.88	108.64	75	32	Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
35.3.5	SnapShot Burst+ >105 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	85	30	Std Plus	30	0.59	5.80	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	Std	0	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
			Cardiac SSBP	CardiacSmall	0.160:1 6.4	0.35	0.625	40	80	ECG Mod 70-225mA Phase 35-85 AvgmA225	30	Std SSB	40	10.57	116.23	75	32	Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
35.3.7	Non ECG Gated		Helical	PedBody	0.984:1 39.375	0.4	0.625	40	80	SmartmA 50-240mA NI=18 AvgmA170	30	Std Plus	30	1.65	17.85	75	32	Weight and height based non ECG gated protocol for cardiac with SmartmA
35.4.1	RC 11.5-14.5kg (25.4-32.0lbs)		Helical	SmallBody	1.375:1 27.5	0.4	5	20	120	65	30	Std Plus	30	1.41	16.23	90	32	Weight and height based routine chest protocol
35.4.3	SnapShot Segment 30-74 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	85	30	Std Plus	30	0.59	5.80	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	Std	0	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
			Cardiac SEG	CardiacSmall	0.160:1 6.4	0.35	0.625	40	80	ECG Mod 70-240mA Phase 40-80 AvgmA240	30	Std SS Seg	40	10.50	115.46	75	32	Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
35.4.4	SnapShot Burst 75-104 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	85	30	Std Plus	30	0.59	5.80	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	Std	0	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
			Cardiac SSB	CardiacSmall	0.160:1 6.4	0.35	0.625	40	80	ECG Mod 70-240mA Phase 40-80 AvgmA240	30	Std SSB	40	10.50	115.46	75	32	Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
35.4.5	SnapShot Burst+ >105 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	85	30	Std Plus	30	0.59	5.80	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	Std	0	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
			Cardiac SSBP	CardiacSmall	0.160:1 6.4	0.35	0.625	40	80	ECG Mod 70-240mA Phase 35-85 AvgmA240	30	Std SSB	40	11.25	123.79	75	32	Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy-cm) ASiR	Scan Length (mm)	Phantom (cm)	Description
35.4.7	Non ECG Gated		Helical	PedBody	0.984:1 39.375	0.4	0.625	40	80	SmartmA 50-280mA NI=18 AvgmA195	30	Std Plus	30	1.90	20.48	75	32	Weight and height based non ECG gated protocol for cardiac with SmartmA
35.5.1	RC 14.5-18.5kg (32.0-40.8lbs)		Helical	SmallBody	1.375:1 27.5	0.4	5	20	120	70	30	Std Plus	30	1.52	17.48	90	32	Weight and height based routine chest protocol
35.5.3	SnapShot Segment 30-74 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	85	30	Std Plus	30	0.59	5.80	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	Std	0	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
			Cardiac SEG	CardiacSmall	0.160:1 6.4	0.35	0.625	40	80	ECG Mod 70-265mA Phase 40-80 AvgmA265	30	Std SS Seg	40	11.53	126.82	75	32	Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
35.5.4	SnapShot Burst 75-104 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	85	30	Std Plus	30	0.59	5.80	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	Std	0	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
			Cardiac SSB	CardiacSmall	0.160:1 6.4	0.35	0.625	40	80	ECG Mod 70-265mA Phase 40-80 AvgmA265	30	Std SSB	40	11.53	126.82	75	32	Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
35.5.5	SnapShot Burst+ >105 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	85	30	Std Plus	30	0.59	5.80	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	Std	0	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
			Cardiac SSBP	CardiacSmall	0.160:1 6.4	0.35	0.625	40	80	ECG Mod 70-265mA Phase 35-85 AvgmA265	30	Std SSB	40	12.40	136.37	75	32	Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
35.5.7	Non ECG Gated		Helical	PedBody	0.984:1 39.375	0.4	0.625	40	80	SmartmA 50-300mA NI=18 AvgmA210	30	Std Plus	30	2.04	22.05	75	32	Weight and height based non ECG gated protocol for cardiac with SmartmA
35.6.1	RC 18.5-22.5kg (40.8-49.6lbs)		Helical	SmallBody	1.375:1 27.5	0.4	5	20	120	75	30	Std Plus	30	1.63	18.73	90	32	Weight and height based routine chest protocol
35.6.3	SnapShot Segment 30-74 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	85	30	Std Plus	30	0.59	5.80	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	Std	0	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy-cm) ASiR	Scan Length (mm)	Phantom (cm)	Description
			Cardiac SEG	CardiacSmall	0.160:1 6.4	0.35	0.625	40	80	ECG Mod 70-280mA Phase 40-80 AvgmA280	30	Std SS Seg	40	12.16	133.72	75	32	Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
35.6.4	SnapShot Burst 75-104 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	85	30	Std Plus	30	0.59	5.80	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	Std	0	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
			Cardiac SSB	CardiacSmall	0.160:1 6.4	0.35	0.625	40	80	ECG Mod 70-280mA Phase 40-80 AvgmA280	30	Std SSB	40	12.16	133.72	75	32	Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
35.6.5	SnapShot Burst+ >105 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	85	30	Std Plus	30	0.59	5.80	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	Std	0	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
			Cardiac SSBP	CardiacSmall	0.160:1 6.4	0.35	0.625	40	80	ECG Mod 70-280mA Phase 35-85 AvgmA280	30	Std SSB	40	13.09	144.01	75	32	Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
35.6.7	Non ECG Gated		Helical	PedBody	0.984:1 39.375	0.4	0.625	40	80	SmartmA 100-320mA NI=18 AvgmA225	30	Std Plus	30	2.19	23.63	75	32	Weight and height based non ECG gated protocol for cardiac with SmartmA
35.7.1	RC 22.5-31.5kg (49.6-69.5lbs)		Helical	LargeBody	1.375:1 55.0	0.4	5	40	120	75	30	Std Plus	30	1.73	22.98	85	32	Weight and height based routine chest protocol
35.7.3	SnapShot Segment 30-74 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	85	30	Std Plus	30	0.59	5.80	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	Std	0	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
			Cardiac SEG	CardiacSmall	0.160:1 6.4	0.35	0.625	40	80	ECG Mod 75-295mA Phase 40-80 AvgmA295	30	Std SS Seg	40	12.81	140.96	75	32	Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
35.7.4	SnapShot Burst 75-104 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	85	30	Std Plus	30	0.59	5.80	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	Std	0	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
			Cardiac SSB	CardiacSmall	0.160:1 6.4	0.35	0.625	40	80	ECG Mod 75-295mA Phase 40-80 AvgmA295	30	Std SSB	40	12.81	140.96	75	32	Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy-cm) ASiR	Scan Length (mm)	Phantom (cm)	Description
35.7.5	SnapShot Burst+ >105 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	85	30	Std Plus	30	0.59	5.80	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	Std	0	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
			Cardiac SSBP	CardiacSmall	0.160:1 6.4	0.35	0.625	40	80	ECG Mod 75-295mA Phase 35-85 AvgmA295	30	Std SSB	40	13.79	151.74	75	32	Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
35.7.7	Non ECG Gated		Helical	PedBody	0.984:1 39.375	0.4	0.625	40	80	SmartmA 100-360mA NI=18 AvgmA250	30	Std Plus	30	2.43	26.25	75	32	Weight and height based non ECG gated protocol for cardiac with SmartmA
35.8.1	RC 31.5-40.5kg (69.5-89.3lbs)		Helical	LargeBody	1.375:1 55.0	0.4	5	40	120	85	30	Std Plus	30	1.96	32.90	120	32	Weight and height based routine chest protocol
35.8.3	SnapShot Segment 30-74 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	100	85	30	Std Plus	30	1.10	10.76	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	100	50	0	Std	0	39.50	19.75	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
			Cardiac SEG	CardiacSmall	0.160:1 6.4	0.35	0.625	40	100	ECG Mod 70-265mA Phase 40-80 AvgmA265	30	Std SS Seg	30	21.40	235.37	75	32	Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
35.8.4	SnapShot Burst 75-104 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	100	85	30	Std Plus	30	1.10	10.76	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	100	50	0	Std	0	39.50	19.75	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
			Cardiac SSB	CardiacSmall	0.160:1 6.4	0.35	0.625	40	100	ECG Mod 70-265mA Phase 40-80 AvgmA265	30	Std SSB	30	21.40	235.37	75	32	Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
35.8.5	SnapShot Burst+ >105 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	100	85	30	Std Plus	30	1.10	10.76	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	100	50	0	Std	0	39.50	19.75	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
			Cardiac SSBP	CardiacSmall	0.160:1 6.4	0.35	0.625	40	100	ECG Mod 70-265mA Phase 35-85 AvgmA265	30	Std SSB	40	23.01	253.11	75	32	Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
35.8.7	Non ECG Gated		Helical	SmallBody	0.984:1 39.375	0.4	0.625	40	100	SmartmA 100-400mA NI=16 AvgmA280	30	Std Plus	30	5.06	54.57	75	32	Weight and height based non ECG gated protocol for cardiac with SmartmA

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy-cm) ASiR	Scan Length (mm)	Phantom (cm)	Description
35.9.1	RC 40.5-55.0kg(89.3-121.3lbs)		Helical	LargeBody	1.375:1 55.0	0.4	5	40	120	90	30	Std Plus	30	2.07	41.06	150	32	Weight and height based routine chest protocol
35.9.3	SnapShot Segment 30-74 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	100	85	30	Std Plus	30	1.10	10.76	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	100	50	0	Std	0	39.50	19.75	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
			Cardiac SEG	CardiacSmall	0.160:1 6.4	0.35	0.625	40	100	ECG Mod 70-295mA Phase 40-80 AvgmA295	30	Std SS Seg	30	23.71	260.86	75	32	Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
35.9.4	SnapShot Burst 75-104 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	100	85	30	Std Plus	30	1.10	10.76	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	100	50	0	Std	0	39.50	19.75	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
			Cardiac SSB	CardiacSmall	0.160:1 6.4	0.35	0.625	40	100	ECG Mod 70-295mA Phase 40-80 AvgmA295	30	Std SSB	30	23.71	260.86	75	32	Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
35.9.5	SnapShot Burst+ >105 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	100	85	30	Std Plus	30	1.10	10.76	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	100	50	0	Std	0	39.50	19.75	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
			Cardiac SSBP	CardiacSmall	0.160:1 6.4	0.35	0.625	40	100	ECG Mod 70-295mA Phase 35-85 AvgmA295	30	Std SSB	40	25.57	281.32	75	32	Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
35.9.7	Non ECG Gated		Helical	SmallBody	0.984:1 39.375	0.4	0.625	40	100	SmartmA 100-440mA NI=16 AvgmA310	30	Std Plus	30	5.60	60.42	75	32	Weight and height based non ECG gated protocol for cardiac with SmartmA

Abdomen

Table 7-4

[Interval]* - For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy-cm) ASiR	Scan Length (mm)	Phantom (cm)	Description
36.1.1	Abd 6.0-7.5kg (13.2-16.5lbs)		Helical	PedBody	1.375:1 27.5	0.4	3.75	20	120	70	30	Std Plus	30	1.52	15.77	78.75	32	Weight and height based routine abdomen protocol
36.2.1	Abd 7.5-9.5kg (16.5-20.9lbs)		Helical	PedBody	1.375:1 27.5	0.4	3.75	20	120	110	30	Std Plus	30	2.39	27.47	90	32	Weight and height based routine abdomen protocol
36.3.1	Abd 9.5-11.5kg (20.9-25.4lbs)		Helical	PedBody	1.375:1 27.5	0.4	3.75	20	120	125	30	Std Plus	30	2.71	34.26	101.25	32	Weight and height based routine abdomen protocol
36.4.1	Abd 11.5-14.5kg (25.4-32.0lbs)		Helical	SmallBody	1.375:1 27.5	0.4	5	20	120	85	30	Std Plus	30	1.84	24.91	110	32	Weight and height based routine abdomen protocol
36.5.1	Abd 14.5-18.5kg (32.0-40.8lbs)		Helical	SmallBody	1.375:1 27.5	0.4	5	20	120	95	30	Std Plus	30	2.06	27.84	110	32	Weight and height based routine abdomen protocol
36.6.1	Abd 18.5-22.5kg (40.8-49.6lbs)		Helical	Medium Body	1.375:1 27.5	0.4	5	20	120	85	30	Std Plus	30	2.18	33.77	130	32	Weight and height based routine abdomen protocol
36.7.1	Abd 22.5-31.5kg (49.6-69.5lbs)		Helical	LargeBody	1.375:1 55.0	0.4	5	40	120	105	30	Std Plus	30	2.42	47.91	150	32	Weight and height based routine abdomen protocol
36.8.1	Abd 31.5-40.5kg (69.5-89.3lbs)		Helical	LargeBody	1.375:1 55.0	0.4	5	40	120	120	30	Std Plus	30	2.77	54.75	150	32	Weight and height based routine abdomen protocol
36.9.1	Abd 40.5-55.0kg (89.3-121.3lbs)		Helical	LargeBody	1.375:1 55.0	0.4	5	40	120	140	30	Std Plus	30	3.23	80.01	200	32	Weight and height based routine abdomen protocol

Miscellaneous

Table 7-5

[Interval]* - For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy-cm) ASiR	Scan Length (mm)	Phantom (cm)	Description
40.1	Pink Zone Routine 0-20 lbs/0-9 kg		Helical	PedBody	1.375:1 55.0	0.5	5	40	80	SmartmA 65-130mA NI=5 Avg mA 70	0	Std	0	0.61	8.70	95	32	Pediatric and Small patient Routine procedure based protocols
40.2	Pink Zone Routine 21-60 lbs/9.1-27.2 kg		Helical	PedBody	1.375:1 55.0	0.5	5	40	100	SmartmA 80-160mA NI=7 Avg mA 70	0	Std	0	1.13	27.47	195	32	Pediatric and Small patient Routine procedure based protocols
40.3	Pink Zone Routine 61-100 lbs/27.3-45.4 kg		Helical	Medium Body	1.375:1 55.0	0.5	5	40	120	SmartmA 95-190mA NI=10 Avg mA 70	0	Std	0	2.09	71.81	295	32	Pediatric and Small patient Routine procedure based protocols
40.4	Pink Zone Routine 101-200 lbs/45.4-90.7 kg		Helical	LargeBody	1.375:1 55.0	0.5	5	40	120	SmartmA 110-220mA NI=12 Avg mA 70	0	Std	0	2.02	89.30	395	32	Pediatric and Small patient Routine procedure based protocols
40.5	Pink Zone Routine >200 lbs/>90.8 kg		Helical	LargeBody	1.375:1 55.0	0.5	5	40	120	SmartmA 125-300mA NI=15 Avg mA 70	0	Std	0	2.02	89.30	395	32	Pediatric and Small patient Routine procedure based protocols
40.7	Green Zone Lower Dose 0-20 lbs/0-9 kg		Helical	PedBody	1.375:1 55.0	0.5	5	40	80	SmartmA 50-100mA NI=7 Avg mA 70	0	Std	0	0.61	8.70	95	32	Pediatric and Small patient Lower Dose procedure based protocols
40.8	Green Zone Lower Dose 21-60 lbs/9.1-27.2 kg		Helical	PedBody	1.375:1 55.0	0.5	5	40	100	SmartmA 60-120mA NI=9 Avg mA 70	0	Std	0	1.13	27.47	195	32	Pediatric and Small patient Lower Dose procedure based protocols
40.9	Green Zone Lower Dose 61-100 lbs/ 27.3-45.4 kg		Helical	Medium Body	1.375:1 55.0	0.5	5	40	120	SmartmA 70-140mA NI=11 Avg mA 70	0	Std	0	2.09	71.81	295	32	Pediatric and Small patient Lower Dose procedure based protocols
40.10	Green Zone Lower Dose 101-200 lbs/ 45.4-90.7 kg		Helical	LargeBody	1.375:1 55.0	0.5	5	40	120	SmartmA 80-160mA NI=13 Avg mA 70	0	Std	0	2.02	89.30	395	32	Pediatric and Small patient Lower Dose procedure based protocols
40.11	Green Zone Lower Dose >200 lbs/ >90.8 kg		Helical	LargeBody	1.375:1 55.0	0.5	5	40	120	SmartmA 90-240mA NI=16 Avg mA 70	0	Std	0	2.02	89.30	395	32	Pediatric and Small patient Lower Dose procedure based protocols
40.13	Grey Zone CT Angio 0-20 lbs/0-9 kg		Helical	PedBody	1.375:1 55.0	0.5	2.5	40	80	SmartmA 100-200mA NI=5 Avg mA 140	0	Std	0	1.22	17.71	97.5	32	Pediatric and Small patient Angio procedure based protocols

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (mm)	Phantom (cm)	Description
40.14	Grey Zone CT Angio 21-60 lbs/9.1-27.2 kg		Helical	SmallBody	1.375:1 55.0	0.5	2.5	40	100	SmartmA 120-240mA NI=9.9 Avg mA 140	0	Std	0	2.26	55.50	197.5	32	Pediatric and Small patient Angio procedure based protocols
40.15	Grey Zone CT Angio 61-100 lbs/ 27.3-45.4 kg		Helical	LargeBody	1.375:1 55.0	0.5	2.5	40	120	SmartmA 120-240mA NI=10 Avg mA 140	0	Std	0	4.03	139.28	297.5	32	Pediatric and Small patient Angio procedure based protocols
40.16	Grey Zone CT Angio 101-200 lbs/ 45.4-90.7 kg		Helical	LargeBody	1.375:1 55.0	0.5	2.5	40	120	SmartmA 120-240mA NI=12 Avg mA 140	0	Std	0	4.03	179.62	397.5	32	Pediatric and Small patient Angio procedure based protocols
40.17	Grey Zone CT Angio >200 lbs/ >90.8 kg		Helical	LargeBody	1.375:1 55.0	0.5	2.5	40	120	SmartmA 120-300mA NI=15 Avg mA 140	0	Std	0	4.03	179.62	397.5	32	Pediatric and Small patient Angio procedure based protocols

Reference Adult Protocols - Installed Options : Sub 0.4sec, VCT85kW and VCT Hi Power

Head

Table 8-1

[Interval]* - For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (s)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (mm)	Phantom (cm)	Description
21.1	Routine Head 1s Group1		Axial	Head	4i	1	5	20	140	200	Std	53.34	320.02	55	16	Routine head protocol for evaluation of the brain for abnormalities.
	Group2		Axial	Head	4i	1	5	20	120	320	Std	61.43	491.41	75	16	
21.2	Routine Head 2s Group1		Axial	Head	4i	2	5	20	140	100	Std	53.34	320.02	55	16	Routine head protocol for evaluation of the brain for abnormalities.
	Group2		Axial	Head	4i	2	5	20	120	160	Std	61.43	491.41	75	16	
21.3	Trauma Head 2.5mm		Axial	Head	8i	0.5	2.5	20	140	560	Std	76.76	1074.71	137.5	16	Emergency head protocol for evaluation of the brain and cranium for abnormalities.
21.4	Routine Head 0.5s Group1		Axial	Head	4i	0.5	5	20	140	400	Std	54.83	328.99	55	16	Routine head protocol for evaluation of the brain for abnormalities.
	Group2		Axial	Head	4i	0.5	5	20	120	600	Std	59.20	473.62	75	16	
21.5	Circle of Willis 0.4s.	MIROI	Axial	Head	1i	1	5 [0]	5	120	50	Std	125.28	62.79	0	16	Test bolus for CT angiography of the head for evaluation of carotid and cerebral vasculature.
			Helical	Head	0.531:1 10.625	0.4	0.625	20	120	300	Std Plus IQE	43.36	421.37	80	16	CT angiography of the head for evaluation of carotid and cerebral vasculature.
21.6	CoW/ Carotid 0.625mm 0.4s.Smart mA	MIROI	Axial	Head	1i	1	5 [0]	5	120	80	Std	180.84	90.42	0	16	Test bolus for CT angiography of the head for evaluation of carotid and cerebral vasculature.
			Helical	Head	0.531:1 10.625	0.4	0.625	20	120	SmartmA 100-650mA NI=8 Avg mA 595	Std Plus IQE	88.41	1657.19	170	16	CT angiography of the head for evaluation of carotid and cerebral vasculature.
21.8	Helical Head	DMPR	Helical	Head	0.531:1 10.625	0.5	0.625	20	120	300	Std Plus IQE	54.20	878.85	145	16	Helical scan mode for evaluation of the brain for cerebral abnormality.
21.10	VHS Head 4D CTA	MIROI	Axial	Head	1i	1	5 [0]	5	120	50	Std	125.58	62.79	0	16	Test bolus for CT angiography of the head for evaluation of carotid and cerebral vasculature.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (s)	Slice Thickness (mm) [Interval] J* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (mm)	Phantom (cm)	Description
			Helical-S	Head	0.984:1 39.375 18 pass	0.4	5 [10]	40	120	300	Std Plus	496.19	6202.39	110	16	CT angiography of the head using Volume Helical scan mode for evaluation of cerebral vasculature.
21.11	VHS Perfusion 350-370 Strength Contrast		Axial	Head	4i	1	5	20	120	335	Std	64.31	900.28	135	16	Non-Enhance Brain, evaluation for hemorrhage or infarction.
		Perfusion	Helical-S/P	Head	0.984:1 39.375 28 pass	0.4	5	40	80	200	Std	186.02	2325.25	110	16	CT Perfusion using Volume Helical scan mode for evaluation of cerebral perfusion over time.
			Axial	Head	4i	1	5	20	120	335	Std	64.31	900.28	135	16	Post Contrast Brain, Evaluation for residual tissue enhancement.
21.12	CT Perfusion 300-320 Strength Contrast		Axial	Head	4i	1	5	20	120	335	Std	64.31	900.28	135	16	Non-Enhance Brain, evaluation for hemorrhage or infarction.
		Perfusion	Cine	Head	8i	1	5 [0]	40	80	200	Std	652.82	2611.27	35	16	CT Perfusion using Cine scan mode for evaluation of cerebral perfusion over time.
			Axial	Head	4i	1	5	20	120	335	Std	64.31	900.28	135	16	Non-Enhance Brain, evaluation for hemorrhage or infarction.
21.13	CT Perfusion 350-370 Strength Contrast		Axial	Head	4i	1	5	20	120	335	Std	64.31	900.28	135	16	Non-Enhance Brain, evaluation for hemorrhage or infarction.
		Perfusion	Cine	Head	8i	1	5 [0]	40	80	200	Std	587.55	2350.19	35	16	CT Perfusion using Cine scan mode for evaluation of cerebral perfusion over time.
			Axial	Head	4i	1	5	20	120	335	Std	64.31	900.28	135	16	Post Contrast Brain, Evaluation for residual tissue enhancement.
21.14	CT Perfusion Brain Tumor - Dual Phase		Axial	Head	4i	1	5	20	120	335	Std	64.31	900.28	135	16	Non-Enhance Brain, evaluation for hemorrhage or infarction.
	Group 1	Perfusion	Cine	Head	8i	1	5 [0]	40	80	200	Std	587.55	2350.19	35	16	CT Perfusion using Cine scan mode for evaluation of cerebral perfusion over time.
	Group 2	Perfusion	Axial	Head	8i	1	5 [0]	40	80	200	Std	130.54	522.15	35	16	Delayed phase CT Perfusion using Axial scan mode for evaluation of Brain tumor permeability.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (s)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (mm)	Phantom (cm)	Description
			Axial	Head	4i	1	5	20	120	335	Std	64.31	900.28	135	16	Post Contrast Brain, Evaluation for residual tissue enhancement.
21.15	CT Perfusion 350-370 Strength Contrast (Axial)		Axial	Head	4i	1	5	20	120	335	Std	64.31	900.28	135	16	Non-Enhance Brain, evaluation for hemorrhage or infarction.
		Perfusion	Axial	Head	8i	1	5 [0]	40	80	200	Std	287.18	1148.73	35	16	CT Perfusion using Axial scan mode for evaluation of cerebral perfusion over time.
			Axial	Head	4i	1	5	20	120	335	Std	64.31	900.28	135	16	Post Contrast Brain, Evaluation for residual tissue enhancement.
21.16	CT Perfusion 350-370 Strength Contrast (Axial Shuttle Mode)		Axial	Head	4i	1	5	20	120	335	Std	64.31	900.28	135	16	Non-Enhance Brain, evaluation for hemorrhage or infarction.
		Perfusion	Axial-S	Head	8i 17 pass	0.4	5	40	80	500	Std	221.91	1775.31	75	16	CT Perfusion using Volume Shuttle (Axial) scan mode for evaluation of cerebral perfusion over time.
			Axial	Head	4i	1	5	20	120	335	Std	64.31	900.28	135	16	Post Contrast Brain, Evaluation for residual tissue enhancement.

Orbit

Table 8-2

[Interval]* - For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy-cm)	Scan Length (cm)	Phantom (cm)	Description
22.1	Sinus Supine Helical + DMPR	DMPR	Helical	Head	0.531:1 10.625	0.4	0.625	20	120	120	Bone Plus IQE	17.34	158.79	74.375	16	Routine Helical Sinus protocol, Evaluation for any abnormality
22.3	Orbits Helical + DMPR	DMPR	Helical	Head	0.531:1 10.625	0.5	0.625	20	120	220	Std Plus IQE	39.75	314.09	61.875	16	Routine Helical Orbit protocol, Evaluation for abnormality soft tissues and bone structures of the face.
22.6	Helical IAC 0.625mm Axial mode - Use Reformat for Coronal		Helical	Head	0.531:1 10.625	0.6	0.625	20	140	270	Std Plus IQE	81.32	454.44	38.75	16	Axial scan mode, Evaluation for inner ear abnormalities in Axial plane
22.8	Axial and Coronal IAC		Axial	Head	16i	1	1.25	20	140	280	Std	74.67	298.68	38.75	16	Helical scan mode, Evaluation for inner ear abnormalities in Axial plane
			Axial	Head	16i	1	1.25	20	140	335	Std	89.34	357.35	38.75	16	Helical scan mode, Evaluation for inner ear abnormalities in Coronal plane

Neck

Table 8-3

[Interval]* - For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (cm)	Phantom (cm)	Description
23.1	C-Spine C5-C7 Axial		Axial	SmallBody	8i	0.5	2.5	20	140	650	Std	35.34	494.83	137.5	32	C-Spine C5-C7 area Axial mode, Evaluation for soft tissue, disc and bone abnormalities.
23.2	C-Spine C1-C4 Axial		Axial	SmallBody	8i	1	2.5	20	120	240	Std	17.90	143.19	77.5	32	C-Spine C1-C4 area Axial mode, Evaluation for soft tissue, disc and bone abnormalities.
23.3	C-Spine Helical + DMPR	DMPR	Helical	SmallBody	0.516:1 20.625	0.8	0.625	40	120	335	Bone Plus IQE	36.23	522.92	110	32	C-Spine Helical mode with DMPR, Evaluation for soft tissue, disc and bone abnormalities.
23.4	C-Spine Helical + DMPR/ SmartmA	DMPR	Helical	SmallBody	0.516:1 20.625	0.8	0.625	40	120	SmartmA 100-650mA NI=11.34 Avg mA 310	Bone Plus IQE	33.52	483.89	110	32	C-Spine Helical mode with DMPR and SmartmA, Evaluation for soft tissue, disc and bone abnormalities.
23.8	Carotid 0.625mm 0.4s. 0.5:1 SmartmA/ SmartPrep		Helical	SmallBody	0.516:1 20.625	0.4	0.625	40	140	SmartmA 100-650mA NI=11 Avg mA 575	Std Plus IQE	44.10	902.52	170	32	Carotid CTA (pitch 0.5:1) with SmartmA and SmartPrep, Evaluation of carotid vasculature.
23.9	Carotid 0.625mm 0.4s. 0.9:1 SmartmA/ SmartPrep		Helical	SmallBody	0.984:1 39.375	0.4	0.625	40	120	SmartmA 80-700mA NI=11 Avg mA 625	Std Plus IQE	17.70	359.81	170	32	Carotid CTA (pitch 0.9:1) with SmartmA and SmartPrep, Evaluation of carotid vasculature.
23.10	Carotid/ CoW 0.625mm 0.4s. AutomA	MIROI	Axial	SmallBody	1i	1	5 [0]	5	120	80	Std	99.00	49.50	0	32	Test bolus for Carotid CTA (pitch 0.5:1) of the head for evaluation of carotid and cerebral vasculature.
			Helical	SmallBody	0.516:1 20.625	0.4	0.625	40	140	AutomA 80-650mA NI=11 Avg mA 600	Std Plus IQE	46.02	941.76	170	32	Carotid and Brain CTA (pitch 0.5:1) with AutomA, Evaluation of carotid and cerebral vasculature.
23.11	Carotid/ CoW 0.625mm 0.4s. SmartmA/ SmartPrep		Helical	SmallBody	0.969:1 19.375	0.4	0.625	20	140	SmartmA 100-650mA NI=11 Avg mA 650	Std Plus IQE	29.19	551.35	170	32	Carotid CTA (pitch 0.9:1) with SmartmA and SmartPrep, Evaluation of carotid and cerebral vasculature.
23.12	Soft Tissue Neck 2.5mm AutomA		Helical	SmallBody	0.984:1 39.375	0.5	2.5	40	120	AutomA 100-335mA NI=9.1 Avg mA 250	Std Plus	8.85	161.79	150	32	Neck with AutomA, Evaluation of soft tissue structures for abnormality

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy-cm)	Scan Length (cm)	Phantom (cm)	Description
23.13	Soft Tissue Neck + DMPR/ SmartmA	DMPR	Helical	SmallBody	0.969:1 19.375	0.5	0.625	20	120	SmartmA 100-600mA NI=12.6 Avg mA 425	Std Plus IQE	16.82	283.95	150	32	Neck with DMPR and SmartmA, Evaluation of soft tissue structures for abnormality

Upper Extremity

Table 8-4

[Interval]* - For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy-cm)	Scan Length (cm)	Phantom (cm)	Description
24.1	Shoulder 1.25mm + DMPR	DMPR	Helical	LargeBody	0.516:1 20.625	0.6	1.25 [0.625]	40	140	600	Std Plus IQE	78.38	1054.65	100	32	Shoulder with DMPR, Evaluation of soft tissue and bone of shoulder area
24.2	Shoulder 1.25mm + DMPR/ SmartmA	DMPR	Helical	LargeBody	0.516:1 20.625	0.6	1.25 [0.625]	40	140	SmartmA 100-650mA NI=20 Avg mA 595	Std Plus IQE	77.73	1045.86	100	32	Shoulder with DMPR and SmartmA, Evaluation of soft tissue and bone of shoulder area

Chest

Table 8-5

[Interval]* - For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (cm)	Phantom (cm)	Description
25.1	Routine Chest 0.4s. 5mm SmartmA		Helical	LargeBody	1.375:1 55.0	0.4	5	40	120	SmartmA 100-650mA NI=13 Avg mA 500	Std Plus	11.52	286.26	200	32	Chest with SmartmA, Evaluation of mediastinum and lungs fields
25.2	Routine Chest 0.4s.+ DMPR/ SmartmA	DMPR	Helical	LargeBody	0.984:1 39.375	0.4	0.625	40	120	SmartmA 100-750mA NI=22 Avg mA 620	Std Plus	19.96	465.64	200	32	Chest with SmartmA and DMPR, Evaluation of mediastinum and lungs fields
25.3	Chest with HiRes 0.4s. SmartmA		Helical	LargeBody	0.969:1 19.375	0.4	5	20	120	SmartmA 80-650mA NI=12 Avg mA 500	Chest Plus	17.99	393.87	200	32	Chest with SmartmA, Evaluation of mediastinum and lung fields with higher resolution images of the lungs
25.4	Hi Res Chest 0.5s. Axial		Axial	LargeBody	1i	0.5	1.25 [10]	1.25	120	380	Bone+	4.48	107.55	230	32	Axial scan mode, Evaluation of mediastinum and lung fields in high resolution mode
25.5	Chest Abd Pelvis 0.4/ 0.5s. SmartmA 5mm (2 groups) Group1		Helical	LargeBody	0.984:1 39.375	0.4	5	40	120	SmartmA 100-650mA NI=13 Avg mA 370	Std Plus	11.91	218.32	150	32	Multi group scan for Chest, Abdomen, Pelvis (0.4/0.5sec) with SmartmA, Evaluation of chest and abdominal anatomy for abnormalities
	Group2		Helical	LargeBody	1.375:1 55.0	0.5	5	40	120	SmartmA 100-650mA NI=13 Avg mA 370	Std Plus	10.66	365.90	295	32	
25.6	Chest Abd Pelvis 0.5s + DMPR SmartPrep	DMPR	Helical	LargeBody	1.375:1 55.0	0.5	0.625	40	120	SmartmA 100-750mA NI=22 Avg mA 540	Std Plus	15.56	775.18	450	32	Helical scan mode (0.5sec) with SmartmA, Evaluation of chest and abdominal anatomy for abnormalities
25.7	Pulmonary Embolis 0.5s.	MIROI	Axial	LargeBody	1i	1	5 [0]	5	120	80	Std	93.89	46.95	0	32	Timing Bolus for Helical scan mode 1.25mm thickness and helical pitch of 0.9:1, evaluation of Chest for pulmonary embolism
			Helical	LargeBody	0.984:1 39.375	0.5	1.25	40	120	580	Std	23.34	497.55	180	32	Helical scan mode 1.25mm thickness and helical pitch of 0.9:1, Evaluation Chest for pulmonary embolism

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy-cm)	Scan Length (cm)	Phantom (cm)	Description
25.8	Pulmonary Embolis 0.5s. SmartmA	MIROI	Axial	LargeBody	1i	1	5 [0]	5	120	80	Std	93.89	46.95	0	32	Timing Bolus for Helical scan mode 1.25mm thickness and helical pitch of 0.9:1 with SmartmA, evaluation of Chest for pulmonary embolism
			Helical	LargeBody	0.984:1 39.375	0.5	1.25	40	120	SmartmA 100-750mA NI=20 Avg mA 500	Std	20.12	428.92	180	32	Helical scan mode 1.25mm thickness and helical pitch of 0.9:1 with SmartmA, Evaluation Chest for pulmonary embolism
25.9	Aortic Dissection 0.625mm + DMPR/ SmartmA/ SmartPrep	DMPR	Helical	LargeBody	1.375:1 55.0	0.5	0.625	40	120	SmartmA 100-750mA NI=22 Avg mA 600	Std Plus	17.29	602.00	300	32	Helical scan mode 0.625mm thickness and helical pitch of 1.375:1 with SmartmA, SmartPrep and DMPR, Evaluation for aortic dissection
25.10	Aortic Dissection 1.25mm + DMPR/ SmartmA/ SmartPrep	DMPR	Helical	LargeBody	1.375:1 55.0	0.5	1.25 [0.8]	40	120	SmartmA 100-750mA NI=20 Avg mA 550	Std Plus	15.85	551.83	300	32	Helical scan mode 1.25mm thickness and helical pitch of 1.375:1 with SmartmA, SmartPrep and DMPR, Evaluation for aortic dissection
25.11	Lung Assessment	Advanced Lung Analysis	Helical	LargeBody	0.969:1 19.375	0.5	1.25	20	120	120	Bone	5.25	114.77	200	32	Helical scan mode for acquisition of data for Advanced Lung Analysis (ALA) software for lung nodule assessment
25.12	Gated Chest Cardiac Helical		Helical	LargeBody	1.375:1 55.0	0.4	5	40	120	60	Std Plus	1.38	13.55	50	32	Localizer scan for Helical Gated scan for evaluation of thoracic vasculature
		MIROI	Axial	LargeBody	1i	1	5 [0]	5	120	60	Std	98.59	49.29	0	32	Timing bolus scan for Helical Gated scan for evaluation of thoracic vasculature
			Cardiac SEG	CardiacLarge	0.160:1 6.4	0.35	0.625	40	120	ECG Mod 185-450mA Phase 70-80 Avg mA 450	Std SS Seg	51.77	1734.24	300	32	Helical Gated scan for evaluation of thoracic vasculature
25.13	Gated Chest SnapShot Pulse		Helical	LargeBody	1.375:1 55.0	0.4	5	40	120	60	Std Plus	1.38	13.55	50	32	Localizer scan for Cine Gated scan for evaluation of thoracic vasculature
		MIROI	Axial	LargeBody	1i	1	5 [0]	5	120	60	Std	98.59	49.29	0	32	Timing bolus scan for Cine Gated scan for evaluation of thoracic vasculature
			Cardiac Pulse Cine	CardiacLarge	56i	0.35	0.625 [35]	40	120	450	Std SS Seg	24.04	757.31	314.375	32	Cine Gated scan for evaluation of thoracic vasculature
25.14	SmartScore - Gated 0.35s	Smart Score	Cine	LargeBody	8i	0.35	2.5	20	120	430	Std Segment	8.99	89.95	97.5	32	Cine gated scan mode for acquisition of data for coronary artery calcification analysis

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy-cm)	Scan Length (cm)	Phantom (cm)	Description
25.15	SnapShot Pulse 30-65 BPM		Helical	LargeBody	1.375:1 55.0	0.4	5	40	120	60	Std Plus	1.38	13.55	50	32	Localizer scan for Prospectively gated Cardiac Cine scan for evaluation of coronary arteries, intended for stable heart rates 30-65 bpm
		MIROI	Axial	LargeBody	1i	1	5 [0]	5	120	60	Std	98.59	49.29	0	32	Timing bolus scan for Prospectively gated Cardiac Cine scan for evaluation of coronary arteries, intended for stable heart rates 30-65 bpm
			Cardiac Pulse Cine	CardiacSmall	56i	0.35	0.625 [35]	40	120	450	Std SS Seg	21.15	222.06	104.375	32	Prospectively gated Cardiac Cine scan for evaluation of coronary arteries, intended for stable heart rates 30-65 bpm
25.16	SnapShot Segment 30-74 BPM		Helical	LargeBody	1.375:1 55.0	0.4	5	40	120	60	Std Plus	1.38	13.55	50	32	Localizer scan for Retrospectively gated Cardiac Helical scan for evaluation of coronary arteries, intended for heart rates 30-74 bpm
		MIROI	Axial	LargeBody	1i	1	5 [0]	5	120	60	Std	98.59	49.29	0	32	Timing bolus scan for Retrospectively gated Cardiac Helical scan for evaluation of coronary arteries, intended for heart rates 30-74 bpm
			Cardiac SEG	CardiacLarge	0.160:1 6.4	0.35	0.625	40	120	ECG Mod 200-450mA Phase 70-80 Avg mA 450	Std SS Seg	55.39	747.80	100	32	Retrospectively gated Cardiac Helical scan for evaluation of coronary arteries, intended for heart rates 30-74 bpm
25.17	SnapShot Burst 75-113 BPM		Helical	LargeBody	1.375:1 55.0	0.4	5	40	120	60	Std Plus	1.38	13.55	50	32	Localizer scan for Retrospectively gated Cardiac Helical scan for evaluation of coronary arteries, intended for heart rates 75-113 bpm
		MIROI	Axial	LargeBody	1i	1	5 [0]	5	120	60	Std	98.59	49.29	0	32	Timing bolus scan for Retrospectively gated Cardiac Helical scan for evaluation of coronary arteries, intended for heart rates 75-113 bpm
			Cardiac SSB	CardiacLarge	0.160:1 6.4	0.35	0.625	40	120	ECG Mod 200-450mA Phase 70-80 Avg mA 450	Std SSB	55.39	747.80	100	32	Retrospectively gated Cardiac Helical scan for evaluation of coronary arteries, intended for heart rates 75-113 bpm
25.18	SnapShot Burst Plus 114+ BPM		Helical	LargeBody	1.375:1 55.0	0.4	5	40	120	60	Std Plus	1.38	13.55	50	32	Localizer scan for Retrospectively gated Cardiac Helical scan for evaluation of coronary arteries, intended for heart rates over 113 bpm

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy-cm)	Scan Length (cm)	Phantom (cm)	Description
		MIROI	Axial	LargeBody	1i	1	5 [0]	5	120	60	Std	98.59	49.29	0	32	Timing bolus scan for Retrospectively gated Cardiac Helical scan for evaluation of coronary arteries, intended for heart rates over 113 bpm
			Cardiac SSBP	CardiacLarge	0.160:1 6.4	0.35	0.625	40	120	ECG Mod 200-450mA Phase 70-80 Avg mA 450	Std SSB	55.39	747.80	100	32	Retrospectively gated Cardiac Helical scan for evaluation of coronary arteries, intended for heart rates over 113 bpm
25.19	Advantage 4D		Helical	LargeBody	1.375:1 55.0	0.6	5	40	120	285	Std Plus	9.85	244.08	200	32	Localizer scan for Advantage 4D series
		Advantage 4D	Cine	LargeBody	8i	1	2.5	20	120	100	Std	50.95	815.24	157.5	32	Cine scan mode for acquisition data for Advantage 4D for retrospective gating
25.20	VHS Chest 4D CTA SmartPrep		Helical-S	LargeBody	1.375:1 55.0 11 pass	0.4	5 [10]	40	120	SmartmA 150-500mA NI=14 Avg mA 350	Std Plus	112.15	3532.61	300	32	Volume Helical scan mode with SmartPrep for time resolved CT angiography of chest vasculature

Abdomen

Table 8-6

[Interval]* - For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy-cm)	Scan Length (cm)	Phantom (cm)	Description
26.1	Abdomen Pelvis 0.5s. 5mm SmartmA		Helical	LargeBody	1.375:1 55.0	0.5	5	40	120	SmartmA 50-650mA NI=11.57 Avg mA 500	Std Plus	14.41	645.73	400	32	Basic Abdomen Pelvis (0.5sec) with Smart mA, Evaluation for abnormalities of abdominal and pelvic structures
26.2	Abdomen Pelvis 0.5s. 5mm + DMPR SmartmA/ SmartPrep	DMPR	Helical	LargeBody	0.984:1 39.375	0.6	0.625	40	120	SmartmA 80-750mA NI=18 Avg mA 520	Std Plus	25.11	1087.62	400	32	Basic Abdomen Pelvis (0.5sec) with Smart mA, SmartPrep and DMPR, Evaluation for abnormalities of abdominal and pelvic structures
26.4	Renal Stone 0.5s. 5mm		Helical	LargeBody	1.375:1 55.0	0.5	5	40	120	380	Std Plus	10.95	490.75	400	32	Helical scan, Evaluation of kidney, ureters and bladder for renal stones
26.5	Renal Stone 0.5s. 5mm SmartmA		Helical	LargeBody	1.375:1 55.0	0.5	5	40	120	SmartmA 50-650mA NI=12 Avg mA 380	Std Plus	10.95	490.75	400	32	Helical scan with SmartmA, Evaluation of kidney, ureters and bladder for renal stones
26.7	AAA 0.5s. + DMPR 0.625mm	MIROI	Axial	LargeBody	1i	1	5 [0]	5	120	80	Std	112.67	56.34	0	32	Timing bolus for Helical scan with DMPR, Evaluation for abdominal aortic aneurysm
		DMPR	Helical	LargeBody	1.375:1 55.0	0.5	0.625	40	140	680	Std Plus	27.76	992.69	309.375	32	Helical scan with DMPR, Evaluation for abdominal aortic aneurysm
26.8	AAA 0.6s. + DMPR 0.625mm SmartmA/ SmartPrep	DMPR	Helical	LargeBody	1.375:1 55.0	0.6	0.625	40	120	SmartmA 100-700mA NI=19 Avg mA 550	Std Plus	19.02	679.85	309.375	32	Helical scan with DMPR, Smart mA and SmartPrep, Evaluation for abdominal aortic aneurysm
26.10	Dual Phase Liver		Helical	LargeBody	1.375:1 55.0	0.6	5	40	120	300	Std Plus	10.37	184.32	130	32	Non contrast evaluation of upper abdominal structures including liver
		MIROI	Axial	LargeBody	1i	1	5 [0]	5	120	80	Std	112.67	56.34	0	32	Timing bolus for Multi-Group acquisition for early arterial and portal venous phase of the liver
	Group1		Helical	LargeBody	1.375:1 55.0	0.5	2.5	40	120	550	Std Plus	15.85	282.44	130	32	Acquisition for early arterial phase of the liver
	Group2		Helical	LargeBody	1.375:1 55.0	0.5	5	40	120	420	Std Plus	12.10	542.41	400	32	Acquisition for portal venous phase of the liver

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy-cm)	Scan Length (cm)	Phantom (cm)	Description
26.11	Tri Phase Liver		Helical	LargeBody	1.375:1 55.0	0.6	5	40	120	300	Std Plus	10.37	184.32	130	32	Non contrast evaluation of upper abdominal structures including liver
		MIROI	Axial	LargeBody	1i	1	5 [0]	5	120	80	Std	112.67	56.34	0	32	Timing bolus for Multi-Group acquisition for early arterial, late arterial and portal venous phase of the liver
	Group1		Helical	LargeBody	1.375:1 55.0	0.5	2.5	40	120	550	Std Plus	15.85	282.44	130	32	Acquisition for early arterial phase of the liver
	Group2		Helical	LargeBody	1.375:1 55.0	0.5	5	40	120	450	Std Plus	12.97	581.15	400	32	Acquisition for late arterial phase of the liver
	Group3		Helical	LargeBody	1.375:1 55.0	0.5	5	40	120	450	Std Plus	12.97	581.15	400	32	Acquisition for portal venous phase of the liver
26.12	Dual Phase Pancreas SmartmA/ SmartPrep		Helical	LargeBody	0.984:1 39.375	0.5	5	40	120	SmartmA 80-650mA NI=12 Avg mA 400	Std Plus	16.10	166.06	70	32	Non contrast evaluation of upper abdominal structures including pancreas
	Group1		Helical	LargeBody	0.984:1 39.375	0.5	2.5	40	120	SmartmA 50-700mA NI=15 Avg mA 520	Std Plus	20.93	383.30	150	32	Acquisition with SmartmA and SmartPrep for arterial phase of the pancreas
	Group2		Helical	LargeBody	0.984:1 39.375	0.5	5	40	120	SmartmA 50-700mA NI=15 Avg mA 450	Std Plus	18.11	639.58	320	32	Acquisition for venous phase of the pancreas
26.13	Dual Phase Liver SmartmA/ SmartPrep		Helical	LargeBody	1.375:1 55.0	0.5	5	40	120	SmartmA 80-500mA NI=12 Avg mA 320	Std Plus	9.22	163.92	130	32	Non contrast evaluation of upper abdominal structures including liver
	Group1		Helical	LargeBody	1.375:1 55.0	0.5	2.5	40	120	SmartmA 50-700mA NI=15 Avg mA 550	Std Plus	15.85	282.44	130	32	Acquisition with SmartmA and SmartPrep for early arterial phase of the liver
	Group2		Helical	LargeBody	0.984:1 39.375	0.5	5	40	120	SmartmA 50-700mA NI=11 Avg mA 450	Std Plus	18.11	784.46	400	32	Acquisition with SmartmA and SmartPrep for portal venous phase of the liver
26.14	Tri Phase Liver SmartmA/ SmartPrep		Helical	LargeBody	0.984:1 39.375	0.5	5	40	120	SmartmA 80-500mA NI=12 Avg mA 400	Std Plus	16.10	262.65	130	32	Non contrast evaluation of upper abdominal structures including liver
	Group1		Helical	LargeBody	1.375:1 55.0	0.5	2.5	40	120	SmartmA 50-700mA NI=15 Avg mA 550	Std Plus	15.85	282.44	130	32	Acquisition with SmartmA and SmartPrep for early arterial phase of the liver
	Group2		Helical	LargeBody	1.375:1 55.0	0.5	5	40	120	SmartmA 50-700mA NI=11 Avg mA 450	Std Plus	12.97	581.15	400	32	Acquisition with SmartmA and SmartPrep for late arterial phase of the liver
	Group3		Helical	LargeBody	1.375:1 55.0	0.5	5	40	120	SmartmA 50-700mA NI=11 Avg mA 450	Std Plus	12.97	581.15	400	32	Acquisition with SmartmA and SmartPrep for portal venous phase of the liver
26.15	VHS Liver Perfusion		Helical	LargeBody	0.984:1 39.375	0.6	5	40	120	170	Std	8.21	133.61	130	32	Non contrast evaluation of upper abdominal structures including liver

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy-cm)	Scan Length (cm)	Phantom (cm)	Description
		Perfusion	Helical-S/P	LargeBody	1.375:1 55.0 21 pass	0.4	5	40	120	300	Std Plus	219.99	3409.79	140	32	Volume Helical scan mode for evaluation of liver tumor perfusion over time
26.16	VHS 4D CTA		Helical	LargeBody	0.984:1 39.375	0.6	5	40	120	340	Std	16.42	267.79	130	32	Localizer scan for Volume Helical Shuttle scan mode for evaluation of abdominal vasculature
		MIROI	Axial	LargeBody	1i	1	5 [0]	5	120	80	Std	112.67	56.34	0	32	Timing bolus scan for Volume Helical Shuttle scan mode for evaluation of abdominal vasculature
			Helical-S	LargeBody	1.375:1 55.0 18 pass	0.5	5 [10]	40	140	SmartmA 150-650mA NI=12 Avg mA 350	Std Plus	325.68	6350.70	180	32	Volume Helical Shuttle scan mode for time resolved CT angiography of abdominal vasculature
26.17	Runoff 0.625mm	MIROI	Axial	LargeBody	1i	1	5 [0]	5	120	80	Std	112.67	56.34	0	32	Timing bolus scan for evaluation of vascular structure of abdomen, femurs and lower extremities (Runoff)
			Helical	LargeBody	1.375:1 55.0	0.5	0.625	40	140	680	Std Plus	27.76	2734.59	936.875	32	Helical scanmode 0.625mm, Evaluation of vascular structure of abdomen, femurs and lower extremities (Runoff)
26.18	Runoff 1.25mm	MIROI	Axial	LargeBody	1i	1	5 [0]	5	120	80	Std	112.67	56.34	0	32	Timing bolus scan for evaluation of vascular structure of abdomen, femurs and lower extremities (Runoff)
			Helical	LargeBody	1.375:1 55.0	0.5	1.25 [0.7]	40	140	580	Std Plus	23.68	2479.30	998.9	32	Helical scanmode 1.25mm, Evaluation of vascular structure of abdomen, femurs and lower extremities (Runoff)
26.19	Runoff 0.625mm SmartmA/ SmartPrep		Helical	LargeBody	1.375:1 55.0	0.5	0.625	40	140	SmartmA 100-700mA NI=19 Avg mA 550	Std Plus	22.45	2211.80	936.875	32	Helical scanmode 0.625mm with SmartmA and SmartPrep, Evaluation of vascular structure of abdomen, femurs and lower extremities (Runoff)
26.20	Runoff 1.25mm SmartmA/ SmartPrep		Helical	LargeBody	1.375:1 55.0	0.5	1.25 [0.7]	40	140	SmartmA 100-700mA NI=20 Avg mA 600	Std Plus	24.49	2564.79	998.9	32	Helical scanmode 1.25mm with SmartmA and SmartPrep, Evaluation of vascular structure of abdomen, femurs and lower extremities (Runoff)
26.22	CT Colonography 1.25mm	Advantage CTC Pro	Helical	LargeBody	1.375:1 55.0	0.5	1.25	40	120	200	Std Plus	5.76	258.04	400	32	Helical scan mode supine acquisition for using Advantage CTC Pro for evaluation of the colon
		Advantage CTC Pro	Helical	LargeBody	1.375:1 55.0	0.5	1.25	40	120	200	Std Plus	5.76	258.04	400	32	Helical scan mode prone acquisition for using Advantage CTC Pro for evaluation of the colon

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy-cm)	Scan Length (cm)	Phantom (cm)	Description
26.41	Body Interventional 5mm/3i		Helical	LargeBody	1.375:1 27.50	0.5	5	20	120	SmartmA 50-650mA NI=11.57 Avg mA 360	Std	11.41	251.58	195	32	Localization scan for CT interventional scan
			SmartView SEG	LargeBody	3i	0.4	5	10	120	80	Std Segment	684.58	684.58	0	32	CT Fluoro scan for interventional scan
26.42	Body Interventional 5mm/1i		Helical	LargeBody	1.375:1 27.50	0.5	5	20	120	SmartmA 50-650mA NI=11.57 Avg mA 360	Std	11.41	251.58	195	32	Localization scan for CT interventional scan
			SmartView SEG	LargeBody	1i	0.4	5	5	120	80	Std Segment	845.04	422.52	0	32	CT Fluoro scan for interventional scan

L-Spine

Table 8-7

[Interval]* - For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy-cm)	Scan Length (cm)	Phantom (cm)	Description
27.1	L-Spine 3 Level Axial Group1		Axial	LargeBody	8i	1	2.5	20	120	320	Std	27.13	54.26	17.5	32	Axial scan mode, Evaluation of soft tissue and bony structures lumbar spine
	Group2		Axial	LargeBody	8i	1	2.5	20	120	320	Std	27.13	54.26	17.5	32	Axial scan mode, Evaluation of soft tissue and bony structures lumbar spine
	Group3		Axial	LargeBody	8i	1	2.5	20	140	360	Std	44.46	88.91	17.5	32	Axial scan mode, Evaluation of soft tissue and bony structures lumbar spine
27.3	L-Spine Survey Helical 2.5mm 0.6s. SmartmA		Helical	LargeBody	0.516:1 20.625	0.6	2.5	40	140	SmartmA 100-700mA NI=14 Avg mA 540	Std Plus	70.54	1478.24	75	32	Helical scan mode 2.5mm (0.6sec), Limited evaluation of lumbar spine
27.4	L-Spine Survey Helical 2.5mm 0.8s. SmartmA		Helical	LargeBody	0.516:1 20.625	0.8	2.5	40	140	SmartmA 100-700mA NI=14 Avg mA 525	Std Plus	91.44	1915.92	75	32	Helical scan mode 2.5mm (0.8sec), Limited evaluation of lumbar spine
27.6	L-Spine Helical 0.625mm + DMPR/ SmartmA	DMPR	Helical	LargeBody	0.516:1 20.625	0.6	0.625	40	140	SmartmA 100-715mA NI=20 Avg mA 695	Std Plus IQE	90.79	1902.55	75	32	Helical scan mode 0.625mm with DMPR and SmartmA, Evaluation of soft tissue and bony structures lumbar spine
27.7	L-Spine Helical 1.25mm + DMPR/ SmartmA - Large Patient	DMPR	Helical	LargeBody	0.516:1 20.625	1	1.25	40	140	SmartmA 100-715mA NI=18 Avg mA 700	Std Plus	152.40	3192.68	75	32	Helical scan mode 1.25mm with DMPR and SmartmA, Evaluation of soft tissue and bony structures lumbar spine

Pelvis

Table 8-8
[Interval]* - For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.
For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (cm)	Phantom (cm)	Description
28.1	Pelvis for Fracture + DMPR/ SmartmA	DMPR	Helical	LargeBody	0.984:1 39.375	0.8	1.25 [0.625]	40	120	SmartmA 100-700mA NI=20 Avg mA 500	Bone Plus IQE	32.20	432.30	101.25	32	Helical scan with DMPR and SmartmA, Evaluation of pelvis bony structures for fracture

Lower Extremity

Table 8-9

[Interval]* - For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy-cm)	Scan Length (cm)	Phantom (cm)	Description
29.1	Lower Extremity Survey 2.5mm		Helical	LargeBody	0.516:1 20.625	0.5	2.5	40	120	250	Std Plus	19.21	181.35	60	32	Helical scan mode 2.5mm, Limited evaluation of lower extremity for abnormalities
29.2	Ankle 0.625 mm Single Joint		Helical	MediumBody	0.531:1 10.625	0.5	0.625	20	120	250	Std Plus IQE	20.72	108.05	35	32	Helical scan mode 0.625mm, Evaluation of soft tissue and bony anatomy of the ankle area
29.3	Ankle 0.625 mm Both Joints		Helical	LargeBody	0.531:1 10.625	0.5	0.625	20	120	300	Std Plus IQE	23.94	124.83	35	32	Helical scan mode 0.625mm, Evaluation of soft tissue and bony anatomy of the ankle area
29.4	Knee 0.625 mm Both Joints		Helical	LargeBody	0.531:1 10.625	0.6	0.625	20	120	Auto mA 100-335mA NI=20 Avg mA 275	Std Plus IQE	26.33	137.27	35	32	Helical scan mode 0.625mm, Evaluation of soft tissue and bony anatomy of both knees

**Reference Pediatric Protocols -
Installed Options : Sub 0.4 sec,
VCT85kW and VCT Hi Power**

Head

Table 9-1

[Interval]* - For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (mm)	Phantom (cm)	Description
31.1	PED HEAD TO 18 Months		Axial	Ped Head	4i	0.5	5	20	120	240	Std	20.43	245.19	115	16	Routine head (0.5s) for infant (up to 18 months) 5mm
31.2	PED HEAD 18 MOS TO 5YRS		Axial	SmallHead	4i	0.5	5	20	120	335	Std	28.52	342.25	115	16	Routine head (0.5s) for children (18 months to 5 years) 5mm
31.3	PED HEAD TO 5YRS TO 18YRS Group1		Axial	SmallHead	8i	1	2.5	20	120	240	Std	40.87	163.46	37.5	16	Routine head (1.0s) for children (5 years to 18 years) 2.5mm
	Group2		Axial	SmallHead	4i	1	5	20	120	180	Std	30.65	245.19	75	16	Routine head (1.0s) for children (5 years to 18 years) 5mm

Orbit

Table 9-2

[Interval]* - For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (mm)	Phantom (cm)	Description
32.1	Sinus Supine Helical + DMPR	DMPR	Helical	SmallHead	0.531:1 10.625	0.4	0.625	20	120	80	Bone Plus IQE	10.26	93.90	74.375	16	Routine Helical scan mode with DMPR, Evaluation of sinus area for abnormality in pediatric patients
32.3	Orbits Helical	DMPR	Helical	SmallHead	0.531:1 10.625	0.4	0.625	20	120	180	Std Plus IQE	23.08	182.43	61.875	16	Helical scan mode, Evaluation of orbital and face structures for abnormality in pediatric patients
32.6	Helical IAC 0.625mm Axial mode - Use Reformat for Coronal		Helical	SmallHead	0.531:1 10.625	0.4	0.625	20	120	160	Std Plus IQE	20.51	114.73	38.75	16	Helical scan mode, Evaluation of inner ear structures for abnormality in pediatric patients

Chest

Table 9-3

[Interval]* - For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (mm)	Phantom (cm)	Description
35.1.1	RC 6.0-7.5 kg (13.2-16.5 lbs)		Helical	PedBody	1.375:1 27.5	0.4	3.75	20	100	80	Std Plus	1.11	11.08	75	32	Weight and height based routine chest protocol
35.1.3	SnapShot Segment 30-74 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	120	Std Plus	0.84	8.19	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	Std	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
			Cardiac SEG	CardiacSmall	0.160:1 6.4	0.35	0.625	40	80	ECG Mod 80-280mA Phase 40-80 AvgmA280	Std SS Seg	12.23	134.55	75	32	Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
35.1.4	SnapShot Burst 75-104 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	120	Std Plus	0.84	8.19	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	Std	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
			Cardiac SSB	CardiacSmall	0.160:1 6.4	0.35	0.625	40	80	ECG Mod 80-280mA Phase 40-80 AvgmA280	Std SSB	12.23	134.55	75	32	Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
35.1.5	SnapShot Burst+ >105 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	120	Std Plus	0.84	8.19	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	Std	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
			Cardiac SSBP	CardiacSmall	0.160:1 6.4	0.35	0.625	40	80	ECG Mod 80-280mA Phase 35-85 AvgmA280	Std SSB	13.12	144.35	75	32	Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
35.1.7	Non ECG Gated		Helical	PedBody	0.984:1 39.375	0.4	0.625	40	80	SmartmA 50-200mA NI=18 AvgmA200	Std Plus	1.95	21.00	75	32	Weight and height based non ECG gated protocol for cardiac with SmartmA
35.2.1	RC 7.5-9.5 kg (16.5-20.9lbs)		Helical	PedBody	1.375:1 27.5	0.4	3.75	20	100	120	Std Plus	1.66	16.62	75	32	Weight and height based routine chest protocol
35.2.3	SnapShot Segment 30-74 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	120	Std Plus	0.84	8.19	50	32	Weight and height based protocol for localization

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (mm)	Phantom (cm)	Description
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	Std	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
			Cardiac SEG	CardiacSmall	0.160:1 6.4	0.35	0.625	40	80	ECG Mod 80-300mA Phase 40-80 AvgmA300	Std SS Seg	13.06	143.68	75	32	Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
35.2.4	SnapShot Burst 75-104 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	120	Std Plus	0.84	8.19	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	Std	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
			Cardiac SSB	CardiacSmall	0.160:1 6.4	0.35	0.625	40	80	ECG Mod 80-300mA Phase 40-80 AvgmA300	Std SSB	13.06	143.68	75	32	Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
35.2.5	SnapShot Burst+ >105 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	120	Std Plus	0.84	8.19	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	Std	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
			Cardiac SSBP	CardiacSmall	0.160:1 6.4	0.35	0.625	40	80	ECG Mod 80-300mA Phase 35-85 AvgmA300	Std SSB	14.04	154.46	75	32	Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
35.2.7	Non ECG Gated		Helical	PedBody	0.984:1 39.375	0.4	0.625	40	80	SmartmA 50-220mA NI=18 AvgmA220	Std Plus	2.14	23.10	75	32	Weight and height based non ECG gated protocol for cardiac with SmartmA
35.3.1	RC 9.5-11.5 kg (20.9-25.4lbs)		Helical	SmallBody	1.375:1 27.5	0.4	3.75	20	120	100	Std Plus	2.17	22.53	78.75	32	Weight and height based routine chest protocol
35.3.3	SnapShot Segment 30-74 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	120	Std Plus	0.84	8.19	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	Std	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
			Cardiac SEG	CardiacSmall	0.160:1 6.4	0.35	0.625	40	80	ECG Mod 100-320mA Phase 40-80 AvgmA320	Std SS Seg	14.05	154.52	75	32	Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
35.3.4	SnapShot Burst 75-104 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	120	Std Plus	0.84	8.19	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	Std	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (mm)	Phantom (cm)	Description
			Cardiac SSB	CardiacSmall	0.160:1 6.4	0.35	0.625	40	80	ECG Mod 100-320mA Phase 40-80 AvgmA320	Std SSB	14.05	154.52	75	32	Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
35.3.5	SnapShot Burst+ >105 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	120	Std Plus	0.84	8.19	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	Std	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
			Cardiac SSBP	CardiacSmall	0.160:1 6.4	0.35	0.625	40	80	ECG Mod 100-320mA Phase 35-85 AvgmA320	Std SSB	15.03	165.30	75	32	Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
35.3.7	Non ECG Gated		Helical	PedBody	0.984:1 39.375	0.4	0.625	40	80	SmartmA 50-240mA NI=18 AvgmA240	Std Plus	2.34	25.20	75	32	Weight and height based non ECG gated protocol for cardiac with SmartmA
35.4.1	RC 11.5-14.5kg (25.4-32.0lbs)		Helical	SmallBody	1.375:1 27.5	0.4	5	20	120	90	Std Plus	1.95	22.47	90	32	Weight and height based routine chest protocol
35.4.3	SnapShot Segment 30-74 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	120	Std Plus	0.84	8.19	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	Std	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
			Cardiac SEG	CardiacSmall	0.160:1 6.4	0.35	0.625	40	80	ECG Mod 100-340mA Phase 40-80 AvgmA340	Std SS Seg	14.87	163.52	75	32	Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
35.4.4	SnapShot Burst 75-104 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	120	Std Plus	0.84	8.19	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	Std	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
			Cardiac SSB	CardiacSmall	0.160:1 6.4	0.35	0.625	40	80	ECG Mod 100-340mA Phase 40-80 AvgmA340	Std SSB	14.87	163.52	75	32	Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
35.4.5	SnapShot Burst+ >105 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	120	Std Plus	0.84	8.19	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	Std	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
			Cardiac SSBP	CardiacSmall	0.160:1 6.4	0.35	0.625	40	80	ECG Mod 100-340mA Phase 35-85 AvgmA340	Std SSB	15.93	175.28	75	32	Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval] (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (mm)	Phantom (cm)	Description
35.4.7	Non ECG Gated		Helical	PedBody	0.984:1 39.375	0.4	0.625	40	80	SmartmA 50-280mA NI=18 AvgmA280	Std Plus	2.72	29.40	75	32	Weight and height based non ECG gated protocol for cardiac with SmartmA
35.5.1	RC 14.5-18.5kg (32.0-40.8lbs)		Helical	SmallBody	1.375:1 27.5	0.4	5	20	120	100	Std Plus	2.17	24.97	90	32	Weight and height based routine chest protocol
35.5.3	SnapShot Segment 30-74 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	120	Std Plus	0.84	8.19	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	Std	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
			Cardiac SEG	CardiacSmall	0.160:1 6.4	0.35	0.625	40	80	ECG Mod 100-380mA Phase 40-80 AvgmA380	Std SS Seg	16.52	181.73	75	32	Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
35.5.4	SnapShot Burst 75-104 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	120	Std Plus	0.84	8.19	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	Std	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
			Cardiac SSB	CardiacSmall	0.160:1 6.4	0.35	0.625	40	80	ECG Mod 100-380mA Phase 40-80 AvgmA380	Std SSB	16.52	181.73	75	32	Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
35.5.5	SnapShot Burst+ >105 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	120	Std Plus	0.84	8.19	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	Std	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
			Cardiac SSBP	CardiacSmall	0.160:1 6.4	0.35	0.625	40	80	ECG Mod 100-380mA Phase 35-85 AvgmA380	Std SSB	17.77	195.45	75	32	Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
35.5.7	Non ECG Gated		Helical	PedBody	0.984:1 39.375	0.4	0.625	40	80	SmartmA 50-300mA NI=18 AvgmA300	Std Plus	2.92	31.50	75	32	Weight and height based non ECG gated protocol for cardiac with SmartmA
35.6.1	RC 18.5-22.5kg (40.8-49.6lbs)		Helical	SmallBody	1.375:1 27.5	0.4	5	20	120	110	Std Plus	2.39	27.47	90	32	Weight and height based routine chest protocol
35.6.3	SnapShot Segment 30-74 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	120	Std Plus	0.84	8.19	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	Std	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval] (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (mm)	Phantom (cm)	Description
			Cardiac SEG	CardiacSmall	0.160:1 6.4	0.35	0.625	40	80	ECG Mod 100-400mA Phase 40-80 AvgmA400	Std SS Seg	17.35	190.80	75	32	Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
35.6.4	SnapShot Burst 75-104 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	120	Std Plus	0.84	8.19	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	Std	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
			Cardiac SSB	CardiacSmall	0.160:1 6.4	0.35	0.625	40	80	ECG Mod 100-400mA Phase 40-80 AvgmA400	Std SSB	17.35	190.80	75	32	Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
35.6.5	SnapShot Burst+ >105 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	120	Std Plus	0.84	8.19	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	Std	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
			Cardiac SSBP	CardiacSmall	0.160:1 6.4	0.35	0.625	40	80	ECG Mod 100-400mA Phase 35-85 AvgmA400	Std SSB	18.68	205.51	75	32	Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
35.6.7	Non ECG Gated		Helical	PedBody	0.984:1 39.375	0.4	0.625	40	80	SmartmA 100-320mA NI=18 AvgmA320	Std Plus	3.11	33.71	75	32	Weight and height based non ECG gated protocol for cardiac with SmartmA
35.7.1	RC 22.5-31.5kg (49.6-69.5lbs)		Helical	LargeBody	1.375:1 55.0	0.4	5	40	120	110	Std Plus	2.54	33.71	85	32	Weight and height based routine chest protocol
35.7.3	SnapShot Segment 30-74 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	120	Std Plus	0.84	8.19	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	Std	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
			Cardiac SEG	CardiacSmall	0.160:1 6.4	0.35	0.625	40	80	ECG Mod 110-420mA Phase 40-80 AvgmA420	Std SS Seg	18.25	200.78	75	32	Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
35.7.4	SnapShot Burst 75-104 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	120	Std Plus	0.84	8.19	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	Std	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
			Cardiac SSB	CardiacSmall	0.160:1 6.4	0.35	0.625	40	80	ECG Mod 110-420mA Phase 40-80 AvgmA420	Std SSB	18.25	200.78	75	32	Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (mm)	Phantom (cm)	Description
35.7.5	SnapShot Burst+ >105 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	120	Std Plus	0.84	8.19	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	Std	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
			Cardiac SSBP	CardiacSmall	0.160:1 6.4	0.35	0.625	40	80	ECG Mod 110-420mA Phase 35-85 AvgmA420	Std SSB	19.63	215.97	75	32	Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
35.7.7	Non ECG Gated		Helical	PedBody	0.984:1 39.375	0.4	0.625	40	80	SmartmA 100-360mA NI=18 AvgmA360	Std Plus	3.50	37.93	75	32	Weight and height based non ECG gated protocol for cardiac with SmartmA
35.8.1	RC 31.5-40.5kg (69.5-89.3lbs)		Helical	LargeBody	1.375:1 55.0	0.4	5	40	120	120	Std Plus	2.77	46.45	120	32	Weight and height based routine chest protocol
35.8.3	SnapShot Segment 30-74 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	100	120	Std Plus	1.55	15.20	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	100	50	Std	39.50	19.75	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
			Cardiac SEG	CardiacSmall	0.160:1 6.4	0.35	0.625	40	100	ECG Mod 100-380mA Phase 40-80 AvgmA380	Std SS Seg	30.66	337.28	75	32	Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
35.8.4	SnapShot Burst 75-104 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	100	120	Std Plus	1.55	15.20	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	100	50	Std	39.50	19.75	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
			Cardiac SSB	CardiacSmall	0.160:1 6.4	0.35	0.625	40	100	ECG Mod 100-380mA Phase 40-80 AvgmA380	Std SSB	30.66	337.28	75	32	Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
35.8.5	SnapShot Burst+ >105 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	100	120	Std Plus	1.55	15.20	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	100	50	Std	39.50	19.75	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
			Cardiac SSBP	CardiacSmall	0.160:1 6.4	0.35	0.625	40	100	ECG Mod 100-380mA Phase 35-85 AvgmA380	Std SSB	32.98	362.75	75	32	Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
35.8.7	Non ECG Gated		Helical	SmallBody	0.984:1 39.375	0.4	0.625	40	100	SmartmA 100-400mA NI=16 AvgmA400	Std Plus	7.22	78.21	75	32	Weight and height based non ECG gated protocol for cardiac with SmartmA

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval] (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (mm)	Phantom (cm)	Description
35.9.1	RC 40.5-55.0kg(89.3-121.3lbs)		Helical	LargeBody	1.375:1 55.0	0.4	5	40	120	130	Std Plus	3.00	59.31	150	32	Weight and height based routine chest protocol
35.9.3	SnapShot Segment 30-74 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	100	120	Std Plus	1.55	15.20	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	100	50	Std	39.50	19.75	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
			Cardiac SEG	CardiacSmall	0.160:1 6.4	0.35	0.625	40	100	ECG Mod 100-420mA Phase 40-80 AvgmA420	Std SS Seg	33.73	371.05	75	32	Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
35.9.4	SnapShot Burst 75-104 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	100	120	Std Plus	1.55	15.20	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	100	50	Std	39.50	19.75	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
			Cardiac SSB	CardiacSmall	0.160:1 6.4	0.35	0.625	40	100	ECG Mod 100-420mA Phase 40-80 AvgmA420	Std SSB	33.73	371.05	75	32	Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
35.9.5	SnapShot Burst+ >105 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	100	120	Std Plus	1.55	15.20	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	100	50	Std	39.50	19.75	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
			Cardiac SSBP	CardiacSmall	0.160:1 6.4	0.35	0.625	40	100	ECG Mod 100-420mA Phase 35-85 AvgmA420	Std SSB	36.38	400.15	75	32	Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
35.9.7	Non ECG Gated		Helical	SmallBody	0.984:1 39.375	0.4	0.625	40	100	SmartmA 100-440mA NI=16 AvgmA440	Std Plus	7.95	86.03	75	32	Weight and height based non ECG gated protocol for cardiac with SmartmA

Abdomen

Table 9-4

[Interval]* - For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (second s)	Slice Thickne ss (mm) [Interval]* (mm)	Beam Collimat ion (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy·c m)	Scan Length (mm)	Phantom (cm)	Description
36.1.1	Abd 6.0-7.5kg (13.2-16.5lbs)		Helical	PedBody	1.375:1 27.5	0.4	3.75	20	120	100	Std Plus	2.17	22.53	78.75	32	Weight and height based routine abdomen protocol
36.2.1	Abd 7.5-9.5kg (16.5-20.9lbs)		Helical	PedBody	1.375:1 27.5	0.4	3.75	20	120	160	Std Plus	3.47	39.95	90	32	Weight and height based routine abdomen protocol
36.3.1	Abd 9.5-11.5kg (20.9-25.4lbs)		Helical	PedBody	1.375:1 27.5	0.4	3.75	20	120	180	Std Plus	3.91	49.34	101.25	32	Weight and height based routine abdomen protocol
36.4.1	Abd 11.5-14.5kg (25.4-32.0lbs)		Helical	SmallBo dy	1.375:1 27.5	0.4	5	20	120	125	Std Plus	2.71	36.64	110	32	Weight and height based routine abdomen protocol
36.5.1	Abd 14.5-18.5kg (32.0-40.8lbs)		Helical	SmallBo dy	1.375:1 27.5	0.4	5	20	120	135	Std Plus	2.93	39.57	110	32	Weight and height based routine abdomen protocol
36.6.1	Abd 18.5-22.5kg (40.8-49.6lbs)		Helical	Medium Body	1.375:1 27.5	0.4	5	20	120	120	Std Plus	3.07	47.68	130	32	Weight and height based routine abdomen protocol
36.7.1	Abd 22.5- 31.5kg(49.6- 69.5lbs)		Helical	LargeBo dy	1.375:1 55.0	0.4	5	40	120	150	Std Plus	3.46	68.44	150	32	Weight and height based routine abdomen protocol
36.8.1	Abd 31.5- 40.5kg(69.5- 89.3lbs)		Helical	LargeBo dy	1.375:1 55.0	0.4	5	40	120	170	Std Plus	3.92	77.56	150	32	Weight and height based routine abdomen protocol
36.9.1	Abd 40.5- 55.0kg(89.3- 121.3lbs)		Helical	LargeBo dy	1.375:1 55.0	0.4	5	40	120	200	Std Plus	4.61	114.30	200	32	Weight and height based routine abdomen protocol

Miscellaneous

Table 9-5

[Interval]* - For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (mm)	Phantom (cm)	Description
40.1	Pink Zone Routine 0-20 lbs/0-9 kg		Helical	PedBody	1.375:1 55.0	0.5	5	40	80	SmartmA 65-130mA NI=5 Avg mA 70	Std	0.61	8.70	95	32	Pediatric and Small patient Routine procedure based protocols
40.2	Pink Zone Routine 21-60 lbs/9.1-27.2 kg		Helical	PedBody	1.375:1 55.0	0.5	5	40	100	SmartmA 80-160mA NI=7 Avg mA 70	Std	1.13	27.47	195	32	Pediatric and Small patient Routine procedure based protocols
40.3	Pink Zone Routine 61-100 lbs/27.3-45.4 kg		Helical	Medium Body	1.375:1 55.0	0.5	5	40	120	SmartmA 95-190mA NI=10 Avg mA 70	Std	2.09	71.81	295	32	Pediatric and Small patient Routine procedure based protocols
40.4	Pink Zone Routine 101-200 lbs/45.4-90.7 kg		Helical	LargeBody	1.375:1 55.0	0.5	5	40	120	SmartmA 110-220mA NI=12 Avg mA 70	Std	2.02	89.30	395	32	Pediatric and Small patient Routine procedure based protocols
40.5	Pink Zone Routine >200 lbs/>90.8 kg		Helical	LargeBody	1.375:1 55.0	0.5	5	40	120	SmartmA 125-300mA NI=15 Avg mA 70	Std	2.02	89.30	395	32	Pediatric and Small patient Routine procedure based protocols
40.7	Green Zone Lower Dose 0-20 lbs/0-9 kg		Helical	PedBody	1.375:1 55.0	0.5	5	40	80	SmartmA 50-100mA NI=7 Avg mA 70	Std	0.61	8.70	95	32	Pediatric and Small patient Lower Dose procedure based protocols
40.8	Green Zone Lower Dose 21-60 lbs/9.1-27.2 kg		Helical	PedBody	1.375:1 55.0	0.5	5	40	100	SmartmA 60-120mA NI=9 Avg mA 70	Std	1.13	27.47	195	32	Pediatric and Small patient Lower Dose procedure based protocols
40.9	Green Zone Lower Dose 61-100 lbs/ 27.3-45.4 kg		Helical	Medium Body	1.375:1 55.0	0.5	5	40	120	SmartmA 70-140mA NI=11 Avg mA 70	Std	2.09	71.81	295	32	Pediatric and Small patient Lower Dose procedure based protocols
40.10	Green Zone Lower Dose 101-200 lbs/ 45.4-90.7 kg		Helical	LargeBody	1.375:1 55.0	0.5	5	40	120	SmartmA 80-160mA NI=13 Avg mA 70	Std	2.02	89.30	395	32	Pediatric and Small patient Lower Dose procedure based protocols
40.11	Green Zone Lower Dose >200 lbs/ >90.8 kg		Helical	LargeBody	1.375:1 55.0	0.5	5	40	120	SmartmA 90-240mA NI=16 Avg mA 70	Std	2.02	89.30	395	32	Pediatric and Small patient Lower Dose procedure based protocols
40.13	Grey Zone CT Angio 0-20 lbs/0-9 kg		Helical	PedBody	1.375:1 55.0	0.5	2.5	40	80	SmartmA 100-200mA NI=5 Avg mA 140	Std	1.22	17.71	97.5	32	Pediatric and Small patient Angio procedure based protocols

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval J* (mm)]	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (mm)	Phantom (cm)	Description
40.14	Grey Zone CT Angio 21-60 lbs/9.1-27.2 kg		Helical	SmallBody	1.375:1 55.0	0.5	2.5	40	100	SmartmA 120-240mA NI=9.9 Avg mA 140	Std	2.26	55.50	197.5	32	Pediatric and Small patient Angio procedure based protocols
40.15	Grey Zone CT Angio 61-100 lbs/ 27.3-45.4 kg		Helical	LargeBody	1.375:1 55.0	0.5	2.5	40	120	SmartmA 120-240mA NI=10 Avg mA 140	Std	4.03	139.28	297.5	32	Pediatric and Small patient Angio procedure based protocols
40.16	Grey Zone CT Angio 101-200 lbs/ 45.4-90.7 kg		Helical	LargeBody	1.375:1 55.0	0.5	2.5	40	120	SmartmA 120-240mA NI=12 Avg mA 140	Std	4.03	179.62	397.5	32	Pediatric and Small patient Angio procedure based protocols
40.17	Grey Zone CT Angio >200 lbs/ >90.8 kg		Helical	LargeBody	1.375:1 55.0	0.5	2.5	40	120	SmartmA 120-300mA NI=15 Avg mA 140	Std	4.03	179.62	397.5	32	Pediatric and Small patient Angio procedure based protocols

Reference Adult Protocols - Installed Options : Sub 0.4sec and VCT85kW

Head

Table 10-1

[Interval]* - For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (s)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (mm)	Phantom (cm)	Description
21.1	Routine Head 1s Group1		Axial	Head	4i	1	5	20	140	200	Std	53.34	320.02	55	16	Routine head protocol for evaluation of the brain for abnormalities.
	Group2		Axial	Head	4i	1	5	20	120	320	Std	61.43	491.41	75	16	
21.2	Routine Head 2s Group1		Axial	Head	4i	2	5	20	140	100	Std	53.34	320.02	55	16	Routine head protocol for evaluation of the brain for abnormalities.
	Group2		Axial	Head	4i	2	5	20	120	160	Std	61.43	491.41	75	16	
21.3	Trauma Head 2.5mm		Axial	Head	8i	0.5	2.5	20	140	560	Std	76.76	1074.71	137.5	16	Emergency head protocol for evaluation of the brain and cranium for abnormalities.
21.4	Routine Head 0.5s Group1		Axial	Head	4i	0.5	5	20	140	400	Std	54.83	328.99	55	16	Routine head protocol for evaluation of the brain for abnormalities.
	Group2		Axial	Head	4i	0.5	5	20	120	600	Std	59.20	473.62	75	16	
21.5	Circle of Willis 0.4s	MIROI	Axial	Head	1i	1	5 [0]	5	120	50	Std	125.58	62.79	0	16	Test bolus for CT angiography of the head for evaluation of carotid and cerebral vasculature.
			Helical	Head	0.531:1 10.625	0.4	0.625	20	120	300	Std Plus IQE	43.36	421.37	80	16	CT angiography of the head for evaluation of carotid and cerebral vasculature.
21.6	CoW/ Carotid 0.625mm 0.4s.Smart mA	MIROI	Axial	Head	1i	1	5 [0]	5	120	80	Std	180.84	90.42	0	16	Test bolus for CT angiography of the head for evaluation of carotid and cerebral vasculature.
			Helical	Head	0.531:1 10.625	0.4	0.625	20	120	SmartmA 100-650mA NI=8 Avg mA 595	Std Plus IQE	88.41	1657.19	170	16	CT angiography of the head for evaluation of carotid and cerebral vasculature.
21.8	Helical Head	DMPR	Helical	Head	0.531:1 10.625	0.5	0.625	20	120	300	Std Plus IQE	54.20	878.85	145	16	Helical scan mode for evaluation of the brain for cerebral abnormality.
21.10	VHS Head 4D CTA	MIROI	Axial	Head	1i	1	5 [0]	5	120	50	Std	125.58	62.79	0	16	Test bolus for CT angiography of the head for evaluation of carotid and cerebral vasculature.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (s)	Slice Thickness (mm) [Interval] J* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (mm)	Phantom (cm)	Description
			Helical-S	Head	0.984:1 39.375 18 pass	0.4	5 [10]	40	120	300	Std Plus	496.19	6202.39	110	16	CT angiography of the head using Volume Helical scan mode for evaluation of cerebral vasculature.
21.11	VHS Perfusion 350-370 Strength Contrast		Axial	Head	4i	1	5	20	120	335	Std	64.31	900.28	135	16	Non-Enhance Brain, evaluation for hemorrhage or infarction.
		Perfusion	Helical-S/P	Head	0.984:1 39.375 28 pass	0.4	5	40	80	200	Std	186.02	2325.25	110	16	CT Perfusion using Volume Helical scan mode for evaluation of cerebral perfusion over time.
			Axial	Head	4i	1	5	20	120	335	Std	64.31	900.28	135	16	Post Contrast Brain, Evaluation for residual tissue enhancement.
21.12	CT Perfusion 300-320 Strength Contrast		Axial	Head	4i	1	5	20	120	335	Std	64.31	900.28	135	16	Non-Enhance Brain, evaluation for hemorrhage or infarction.
		Perfusion	Cine	Head	8i	1	5 [0]	40	80	200	Std	652.82	2611.27	35	16	CT Perfusion using Cine scan mode for evaluation of cerebral perfusion over time.
			Axial	Head	4i	1	5	20	120	335	Std	64.31	900.28	135	16	Non-Enhance Brain, evaluation for hemorrhage or infarction.
21.13	CT Perfusion 350-370 Strength Contrast		Axial	Head	4i	1	5	20	120	335	Std	64.31	900.28	135	16	Non-Enhance Brain, evaluation for hemorrhage or infarction.
		Perfusion	Cine	Head	8i	1	5 [0]	40	80	200	Std	587.55	2350.19	35	16	CT Perfusion using Cine scan mode for evaluation of cerebral perfusion over time.
			Axial	Head	4i	1	5	20	120	335	Std	64.31	900.28	135	16	Post Contrast Brain, Evaluation for residual tissue enhancement.
21.14	CT Perfusion Brain Tumor - Dual Phase		Axial	Head	4i	1	5	20	120	335	Std	64.31	900.28	135	16	Non-Enhance Brain, evaluation for hemorrhage or infarction.
	Group 1	Perfusion	Cine	Head	8i	1	5 [0]	40	80	200	Std	587.55	2350.19	35	16	CT Perfusion using Cine scan mode for evaluation of cerebral perfusion over time.
	Group 2	Perfusion	Axial	Head	8i	1	5 [0]	40	80	200	Std	130.54	522.15	35	16	Delayed phase CT Perfusion using Axial scan mode for evaluation of Brain tumor permeability.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (s)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (mm)	Phantom (cm)	Description
			Axial	Head	4i	1	5	20	120	335	Std	64.31	900.28	135	16	Post Contrast Brain, Evaluation for residual tissue enhancement.
21.15	CT Perfusion 350-370 Strength Contrast (Axial)		Axial	Head	4i	1	5	20	120	335	Std	64.31	900.28	135	16	Non-Enhance Brain, evaluation for hemorrhage or infarction.
		Perfusion	Axial	Head	8i	1	5 [0]	40	80	200	Std	287.18	1148.73	35	16	CT Perfusion using Axial scan mode for evaluation of cerebral perfusion over time.
			Axial	Head	4i	1	5	20	120	335	Std	64.31	900.28	135	16	Post Contrast Brain, Evaluation for residual tissue enhancement.
21.16	CT Perfusion 350-370 Strength Contrast (Axial Shuttle Mode)		Axial	Head	4i	1	5	20	120	335	Std	64.31	900.28	135	16	Non-Enhance Brain, evaluation for hemorrhage or infarction.
		Perfusion	Axial-S	Head	8i 17 pass	0.4	5	40	80	500	Std	221.91	1775.31	75	16	CT Perfusion using Volume Shuttle (Axial) scan mode for evaluation of cerebral perfusion over time.
			Axial	Head	4i	1	5	20	120	335	Std	64.31	900.28	135	16	Post Contrast Brain, Evaluation for residual tissue enhancement.

Orbit

Table 10-2

[Interval]* - For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy-cm)	Scan Length (cm)	Phantom (cm)	Description
22.1	Sinus Supine Helical + DMPR	DMPR	Helical	Head	0.531:1 10.625	0.4	0.625	20	120	120	Bone Plus IQE	17.34	158.79	74.375	16	Routine Helical Sinus protocol, Evaluation for any abnormality
22.3	Orbits Helical + DMPR	DMPR	Helical	Head	0.531:1 10.625	0.5	0.625	20	120	220	Std Plus IQE	39.75	314.09	61.875	16	Routine Helical Orbit protocol, Evaluation for abnormality soft tissues and bone structures of the face.
22.6	Helical IAC 0.625mm Axial mode - Use Reformat for Coronal		Helical	Head	0.531:1 10.625	0.6	0.625	20	140	270	Std Plus IQE	81.32	454.44	38.75	16	Axial scan mode, Evaluation for inner ear abnormalities in Axial plane
22.8	Axial and Coronal IAC		Axial	Head	16i	1	1.25	20	140	280	Std	74.67	298.68	38.75	16	Helical scan mode, Evaluation for inner ear abnormalities in Axial plane
			Axial	Head	16i	1	1.25	20	140	335	Std	89.34	357.35	38.75	16	Helical scan mode, Evaluation for inner ear abnormalities in Coronal plane

Neck

Table 10-3

[Interval]* - For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (cm)	Phantom (cm)	Description
23.1	C-Spine C5-C7 Axial		Axial	SmallBody	8i	0.5	2.5	20	140	610	Std	33.17	464.38	137.5	32	C-Spine C5-C7 area Axial mode, Evaluation for soft tissue, disc and bone abnormalities.
23.2	C-Spine C1-C4 Axial		Axial	SmallBody	8i	1	2.5	20	120	240	Std	17.90	143.19	77.5	32	C-Spine C1-C4 area Axial mode, Evaluation for soft tissue, disc and bone abnormalities.
23.3	C-Spine Helical + DMPR	DMPR	Helical	SmallBody	0.516:1 20.625	0.8	0.625	40	120	335	Bone Plus IQE	36.23	522.92	110	32	C-Spine Helical mode with DMPR, Evaluation for soft tissue, disc and bone abnormalities.
23.4	C-Spine Helical + DMPR/ SmartmA	DMPR	Helical	SmallBody	0.516:1 20.625	0.8	0.625	40	120	SmartmA 100-650mA NI=11.34 Avg mA 310	Bone Plus IQE	33.52	483.89	110	32	C-Spine Helical mode with DMPR and SmartmA, Evaluation for soft tissue, disc and bone abnormalities.
23.8	Carotid 0.625mm 0.4s. 0.5:1 SmartmA/ SmartPrep		Helical	SmallBody	0.516:1 20.625	0.4	0.625	40	140	SmartmA 100-610mA NI=11 Avg mA 575	Std Plus IQE	44.10	902.52	170	32	Carotid CTA (pitch 0.5:1) with SmartmA and SmartPrep, Evaluation of carotid vasculature.
23.9	Carotid 0.625mm 0.4s. 0.9:1 SmartmA/ SmartPrep		Helical	SmallBody	0.984:1 39.375	0.4	0.625	40	120	SmartmA 80-700mA NI=11 Avg mA 625	Std Plus IQE	17.70	359.81	170	32	Carotid CTA (pitch 0.9:1) with SmartmA and SmartPrep, Evaluation of carotid vasculature.
23.10	Carotid/ CoW 0.625mm 0.4s. AutomA	MIROI	Axial	SmallBody	1i	1	5 [0]	5	120	80	Std	99.00	49.50	0	32	Test bolus for Carotid CTA (pitch 0.5:1) of the head for evaluation of carotid and cerebral vasculature.
			Helical	SmallBody	0.516:1 20.625	0.4	0.625	40	140	AutomA 80-610mA NI=11 Avg mA 600	Std Plus IQE	46.02	941.76	170	32	Carotid and Brain CTA (pitch 0.5:1) with AutomA, Evaluation of carotid and cerebral vasculature.
23.11	Carotid/ CoW 0.625mm 0.4s. SmartmA/ SmartPrep		Helical	SmallBody	0.969:1 19.375	0.4	0.625	20	140	SmartmA 100-610mA NI=11 Avg mA 610	Std Plus IQE	27.39	517.42	170	32	Carotid CTA (pitch 0.9:1) with SmartmA and SmartPrep, Evaluation of carotid and cerebral vasculature.
23.12	Soft Tissue Neck 2.5mm AutomA		Helical	SmallBody	0.984:1 39.375	0.5	2.5	40	120	AutomA 100-335mA NI=9.1 Avg mA 250	Std Plus	8.85	161.79	150	32	Neck with AutomA, Evaluation of soft tissue structures for abnormality

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy-cm)	Scan Length (cm)	Phantom (cm)	Description
23.13	Soft Tissue Neck + DMPR/ SmartmA	DMPR	Helical	SmallBody	0.969:1 19.375	0.5	0.625	20	120	SmartmA 100-600mA NI=12.6 Avg mA 425	Std Plus IQE	16.82	283.95	150	32	Neck with DMPR and SmartmA, Evaluation of soft tissue structures for abnormality

Upper Extremity

Table 10-4

[Interval]* - For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy-cm)	Scan Length (cm)	Phantom (cm)	Description
24.1	Shoulder 1.25mm + DMPR	DMPR	Helical	LargeBody	0.516:1 20.625	0.6	1.25 [0.625]	40	140	600	Std Plus IQE	78.38	1054.65	100	32	Shoulder with DMPR, Evaluation of soft tissue and bone of shoulder area
24.2	Shoulder 1.25mm + DMPR/ SmartmA	DMPR	Helical	LargeBody	0.516:1 20.625	0.6	1.25 [0.625]	40	140	SmartmA 100-650mA NI=20 Avg mA 595	Std Plus IQE	77.73	1045.86	100	32	Shoulder with DMPR and SmartmA, Evaluation of soft tissue and bone of shoulder area

Chest

Table 10-5

[Interval]* - For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (second s)	Slice Thickne ss (mm) [Interval]* (mm)	Beam Collimat ion (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy·c m)	Scan Length (cm)	Phantom (cm)	Description
25.1	Routine Chest 0.4s. 5mm SmartmA		Helical	LargeBody	1.375:1 55.0	0.4	5	40	120	SmartmA 100-650mA NI=13 Avg mA 500	Std Plus	11.52	286.26	200	32	Chest with SmartmA, Evaluation of mediastinum and lungs fields
25.2	Routine Chest 0.4s.+ DMPR/ SmartmA	DMPR	Helical	LargeBody	0.984:1 39.375	0.4	0.625	40	120	SmartmA 100-700mA NI=22 Avg mA 620	Std Plus	19.96	465.64	200	32	Chest with SmartmA and DMPR, Evaluation of mediastinum and lungs fields
25.3	Chest with HiRes 0.4s. SmartmA		Helical	LargeBody	0.969:1 19.375	0.4	5	20	120	SmartmA 80-650mA NI=12 Avg mA 500	Chest Plus	17.99	393.87	200	32	Chest with SmartmA, Evaluation of mediastinum and lung fields with higher resolution images of the lungs
25.4	Hi Res Chest 0.5s. Axial		Axial	LargeBody	1i	0.5	1.25 [10]	1.25	120	380	Bone+	4.48	107.55	230	32	Axial scan mode, Evaluation of mediastinum and lung fields in high resolution mode
25.5	Chest Abd Pelvis 0.4/ 0.5s. SmartmA 5mm (2 groups) Group1		Helical	LargeBody	0.984:1 39.375	0.4	5	40	120	SmartmA 100-650mA NI=13 Avg mA 370	Std Plus	11.91	218.32	150	32	Multi group scan for Chest, Abdomen, Pelvis (0.4/0.5 sec) with SmartmA, Evaluation of chest and abdominal anatomy for abnormalities
	Group2		Helical	LargeBody	1.375:1 55.0	0.5	5	40	120	SmartmA 100-650mA NI=13 Avg mA 370	Std Plus	10.66	365.90	295	32	
25.6	Chest Abd Pelvis 0.5s + DMPR SmartPrep	DMPR	Helical	LargeBody	1.375:1 55.0	0.5	0.625	40	120	SmartmA 100-700mA NI=22 Avg mA 540	Std Plus	15.56	775.18	450	32	Helical scan mode (0.5sec) with SmartmA, Evaluation of chest and abdominal anatomy for abnormalities
25.7	Pulmonary Embolis 0.5s.	MIROI	Axial	LargeBody	1i	1	5 [0]	5	120	80	Std	93.89	46.95	0	32	Timing Bolus for Helical scan mode 1.25mm thickness and helical pitch of 0.9:1, evaluation of Chest for pulmonary embolism
			Helical	LargeBody	0.984:1 39.375	0.5	1.25	40	120	580	Std	23.34	497.55	180	32	Helical scan mode 1.25mm thickness and helical pitch of 0.9:1, Evaluation Chest for pulmonary embolism

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy-cm)	Scan Length (cm)	Phantom (cm)	Description
25.8	Pulmonary Embolis 0.5s. SmartmA	MIROI	Axial	LargeBody	1i	1	5 [0]	5	120	80	Std	93.89	46.95	0	32	Timing Bolus for Helical scan mode 1.25mm thickness and helical pitch of 0.9:1 with SmartmA, evaluation of Chest for pulmonary embolism
			Helical	LargeBody	0.984:1 39.375	0.5	1.25	40	120	SmartmA 100-700mA NI=20 Avg mA 500	Std	20.12	428.92	180	32	Helical scan mode 1.25mm thickness and helical pitch of 0.9:1 with SmartmA, Evaluation Chest for pulmonary embolism
25.9	Aortic Dissection 0.625mm + DMPR/ SmartmA/ SmartPrep	DMPR	Helical	LargeBody	1.375:1 55.0	0.5	0.625	40	120	SmartmA 100-700mA NI=22 Avg mA 600	Std Plus	17.29	602.00	300	32	Helical scan mode 0.625mm thickness and helical pitch of 1.375:1 with SmartmA, SmartPrep and DMPR, Evaluation for aortic dissection
25.10	Aortic Dissection 1.25mm + DMPR/ SmartmA/ SmartPrep	DMPR	Helical	LargeBody	1.375:1 55.0	0.5	1.25 [0.8]	40	120	SmartmA 100-700mA NI=20 Avg mA 550	Std Plus	15.85	551.83	300	32	Helical scan mode 1.25mm thickness and helical pitch of 1.375:1 with SmartmA, SmartPrep and DMPR, Evaluation for aortic dissection
25.11	Lung Assessment	Advanced Lung Analysis	Helical	LargeBody	0.969:1 19.375	0.5	1.25	20	120	120	Bone	5.25	114.77	200	32	Helical scan mode for acquisition of data for Advanced Lung Analysis (ALA) software for lung nodule assessment
25.12	Gated Chest Cardiac Helical		Helical	LargeBody	1.375:1 55.0	0.4	5	40	120	60	Std Plus	1.38	13.55	50	32	Localizer scan for Helical Gated scan for evaluation of thoracic vasculature
		MIROI	Axial	LargeBody	1i	1	5 [0]	5	120	60	Std	98.59	49.29	0	32	Timing bolus scan for Helical Gated scan for evaluation of thoracic vasculature
			Cardiac SEG	CardiacLarge	0.160:1 6.4	0.35	0.625	40	120	ECG Mod 185-450mA Phase 70-80 Avg mA 450	Std SS Seg	51.77	1734.24	300	32	Helical Gated scan for evaluation of thoracic vasculature
25.13	Gated Chest SnapShot Pulse		Helical	LargeBody	1.375:1 55.0	0.4	5	40	120	60	Std Plus	1.38	13.55	50	32	Localizer scan for Cine Gated scan for evaluation of thoracic vasculature
		MIROI	Axial	LargeBody	1i	1	5 [0]	5	120	60	Std	98.59	49.29	0	32	Timing bolus scan for Cine Gated scan for evaluation of thoracic vasculature
			Cardiac Pulse Cine	CardiacLarge	56i	0.35	0.625 [35]	40	120	450	Std SS Seg	24.04	757.31	314.375	32	Cine Gated scan for evaluation of thoracic vasculature
25.14	SmartScore - Gated 0.35s	Smart Score	Cine	LargeBody	8i	0.35	2.5	20	120	430	Std Segment	8.99	89.95	97.5	32	Cine gated scan mode for acquisition of data for coronary artery calcification analysis

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy-cm)	Scan Length (cm)	Phantom (cm)	Description
25.15	SnapShot Pulse 30-65 BPM		Helical	LargeBody	1.375:1 55.0	0.4	5	40	120	60	Std Plus	1.38	13.55	50	32	Localizer scan for Prospectively gated Cardiac Cine scan for evaluation of coronary arteries, intended for stable heart rates 30-65 bpm
		MIROI	Axial	LargeBody	1i	1	5 [0]	5	120	60	Std	98.59	49.29	0	32	Timing bolus scan for Prospectively gated Cardiac Cine scan for evaluation of coronary arteries, intended for stable heart rates 30-65 bpm
			Cardiac Pulse Cine	CardiacSmall	56i	0.35	0.625 [35]	40	120	450	Std SS Seg	21.15	222.06	104.375	32	Prospectively gated Cardiac Cine scan for evaluation of coronary arteries, intended for stable heart rates 30-65 bpm
25.16	SnapShot Segment 30-74 BPM		Helical	LargeBody	1.375:1 55.0	0.4	5	40	120	60	Std Plus	1.38	13.55	50	32	Localizer scan for Retrospectively gated Cardiac Helical scan for evaluation of coronary arteries, intended for heart rates 30-74 bpm
		MIROI	Axial	LargeBody	1i	1	5 [0]	5	120	60	Std	98.59	49.29	0	32	Timing bolus scan for Retrospectively gated Cardiac Helical scan for evaluation of coronary arteries, intended for heart rates 30-74 bpm
			Cardiac SEG	CardiacLarge	0.160:1 6.4	0.35	0.625	40	120	ECG Mod 200-450mA Phase 70-80 Avg mA 450	Std SS Seg	55.39	747.80	100	32	Retrospectively gated Cardiac Helical scan for evaluation of coronary arteries, intended for heart rates 30-74 bpm
25.17	SnapShot Burst 75-113 BPM		Helical	LargeBody	1.375:1 55.0	0.4	5	40	120	60	Std Plus	1.38	13.55	50	32	Localizer scan for Retrospectively gated Cardiac Helical scan for evaluation of coronary arteries, intended for heart rates 75-113 bpm
		MIROI	Axial	LargeBody	1i	1	5 [0]	5	120	60	Std	98.59	49.29	0	32	Timing bolus scan for Retrospectively gated Cardiac Helical scan for evaluation of coronary arteries, intended for heart rates 75-113 bpm
			Cardiac SSB	CardiacLarge	0.160:1 6.4	0.35	0.625	40	120	ECG Mod 200-450mA Phase 70-80 Avg mA 450	Std SSB	55.39	747.80	100	32	Retrospectively gated Cardiac Helical scan for evaluation of coronary arteries, intended for heart rates 75-113 bpm
25.18	SnapShot Burst Plus 114+ BPM		Helical	LargeBody	1.375:1 55.0	0.4	5	40	120	60	Std Plus	1.38	13.55	50	32	Localizer scan for Retrospectively gated Cardiac Helical scan for evaluation of coronary arteries, intended for heart rates over 113 bpm

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy-cm)	Scan Length (cm)	Phantom (cm)	Description
		MIROI	Axial	LargeBody	1i	1	5 [0]	5	120	60	Std	98.59	49.29	0	32	Timing bolus scan for Retrospectively gated Cardiac Helical scan for evaluation of coronary arteries, intended for heart rates over 113 bpm
			Cardiac SSBP	CardiacLarge	0.160:1 6.4	0.35	0.625	40	120	ECG Mod 200-450mA Phase 70-80 Avg mA 450	Std SSB	55.39	747.80	100	32	Retrospectively gated Cardiac Helical scan for evaluation of coronary arteries, intended for heart rates over 113 bpm
25.19	Advantage 4D		Helical	LargeBody	1.375:1 55.0	0.6	5	40	120	285	Std Plus	9.85	244.08	200	32	Localizer scan for Advantage 4D series
		Advantage 4D	Cine	LargeBody	8i	1	2.5	20	120	100	Std	50.95	815.24	157.5	32	Cine scan mode for acquisition data for Advantage 4D for retrospective gating
25.20	VHS Chest 4D CTA SmartPrep		Helical-S	LargeBody	1.375:1 55.0 11 pass	0.4	5 [10]	40	120	SmartmA 150-500mA NI=14 Avg mA 350	Std Plus	112.15	3532.61	300	32	Volume Helical scan mode with SmartPrep for time resolved CT angiography of chest vasculature

Abdomen

Table 10-6

[Interval]* - For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy-cm)	Scan Length (cm)	Phantom (cm)	Description
26.1	Abdomen Pelvis 0.5s. 5mm SmartmA		Helical	LargeBody	1.375:1 55.0	0.5	5	40	120	SmartmA 50-650mA NI=11.57 Avg mA 500	Std Plus	14.41	645.73	400	32	Basic Abdomen Pelvis (0.5sec) with Smart mA, Evaluation for abnormalities of abdominal and pelvic structures
26.2	Abdomen Pelvis 0.5s. 5mm + DMPR SmartmA/ SmartPrep	DMPR	Helical	LargeBody	0.984:1 39.375	0.6	0.625	40	120	SmartmA 80-700mA NI=18 Avg mA 520	Std Plus	25.11	1087.62	400	32	Basic Abdomen Pelvis (0.5sec) with Smart mA, SmartPrep and DMPR, Evaluation for abnormalities of abdominal and pelvic structures
26.4	Renal Stone 0.5s. 5mm		Helical	LargeBody	1.375:1 55.0	0.5	5	40	120	380	Std Plus	10.95	490.75	400	32	Helical scan, Evaluation of kidney, ureters and bladder for renal stones
26.5	Renal Stone 0.5s. 5mm SmartmA		Helical	LargeBody	1.375:1 55.0	0.5	5	40	120	SmartmA 50-650mA NI=12 Avg mA 380	Std Plus	10.95	490.75	400	32	Helical scan with SmartmA, Evaluation of kidney, ureters and bladder for renal stones
26.7	AAA 0.5s. + DMPR 0.625mm	MIROI	Axial	LargeBody	1i	1	5 [0]	5	120	80	Std	112.67	56.34	0	32	Timing bolus for Helical scan with DMPR, Evaluation for abdominal aortic aneurysm
		DMPR	Helical	LargeBody	1.375:1 55.0	0.5	0.625	40	140	610	Std Plus	24.90	890.50	309.375	32	Helical scan with DMPR, Evaluation for abdominal aortic aneurysm
26.8	AAA 0.5s. + DMPR 0.625mm SmartmA/ SmartPrep	DMPR	Helical	LargeBody	1.375:1 55.0	0.6	0.625	40	120	SmartmA 100-700mA NI=19 Avg mA 550	Std Plus	19.02	679.85	309.375	32	Helical scan with DMPR, Smart mA and SmartPrep, Evaluation for abdominal aortic aneurysm
26.10	Dual Phase Liver		Helical	LargeBody	1.375:1 55.0	0.6	5	40	120	300	Std Plus	10.37	184.32	130	32	Non contrast evaluation of upper abdominal structures including liver
		MIROI	Axial	LargeBody	1i	1	5 [0]	5	120	80	Std	112.67	56.34	0	32	Timing bolus for Multi-Group acquisition for early arterial and portal venous phase of the liver
	Group1		Helical	LargeBody	1.375:1 55.0	0.5	2.5	40	120	550	Std Plus	15.85	282.44	130	32	Acquisition for early arterial phase of the liver
	Group2		Helical	LargeBody	1.375:1 55.0	0.5	5	40	120	420	Std Plus	12.10	542.41	400	32	Acquisition for portal venous phase of the liver

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy-cm)	Scan Length (cm)	Phantom (cm)	Description
26.11	Tri Phase Liver		Helical	LargeBody	1.375:1 55.0	0.6	5	40	120	300	Std Plus	10.37	184.32	130	32	Non contrast evaluation of upper abdominal structures including liver
		MIROI	Axial	LargeBody	1i	1	5 [0]	5	120	80	Std	112.67	56.34	0	32	Timing bolus for Multi-Group acquisition for early arterial, late arterial and portal venous phase of the liver
	Group1		Helical	LargeBody	1.375:1 55.0	0.5	2.5	40	120	550	Std Plus	15.85	282.44	130	32	Acquisition for early arterial phase of the liver
	Group2		Helical	LargeBody	1.375:1 55.0	0.5	5	40	120	450	Std Plus	12.97	581.15	400	32	Acquisition for late arterial phase of the liver
	Group3		Helical	LargeBody	1.375:1 55.0	0.5	5	40	120	450	Std Plus	12.97	581.15	400	32	Acquisition for portal venous phase of the liver
26.12	Dual Phase Pancreas SmartmA/ SmartPrep		Helical	LargeBody	0.984:1 39.375	0.5	5	40	120	SmartmA 80-650mA NI=12 Avg mA 400	Std Plus	16.10	166.06	70	32	Non contrast evaluation of upper abdominal structures including pancreas
	Group1		Helical	LargeBody	0.984:1 39.375	0.5	2.5	40	120	SmartmA 50-700mA NI=15 Avg mA 520	Std Plus	20.93	383.30	150	32	Acquisition with SmartmA and SmartPrep for arterial phase of the pancreas
	Group2		Helical	LargeBody	0.984:1 39.375	0.5	5	40	120	SmartmA 50-700mA NI=15 Avg mA 450	Std Plus	18.11	639.58	320	32	Acquisition for venous phase of the pancreas
26.13	Dual Phase Liver SmartmA/ SmartPrep		Helical	LargeBody	1.375:1 55.0	0.5	5	40	120	SmartmA 80-500mA NI=12 Avg mA 320	Std Plus	9.22	163.92	130	32	Non contrast evaluation of upper abdominal structures including liver
	Group1		Helical	LargeBody	1.375:1 55.0	0.5	2.5	40	120	SmartmA 50-700mA NI=15 Avg mA 550	Std Plus	15.85	282.44	130	32	Acquisition with SmartmA and SmartPrep for early arterial phase of the liver
	Group2		Helical	LargeBody	0.984:1 39.375	0.5	5	40	120	SmartmA 50-700mA NI=11 Avg mA 450	Std Plus	18.11	784.46	400	32	Acquisition with SmartmA and SmartPrep for portal venous phase of the liver
26.14	Tri Phase Liver SmartmA/ SmartPrep		Helical	LargeBody	0.984:1 39.375	0.5	5	40	120	SmartmA 80-500mA NI=12 Avg mA 400	Std Plus	16.10	262.65	130	32	Non contrast evaluation of upper abdominal structures including liver
	Group1		Helical	LargeBody	1.375:1 55.0	0.5	2.5	40	120	SmartmA 50-700mA NI=15 Avg mA 550	Std Plus	15.85	282.44	130	32	Acquisition with SmartmA and SmartPrep for early arterial phase of the liver
	Group2		Helical	LargeBody	1.375:1 55.0	0.5	5	40	120	SmartmA 50-700mA NI=11 Avg mA 450	Std Plus	12.97	581.15	400	32	Acquisition with SmartmA and SmartPrep for late arterial phase of the liver
	Group3		Helical	LargeBody	1.375:1 55.0	0.5	5	40	120	SmartmA 50-700mA NI=11 Avg mA 450	Std Plus	12.97	581.15	400	32	Acquisition with SmartmA and SmartPrep for portal venous phase of the liver
26.15	VHS Liver Perfusion		Helical	LargeBody	0.984:1 39.375	0.6	5	40	120	170	Std	8.21	133.61	130	32	Non contrast evaluation of upper abdominal structures including liver

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy-cm)	Scan Length (cm)	Phantom (cm)	Description
		Perfusion	Helical-S/P	LargeBody	1.375:1 55.0 21 pass	0.4	5	40	120	300	Std Plus	219.99	3409.79	140	32	Volume Helical scan mode for evaluation of liver tumor perfusion over time
26.16	VHS 4D CTA		Helical	LargeBody	0.984:1 39.375	0.6	5	40	120	340	Std	16.42	267.79	130	32	Localizer scan for Volume Helical Shuttle scan mode for evaluation of abdominal vasculature
		MIROI	Axial	LargeBody	1i	1	5 [0]	5	120	80	Std	112.67	56.34	0	32	Timing bolus scan for Volume Helical Shuttle scan mode for evaluation of abdominal vasculature
			Helical-S	LargeBody	1.375:1 55.0 18 pass	0.5	5 [10]	40	140	SmartmA 150-610mA NI=12 Avg mA 350	Std Plus	325.68	6350.70	180	32	Volume Helical Shuttle scan mode for time resolved CT angiography of abdominal vasculature
26.17	Runoff 0.625mm	MIROI	Axial	LargeBody	1i	1	5 [0]	5	120	80	Std	112.67	56.34	0	32	Timing bolus scan for evaluation of vascular structure of abdomen, femurs and lower extremities (Runoff)
			Helical	LargeBody	1.375:1 55.0	0.5	0.625	40	140	610	Std Plus	24.90	2453.08	936.875	32	Helical scanmode 0.625mm, Evaluation of vascular structure of abdomen, femurs and lower extremities (Runoff)
26.18	Runoff 1.25mm	MIROI	Axial	LargeBody	1i	1	5 [0]	5	120	80	Std	112.67	56.34	0	32	Timing bolus scan for evaluation of vascular structure of abdomen, femurs and lower extremities (Runoff)
			Helical	LargeBody	1.375:1 55.0	0.5	1.25 [0.7]	40	140	580	Std Plus	23.68	2479.30	998.9	32	Helical scanmode 1.25mm, Evaluation of vascular structure of abdomen, femurs and lower extremities (Runoff)
26.19	Runoff 0.625mm SmartmA/ SmartPrep		Helical	LargeBody	1.375:1 55.0	0.5	0.625	40	140	SmartmA 100-610mA NI=19 Avg mA 550	Std Plus	22.45	2211.80	936.875	32	Helical scanmode 0.625mm with SmartmA and SmartPrep, Evaluation of vascular structure of abdomen, femurs and lower extremities (Runoff)
26.20	Runoff 1.25mm SmartmA/ SmartPrep		Helical	LargeBody	1.375:1 55.0	0.5	1.25 [0.7]	40	140	SmartmA 100-610mA NI=20 Avg mA 600	Std Plus	24.49	2564.79	998.9	32	Helical scanmode 1.25mm with SmartmA and SmartPrep, Evaluation of vascular structure of abdomen, femurs and lower extremities (Runoff)
26.22	CT Colonography 1.25mm	Advantage CTC Pro	Helical	LargeBody	1.375:1 55.0	0.5	1.25	40	120	200	Std Plus	5.76	258.04	400	32	Helical scan mode supine acquisition for using Advantage CTC Pro for evaluation of the colon
		Advantage CTC Pro	Helical	LargeBody	1.375:1 55.0	0.5	1.25	40	120	200	Std Plus	5.76	258.04	400	32	Helical scan mode prone acquisition for using Advantage CTC Pro for evaluation of the colon

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy-cm)	Scan Length (cm)	Phantom (cm)	Description
26.41	Body Interventional 5mm/3i		Helical	LargeBody	1.375:1 27.50	0.5	5	20	120	SmartmA 50-650mA NI=11.57 Avg mA 360	Std	11.41	251.58	195	32	Localization scan for CT interventional scan
			SmartView SEG	LargeBody	3i	0.4	5	10	120	80	Std Segment	684.58	684.58	0	32	CT Fluoro scan for interventional scan
26.42	Body Interventional 5mm/1i		Helical	LargeBody	1.375:1 27.50	0.5	5	20	120	SmartmA 50-650mA NI=11.57 Avg mA 360	Std	11.41	251.58	195	32	Localization scan for CT interventional scan
			SmartView SEG	LargeBody	1i	0.4	5	5	120	80	Std Segment	845.04	422.52	0	32	CT Fluoro scan for interventional scan

L-Spine

Table 10-7

[Interval]* - For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy-cm)	Scan Length (cm)	Phantom (cm)	Description
27.1	L-Spine 3 Level Axial Group1		Axial	LargeBody	8i	1	2.5	20	120	320	Std	27.13	54.26	17.5	32	Axial scan mode, Evaluation of soft tissue and bony structures lumbar spine
	Group2		Axial	LargeBody	8i	1	2.5	20	120	320	Std	27.13	54.26	17.5	32	Axial scan mode, Evaluation of soft tissue and bony structures lumbar spine
	Group3		Axial	LargeBody	8i	1	2.5	20	140	360	Std	44.46	88.91	17.5	32	Axial scan mode, Evaluation of soft tissue and bony structures lumbar spine
27.3	L-Spine Survey Helical 2.5mm 0.6s. SmartmA		Helical	LargeBody	0.516:1 20.625	0.6	2.5	40	140	SmartmA 100-610mA NI=14 Avg mA 540	Std Plus	70.54	1478.24	75	32	Helical scan mode 2.5mm (0.6sec), Limited evaluation of lumbar spine
27.4	L-Spine Survey Helical 2.5mm 0.8s. SmartmA		Helical	LargeBody	0.516:1 20.625	0.8	2.5	40	140	SmartmA 100-610mA NI=14 Avg mA 525	Std Plus	91.44	1915.92	75	32	Helical scan mode 2.5mm (0.8sec), Limited evaluation of lumbar spine
27.6	L-Spine Helical 0.625mm + DMPR/ SmartmA	DMPR	Helical	LargeBody	0.516:1 20.625	0.6	0.625	40	140	SmartmA 100-610mA NI=20 Avg mA 610	Std Plus IQE	79.69	1669.87	75	32	Helical scan mode 0.625mm with DMPR and SmartmA, Evaluation of soft tissue and bony structures lumbar spine
27.7	L-Spine Helical 1.25mm + DMPR/ SmartmA - Large Patient	DMPR	Helical	LargeBody	0.516:1 20.625	1	1.25	40	140	SmartmA 100-610mA NI=18 Avg mA 610	Std Plus	132.81	2782.19	75	32	Helical scan mode 1.25mm with DMPR and SmartmA, Evaluation of soft tissue and bony structures lumbar spine

Pelvis

Table 10-8

[Interval]* - For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (cm)	Phantom (cm)	Description
28.1	Pelvis for Fracture + DMPR/ SmartmA	DMPR	Helical	LargeBody	0.984:1 39.375	0.8	1.25 [0.625]	40	120	SmartmA 100-700mA NI=20 Avg mA 500	Bone Plus IQE	32.20	432.30	101.25	32	Helical scan with DMPR and SmartmA, Evaluation of pelvis bony structures for fracture

Lower Extremity

Table 10-9

[Interval]* - For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy-cm)	Scan Length (cm)	Phantom (cm)	Description
29.1	Lower Extremity Survey 2.5mm		Helical	LargeBody	0.516:1 20.625	0.5	2.5	40	120	250	Std Plus	19.21	181.35	60	32	Helical scan mode 2.5mm, Limited evaluation of lower extremity for abnormalities
29.2	Ankle 0.625 mm Single Joint		Helical	MediumBody	0.531:1 10.625	0.5	0.625	20	120	250	Std Plus IQE	20.72	108.05	35	32	Helical scan mode 0.625mm, Evaluation of soft tissue and bony anatomy of the ankle area
29.3	Ankle 0.625 mm Both Joints		Helical	LargeBody	0.531:1 10.625	0.5	0.625	20	120	300	Std Plus IQE	23.94	124.83	35	32	Helical scan mode 0.625mm, Evaluation of soft tissue and bony anatomy of the ankle area
29.4	Knee 0.625 mm Both Joints		Helical	LargeBody	0.531:1 10.625	0.6	0.625	20	120	Auto mA 100-335mA NI=20 Avg mA 275	Std Plus IQE	26.33	137.27	35	32	Helical scan mode 0.625mm, Evaluation of soft tissue and bony anatomy of both knees

Reference Pediatric Protocols - Installed Options : Sub 0.4 sec and VCT85kW

Head

Table 11-1

[Interval]* - For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (mm)	Phantom (cm)	Description
31.1	PED HEAD TO 18 Months		Axial	Ped Head	4i	0.5	5	20	120	240	Std	20.43	245.19	115	16	Routine head (0.5s) for infant (up to 18 months) 5mm
31.2	PED HEAD 18 MOS TO 5YRS		Axial	SmallHead	4i	0.5	5	20	120	335	Std	28.52	342.25	115	16	Routine head (0.5s) for children (18 months to 5 years) 5mm
31.3	PED HEAD TO 5YRS TO 18YRS Group1		Axial	SmallHead	8i	1	2.5	20	120	240	Std	40.87	163.46	37.5	16	Routine head (1.0s) for children (5 years to 18 years) 2.5mm
	Group2		Axial	SmallHead	4i	1	5	20	120	180	Std	30.65	245.19	75	16	Routine head (1.0s) for children (5 years to 18 years) 5mm

Orbit

Table 11-2

[Interval]* - For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (mm)	Phantom (cm)	Description
32.1	Sinus Supine Helical + DMPR	DMPR	Helical	SmallHead	0.531:1 10.625	0.4	0.625	20	120	80	Bone Plus IQE	10.26	93.90	74.375	16	Routine Helical scan mode with DMPR, Evaluation of sinus area for abnormality in pediatric patients
32.3	Orbits Helical	DMPR	Helical	SmallHead	0.531:1 10.625	0.4	0.625	20	120	180	Std Plus IQE	23.08	182.43	61.875	16	Helical scan mode, Evaluation of orbital and face structures for abnormality in pediatric patients
32.6	Helical IAC 0.625mm Axial mode - Use Reformat for Coronal		Helical	SmallHead	0.531:1 10.625	0.4	0.625	20	120	160	Std Plus IQE	20.51	114.73	38.75	16	Helical scan mode, Evaluation of inner ear structures for abnormality in pediatric patients

Chest

Table 11-3

[Interval]* - For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (mm)	Phantom (cm)	Description
35.1.1	RC 6.0-7.5 kg (13.2-16.5 lbs)		Helical	PedBody	1.375:1 27.5	0.4	3.75	20	100	80	Std Plus	1.11	11.08	75	32	Weight and height based routine chest protocol
35.1.3	SnapShot Segment 30-74 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	120	Std Plus	0.84	8.19	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	Std	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
			Cardiac SEG	CardiacSmall	0.160:1 6.4	0.35	0.625	40	80	ECG Mod 80-280mA Phase 40-80 AvgmA280	Std SS Seg	12.23	134.55	75	32	Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
35.1.4	SnapShot Burst 75-104 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	120	Std Plus	0.84	8.19	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	Std	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
			Cardiac SSB	CardiacSmall	0.160:1 6.4	0.35	0.625	40	80	ECG Mod 80-280mA Phase 40-80 AvgmA280	Std SSB	12.23	134.55	75	32	Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
35.1.5	SnapShot Burst+ >105 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	120	Std Plus	0.84	8.19	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	Std	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
			Cardiac SSBP	CardiacSmall	0.160:1 6.4	0.35	0.625	40	80	ECG Mod 80-280mA Phase 35-85 AvgmA280	Std SSB	13.12	144.35	75	32	Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
35.1.7	Non ECG Gated		Helical	PedBody	0.984:1 39.375	0.4	0.625	40	80	SmartmA 50-200mA NI=18 AvgmA200	Std Plus	1.95	21.00	75	32	Weight and height based non ECG gated protocol for cardiac with SmartmA
35.2.1	RC 7.5-9.5 kg (16.5-20.9lbs)		Helical	PedBody	1.375:1 27.5	0.4	3.75	20	100	120	Std Plus	1.66	16.62	75	32	Weight and height based routine chest protocol
35.2.3	SnapShot Segment 30-74 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	120	Std Plus	0.84	8.19	50	32	Weight and height based protocol for localization

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (mm)	Phantom (cm)	Description
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	Std	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
			Cardiac SEG	CardiacSmall	0.160:1 6.4	0.35	0.625	40	80	ECG Mod 80-300mA Phase 40-80 AvgmA300	Std SS Seg	13.06	143.68	75	32	Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
35.2.4	SnapShot Burst 75-104 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	120	Std Plus	0.84	8.19	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	Std	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
			Cardiac SSB	CardiacSmall	0.160:1 6.4	0.35	0.625	40	80	ECG Mod 80-300mA Phase 40-80 AvgmA300	Std SSB	13.06	143.68	75	32	Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
35.2.5	SnapShot Burst+ >105 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	120	Std Plus	0.84	8.19	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	Std	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
			Cardiac SSBP	CardiacSmall	0.160:1 6.4	0.35	0.625	40	80	ECG Mod 80-300mA Phase 35-85 AvgmA300	Std SSB	14.04	154.46	75	32	Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
35.2.7	Non ECG Gated		Helical	PedBody	0.984:1 39.375	0.4	0.625	40	80	SmartmA 50-220mA NI=18 AvgmA220	Std Plus	2.14	23.10	75	32	Weight and height based non ECG gated protocol for cardiac with SmartmA
35.3.1	RC 9.5-11.5 kg (20.9-25.4lbs)		Helical	SmallBody	1.375:1 27.5	0.4	3.75	20	120	100	Std Plus	2.17	22.53	78.75	32	Weight and height based routine chest protocol
35.3.3	SnapShot Segment 30-74 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	120	Std Plus	0.84	8.19	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	Std	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
			Cardiac SEG	CardiacSmall	0.160:1 6.4	0.35	0.625	40	80	ECG Mod 100-320mA Phase 40-80 AvgmA320	Std SS Seg	14.05	154.52	75	32	Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
35.3.4	SnapShot Burst 75-104 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	120	Std Plus	0.84	8.19	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	Std	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (mm)	Phantom (cm)	Description
			Cardiac SSB	CardiacSmall	0.160:1 6.4	0.35	0.625	40	80	ECG Mod 100-320mA Phase 40-80 AvgmA320	Std SSB	14.05	154.52	75	32	Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
35.3.5	SnapShot Burst+ >105 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	120	Std Plus	0.84	8.19	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	Std	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
			Cardiac SSBP	CardiacSmall	0.160:1 6.4	0.35	0.625	40	80	ECG Mod 100-320mA Phase 35-85 AvgmA320	Std SSB	15.03	165.30	75	32	Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
35.3.7	Non ECG Gated		Helical	PedBody	0.984:1 39.375	0.4	0.625	40	80	SmartmA 50-240mA NI=18 AvgmA240	Std Plus	2.34	25.20	75	32	Weight and height based non ECG gated protocol for cardiac with SmartmA
35.4.1	RC 11.5-14.5kg (25.4-32.0lbs)		Helical	SmallBody	1.375:1 27.5	0.4	5	20	120	90	Std Plus	1.95	22.47	90	32	Weight and height based routine chest protocol
35.4.3	SnapShot Segment 30-74 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	120	Std Plus	0.84	8.19	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	Std	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
			Cardiac SEG	CardiacSmall	0.160:1 6.4	0.35	0.625	40	80	ECG Mod 100-340mA Phase 40-80 AvgmA340	Std SS Seg	14.87	163.52	75	32	Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
35.4.4	SnapShot Burst 75-104 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	120	Std Plus	0.84	8.19	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	Std	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
			Cardiac SSB	CardiacSmall	0.160:1 6.4	0.35	0.625	40	80	ECG Mod 100-340mA Phase 40-80 AvgmA340	Std SSB	14.87	163.52	75	32	Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
35.4.5	SnapShot Burst+ >105 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	120	Std Plus	0.84	8.19	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	Std	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
			Cardiac SSBP	CardiacSmall	0.160:1 6.4	0.35	0.625	40	80	ECG Mod 100-340mA Phase 35-85 AvgmA340	Std SSB	15.93	175.28	75	32	Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval] (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (mm)	Phantom (cm)	Description
35.4.7	Non ECG Gated		Helical	PedBody	0.984:1 39.375	0.4	0.625	40	80	SmartmA 50-280mA NI=18 AvgmA280	Std Plus	2.72	29.40	75	32	Weight and height based non ECG gated protocol for cardiac with SmartmA
35.5.1	RC 14.5-18.5kg (32.0-40.8lbs)		Helical	SmallBody	1.375:1 27.5	0.4	5	20	120	100	Std Plus	2.17	24.97	90	32	Weight and height based routine chest protocol
35.5.3	SnapShot Segment 30-74 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	120	Std Plus	0.84	8.19	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	Std	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
			Cardiac SEG	CardiacSmall	0.160:1 6.4	0.35	0.625	40	80	ECG Mod 100-380mA Phase 40-80 AvgmA380	Std SS Seg	16.52	181.73	75	32	Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
35.5.4	SnapShot Burst 75-104 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	120	Std Plus	0.84	8.19	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	Std	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
			Cardiac SSB	CardiacSmall	0.160:1 6.4	0.35	0.625	40	80	ECG Mod 100-380mA Phase 40-80 AvgmA380	Std SSB	16.52	181.73	75	32	Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
35.5.5	SnapShot Burst+ >105 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	120	Std Plus	0.84	8.19	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	Std	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
			Cardiac SSBP	CardiacSmall	0.160:1 6.4	0.35	0.625	40	80	ECG Mod 100-380mA Phase 35-85 AvgmA380	Std SSB	17.77	195.45	75	32	Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
35.5.7	Non ECG Gated		Helical	PedBody	0.984:1 39.375	0.4	0.625	40	80	SmartmA 50-300mA NI=18 AvgmA300	Std Plus	2.92	31.50	75	32	Weight and height based non ECG gated protocol for cardiac with SmartmA
35.6.1	RC 18.5-22.5kg (40.8-49.6lbs)		Helical	SmallBody	1.375:1 27.5	0.4	5	20	120	110	Std Plus	2.39	27.47	90	32	Weight and height based routine chest protocol
35.6.3	SnapShot Segment 30-74 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	120	Std Plus	0.84	8.19	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	Std	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval] (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (mm)	Phantom (cm)	Description
			Cardiac SEG	CardiacSmall	0.160:1 6.4	0.35	0.625	40	80	ECG Mod 100-400mA Phase 40-80 AvgmA400	Std SS Seg	17.35	190.80	75	32	Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
35.6.4	SnapShot Burst 75-104 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	120	Std Plus	0.84	8.19	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	Std	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
			Cardiac SSB	CardiacSmall	0.160:1 6.4	0.35	0.625	40	80	ECG Mod 100-400mA Phase 40-80 AvgmA400	Std SSB	17.35	190.80	75	32	Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
35.6.5	SnapShot Burst+ >105 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	120	Std Plus	0.84	8.19	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	Std	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
			Cardiac SSBP	CardiacSmall	0.160:1 6.4	0.35	0.625	40	80	ECG Mod 100-400mA Phase 35-85 AvgmA400	Std SSB	18.68	205.51	75	32	Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
35.6.7	Non ECG Gated		Helical	PedBody	0.984:1 39.375	0.4	0.625	40	80	SmartmA 100-320mA NI=18 AvgmA320	Std Plus	3.11	33.71	75	32	Weight and height based non ECG gated protocol for cardiac with SmartmA
35.7.1	RC 22.5-31.5kg (49.6-69.5lbs)		Helical	LargeBody	1.375:1 55.0	0.4	5	40	120	110	Std Plus	2.54	33.71	85	32	Weight and height based routine chest protocol
35.7.3	SnapShot Segment 30-74 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	120	Std Plus	0.84	8.19	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	Std	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
			Cardiac SEG	CardiacSmall	0.160:1 6.4	0.35	0.625	40	80	ECG Mod 110-420mA Phase 40-80 AvgmA420	Std SS Seg	18.25	200.78	75	32	Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
35.7.4	SnapShot Burst 75-104 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	120	Std Plus	0.84	8.19	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	Std	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
			Cardiac SSB	CardiacSmall	0.160:1 6.4	0.35	0.625	40	80	ECG Mod 110-420mA Phase 40-80 AvgmA420	Std SSB	18.25	200.78	75	32	Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (mm)	Phantom (cm)	Description
35.7.5	SnapShot Burst+ >105 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	120	Std Plus	0.84	8.19	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	Std	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
			Cardiac SSBP	CardiacSmall	0.160:1 6.4	0.35	0.625	40	80	ECG Mod 110-420mA Phase 35-85 AvgmA420	Std SSB	19.63	215.97	75	32	Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
35.7.7	Non ECG Gated		Helical	PedBody	0.984:1 39.375	0.4	0.625	40	80	SmartmA 100-360mA NI=18 AvgmA360	Std Plus	3.50	37.93	75	32	Weight and height based non ECG gated protocol for cardiac with SmartmA
35.8.1	RC 31.5-40.5kg (69.5-89.3lbs)		Helical	LargeBody	1.375:1 55.0	0.4	5	40	120	120	Std Plus	2.77	46.45	120	32	Weight and height based routine chest protocol
35.8.3	SnapShot Segment 30-74 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	100	120	Std Plus	1.55	15.20	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	100	50	Std	39.50	19.75	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
			Cardiac SEG	CardiacSmall	0.160:1 6.4	0.35	0.625	40	100	ECG Mod 100-380mA Phase 40-80 AvgmA380	Std SS Seg	30.66	337.28	75	32	Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
35.8.4	SnapShot Burst 75-104 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	100	120	Std Plus	1.55	15.20	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	100	50	Std	39.50	19.75	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
			Cardiac SSB	CardiacSmall	0.160:1 6.4	0.35	0.625	40	100	ECG Mod 100-380mA Phase 40-80 AvgmA380	Std SSB	30.66	337.28	75	32	Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
35.8.5	SnapShot Burst+ >105 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	100	120	Std Plus	1.55	15.20	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	100	50	Std	39.50	19.75	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
			Cardiac SSBP	CardiacSmall	0.160:1 6.4	0.35	0.625	40	100	ECG Mod 100-380mA Phase 35-85 AvgmA380	Std SSB	32.98	362.75	75	32	Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
35.8.7	Non ECG Gated		Helical	SmallBody	0.984:1 39.375	0.4	0.625	40	100	SmartmA 100-400mA NI=16 AvgmA400	Std Plus	7.22	78.21	75	32	Weight and height based non ECG gated protocol for cardiac with SmartmA

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval] (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (mm)	Phantom (cm)	Description
35.9.1	RC 40.5-55.0kg(89.3-121.3lbs)		Helical	LargeBody	1.375:1 55.0	0.4	5	40	120	130	Std Plus	3.00	59.31	150	32	Weight and height based routine chest protocol
35.9.3	SnapShot Segment 30-74 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	100	120	Std Plus	1.55	15.20	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	100	50	Std	39.50	19.75	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
			Cardiac SEG	CardiacSmall	0.160:1 6.4	0.35	0.625	40	100	ECG Mod 100-420mA Phase 40-80 AvgmA420	Std SS Seg	33.73	371.05	75	32	Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
35.9.4	SnapShot Burst 75-104 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	100	120	Std Plus	1.55	15.20	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	100	50	Std	39.50	19.75	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
			Cardiac SSB	CardiacSmall	0.160:1 6.4	0.35	0.625	40	100	ECG Mod 100-420mA Phase 40-80 AvgmA420	Std SSB	33.73	371.05	75	32	Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
35.9.5	SnapShot Burst+ >105 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	100	120	Std Plus	1.55	15.20	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	100	50	Std	39.50	19.75	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
			Cardiac SSBP	CardiacSmall	0.160:1 6.4	0.35	0.625	40	100	ECG Mod 100-420mA Phase 35-85 AvgmA420	Std SSB	36.38	400.15	75	32	Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
35.9.7	Non ECG Gated		Helical	SmallBody	0.984:1 39.375	0.4	0.625	40	100	SmartmA 100-440mA NI=16 AvgmA440	Std Plus	7.95	86.03	75	32	Weight and height based non ECG gated protocol for cardiac with SmartmA

Abdomen

Table 11-4

[Interval]* - For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (second s)	Slice Thickne ss (mm) [Interval]* (mm)	Beam Collimat ion (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy·c m)	Scan Length (mm)	Phantom (cm)	Description
36.1.1	Abd 6.0-7.5kg (13.2-16.5lbs)		Helical	PedBody	1.375:1 27.5	0.4	3.75	20	120	100	Std Plus	2.17	22.53	78.75	32	Weight and height based routine abdomen protocol
36.2.1	Abd 7.5-9.5kg (16.5-20.9lbs)		Helical	PedBody	1.375:1 27.5	0.4	3.75	20	120	160	Std Plus	3.47	39.95	90	32	Weight and height based routine abdomen protocol
36.3.1	Abd 9.5-11.5kg (20.9-25.4lbs)		Helical	PedBody	1.375:1 27.5	0.4	3.75	20	120	180	Std Plus	3.91	49.34	101.25	32	Weight and height based routine abdomen protocol
36.4.1	Abd 11.5-14.5kg (25.4-32.0lbs)		Helical	SmallBo dy	1.375:1 27.5	0.4	5	20	120	125	Std Plus	2.71	36.64	110	32	Weight and height based routine abdomen protocol
36.5.1	Abd 14.5-18.5kg (32.0-40.8lbs)		Helical	SmallBo dy	1.375:1 27.5	0.4	5	20	120	135	Std Plus	2.93	39.57	110	32	Weight and height based routine abdomen protocol
36.6.1	Abd 18.5-22.5kg (40.8-49.6lbs)		Helical	Medium Body	1.375:1 27.5	0.4	5	20	120	120	Std Plus	3.07	47.68	130	32	Weight and height based routine abdomen protocol
36.7.1	Abd 22.5- 31.5kg(49.6- 69.5lbs)		Helical	LargeBo dy	1.375:1 55.0	0.4	5	40	120	150	Std Plus	3.46	68.44	150	32	Weight and height based routine abdomen protocol
36.8.1	Abd 31.5- 40.5kg(69.5- 89.3lbs)		Helical	LargeBo dy	1.375:1 55.0	0.4	5	40	120	170	Std Plus	3.92	77.56	150	32	Weight and height based routine abdomen protocol
36.9.1	Abd 40.5- 55.0kg(89.3- 121.3lbs)		Helical	LargeBo dy	1.375:1 55.0	0.4	5	40	120	200	Std Plus	4.61	114.30	200	32	Weight and height based routine abdomen protocol

Miscellaneous

Table 11-5

[Interval]* - For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (mm)	Phantom (cm)	Description
40.1	Pink Zone Routine 0-20 lbs/0-9 kg		Helical	PedBody	1.375:1 55.0	0.5	5	40	80	SmartmA 65-130mA NI=5 Avg mA 70	Std	0.61	8.70	95	32	Pediatric and Small patient Routine procedure based protocols
40.2	Pink Zone Routine 21-60 lbs/9.1-27.2 kg		Helical	PedBody	1.375:1 55.0	0.5	5	40	100	SmartmA 80-160mA NI=7 Avg mA 70	Std	1.13	27.47	195	32	Pediatric and Small patient Routine procedure based protocols
40.3	Pink Zone Routine 61-100 lbs/27.3-45.4 kg		Helical	Medium Body	1.375:1 55.0	0.5	5	40	120	SmartmA 95-190mA NI=10 Avg mA 70	Std	2.09	71.81	295	32	Pediatric and Small patient Routine procedure based protocols
40.4	Pink Zone Routine 101-200 lbs/45.4-90.7 kg		Helical	LargeBody	1.375:1 55.0	0.5	5	40	120	SmartmA 110-220mA NI=12 Avg mA 70	Std	2.02	89.30	395	32	Pediatric and Small patient Routine procedure based protocols
40.5	Pink Zone Routine >200 lbs/>90.8 kg		Helical	LargeBody	1.375:1 55.0	0.5	5	40	120	SmartmA 125-300mA NI=15 Avg mA 70	Std	2.02	89.30	395	32	Pediatric and Small patient Routine procedure based protocols
40.7	Green Zone Lower Dose 0-20 lbs/0-9 kg		Helical	PedBody	1.375:1 55.0	0.5	5	40	80	SmartmA 50-100mA NI=7 Avg mA 70	Std	0.61	8.70	95	32	Pediatric and Small patient Lower Dose procedure based protocols
40.8	Green Zone Lower Dose 21-60 lbs/9.1-27.2 kg		Helical	PedBody	1.375:1 55.0	0.5	5	40	100	SmartmA 60-120mA NI=9 Avg mA 70	Std	1.13	27.47	195	32	Pediatric and Small patient Lower Dose procedure based protocols
40.9	Green Zone Lower Dose 61-100 lbs/27.3-45.4 kg		Helical	Medium Body	1.375:1 55.0	0.5	5	40	120	SmartmA 70-140mA NI=11 Avg mA 70	Std	2.09	71.81	295	32	Pediatric and Small patient Lower Dose procedure based protocols
40.10	Green Zone Lower Dose 101-200 lbs/45.4-90.7 kg		Helical	LargeBody	1.375:1 55.0	0.5	5	40	120	SmartmA 80-160mA NI=13 Avg mA 70	Std	2.02	89.30	395	32	Pediatric and Small patient Lower Dose procedure based protocols
40.11	Green Zone Lower Dose >200 lbs/>90.8 kg		Helical	LargeBody	1.375:1 55.0	0.5	5	40	120	SmartmA 90-240mA NI=16 Avg mA 70	Std	2.02	89.30	395	32	Pediatric and Small patient Lower Dose procedure based protocols
40.13	Grey Zone CT Angio 0-20 lbs/0-9 kg		Helical	PedBody	1.375:1 55.0	0.5	2.5	40	80	SmartmA 100-200mA NI=5 Avg mA 140	Std	1.22	17.71	97.5	32	Pediatric and Small patient Angio procedure based protocols

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval J* (mm)]	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (mm)	Phantom (cm)	Description
40.14	Grey Zone CT Angio 21-60 lbs/9.1-27.2 kg		Helical	SmallBody	1.375:1 55.0	0.5	2.5	40	100	SmartmA 120-240mA NI=9.9 Avg mA 140	Std	2.26	55.50	197.5	32	Pediatric and Small patient Angio procedure based protocols
40.15	Grey Zone CT Angio 61-100 lbs/ 27.3-45.4 kg		Helical	LargeBody	1.375:1 55.0	0.5	2.5	40	120	SmartmA 120-240mA NI=10 Avg mA 140	Std	4.03	139.28	297.5	32	Pediatric and Small patient Angio procedure based protocols
40.16	Grey Zone CT Angio 101-200 lbs/ 45.4-90.7 kg		Helical	LargeBody	1.375:1 55.0	0.5	2.5	40	120	SmartmA 120-240mA NI=12 Avg mA 140	Std	4.03	179.62	397.5	32	Pediatric and Small patient Angio procedure based protocols
40.17	Grey Zone CT Angio >200 lbs/ >90.8 kg		Helical	LargeBody	1.375:1 55.0	0.5	2.5	40	120	SmartmA 120-300mA NI=15 Avg mA 140	Std	4.03	179.62	397.5	32	Pediatric and Small patient Angio procedure based protocols

Reference Adult Protocols - Installed

Options : Sub 0.4sec

Head

Table 12-1

[Interval]* - For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (s)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (mm)	Phantom (cm)	Description
21.1	Routine Head 1s Group1		Axial	Head	4i	1	5	20	140	200	Std	53.34	320.02	55	16	Routine head protocol for evaluation of the brain for abnormalities.
	Group2		Axial	Head	4i	1	5	20	120	320	Std	61.43	491.41	75	16	
21.2	Routine Head 2s Group1		Axial	Head	4i	2	5	20	140	100	Std	53.34	320.02	55	16	Routine head protocol for evaluation of the brain for abnormalities.
	Group2		Axial	Head	4i	2	5	20	120	160	Std	61.43	491.41	75	16	
21.3	Trauma Head 2.5mm		Axial	Head	8i	0.5	2.5	20	140	515	Std	70.60	988.35	137.5	16	Emergency head protocol for evaluation of the brain and cranium for abnormalities.
21.4	Routine Head 0.5s Group1		Axial	Head	4i	0.5	5	20	140	400	Std	54.83	328.99	55	16	Routine head protocol for evaluation of the brain for abnormalities.
	Group2		Axial	Head	4i	0.5	5	20	120	600	Std	59.20	473.62	75	16	
21.5	Circle of Willis 0.4s	MIROI	Axial	Head	1i	1	5 [0]	5	120	50	Std	125.58	62.79	0	16	Test bolus for CT angiography of the head for evaluation of carotid and cerebral vasculature.
			Helical	Head	0.531:1 10.625	0.4	0.625	20	120	300	Std Plus IQE	43.36	421.37	80	16	CT angiography of the head for evaluation of carotid and cerebral vasculature.
21.6	CoW/ Carotid 0.625mm 0.4s.Smart mA	MIROI	Axial	Head	1i	1	5 [0]	5	120	80	Std	180.84	90.42	0	16	Test bolus for CT angiography of the head for evaluation of carotid and cerebral vasculature.
			Helical	Head	0.531:1 10.625	0.4	0.625	20	120	SmartmA 100-600mA NI=8 Avg mA 595	Std Plus IQE	88.41	1657.19	170	16	CT angiography of the head for evaluation of carotid and cerebral vasculature.
21.8	Helical Head	DMPR	Helical	Head	0.531:1 10.625	0.5	0.625	20	120	300	Std Plus IQE	54.20	878.85	145	16	Helical scan mode for evaluation of the brain for cerebral abnormality.
21.10	VHS Head 4D CTA	MIROI	Axial	Head	1i	1	5 [0]	5	120	50	Std	125.58	62.79	0	16	Test bolus for CT angiography of the head for evaluation of carotid and cerebral vasculature.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (s)	Slice Thickness (mm) [Interval] J* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (mm)	Phantom (cm)	Description
			Helical-S	Head	0.984:1 39.375 18 pass	0.4	5 [10]	40	120	300	Std Plus	496.19	6202.39	110	16	CT angiography of the head using Volume Helical scan mode for evaluation of cerebral vasculature.
21.11	VHS Perfusion 350-370 Strength Contrast		Axial	Head	4i	1	5	20	120	335	Std	64.31	900.28	135	16	Non-Enhance Brain, evaluation for hemorrhage or infarction.
		Perfusion	Helical-S/P	Head	0.984:1 39.375 28 pass	0.4	5 [10]	40	80	200	Std	186.02	2325.25	110	16	CT Perfusion using Volume Helical scan mode for evaluation of cerebral perfusion over time.
			Axial	Head	4i	1	5	20	120	335	Std	64.31	900.28	135	16	Post Contrast Brain, Evaluation for residual tissue enhancement.
21.12	CT Perfusion 300-320 Strength Contrast		Axial	Head	4i	1	5	20	120	335	Std	64.31	900.28	135	16	Non-Enhance Brain, evaluation for hemorrhage or infarction.
		Perfusion	Cine	Head	8i	1	5 [0]	40	80	200	Std	652.82	2611.27	35	16	CT Perfusion using Cine scan mode for evaluation of cerebral perfusion over time.
			Axial	Head	4i	1	5	20	120	335	Std	64.31	900.28	135	16	Non-Enhance Brain, evaluation for hemorrhage or infarction.
21.13	CT Perfusion 350-370 Strength Contrast		Axial	Head	4i	1	5	20	120	335	Std	64.31	900.28	135	16	Non-Enhance Brain, evaluation for hemorrhage or infarction.
		Perfusion	Cine	Head	8i	1	5 [0]	40	80	200	Std	587.55	2350.19	35	16	CT Perfusion using Cine scan mode for evaluation of cerebral perfusion over time.
			Axial	Head	4i	1	5	20	120	335	Std	64.31	900.28	135	16	Post Contrast Brain, Evaluation for residual tissue enhancement.
21.14	CT Perfusion Brain Tumor - Dual Phase		Axial	Head	4i	1	5	20	120	335	Std	64.31	900.28	135	16	Non-Enhance Brain, evaluation for hemorrhage or infarction.
	Group 1	Perfusion	Cine	Head	8i	1	5 [0]	40	80	200	Std	587.55	2350.19	35	16	CT Perfusion using Cine scan mode for evaluation of cerebral perfusion over time.
	Group 2	Perfusion	Axial	Head	8i	1	5 [0]	40	80	200	Std	130.54	522.15	35	16	Delayed phase CT Perfusion using Axial scan mode for evaluation of Brain tumor permeability.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (s)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (mm)	Phantom (cm)	Description
			Axial	Head	4i	1	5	20	120	335	Std	64.31	900.28	135	16	Post Contrast Brain, Evaluation for residual tissue enhancement.
21.15	CT Perfusion 350-370 Strength Contrast (Axial)		Axial	Head	4i	1	5	20	120	335	Std	64.31	900.28	135	16	Non-Enhance Brain, evaluation for hemorrhage or infarction.
		Perfusion	Axial	Head	8i	1	5 [0]	40	80	200	Std	287.18	1148.73	35	16	CT Perfusion using Axial scan mode for evaluation of cerebral perfusion over time.
			Axial	Head	4i	1	5	20	120	335	Std	64.31	900.28	135	16	Post Contrast Brain, Evaluation for residual tissue enhancement.
21.16	CT Perfusion 350-370 Strength Contrast (Axial Shuttle Mode)		Axial	Head	4i	1	5	20	120	335	Std	64.31	900.28	135	16	Non-Enhance Brain, evaluation for hemorrhage or infarction.
		Perfusion	Axial-S	Head	8i 17 pass	0.4	5	40	80	500	Std	221.91	1775.31	75	16	CT Perfusion using Volume Shuttle (Axial) scan mode for evaluation of cerebral perfusion over time.
			Axial	Head	4i	1	5	20	120	335	Std	64.31	900.28	135	16	Post Contrast Brain, Evaluation for residual tissue enhancement.

Orbit

Table 12-2

[Interval]* - For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy-cm)	Scan Length (cm)	Phantom (cm)	Description
22.1	Sinus Supine Helical + DMPR	DMPR	Helical	Head	0.531:1 10.625	0.4	0.625	20	120	120	Bone Plus IQE	17.34	158.79	74.375	16	Routine Helical Sinus protocol, Evaluation for any abnormality
22.3	Orbits Helical + DMPR	DMPR	Helical	Head	0.531:1 10.625	0.5	0.625	20	120	220	Std Plus IQE	39.75	314.09	61.875	16	Routine Helical Orbit protocol, Evaluation for abnormality soft tissues and bone structures of the face.
22.6	Helical IAC 0.625mm Axial mode - Use Reformat for Coronal		Helical	Head	0.531:1 10.625	0.6	0.625	20	140	270	Std Plus IQE	81.32	454.44	38.75	16	Axial scan mode, Evaluation for inner ear abnormalities in Axial plane
22.8	Axial and Coronal IAC		Axial	Head	16i	1	1.25	20	140	280	Std	74.67	298.68	38.75	16	Helical scan mode, Evaluation for inner ear abnormalities in Axial plane
			Axial	Head	16i	1	1.25	20	140	335	Std	89.34	357.35	38.75	16	Helical scan mode, Evaluation for inner ear abnormalities in Coronal plane

Neck

Table 12-3

[Interval]* - For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (cm)	Phantom (cm)	Description
23.1	C-Spine C5-C7 Axial		Axial	SmallBody	8i	0.5	2.5	20	140	515	Std	28.00	392.06	137.5	32	C-Spine C5-C7 area Axial mode, Evaluation for soft tissue, disc and bone abnormalities.
23.2	C-Spine C1-C4 Axial		Axial	SmallBody	8i	1	2.5	20	120	240	Std	17.90	143.19	77.5	32	C-Spine C1-C4 area Axial mode, Evaluation for soft tissue, disc and bone abnormalities.
23.3	C-Spine Helical + DMPR	DMPR	Helical	SmallBody	0.516:1 20.625	0.8	0.625	40	120	335	Bone Plus IQE	36.23	522.92	110	32	C-Spine Helical mode with DMPR, Evaluation for soft tissue, disc and bone abnormalities.
23.4	C-Spine Helical + DMPR/ SmartmA	DMPR	Helical	SmallBody	0.516:1 20.625	0.8	0.625	40	120	SmartmA 100-600mA NI=11.34 Avg mA 310	Bone Plus IQE	33.52	483.89	110	32	C-Spine Helical mode with DMPR and SmartmA, Evaluation for soft tissue, disc and bone abnormalities.
23.8	Carotid 0.625mm 0.4s. 0.5:1 SmartmA/ SmartPrep		Helical	SmallBody	0.516:1 20.625	0.4	0.625	40	140	SmartmA 100-515mA NI=11 Avg mA 515	Std Plus IQE	39.50	808.35	170	32	Carotid CTA (pitch 0.5:1) with SmartmA and SmartPrep, Evaluation of carotid vasculature.
23.9	Carotid 0.625mm 0.4s. 0.9:1 SmartmA/ SmartPrep		Helical	SmallBody	0.984:1 39.375	0.4	0.625	40	120	SmartmA 80-600mA NI=11 Avg mA 600	Std Plus IQE	16.99	345.42	170	32	Carotid CTA (pitch 0.9:1) with SmartmA and SmartPrep, Evaluation of carotid vasculature.
23.10	Carotid/ CoW 0.625mm 0.4s. AutomA	MIROI	Axial	SmallBody	1i	1	5 [0]	5	120	80	Std	99.00	49.50	0	32	Test bolus for Carotid CTA (pitch 0.5:1) of the head for evaluation of carotid and cerebral vasculature.
			Helical	SmallBody	0.516:1 20.625	0.4	0.625	40	140	AutomA 80-515mA NI=11 Avg mA 515	Std Plus IQE	39.50	808.35	170	32	Carotid and Brain CTA (pitch 0.5:1) with AutomA, Evaluation of carotid and cerebral vasculature.
23.11	Carotid/ CoW 0.625mm 0.4s. SmartmA/ SmartPrep		Helical	SmallBody	0.969:1 19.375	0.4	0.625	20	140	SmartmA 100-515mA NI=11 Avg mA 515	Std Plus IQE	23.13	436.84	170	32	Carotid CTA (pitch 0.9:1) with SmartmA and SmartPrep, Evaluation of carotid and cerebral vasculature.
23.12	Soft Tissue Neck 2.5mm AutomA		Helical	SmallBody	0.984:1 39.375	0.5	2.5	40	120	AutomA 100-335mA NI=9.1 Avg mA 250	Std Plus	8.85	161.79	150	32	Neck with AutomA, Evaluation of soft tissue structures for abnormality

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy-cm)	Scan Length (cm)	Phantom (cm)	Description
23.13	Soft Tissue Neck + DMPR/ SmartmA	DMPR	Helical	SmallBody	0.969:1 19.375	0.5	0.625	20	120	SmartmA 100-600mA NI=12.6 Avg mA 425	Std Plus IQE	16.82	283.95	150	32	Neck with DMPR and SmartmA, Evaluation of soft tissue structures for abnormality

Upper Extremity

Table 12-4

[Interval]* - For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy-cm)	Scan Length (cm)	Phantom (cm)	Description
24.1	Shoulder 1.25mm + DMPR	DMPR	Helical	LargeBody	0.516:1 20.625	0.6	1.25 [0.625]	40	140	515	Std Plus IQE	67.28	905.24	100	32	Shoulder with DMPR, Evaluation of soft tissue and bone of shoulder area
24.2	Shoulder 1.25mm + DMPR/ SmartmA	DMPR	Helical	LargeBody	0.516:1 20.625	0.6	1.25 [0.625]	40	140	SmartmA 100-515mA NI=20 Avg mA 515	Std Plus IQE	67.28	905.24	100	32	Shoulder with DMPR and SmartmA, Evaluation of soft tissue and bone of shoulder area

Chest

Table 12-5

[Interval]* - For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (second s)	Slice Thickne ss (mm) [Interval]* (mm)	Beam Collimat ion (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy·c m)	Scan Length (cm)	Phantom (cm)	Description
25.1	Routine Chest 0.4s. 5mm SmartmA		Helical	LargeBody	1.375:1 55.0	0.4	5	40	120	SmartmA 100-600mA NI=13 Avg mA 500	Std Plus	11.52	286.26	200	32	Chest with SmartmA, Evaluation of mediastinum and lungs fields
25.2	Routine Chest 0.4s.+ DMPR/ SmartmA	DMPR	Helical	LargeBody	0.984:1 39.375	0.4	0.625	40	120	SmartmA 100-600mA NI=22 Avg mA 600	Std Plus	19.32	450.62	200	32	Chest with SmartmA and DMPR, Evaluation of mediastinum and lungs fields
25.3	Chest with HiRes 0.4s. SmartmA		Helical	LargeBody	0.969:1 19.375	0.4	5	20	120	SmartmA 80-600mA NI=12 Avg mA 500	Chest Plus	17.99	393.87	200	32	Chest with SmartmA, Evaluation of mediastinum and lung fields with higher resolution images of the lungs
25.4	Hi Res Chest 0.5s. Axial		Axial	LargeBody	1i	0.5	1.25 [10]	1.25	120	380	Bone+	4.48	107.55	230	32	Axial scan mode, Evaluation of mediastinum and lung fields in high resolution mode
25.5	Chest Abd Pelvis 0.4/ 0.5s. SmartmA 5mm (2 groups) Group1		Helical	LargeBody	0.984:1 39.375	0.4	5	40	120	SmartmA 100-600mA NI=13 Avg mA 370	Std Plus	11.91	218.32	150	32	Multi group scan for Chest, Abdomen, Pelvis (0.4/0.5 sec) with SmartmA, Evaluation of chest and abdominal anatomy for abnormalities
	Group2		Helical	LargeBody	1.375:1 55.0	0.5	5	40	120	SmartmA 100-600mA NI=13 Avg mA 370	Std Plus	10.66	365.90	295	32	
25.6	Chest Abd Pelvis 0.5s + DMPR SmartPrep	DMPR	Helical	LargeBody	1.375:1 55.0	0.5	0.625	40	120	SmartmA 100-600mA NI=22 Avg mA 540	Std Plus	15.56	775.18	450	32	Helical scan mode (0.5sec) with SmartmA, Evaluation of chest and abdominal anatomy for abnormalities
25.7	Pulmonary Embolis 0.5s.	MIROI	Axial	LargeBody	1i	1	5 [0]	5	120	80	Std	93.89	46.95	0	32	Timing Bolus for Helical scan mode 1.25mm thickness and helical pitch of 0.9:1, evaluation of Chest for pulmonary embolism
			Helical	LargeBody	0.984:1 39.375	0.5	1.25	40	120	580	Std	23.34	497.55	180	32	Helical scan mode 1.25mm thickness and helical pitch of 0.9:1, Evaluation Chest for pulmonary embolism

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy-cm)	Scan Length (cm)	Phantom (cm)	Description
25.8	Pulmonary Embolis 0.5s. SmartmA	MIROI	Axial	LargeBody	1i	1	5 [0]	5	120	80	Std	93.89	46.95	0	32	Timing Bolus for Helical scan mode 1.25mm thickness and helical pitch of 0.9:1 with SmartmA, evaluation of Chest for pulmonary embolism
			Helical	LargeBody	0.984:1 39.375	0.5	1.25	40	120	SmartmA 100-600mA NI=20 Avg mA 500	Std	20.12	428.92	180	32	Helical scan mode 1.25mm thickness and helical pitch of 0.9:1 with SmartmA, Evaluation Chest for pulmonary embolism
25.9	Aortic Dissection 0.625mm + DMPR/ SmartmA/ SmartPrep	DMPR	Helical	LargeBody	1.375:1 55.0	0.5	0.625	40	120	SmartmA 100-600mA NI=22 Avg mA 600	Std Plus	17.29	602.00	300	32	Helical scan mode 0.625mm thickness and helical pitch of 1.375:1 with SmartmA, SmartPrep and DMPR, Evaluation for aortic dissection
25.10	Aortic Dissection 1.25mm + DMPR/ SmartmA/ SmartPrep	DMPR	Helical	LargeBody	1.375:1 55.0	0.5	1.25 [0.8]	40	120	SmartmA 100-600mA NI=20 Avg mA 550	Std Plus	15.85	551.83	300	32	Helical scan mode 1.25mm thickness and helical pitch of 1.375:1 with SmartmA, SmartPrep and DMPR, Evaluation for aortic dissection
25.11	Lung Assessment	Advanced Lung Analysis	Helical	LargeBody	0.969:1 19.375	0.5	1.25	20	120	120	Bone	5.25	114.77	200	32	Helical scan mode for acquisition of data for Advanced Lung Analysis (ALA) software for lung nodule assessment
25.12	Gated Chest Cardiac Helical		Helical	LargeBody	1.375:1 55.0	0.4	5	40	120	60	Std Plus	1.38	13.55	50	32	Localizer scan for Helical Gated scan for evaluation of thoracic vasculature
		MIROI	Axial	LargeBody	1i	1	5 [0]	5	120	60	Std	98.59	49.29	0	32	Timing bolus scan for Helical Gated scan for evaluation of thoracic vasculature
			Cardiac SEG	CardiacLarge	0.160:1 6.4	0.35	0.625	40	120	ECG Mod 185-450mA Phase 70-80 Avg mA 450	Std SS Seg	51.77	1734.24	300	32	Helical Gated scan for evaluation of thoracic vasculature
25.13	Gated Chest SnapShot Pulse		Helical	LargeBody	1.375:1 55.0	0.4	5	40	120	60	Std Plus	1.38	13.55	50	32	Localizer scan for Cine Gated scan for evaluation of thoracic vasculature
		MIROI	Axial	LargeBody	1i	1	5 [0]	5	120	60	Std	98.59	49.29	0	32	Timing bolus scan for Cine Gated scan for evaluation of thoracic vasculature
			Cardiac Pulse Cine	CardiacLarge	56i	0.35	0.625 [35]	40	120	450	Std SS Seg	24.04	757.31	314.375	32	Cine Gated scan for evaluation of thoracic vasculature
25.14	SmartScore - Gated 0.35s	Smart Score	Cine	LargeBody	8i	0.35	2.5	20	120	430	Std Segment	8.99	89.95	97.5	32	Cine gated scan mode for acquisition of data for coronary artery calcification analysis

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy-cm)	Scan Length (cm)	Phantom (cm)	Description
25.15	SnapShot Pulse 30-65 BPM		Helical	LargeBody	1.375:1 55.0	0.4	5	40	120	60	Std Plus	1.38	13.55	50	32	Localizer scan for Prospectively gated Cardiac Cine scan for evaluation of coronary arteries, intended for stable heart rates 30-65 bpm
		MIROI	Axial	LargeBody	1i	1	5 [0]	5	120	60	Std	98.59	49.29	0	32	Timing bolus scan for Prospectively gated Cardiac Cine scan for evaluation of coronary arteries, intended for stable heart rates 30-65 bpm
			Cardiac Pulse Cine	CardiacSmall	56i	0.35	0.625 [35]	40	120	450	Std SS Seg	21.15	222.06	104.375	32	Prospectively gated Cardiac Cine scan for evaluation of coronary arteries, intended for stable heart rates 30-65 bpm
25.16	SnapShot Segment 30-74 BPM		Helical	LargeBody	1.375:1 55.0	0.4	5	40	120	60	Std Plus	1.38	13.55	50	32	Localizer scan for Retrospectively gated Cardiac Helical scan for evaluation of coronary arteries, intended for heart rates 30-74 bpm
		MIROI	Axial	LargeBody	1i	1	5 [0]	5	120	60	Std	98.59	49.29	0	32	Timing bolus scan for Retrospectively gated Cardiac Helical scan for evaluation of coronary arteries, intended for heart rates 30-74 bpm
			Cardiac SEG	CardiacLarge	0.160:1 6.4	0.35	0.625	40	120	ECG Mod 200-450mA Phase 70-80 Avg mA 450	Std SS Seg	55.39	747.80	100	32	Retrospectively gated Cardiac Helical scan for evaluation of coronary arteries, intended for heart rates 30-74 bpm
25.17	SnapShot Burst 75-113 BPM		Helical	LargeBody	1.375:1 55.0	0.4	5	40	120	60	Std Plus	1.38	13.55	50	32	Localizer scan for Retrospectively gated Cardiac Helical scan for evaluation of coronary arteries, intended for heart rates 75-113 bpm
		MIROI	Axial	LargeBody	1i	1	5 [0]	5	120	60	Std	98.59	49.29	0	32	Timing bolus scan for Retrospectively gated Cardiac Helical scan for evaluation of coronary arteries, intended for heart rates 75-113 bpm
			Cardiac SSB	CardiacLarge	0.160:1 6.4	0.35	0.625	40	120	ECG Mod 200-450mA Phase 70-80 Avg mA 450	Std SSB	55.39	747.80	100	32	Retrospectively gated Cardiac Helical scan for evaluation of coronary arteries, intended for heart rates 75-113 bpm
25.18	SnapShot Burst Plus 114+ BPM		Helical	LargeBody	1.375:1 55.0	0.4	5	40	120	60	Std Plus	1.38	13.55	50	32	Localizer scan for Retrospectively gated Cardiac Helical scan for evaluation of coronary arteries, intended for heart rates over 113 bpm

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy-cm)	Scan Length (cm)	Phantom (cm)	Description
		MIROI	Axial	LargeBody	1i	1	5 [0]	5	120	60	Std	98.59	49.29	0	32	Timing bolus scan for Retrospectively gated Cardiac Helical scan for evaluation of coronary arteries, intended for heart rates over 113 bpm
			Cardiac SSBP	CardiacLarge	0.160:1 6.4	0.35	0.625	40	120	ECG Mod 200-450mA Phase 70-80 Avg mA 450	Std SSB	55.39	747.80	100	32	Retrospectively gated Cardiac Helical scan for evaluation of coronary arteries, intended for heart rates over 113 bpm
25.19	Advantage 4D		Helical	LargeBody	1.375:1 55.0	0.6	5	40	120	285	Std Plus	9.85	244.08	200	32	Localizer scan for Advantage 4D series
		Advantage 4D	Cine	LargeBody	8i	1	2.5	20	120	100	Std	50.95	815.24	157.5	32	Cine scan mode for acquisition data for Advantage 4D for retrospective gating
25.20	VHS Chest 4D CTA SmartPrep		Helical-S	LargeBody	1.375:1 55.0 11 pass	0.4	5 [10]	40	120	SmartmA 150-500mA NI=14 Avg mA 350	Std Plus	112.15	3532.61	300	32	Volume Helical scan mode with SmartPrep for time resolved CT angiography of chest vasculature

Abdomen

Table 12-6

[Interval]* - For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (cm)	Phantom (cm)	Description
26.1	Abdomen Pelvis 0.5s. 5mm SmartmA		Helical	LargeBody	1.375:1 55.0	0.5	5	40	120	SmartmA 50-600mA NI=11.57 Avg mA 500	Std Plus	14.41	645.73	400	32	Basic Abdomen Pelvis (0.5sec) with Smart mA, Evaluation for abnormalities of abdominal and pelvic structures
26.2	Abdomen Pelvis 0.5s. 5mm + DMPR SmartmA/ SmartPrep	DMPR	Helical	LargeBody	0.984:1 39.375	0.6	0.625	40	120	SmartmA 80-600mA NI=18 Avg mA 520	Std Plus	25.11	1087.62	400	32	Basic Abdomen Pelvis (0.5sec) with Smart mA, SmartPrep and DMPR, Evaluation for abnormalities of abdominal and pelvic structures
26.4	Renal Stone 0.5s. 5mm		Helical	LargeBody	1.375:1 55.0	0.5	5	40	120	380	Std Plus	10.95	490.75	400	32	Helical scan, Evaluation of kidney, ureters and bladder for renal stones
26.5	Renal Stone 0.5s. 5mm SmartmA		Helical	LargeBody	1.375:1 55.0	0.5	5	40	120	SmartmA 50-600mA NI=12 Avg mA 380	Std Plus	10.95	490.75	400	32	Helical scan with SmartmA, Evaluation of kidney, ureters and bladder for renal stones
26.7	AAA 0.5s. + DMPR 0.625mm	MIROI	Axial	LargeBody	1i	1	5 [0]	5	120	80	Std	112.67	56.34	0	32	Timing bolus for Helical scan with DMPR, Evaluation for abdominal aortic aneurysm
		DMPR	Helical	LargeBody	1.375:1 55.0	0.5	0.625	40	140	515	Std Plus	21.02	751.82	309.375	32	Helical scan with DMPR, Evaluation for abdominal aortic aneurysm
26.8	AAA 0.6s. + DMPR 0.625mm SmartmA/ SmartPrep	DMPR	Helical	LargeBody	1.375:1 55.0	0.6	0.625	40	120	SmartmA 100-600mA NI=19 Avg mA 550	Std Plus	19.02	679.85	309.375	32	Helical scan with DMPR, Smart mA and SmartPrep, Evaluation for abdominal aortic aneurysm
26.10	Dual Phase Liver		Helical	LargeBody	1.375:1 55.0	0.6	5	40	120	300	Std Plus	10.37	184.32	130	32	Non contrast evaluation of upper abdominal structures including liver
		MIROI	Axial	LargeBody	1i	1	5 [0]	5	120	80	Std	112.67	56.34	0	32	Timing bolus for Multi-Group acquisition for early arterial and portal venous phase of the liver
	Group1		Helical	LargeBody	1.375:1 55.0	0.5	2.5	40	120	550	Std Plus	15.85	282.44	130	32	Acquisition for early arterial phase of the liver
	Group2		Helical	LargeBody	1.375:1 55.0	0.5	5	40	120	420	Std Plus	12.10	542.41	400	32	Acquisition for portal venous phase of the liver

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy-cm)	Scan Length (cm)	Phantom (cm)	Description
26.11	Tri Phase Liver		Helical	LargeBody	1.375:1 55.0	0.6	5	40	120	300	Std Plus	10.37	184.32	130	32	Non contrast evaluation of upper abdominal structures including liver
		MIROI	Axial	LargeBody	1i	1	5 [0]	5	120	80	Std	112.67	56.34	0	32	Timing bolus for Multi-Group acquisition for early arterial, late arterial and portal venous phase of the liver
	Group1		Helical	LargeBody	1.375:1 55.0	0.5	2.5	40	120	550	Std Plus	15.85	282.44	130	32	Acquisition for early arterial phase of the liver
	Group2		Helical	LargeBody	1.375:1 55.0	0.5	5	40	120	450	Std Plus	12.97	581.15	400	32	Acquisition for late arterial phase of the liver
	Group3		Helical	LargeBody	1.375:1 55.0	0.5	5	40	120	450	Std Plus	12.97	581.15	400	32	Acquisition for portal venous phase of the liver
26.12	Dual Phase Pancreas SmartmA/ SmartPrep		Helical	LargeBody	0.984:1 39.375	0.5	5	40	120	SmartmA 80-600mA NI=12 Avg mA 400	Std Plus	16.10	166.06	70	32	Non contrast evaluation of upper abdominal structures including pancreas
	Group1		Helical	LargeBody	0.984:1 39.375	0.5	2.5	40	120	SmartmA 50-600mA NI=15 Avg mA 520	Std Plus	20.93	383.30	150	32	Acquisition with SmartmA and SmartPrep for arterial phase of the pancreas
	Group2		Helical	LargeBody	0.984:1 39.375	0.5	5	40	120	SmartmA 50-600mA NI=15 Avg mA 450	Std Plus	18.11	639.58	320	32	Acquisition for venous phase of the pancreas
26.13	Dual Phase Liver SmartmA/ SmartPrep		Helical	LargeBody	1.375:1 55.0	0.5	5	40	120	SmartmA 80-500mA NI=12 Avg mA 320	Std Plus	9.22	163.92	130	32	Non contrast evaluation of upper abdominal structures including liver
	Group1		Helical	LargeBody	1.375:1 55.0	0.5	2.5	40	120	SmartmA 50-600mA NI=15 Avg mA 550	Std Plus	15.85	282.44	130	32	Acquisition with SmartmA and SmartPrep for early arterial phase of the liver
	Group2		Helical	LargeBody	0.984:1 39.375	0.5	5	40	120	SmartmA 50-600mA NI=11 Avg mA 450	Std Plus	18.11	784.46	400	32	Acquisition with SmartmA and SmartPrep for portal venous phase of the liver
26.14	Tri Phase Liver SmartmA/ SmartPrep		Helical	LargeBody	0.984:1 39.375	0.5	5	40	120	SmartmA 80-500mA NI=12 Avg mA 400	Std Plus	16.10	262.65	130	32	Non contrast evaluation of upper abdominal structures including liver
	Group1		Helical	LargeBody	1.375:1 55.0	0.5	2.5	40	120	SmartmA 50-600mA NI=15 Avg mA 550	Std Plus	15.85	282.44	130	32	Acquisition with SmartmA and SmartPrep for early arterial phase of the liver
	Group2		Helical	LargeBody	1.375:1 55.0	0.5	5	40	120	SmartmA 50-600mA NI=11 Avg mA 450	Std Plus	12.97	581.15	400	32	Acquisition with SmartmA and SmartPrep for late arterial phase of the liver
	Group3		Helical	LargeBody	1.375:1 55.0	0.5	5	40	120	SmartmA 50-600mA NI=11 Avg mA 450	Std Plus	12.97	581.15	400	32	Acquisition with SmartmA and SmartPrep for portal venous phase of the liver
26.15	VHS Liver Perfusion		Helical	LargeBody	0.984:1 39.375	0.6	5	40	120	170	Std	8.21	133.61	130	32	Non contrast evaluation of upper abdominal structures including liver

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (cm)	Phantom (cm)	Description
		Perfusion	Helical-S/P	LargeBody	1.375:1 55.0 21 pass	0.4	5	40	120	300	Std Plus	219.99	3409.79	140	32	Volume Helical scan mode for evaluation of liver tumor perfusion over time
26.16	VHS 4D CTA		Helical	LargeBody	0.984:1 39.375	0.6	5	40	120	340	Std	16.42	267.79	130	32	Localizer scan for Volume Helical Shuttle scan mode for evaluation of abdominal vasculature
		MIROI	Axial	LargeBody	1i	1	5 [0]	5	120	80	Std	112.67	56.34	0	32	Timing bolus scan for Volume Helical Shuttle scan mode for evaluation of abdominal vasculature
			Helical-S	LargeBody	1.375:1 55.0 18 pass	0.5	5 [10]	40	140	SmartmA 150-515mA NI=12 Avg mA 350	Std Plus	325.68	6350.70	180	32	Volume Helical Shuttle scan mode for time resolved CT angiography of abdominal vasculature
26.17	Runoff 0.625mm	MIROI	Axial	LargeBody	1i	1	5 [0]	5	120	80	Std	112.67	56.34	0	32	Timing bolus scan for evaluation of vascular structure of abdomen, femurs and lower extremities (Runoff)
			Helical	LargeBody	1.375:1 55.0	0.5	0.625	40	140	515	Std Plus	21.02	2071.05	936.875	32	Helical scanmode 0.625mm, Evaluation of vascular structure of abdomen, femurs and lower extremities (Runoff)
26.18	Runoff 1.25mm	MIROI	Axial	LargeBody	1i	1	5 [0]	5	120	80	Std	112.67	56.34	0	32	Timing bolus scan for evaluation of vascular structure of abdomen, femurs and lower extremities (Runoff)
			Helical	LargeBody	1.375:1 55.0	0.5	1.25 [0.7]	40	140	515	Std Plus	21.02	2201.44	998.9	32	Helical scanmode 1.25mm, Evaluation of vascular structure of abdomen, femurs and lower extremities (Runoff)
26.19	Runoff 0.625mm SmartmA/ SmartPrep		Helical	LargeBody	1.375:1 55.0	0.5	0.625	40	140	SmartmA 100-515mA NI=19 Avg mA 515	Std Plus	21.02	2071.05	936.875	32	Helical scanmode 0.625mm with SmartmA and SmartPrep, Evaluation of vascular structure of abdomen, femurs and lower extremities (Runoff)
26.20	Runoff 1.25mm SmartmA/ SmartPrep		Helical	LargeBody	1.375:1 55.0	0.5	1.25 [0.7]	40	140	SmartmA 100-515mA NI=20 Avg mA 515	Std Plus	21.02	2201.44	998.9	32	Helical scanmode 1.25mm with SmartmA and SmartPrep, Evaluation of vascular structure of abdomen, femurs and lower extremities (Runoff)
26.22	CT Colonography 1.25mm	Advantage CTC Pro	Helical	LargeBody	1.375:1 55.0	0.5	1.25	40	120	200	Std Plus	5.76	258.04	400	32	Helical scan mode supine acquisition for using Advantage CTC Pro for evaluation of the colon
		Advantage CTC Pro	Helical	LargeBody	1.375:1 55.0	0.5	1.25	40	120	200	Std Plus	5.76	258.04	400	32	Helical scan mode prone acquisition for using Advantage CTC Pro for evaluation of the colon

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy-cm)	Scan Length (cm)	Phantom (cm)	Description
26.41	Body Interventional 5mm/3i		Helical	LargeBody	1.375:1 27.50	0.5	5	20	120	SmartmA 50-600mA NI=11.57 Avg mA 360	Std	11.41	251.58	195	32	Localization scan for CT interventional scan
			SmartView SEG	LargeBody	3i	0.4	5	10	120	80	Std Segment	684.58	684.58	0	32	CT Fluoro scan for interventional scan
26.42	Body Interventional 5mm/1i		Helical	LargeBody	1.375:1 27.50	0.5	5	20	120	SmartmA 50-600mA NI=11.57 Avg mA 360	Std	11.41	251.58	195	32	Localization scan for CT interventional scan
			SmartView SEG	LargeBody	1i	0.4	5	5	120	80	Std Segment	845.04	422.52	0	32	CT Fluoro scan for interventional scan

L-Spine

Table 12-7

[Interval]* - For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy-cm)	Scan Length (cm)	Phantom (cm)	Description
27.1	L-Spine 3 Level Axial Group1		Axial	LargeBody	8i	1	2.5	20	120	320	Std	27.13	54.26	17.5	32	Axial scan mode, Evaluation of soft tissue and bony structures lumbar spine
	Group2		Axial	LargeBody	8i	1	2.5	20	120	320	Std	27.13	54.26	17.5	32	Axial scan mode, Evaluation of soft tissue and bony structures lumbar spine
	Group3		Axial	LargeBody	8i	1	2.5	20	140	360	Std	44.46	88.91	17.5	32	Axial scan mode, Evaluation of soft tissue and bony structures lumbar spine
27.3	L-Spine Survey Helical 2.5mm 0.6s. SmartmA		Helical	LargeBody	0.516:1 20.625	0.6	2.5	40	140	SmartmA 100-515mA NI=14 Avg mA 515	Std Plus	67.28	1409.81	75	32	Helical scan mode 2.5mm (0.6sec), Limited evaluation of lumbar spine
27.4	L-Spine Survey Helical 2.5mm 0.8s. SmartmA		Helical	LargeBody	0.516:1 20.625	0.8	2.5	40	140	SmartmA 100-515mA NI=14 Avg mA 515	Std Plus	89.70	1879.43	75	32	Helical scan mode 2.5mm (0.8sec), Limited evaluation of lumbar spine
27.6	L-Spine Helical 0.625mm + DMPR/ SmartmA	DMPR	Helical	LargeBody	0.516:1 20.625	0.6	0.625	40	140	SmartmA 100-515mA NI=20 Avg mA 515	Std Plus IQE	67.28	1409.81	75	32	Helical scan mode 0.625mm with DMPR and SmartmA, Evaluation of soft tissue and bony structures lumbar spine
27.7	L-Spine Helical 1.25mm + DMPR/ SmartmA - Large Patient	DMPR	Helical	LargeBody	0.516:1 20.625	1	1.25	40	140	SmartmA 100-515mA NI=18 Avg mA 515	Std Plus	112.13	2348.90	75	32	Helical scan mode 1.25mm with DMPR and SmartmA, Evaluation of soft tissue and bony structures lumbar spine

Pelvis

Table 12-8

[Interval]* - For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.
For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (second s)	Slice Thickne ss (mm) [Interval]* (mm)	Beam Collimat ion (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy·c m)	Scan Length (cm)	Phantom (cm)	Description
28.1	Pelvis for Fracture + DMPR/ SmartmA	DMPR	Helical	LargeBody	0.984:1 39.375	0.8	1.25 [0.625]	40	120	SmartmA 100-600mA NI=20 Avg mA 500	Bone Plus IQE	32.20	432.30	101.25	32	Helical scan with DMPR and SmartmA, Evaluation of pelvis bony structures for fracture

Lower Extremity

Table 12-9

[Interval]* - For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy-cm)	Scan Length (cm)	Phantom (cm)	Description
29.1	Lower Extremity Survey 2.5mm		Helical	LargeBody	0.516:1 20.625	0.5	2.5	40	120	250	Std Plus	19.21	181.35	60	32	Helical scan mode 2.5mm, Limited evaluation of lower extremity for abnormalities
29.2	Ankle 0.625 mm Single Joint		Helical	MediumBody	0.531:1 10.625	0.5	0.625	20	120	250	Std Plus IQE	20.72	108.05	35	32	Helical scan mode 0.625mm, Evaluation of soft tissue and bony anatomy of the ankle area
29.3	Ankle 0.625 mm Both Joints		Helical	LargeBody	0.531:1 10.625	0.5	0.625	20	120	300	Std Plus IQE	23.94	124.83	35	32	Helical scan mode 0.625mm, Evaluation of soft tissue and bony anatomy of the ankle area
29.4	Knee 0.625 mm Both Joints		Helical	LargeBody	0.531:1 10.625	0.6	0.625	20	120	Auto mA 100-335mA NI=20 Avg mA 275	Std Plus IQE	26.33	137.27	35	32	Helical scan mode 0.625mm, Evaluation of soft tissue and bony anatomy of both knees

Reference Pediatric Protocols - Installed Options : Sub 0.4 sec

Head

Table 13-1

[Interval]* - For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (mm)	Phantom (cm)	Description
31.1	PED HEAD TO 18 Months		Axial	Ped Head	4i	0.5	5	20	120	240	Std	20.43	245.19	115	16	Routine head (0.5s) for infant (up to 18 months) 5mm
31.2	PED HEAD 18 MOS TO 5YRS		Axial	SmallHead	4i	0.5	5	20	120	335	Std	28.52	342.25	115	16	Routine head (0.5s) for children (18 months to 5 years) 5mm
31.3	PED HEAD TO 5YRS TO 18YRS Group1		Axial	SmallHead	8i	1	2.5	20	120	240	Std	40.87	163.46	37.5	16	Routine head (1.0s) for children (5 years to 18 years) 2.5mm
	Group2		Axial	SmallHead	4i	1	5	20	120	180	Std	30.65	245.19	75	16	Routine head (1.0s) for children (5 years to 18 years) 5mm

Orbit

Table 13-2

[Interval]* - For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (mm)	Phantom (cm)	Description
32.1	Sinus Supine Helical + DMPR	DMPR	Helical	SmallHead	0.531:1 10.625	0.4	0.625	20	120	80	Bone Plus IQE	10.26	93.90	74.375	16	Routine Helical scan mode with DMPR, Evaluation of sinus area for abnormality in pediatric patients
32.3	Orbits Helical	DMPR	Helical	SmallHead	0.531:1 10.625	0.4	0.625	20	120	180	Std Plus IQE	23.08	182.43	61.875	16	Helical scan mode, Evaluation of orbital and face structures for abnormality in pediatric patients
32.6	Helical IAC 0.625mm Axial mode - Use Reformat for Coronal		Helical	SmallHead	0.531:1 10.625	0.4	0.625	20	120	160	Std Plus IQE	20.51	114.73	38.75	16	Helical scan mode, Evaluation of inner ear structures for abnormality in pediatric patients

Chest

Table 13-3

[Interval]* - For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (mm)	Phantom (cm)	Description
35.1.1	RC 6.0-7.5 kg (13.2-16.5 lbs)		Helical	PedBody	1.375:1 27.5	0.4	3.75	20	100	80	Std Plus	1.11	11.08	75	32	Weight and height based routine chest protocol
35.1.3	SnapShot Segment 30-74 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	120	Std Plus	0.84	8.19	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	Std	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
			Cardiac SEG	CardiacSmall	0.160:1 6.4	0.35	0.625	40	80	ECG Mod 80-280mA Phase 40-80 AvgmA280	Std SS Seg	12.23	134.55	75	32	Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
35.1.4	SnapShot Burst 75-104 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	120	Std Plus	0.84	8.19	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	Std	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
			Cardiac SSB	CardiacSmall	0.160:1 6.4	0.35	0.625	40	80	ECG Mod 80-280mA Phase 40-80 AvgmA280	Std SSB	12.23	134.55	75	32	Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
35.1.5	SnapShot Burst+ >105 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	120	Std Plus	0.84	8.19	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	Std	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
			Cardiac SSBP	CardiacSmall	0.160:1 6.4	0.35	0.625	40	80	ECG Mod 80-280mA Phase 35-85 AvgmA280	Std SSB	13.12	144.35	75	32	Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
35.1.7	Non ECG Gated		Helical	PedBody	0.984:1 39.375	0.4	0.625	40	80	SmartmA 50-200mA NI=18 AvgmA200	Std Plus	1.95	21.00	75	32	Weight and height based non ECG gated protocol for cardiac with SmartmA
35.2.1	RC 7.5-9.5 kg (16.5-20.9lbs)		Helical	PedBody	1.375:1 27.5	0.4	3.75	20	100	120	Std Plus	1.66	16.62	75	32	Weight and height based routine chest protocol
35.2.3	SnapShot Segment 30-74 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	120	Std Plus	0.84	8.19	50	32	Weight and height based protocol for localization

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (mm)	Phantom (cm)	Description
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	Std	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
			Cardiac SEG	CardiacSmall	0.160:1 6.4	0.35	0.625	40	80	ECG Mod 80-300mA Phase 40-80 AvgmA300	Std SS Seg	13.06	143.68	75	32	Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
35.2.4	SnapShot Burst 75-104 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	120	Std Plus	0.84	8.19	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	Std	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
			Cardiac SSB	CardiacSmall	0.160:1 6.4	0.35	0.625	40	80	ECG Mod 80-300mA Phase 40-80 AvgmA300	Std SSB	13.06	143.68	75	32	Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
35.2.5	SnapShot Burst+ >105 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	120	Std Plus	0.84	8.19	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	Std	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
			Cardiac SSBP	CardiacSmall	0.160:1 6.4	0.35	0.625	40	80	ECG Mod 80-300mA Phase 35-85 AvgmA300	Std SSB	14.04	154.46	75	32	Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
35.2.7	Non ECG Gated		Helical	PedBody	0.984:1 39.375	0.4	0.625	40	80	SmartmA 50-220mA NI=18 AvgmA220	Std Plus	2.14	23.10	75	32	Weight and height based non ECG gated protocol for cardiac with SmartmA
35.3.1	RC 9.5-11.5 kg (20.9-25.4lbs)		Helical	SmallBody	1.375:1 27.5	0.4	3.75	20	120	100	Std Plus	2.17	22.53	78.75	32	Weight and height based routine chest protocol
35.3.3	SnapShot Segment 30-74 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	120	Std Plus	0.84	8.19	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	Std	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
			Cardiac SEG	CardiacSmall	0.160:1 6.4	0.35	0.625	40	80	ECG Mod 100-320mA Phase 40-80 AvgmA320	Std SS Seg	14.05	154.52	75	32	Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
35.3.4	SnapShot Burst 75-104 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	120	Std Plus	0.84	8.19	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	Std	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (mm)	Phantom (cm)	Description
			Cardiac SSB	CardiacSmall	0.160:1 6.4	0.35	0.625	40	80	ECG Mod 100-320mA Phase 40-80 AvgmA320	Std SSB	14.05	154.52	75	32	Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
35.3.5	SnapShot Burst+ >105 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	120	Std Plus	0.84	8.19	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	Std	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
			Cardiac SSBP	CardiacSmall	0.160:1 6.4	0.35	0.625	40	80	ECG Mod 100-320mA Phase 35-85 AvgmA320	Std SSB	15.03	165.30	75	32	Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
35.3.7	Non ECG Gated		Helical	PedBody	0.984:1 39.375	0.4	0.625	40	80	SmartmA 50-240mA NI=18 AvgmA240	Std Plus	2.34	25.20	75	32	Weight and height based non ECG gated protocol for cardiac with SmartmA
35.4.1	RC 11.5-14.5kg (25.4-32.0lbs)		Helical	SmallBody	1.375:1 27.5	0.4	5	20	120	90	Std Plus	1.95	22.47	90	32	Weight and height based routine chest protocol
35.4.3	SnapShot Segment 30-74 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	120	Std Plus	0.84	8.19	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	Std	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
			Cardiac SEG	CardiacSmall	0.160:1 6.4	0.35	0.625	40	80	ECG Mod 100-340mA Phase 40-80 AvgmA340	Std SS Seg	14.87	163.52	75	32	Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
35.4.4	SnapShot Burst 75-104 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	120	Std Plus	0.84	8.19	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	Std	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
			Cardiac SSB	CardiacSmall	0.160:1 6.4	0.35	0.625	40	80	ECG Mod 100-340mA Phase 40-80 AvgmA340	Std SSB	14.87	163.52	75	32	Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
35.4.5	SnapShot Burst+ >105 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	120	Std Plus	0.84	8.19	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	Std	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
			Cardiac SSBP	CardiacSmall	0.160:1 6.4	0.35	0.625	40	80	ECG Mod 100-340mA Phase 35-85 AvgmA340	Std SSB	15.93	175.28	75	32	Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval] (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (mm)	Phantom (cm)	Description
35.4.7	Non ECG Gated		Helical	PedBody	0.984:1 39.375	0.4	0.625	40	80	SmartmA 50-280mA NI=18 AvgmA280	Std Plus	2.72	29.40	75	32	Weight and height based non ECG gated protocol for cardiac with SmartmA
35.5.1	RC 14.5-18.5kg (32.0-40.8lbs)		Helical	SmallBody	1.375:1 27.5	0.4	5	20	120	100	Std Plus	2.17	24.97	90	32	Weight and height based routine chest protocol
35.5.3	SnapShot Segment 30-74 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	120	Std Plus	0.84	8.19	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	Std	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
			Cardiac SEG	CardiacSmall	0.160:1 6.4	0.35	0.625	40	80	ECG Mod 100-380mA Phase 40-80 AvgmA380	Std SS Seg	16.52	181.73	75	32	Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
35.5.4	SnapShot Burst 75-104 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	120	Std Plus	0.84	8.19	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	Std	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
			Cardiac SSB	CardiacSmall	0.160:1 6.4	0.35	0.625	40	80	ECG Mod 100-380mA Phase 40-80 AvgmA380	Std SSB	16.52	181.73	75	32	Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
35.5.5	SnapShot Burst+ >105 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	120	Std Plus	0.84	8.19	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	Std	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
			Cardiac SSBP	CardiacSmall	0.160:1 6.4	0.35	0.625	40	80	ECG Mod 100-380mA Phase 35-85 AvgmA380	Std SSB	17.77	195.45	75	32	Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
35.5.7	Non ECG Gated		Helical	PedBody	0.984:1 39.375	0.4	0.625	40	80	SmartmA 50-300mA NI=18 AvgmA300	Std Plus	2.92	31.50	75	32	Weight and height based non ECG gated protocol for cardiac with SmartmA
35.6.1	RC 18.5-22.5kg (40.8-49.6lbs)		Helical	SmallBody	1.375:1 27.5	0.4	5	20	120	110	Std Plus	2.39	27.47	90	32	Weight and height based routine chest protocol
35.6.3	SnapShot Segment 30-74 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	120	Std Plus	0.84	8.19	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	Std	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (mm)	Phantom (cm)	Description
			Cardiac SEG	CardiacSmall	0.160:1 6.4	0.35	0.625	40	80	ECG Mod 100-400mA Phase 40-80 AvgmA400	Std SS Seg	17.35	190.80	75	32	Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
35.6.4	SnapShot Burst 75-104 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	120	Std Plus	0.84	8.19	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	Std	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
			Cardiac SSB	CardiacSmall	0.160:1 6.4	0.35	0.625	40	80	ECG Mod 100-400mA Phase 40-80 AvgmA400	Std SSB	17.35	190.80	75	32	Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
35.6.5	SnapShot Burst+ >105 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	120	Std Plus	0.84	8.19	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	Std	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
			Cardiac SSBP	CardiacSmall	0.160:1 6.4	0.35	0.625	40	80	ECG Mod 100-400mA Phase 35-85 AvgmA400	Std SSB	18.68	205.51	75	32	Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
35.6.7	Non ECG Gated		Helical	PedBody	0.984:1 39.375	0.4	0.625	40	80	SmartmA 100-320mA NI=18 AvgmA320	Std Plus	3.11	33.71	75	32	Weight and height based non ECG gated protocol for cardiac with SmartmA
35.7.1	RC 22.5-31.5kg (49.6-69.5lbs)		Helical	LargeBody	1.375:1 55.0	0.4	5	40	120	110	Std Plus	2.54	33.71	85	32	Weight and height based routine chest protocol
35.7.3	SnapShot Segment 30-74 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	120	Std Plus	0.84	8.19	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	Std	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
			Cardiac SEG	CardiacSmall	0.160:1 6.4	0.35	0.625	40	80	ECG Mod 110-420mA Phase 40-80 AvgmA420	Std SS Seg	18.25	200.78	75	32	Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
35.7.4	SnapShot Burst 75-104 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	120	Std Plus	0.84	8.19	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	Std	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
			Cardiac SSB	CardiacSmall	0.160:1 6.4	0.35	0.625	40	80	ECG Mod 110-420mA Phase 40-80 AvgmA420	Std SSB	18.25	200.78	75	32	Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (mm)	Phantom (cm)	Description
35.7.5	SnapShot Burst+ >105 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	120	Std Plus	0.84	8.19	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	Std	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
			Cardiac SSBP	CardiacSmall	0.160:1 6.4	0.35	0.625	40	80	ECG Mod 110-420mA Phase 35-85 AvgmA 420	Std SSB	19.63	215.97	75	32	Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
35.7.7	Non ECG Gated		Helical	PedBody	0.984:1 39.375	0.4	0.625	40	80	SmartmA 100-360mA NI=18 AvgmA360	Std Plus	3.50	37.93	75	32	Weight and height based non ECG gated protocol for cardiac with SmartmA
35.8.1	RC 31.5-40.5kg (69.5-89.3lbs)		Helical	LargeBody	1.375:1 55.0	0.4	5	40	120	120	Std Plus	2.77	46.45	120	32	Weight and height based routine chest protocol
35.8.3	SnapShot Segment 30-74 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	100	120	Std Plus	1.55	15.20	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	100	50	Std	39.50	19.75	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
			Cardiac SEG	CardiacSmall	0.160:1 6.4	0.35	0.625	40	100	ECG Mod 100-380mA Phase 40-80 AvgmA380	Std SS Seg	30.66	337.28	75	32	Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
35.8.4	SnapShot Burst 75-104 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	100	120	Std Plus	1.55	15.20	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	100	50	Std	39.50	19.75	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
			Cardiac SSB	CardiacSmall	0.160:1 6.4	0.35	0.625	40	100	ECG Mod 100-380mA Phase 40-80 AvgmA380	Std SSB	30.66	337.28	75	32	Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
35.8.5	SnapShot Burst+ >105 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	100	120	Std Plus	1.55	15.20	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	100	50	Std	39.50	19.75	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
			Cardiac SSBP	CardiacSmall	0.160:1 6.4	0.35	0.625	40	100	ECG Mod 100-380mA Phase 35-85 AvgmA380	Std SSB	32.98	362.75	75	32	Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
35.8.7	Non ECG Gated		Helical	SmallBody	0.984:1 39.375	0.4	0.625	40	100	SmartmA 100-400mA NI=16 AvgmA400	Std Plus	7.22	78.21	75	32	Weight and height based non ECG gated protocol for cardiac with SmartmA

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval] (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (mm)	Phantom (cm)	Description
35.9.1	RC 40.5-55.0kg(89.3-121.3lbs)		Helical	LargeBody	1.375:1 55.0	0.4	5	40	120	130	Std Plus	3.00	59.31	150	32	Weight and height based routine chest protocol
35.9.3	SnapShot Segment 30-74 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	100	120	Std Plus	1.55	15.20	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	100	50	Std	39.50	19.75	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
			Cardiac SEG	CardiacSmall	0.160:1 6.4	0.35	0.625	40	100	ECG Mod 100-420mA Phase 40-80 AvgmA420	Std SS Seg	33.73	371.05	75	32	Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
35.9.4	SnapShot Burst 75-104 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	100	120	Std Plus	1.55	15.20	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	100	50	Std	39.50	19.75	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
			Cardiac SSB	CardiacSmall	0.160:1 6.4	0.35	0.625	40	100	ECG Mod 100-420mA Phase 40-80 AvgmA420	Std SSB	33.73	371.05	75	32	Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
35.9.5	SnapShot Burst+ >105 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	100	120	Std Plus	1.55	15.20	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	100	50	Std	39.50	19.75	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
			Cardiac SSBP	CardiacSmall	0.160:1 6.4	0.35	0.625	40	100	ECG Mod 100-420mA Phase 35-85 AvgmA420	Std SSB	36.38	400.15	75	32	Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
35.9.7	Non ECG Gated		Helical	SmallBody	0.984:1 39.375	0.4	0.625	40	100	SmartmA 100-440mA NI=16 AvgmA440	Std Plus	7.95	86.03	75	32	Weight and height based non ECG gated protocol for cardiac with SmartmA

Abdomen

Table 13-4

[Interval]* - For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (second s)	Slice Thickne ss (mm) [Interval]* (mm)	Beam Collimat ion (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy·c m)	Scan Length (mm)	Phantom (cm)	Description
36.1.1	Abd 6.0-7.5kg (13.2-16.5lbs)		Helical	PedBody	1.375:1 27.5	0.4	3.75	20	120	100	Std Plus	2.17	22.53	78.75	32	Weight and height based routine abdomen protocol
36.2.1	Abd 7.5-9.5kg (16.5-20.9lbs)		Helical	PedBody	1.375:1 27.5	0.4	3.75	20	120	160	Std Plus	3.47	39.95	90	32	Weight and height based routine abdomen protocol
36.3.1	Abd 9.5-11.5kg (20.9-25.4lbs)		Helical	PedBody	1.375:1 27.5	0.4	3.75	20	120	180	Std Plus	3.91	49.34	101.25	32	Weight and height based routine abdomen protocol
36.4.1	Abd 11.5-14.5kg (25.4-32.0lbs)		Helical	SmallBo dy	1.375:1 27.5	0.4	5	20	120	125	Std Plus	2.71	36.64	110	32	Weight and height based routine abdomen protocol
36.5.1	Abd 14.5-18.5kg (32.0-40.8lbs)		Helical	SmallBo dy	1.375:1 27.5	0.4	5	20	120	135	Std Plus	2.93	39.57	110	32	Weight and height based routine abdomen protocol
36.6.1	Abd 18.5-22.5kg (40.8-49.6lbs)		Helical	Medium Body	1.375:1 27.5	0.4	5	20	120	120	Std Plus	3.07	47.68	130	32	Weight and height based routine abdomen protocol
36.7.1	Abd 22.5- 31.5kg(49.6- 69.5lbs)		Helical	LargeBo dy	1.375:1 55.0	0.4	5	40	120	150	Std Plus	3.46	68.44	150	32	Weight and height based routine abdomen protocol
36.8.1	Abd 31.5- 40.5kg(69.5- 89.3lbs)		Helical	LargeBo dy	1.375:1 55.0	0.4	5	40	120	170	Std Plus	3.92	77.56	150	32	Weight and height based routine abdomen protocol
36.9.1	Abd 40.5- 55.0kg(89.3- 121.3lbs)		Helical	LargeBo dy	1.375:1 55.0	0.4	5	40	120	200	Std Plus	4.61	114.30	200	32	Weight and height based routine abdomen protocol

Miscellaneous

Table 13-5

[Interval]* - For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (mm)	Phantom (cm)	Description
40.1	Pink Zone Routine 0-20 lbs/0-9 kg		Helical	PedBody	1.375:1 55.0	0.5	5	40	80	SmartmA 65-130mA NI=5 Avg mA 70	Std	0.61	8.70	95	32	Pediatric and Small patient Routine procedure based protocols
40.2	Pink Zone Routine 21-60 lbs/9.1-27.2 kg		Helical	PedBody	1.375:1 55.0	0.5	5	40	100	SmartmA 80-160mA NI=7 Avg mA 70	Std	1.13	27.47	195	32	Pediatric and Small patient Routine procedure based protocols
40.3	Pink Zone Routine 61-100 lbs/27.3-45.4 kg		Helical	Medium Body	1.375:1 55.0	0.5	5	40	120	SmartmA 95-190mA NI=10 Avg mA 70	Std	2.09	71.81	295	32	Pediatric and Small patient Routine procedure based protocols
40.4	Pink Zone Routine 101-200 lbs/45.4-90.7 kg		Helical	LargeBody	1.375:1 55.0	0.5	5	40	120	SmartmA 110-220mA NI=12 Avg mA 70	Std	2.02	89.30	395	32	Pediatric and Small patient Routine procedure based protocols
40.5	Pink Zone Routine >200 lbs/>90.8 kg		Helical	LargeBody	1.375:1 55.0	0.5	5	40	120	SmartmA 125-300mA NI=15 Avg mA 70	Std	2.02	89.30	395	32	Pediatric and Small patient Routine procedure based protocols
40.7	Green Zone Lower Dose 0-20 lbs/0-9 kg		Helical	PedBody	1.375:1 55.0	0.5	5	40	80	SmartmA 50-100mA NI=7 Avg mA 70	Std	0.61	8.70	95	32	Pediatric and Small patient Lower Dose procedure based protocols
40.8	Green Zone Lower Dose 21-60 lbs/9.1-27.2 kg		Helical	PedBody	1.375:1 55.0	0.5	5	40	100	SmartmA 60-120mA NI=9 Avg mA 70	Std	1.13	27.47	195	32	Pediatric and Small patient Lower Dose procedure based protocols
40.9	Green Zone Lower Dose 61-100 lbs/ 27.3-45.4 kg		Helical	Medium Body	1.375:1 55.0	0.5	5	40	120	SmartmA 70-140mA NI=11 Avg mA 70	Std	2.09	71.81	295	32	Pediatric and Small patient Lower Dose procedure based protocols
40.10	Green Zone Lower Dose 101-200 lbs/ 45.4-90.7 kg		Helical	LargeBody	1.375:1 55.0	0.5	5	40	120	SmartmA 80-160mA NI=13 Avg mA 70	Std	2.02	89.30	395	32	Pediatric and Small patient Lower Dose procedure based protocols
40.11	Green Zone Lower Dose >200 lbs/ >90.8 kg		Helical	LargeBody	1.375:1 55.0	0.5	5	40	120	SmartmA 90-240mA NI=16 Avg mA 70	Std	2.02	89.30	395	32	Pediatric and Small patient Lower Dose procedure based protocols
40.13	Grey Zone CT Angio 0-20 lbs/0-9 kg		Helical	PedBody	1.375:1 55.0	0.5	2.5	40	80	SmartmA 100-200mA NI=5 Avg mA 140	Std	1.22	17.71	97.5	32	Pediatric and Small patient Angio procedure based protocols

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval J* (mm)]	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (mm)	Phantom (cm)	Description
40.14	Grey Zone CT Angio 21-60 lbs/9.1-27.2 kg		Helical	SmallBody	1.375:1 55.0	0.5	2.5	40	100	SmartmA 120-240mA NI=9.9 Avg mA 140	Std	2.26	55.50	197.5	32	Pediatric and Small patient Angio procedure based protocols
40.15	Grey Zone CT Angio 61-100 lbs/ 27.3-45.4 kg		Helical	LargeBody	1.375:1 55.0	0.5	2.5	40	120	SmartmA 120-240mA NI=10 Avg mA 140	Std	4.03	139.28	297.5	32	Pediatric and Small patient Angio procedure based protocols
40.16	Grey Zone CT Angio 101-200 lbs/ 45.4-90.7 kg		Helical	LargeBody	1.375:1 55.0	0.5	2.5	40	120	SmartmA 120-240mA NI=12 Avg mA 140	Std	4.03	179.62	397.5	32	Pediatric and Small patient Angio procedure based protocols
40.17	Grey Zone CT Angio >200 lbs/ >90.8 kg		Helical	LargeBody	1.375:1 55.0	0.5	2.5	40	120	SmartmA 120-300mA NI=15 Avg mA 140	Std	4.03	179.62	397.5	32	Pediatric and Small patient Angio procedure based protocols

Reference Adult Protocols - Installed Options : ASiR, VCT85kW and VCT Hi Power

Head

Table 14-1

[Interval]* - For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (s)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (mm)	Phantom (cm)	Description
21.1	Routine Head 1s Group1		Axial	Head	4i	1	5	20	140	120	40	Std	40	32.00	192.01	55	16	Routine head protocol for evaluation of the brain for abnormalities.
	Group2		Axial	Head	4i	1	5	20	120	190	40	Std	40	36.47	291.78	75	16	
21.2	Routine Head 2s Group1		Axial	Head	4i	2	5	20	140	60	40	Std	40	32.00	192.01	55	16	Routine head protocol for evaluation of the brain for abnormalities.
	Group2		Axial	Head	4i	2	5	20	120	95	40	Std	40	36.47	291.78	75	16	
21.3	Trauma Head 2.5mm		Axial	Head	8i	0.5	2.5	20	140	335	40	Std	40	44.67	625.37	137.5	16	Emergency head protocol for evaluation of the brain and cranium for abnormalities.
21.4	Routine Head 0.5s Group1		Axial	Head	4i	0.5	5	20	140	240	40	Std	40	32.00	192.01	55	16	Routine head protocol for evaluation of the brain for abnormalities.
	Group2		Axial	Head	4i	0.5	5	20	120	360	40	Std	40	35.52	284.17	75	16	
21.5	Circle of Willis 0.4s.	MIROI	Axial	Head	1i	1	5 [0]	5	120	50	0	Std	0	125.58	62.79	0	16	Test bolus for CT angiography of the head for evaluation of carotid and cerebral vasculature.
			Helical	Head	0.531:1 10.625	0.4	0.625	20	120	180	40	Std Plus IQE	50	26.02	252.82	80	16	CT angiography of the head for evaluation of carotid and cerebral vasculature.
21.6	CoW/ Carotid 0.625mm 0.4s.Smart mA	MIROI	Axial	Head	1i	1	5 [0]	5	120	80	0	Std	0	180.84	90.42	0	16	Test bolus for CT angiography of the head for evaluation of carotid and cerebral vasculature.
			Helical	Head	0.531:1 10.625	0.4	0.625	20	120	SmartmA 100-600mA NI=8 Avg mA 415	30	Std Plus IQE	40	61.66	1155.86	170	16	CT angiography of the head for evaluation of carotid and cerebral vasculature.
21.8	Helical Head	DMPR	Helical	Head	0.531:1 10.625	0.5	0.625	20	120	180	40	Std Plus IQE	40	32.52	527.31	145	16	Helical scan mode for evaluation of the brain for cerebral abnormality.
21.10	VHS Head 4D CTA	MIROI	Axial	Head	1i	1	5 [0]	5	120	50	0	Std	0	125.58	62.79	0	16	Test bolus for CT angiography of the head for evaluation of carotid and cerebral vasculature.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (s)	Slice Thickness (mm) [Interval] J* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (mm)	Phantom (cm)	Description
			Helical-S	Head	0.984:1 39.375 18 pass	0.4	5 [10]	40	120	150	50	Std Plus	50	248.10	3101.19	110	16	CT angiography of the head using Volume Helical scan mode for evaluation of cerebral vasculature.
21.11	VHS Perfusion 350-370 Strength Contrast		Axial	Head	4i	1	5	20	120	200	40	Std	40	38.39	537.48	135	16	Non-Enhance Brain, evaluation for hemorrhage or infarction.
		Perfusion	Helical-S/P	Head	0.984:1 39.375 28 pass	0.4	5	40	80	200	0	Std	50	186.02	2325.25	110	16	CT Perfusion using Volume Helical scan mode for evaluation of cerebral perfusion over time.
			Axial	Head	4i	1	5	20	120	200	40	Std	40	38.39	537.48	135	16	Post Contrast Brain, Evaluation for residual tissue enhancement.
21.12	CT Perfusion 300-320 Strength Contrast		Axial	Head	4i	1	5	20	120	200	40	Std	40	38.39	537.48	135	16	Non-Enhance Brain, evaluation for hemorrhage or infarction.
		Perfusion	Cine	Head	8i	1	5 [0]	40	80	200	0	Std	0	652.82	2611.27	35	16	CT Perfusion using Cine scan mode for evaluation of cerebral perfusion over time.
			Axial	Head	4i	1	5	20	120	200	40	Std	40	38.39	537.48	135	16	Non-Enhance Brain, evaluation for hemorrhage or infarction.
21.13	CT Perfusion 350-370 Strength Contrast		Axial	Head	4i	1	5	20	120	200	40	Std	40	38.39	537.48	135	16	Non-Enhance Brain, evaluation for hemorrhage or infarction.
		Perfusion	Cine	Head	8i	1	5	40	80	200	0	Std	0	587.55	2350.19	35	16	CT Perfusion using Cine scan mode for evaluation of cerebral perfusion over time.
			Axial	Head	4i	1	5	20	120	200	40	Std	40	38.39	537.48	135	16	Post Contrast Brain, Evaluation for residual tissue enhancement.
21.14	CT Perfusion Brain Tumor - Dual Phase		Axial	Head	4i	1	5	20	120	200	40	Std	40	38.39	537.48	135	16	Non-Enhance Brain, evaluation for hemorrhage or infarction.
	Group 1	Perfusion	Cine	Head	8i	1	5 [0]	40	80	200	0	Std	0	587.55	2350.19	35	16	CT Perfusion using Cine scan mode for evaluation of cerebral perfusion over time.
	Group 2	Perfusion	Axial	Head	8i	1	5 [0]	40	80	200	0	Std	0	130.54	522.15	35	16	Delayed phase CT Perfusion using Axial scan mode for evaluation of Brain tumor permeability.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (s)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (mm)	Phantom (cm)	Description
			Axial	Head	4i	1	5	20	120	200	40	Std	40	38.39	537.48	135	16	Post Contrast Brain, Evaluation for residual tissue enhancement.
21.15	CT Perfusion 350-370 Strength Contrast (Axial)		Axial	Head	4i	1	5	20	120	200	40	Std	40	38.39	537.48	135	16	Non-Enhance Brain, evaluation for hemorrhage or infarction.
		Perfusion	Axial	Head	8i	1	5	40	80	200	0	Std	0	287.18	1148.73	35	16	CT Perfusion using Axial scan mode for evaluation of cerebral perfusion over time.
			Axial	Head	4i	1	5	20	120	200	40	Std	40	38.39	537.48	135	16	Post Contrast Brain, Evaluation for residual tissue enhancement.
21.16	CT Perfusion 350-370 Strength Contrast (Axial Shuttle Mode)		Axial	Head	4i	1	5	20	120	200	40	Std	40	38.39	537.48	135	16	Non-Enhance Brain, evaluation for hemorrhage or infarction.
		Perfusion	Axial-S	Head	8i 17 pass	0.4	5	40	80	500	0	Std	0	221.91	1775.31	75	16	CT Perfusion using Volume Shuttle (Axial) scan mode for evaluation of cerebral perfusion over time.
			Axial	Head	4i	1	5	20	120	200	40	Std	40	38.39	537.48	135	16	Post Contrast Brain, Evaluation for residual tissue enhancement.

Orbit

Table 14-2

[Interval]* - For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy-cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
22.1	Sinus Supine Helical + DMPPR	DMPPR	Helical	Head	0.531:1 10.625	0.4	0.625	20	120	85	30	Bone Plus IQE	50	12.29	112.48	74.375	16	Routine Helical Sinus protocol, Evaluation for any abnormality
22.3	Orbits Helical + DMPPR	DMPPR	Helical	Head	0.531:1 10.625	0.5	0.625	20	120	155	30	Std Plus IQE	40	28.00	221.29	61.875	16	Routine Helical Orbit protocol, Evaluation for abnormality soft tissues and bone structures of the face.
22.6	Helical IAC 0.625mm Axial mode - Use Reformat for Coronal		Helical	Head	0.531:1 10.625	0.6	0.625	20	140	190	30	Std Plus IQE	40	57.23	319.79	38.75	16	Axial scan mode, Evaluation for inner ear abnormalities in Axial plane
22.8	Axial and Coronal IAC		Axial	Head	16i	1	1.25	20	140	195	30	Std	40	52.00	208.01	38.75	16	Helical scan mode, Evaluation for inner ear abnormalities in Axial plane
			Axial	Head	16i	1	1.25	20	140	235	30	Std	40	62.67	250.68	38.75	16	Helical scan mode, Evaluation for inner ear abnormalities in Coronal plane

Neck

Table 14-3

[Interval]* - For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
23.1	C-Spine C5-C7 Axial		Axial	SmallBody	8i	0.5	2.5	20	140	490	20	Std	40	26.64	373.03	137.5	32	C-Spine C5-C7 area Axial mode, Evaluation for soft tissue, disc and bone abnormalities.
23.2	C-Spine C1-C4 Axial		Axial	SmallBody	8i	1	2.5	20	120	190	20	Std	40	14.17	113.36	77.5	32	C-Spine C1-C4 area Axial mode, Evaluation for soft tissue, disc and bone abnormalities.
23.3	C-Spine Helical + DMPR	DMPR	Helical	SmallBody	0.516:1 20.625	0.8	0.625	40	120	270	20	Bone Plus IQE	40	29.20	421.46	110	32	C-Spine Helical mode with DMPR, Evaluation for soft tissue, disc and bone abnormalities.
23.4	C-Spine Helical + DMPR/ SmartmA	DMPR	Helical	SmallBody	0.516:1 20.625	0.8	0.625	40	120	SmartmA 100-600mA NI=11.34 Avg mA 250	20	Bone Plus IQE	40	27.04	390.24	110	32	C-Spine Helical mode with DMPR and SmartmA, Evaluation for soft tissue, disc and bone abnormalities.
23.8	Carotid 0.625mm 0.4s. 0.5:1 SmartmA/ SmartPrep		Helical	SmallBody	0.516:1 20.625	0.4	0.625	40	140	SmartmA 100-610mA NI=11 Avg mA 345	40	Std Plus IQE	50	24.46	541.51	170	32	Carotid CTA (pitch 0.5:1) with SmartmA and SmartPrep, Evaluation of carotid vasculature.
23.9	Carotid 0.625mm 0.4s. 0.9:1 SmartmA/ SmartPrep		Helical	SmallBody	0.984:1 39.375	0.4	0.625	40	120	SmartmA 80-610mA NI=11 Avg mA 375	40	Std Plus IQE	50	10.62	215.89	170	32	Carotid CTA (pitch 0.9:1) with SmartmA and SmartPrep, Evaluation of carotid vasculature.
23.10	Carotid/ CoW 0.625mm 0.4s. AutomA	MIROI	Axial	SmallBody	1i	1	5 [0]	5	120	80	0	Std	0	99.00	49.50	0	32	Test bolus for Carotid CTA (pitch 0.5:1) of the head for evaluation of carotid and cerebral vasculature.
			Helical	SmallBody	0.516:1 20.625	0.4	0.625	40	140	AutomA 80-515mA NI=11 Avg mA 310	40	Std Plus IQE	50	23.78	486.15	170	32	Carotid and Brain CTA (pitch 0.5:1) with AutomA, Evaluation of carotid and cerebral vasculature.
23.11	Carotid/ CoW 0.625mm 0.4s. Smart mA/ SmartPrep		Helical	SmallBody	0.969:1 19.375	0.4	0.625	20	140	SmartmA 100-610mA NI=11 Avg mA 360	40	Std Plus IQE	50	16.17	305.36	170	32	Carotid CTA (pitch 0.9:1) with SmartmA and SmartPrep, Evaluation of carotid and cerebral vasculature.
23.12	Soft Tissue Neck 2.5mm AutomA		Helical	SmallBody	0.984:1 39.375	0.5	2.5	40	120	AutomA 100-335mA NI=9.1 Avg mA 175	30	Std Plus	30	6.20	113.26	150	32	Neck with AutomA, Evaluation of soft tissue structures for abnormality

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
23.13	Soft Tissue Neck + DMPR/ SmartmA	DMPR	Helical	SmallBody	0.969:1 19.375	0.5	0.625	20	120	SmartmA 100-600mA NI=12.6 Avg mA 255	40	Std Plus IQE	50	9.82	165.46	150	32	Neck with DMPR and SmartmA, Evaluation of soft tissue structures for abnormality

Upper Extremity

Table 14-4

[Interval]* - For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
24.1	Shoulder 1.25mm + DMPR	DMPR	Helical	LargeBody	0.516:1 20.625	0.6	1.25 [0.625]	40	140	215	40	Std Plus IQE	40	28.09	377.41	100	32	Shoulder with DMPR, Evaluation of soft tissue and bone of shoulder area
24.2	Shoulder 1.25mm + DMPR/ SmartmA	DMPR	Helical	LargeBody	0.516:1 20.625	0.6	1.25 [0.625]	40	140	SmartmA 100-610mA NI=20 Avg mA 270	40	Std Plus IQE	40	35.27	473.96	100	32	Shoulder with DMPR and SmartmA, Evaluation of soft tissue and bone of shoulder area

Chest

Table 14-5

[Interval]* - For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
25.1	Routine Chest 0.4s. 5mm SmartmA		Helical	LargeBody	1.375:1 55.0	0.4	5	40	120	SmartmA 100-650mA NI=13 Avg mA 250	50	Std Plus	50	5.76	142.87	200	32	Chest with SmartmA, Evaluation of mediastinum and lungs fields
25.2	Routine Chest 0.4s.+ DMPR/ SmartmA	DMPR	Helical	LargeBody	0.984:1 39.375	0.4	0.625	40	120	SmartmA 100-700mA NI=22 Avg mA 310	50	Std Plus	50	9.98	232.46	200	32	Chest with SmartmA and DMPR, Evaluation of mediastinum and lungs fields
25.3	Chest with HiRes 0.4s. SmartmA		Helical	LargeBody	0.969:1 19.375	0.4	5	20	120	SmartmA 80-650mA NI=12 Avg mA 300	40	Chest Plus	40	10.50	229.59	200	32	Chest with SmartmA, Evaluation of mediastinum and lung fields with higher resolution images of the lungs
25.4	Hi Res Chest 0.5s. Axial		Axial	LargeBody	1i	0.5	1.25 [10]	1.25	120	190	50	Bone+	50	2.22	53.35	230	32	Axial scan mode, Evaluation of mediastinum and lung fields in high resolution mode
25.5	Chest Abd Pelvis 0.4/ 0.5s. SmartmA 5mm (2 groups) Group1		Helical	LargeBody	0.984:1 39.375	0.4	5	40	120	SmartmA 100-650mA NI=13 Avg mA 185	50	Std Plus	50	5.96	108.95	150	32	Multi group scan for Chest, Abdomen, Pelvis (0.4/0.5 sec) with SmartmA, Evaluation of chest and abdominal anatomy for abnormalities
	Group2		Helical	LargeBody	1.375:1 55.0	0.5	5	40	120	SmartmA 100-650mA NI=13 Avg mA 185	50	Std Plus	50	5.33	182.72	295	32	
25.6	Chest Abd Pelvis 0.5s + DMPR SmartPrep	DMPR	Helical	LargeBody	1.375:1 55.0	0.5	0.625	40	120	SmartmA 100-700mA NI=22 Avg mA 270	50	Std Plus	50	7.78	387.24	450	32	Helical scan mode (0.5sec) with SmartmA, Evaluation of chest and abdominal anatomy for abnormalities
25.7	Pulmonary Embolis 0.5s.	MIROI	Axial	LargeBody	1i	1	5 [0]	5	120	80	0	Std	0	93.89	46.95	0	32	Timing Bolus for Helical scan mode 1.25mm thickness and helical pitch of 0.9:1, evaluation of Chest for pulmonary embolism
			Helical	LargeBody	0.984:1 39.375	0.5	1.25	40	120	350	40	Std	40	14.09	300.25	180	32	Helical scan mode 1.25mm thickness and helical pitch of 0.9:1, Evaluation Chest for pulmonary embolism

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
25.8	Pulmonary Embolis 0.5s. SmartmA	MIROI	Axial	LargeBody	1i	1	5 [0]	5	120	80	0	Std	0	93.89	46.95	0	32	Timing Bolus for Helical scan mode 1.25mm thickness and helical pitch of 0.9:1 with SmartmA, evaluation of Chest for pulmonary embolism
			Helical	LargeBody	0.984:1 39.375	0.5	1.25	40	120	SmartmA 100-600mA NI=20 Avg mA 300	40	Std	40	12.07	256.93	180	32	Helical scan mode 1.25mm thickness and helical pitch of 0.9:1 with SmartmA, Evaluation Chest for pulmonary embolism
25.9	Aortic Dissection 0.625mm + DMPR/ SmartmA/ SmartPrep	DMPR	Helical	LargeBody	1.375:1 55.0	0.5	0.625	40	120	SmartmA 100-700mA NI=22 Avg mA 360	40	Std Plus	40	10.37	361.20	300	32	Helical scan mode 0.625mm thickness and helical pitch of 1.375:1 with SmartmA, SmartPrep and DMPR, Evaluation for aortic dissection
25.10	Aortic Dissection 1.25mm + DMPR/ SmartmA/ SmartPrep	DMPR	Helical	LargeBody	1.375:1 55.0	0.5	1.25 [0.8]	40	120	SmartmA 100-700mA NI=20 Avg mA 330	40	Std Plus	40	9.51	330.68	300	32	Helical scan mode 1.25mm thickness and helical pitch of 1.375:1 with SmartmA, SmartPrep and DMPR, Evaluation for aortic dissection
25.11	Lung Assessment	Advanced Lung Analysis	Helical	LargeBody	0.969:1 19.375	0.5	1.25	20	120	120	0	Bone	0	5.25	114.77	200	32	Helical scan mode for acquisition of data for Advanced Lung Analysis (ALA) software for lung nodule assessment
25.12	Gated Chest Cardiac Helical		Helical	LargeBody	1.375:1 55.0	0.4	5	40	120	60	0	Std Plus	0	1.38	13.55	50	32	Localizer scan for Helical Gated scan for evaluation of thoracic vasculature
		MIROI	Axial	LargeBody	1i	1	5 [0]	5	120	60	0	Std	0	98.59	49.29	0	32	Timing bolus scan for Helical Gated scan for evaluation of thoracic vasculature
			Cardiac SEG	CardiacLarge	0.180:1 7.2	0.4	0.625	40	120	ECG Mod 120-235mA Phase 70-80 Avg mA 235	40	Std SS Seg	40	30.28	1014.47	300	32	Helical Gated scan for evaluation of thoracic vasculature
25.13	SmartScore - Gated 0.4s	Smart Score	Cine	LargeBody	8i	0.4	2.5	20	120	375	0	Std Segment	0	9.15	91.51	97.5	32	Cine gated scan mode for acquisition of data for coronary artery calcification analysis
25.14	SnapShot Segment 30-74 BPM		Helical	LargeBody	1.375:1 55.0	0.4	5	40	120	60	0	Std Plus	0	1.38	13.55	50	32	Localizer scan for Retrospectively gated Cardiac Helical scan for evaluation of coronary arteries, intended for heart rates 30-74 bpm

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy-cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
		MIROI	Axial	LargeBody	1i	1	5 [0]	5	120	60	0	Std	0	98.59	49.29	0	32	Timing bolus scan for Retrospectively gated Cardiac Helical scan for evaluation of coronary arteries, intended for heart rates 30-74 bpm
			Cardiac SEG	CardiacLarge	0.180:1 7.2	0.4	0.625	40	120	ECG Mod 120-235mA Phase 70-80 Avg mA 235	40	Std SS Seg	40	31.29	422.36	100	32	Retrospectively gated Cardiac Helical scan for evaluation of coronary arteries, intended for heart rates 30-74 bpm
25.15	SnapShot Burst 75-113 BPM		Helical	LargeBody	1.375:1 55.0	0.4	5	40	120	60	0	Std Plus	0	1.38	13.55	50	32	Localizer scan for Retrospectively gated Cardiac Helical scan for evaluation of coronary arteries, intended for heart rates 75-113 bpm
		MIROI	Axial	LargeBody	1i	1	5 [0]	5	120	60	0	Std	0	98.59	49.29	0	32	Timing bolus scan for Retrospectively gated Cardiac Helical scan for evaluation of coronary arteries, intended for heart rates 75-113 bpm
			Cardiac SSB	CardiacLarge	0.180:1 7.2	0.4	0.625	40	120	ECG Mod 120-235mA Phase 70-80 Avg mA 235	40	Std SSB	40	31.29	422.36	100	32	Retrospectively gated Cardiac Helical scan for evaluation of coronary arteries, intended for heart rates 75-113 bpm
25.16	SnapShot Burst Plus 114+ BPM		Helical	LargeBody	1.375:1 55.0	0.4	5	40	120	60	0	Std Plus	0	1.38	13.55	50	32	Localizer scan for Retrospectively gated Cardiac Helical scan for evaluation of coronary arteries, intended for heart rates over 113 bpm
		MIROI	Axial	LargeBody	1i	1	5 [0]	5	120	60	0	Std	0	98.59	49.29	0	32	Timing bolus scan for Retrospectively gated Cardiac Helical scan for evaluation of coronary arteries, intended for heart rates over 113 bpm
			Cardiac SSBP	CardiacLarge	0.180:1 7.2	0.4	0.625	40	120	ECG Mod 120-235mA Phase 70-80 Avg mA 235	40	Std SSB	40	31.29	422.36	100	32	Retrospectively gated Cardiac Helical scan for evaluation of coronary arteries, intended for heart rates over 113 bpm
25.17	Advantage 4D		Helical	LargeBody	1.375:1 55.0	0.6	5	40	120	170	40	Std Plus	40	5.88	145.59	200	32	Localizer scan for Advantage 4D series
		Advantage 4D	Cine	LargeBody	8i	1	2.5	20	120	70	30	Std	30	35.67	570.67	157.5	32	Cine scan mode for acquisition data for Advantage 4D for retrospective gating
25.18	VHS Chest 4D CTA SmartPrep		Helical-S	LargeBody	1.375:1 55.0 11 pass	0.4	5 [10]	40	120	SmartmA 100-600mA NI=12 Avg mA 360	40	Std Plus	40	115.35	3633.55	300	32	Volume Helical scan mode with SmartPrep for time resolved CT angiography of chest vasculature

Abdomen

Table 14-6

[Interval]* - For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
26.1	Abdomen Pelvis 0.5s. 5mm SmartmA		Helical	LargeBody	1.375:1 55.0	0.5	5	40	120	SmartmA 50-650mA NI=11.57 Avg mA 250	50	Std Plus	50	7.20	322.54	400	32	Basic Abdomen Pelvis (0.5sec) with Smart mA, Evaluation for abnormalities of abdominal and pelvic structures
26.2	Abdomen Pelvis 0.5s. 5mm + DMPR SmartmA/ SmartPrep	DMPR	Helical	LargeBody	0.984:1 39.375	0.6	0.625	40	120	SmartmA 80-700mA NI=11.57 Avg mA 260	50	Std Plus	50	12.56	543.37	400	32	Basic Abdomen Pelvis (0.5sec) with Smart mA, SmartPrep and DMPR, Evaluation for abnormalities of abdominal and pelvic structures
26.4	Renal Stone 0.5s. 5mm		Helical	LargeBody	1.375:1 55.0	0.5	5	40	120	190	50	Std Plus	50	5.47	245.13	400	32	Helical scan, Evaluation of kidney, ureters and bladder for renal stones
26.5	Renal Stone 0.5s. 5mm SmartmA		Helical	LargeBody	1.375:1 55.0	0.5	5	40	120	SmartmA 50-650mA NI=12 Avg mA 190	50	Std Plus	50	5.47	245.13	400	32	Helical scan with SmartmA, Evaluation of kidney, ureters and bladder for renal stones
26.7	AAA 0.5s. + DMPR 0.625mm	MIROI	Axial	LargeBody	1i	1	5 [0]	5	120	80	0	Std	0	112.67	56.34	0	32	Timing bolus for Helical scan with DMPR, Evaluation for abdominal aortic aneurysm
		DMPR	Helical	LargeBody	1.375:1 55.0	0.5	0.625	40	140	255	50	Std Plus	50	10.41	371.80	309.375	32	Helical scan with DMPR, Evaluation for abdominal aortic aneurysm
26.8	AAA 0.6s. + DMPR 0.625mm SmartmA/ SmartPrep	DMPR	Helical	LargeBody	1.375:1 55.0	0.6	0.625	40	120	SmartmA 100-700mA NI=19 Avg mA 275	50	Std Plus	50	9.51	339.51	309.375	32	Helical scan with DMPR, Smart mA and SmartPrep, Evaluation for abdominal aortic aneurysm
26.10	Dual Phase Liver		Helical	LargeBody	1.375:1 55.0	0.6	5	40	120	150	50	Std Plus	50	5.19	92.16	130	32	Non contrast evaluation of upper abdominal structures including liver
		MIROI	Axial	LargeBody	1i	1	5 [0]	5	120	80	0	Std	0	112.67	56.34	0	32	Timing bolus for Multi-Group acquisition for early arterial and portal venous phase of the liver
	Group1		Helical	LargeBody	1.375:1 55.0	0.5	2.5	40	120	275	50	Std Plus	50	7.92	140.87	130	32	Acquisition for early arterial phase of the liver
	Group2		Helical	LargeBody	1.375:1 55.0	0.5	5	40	120	210	50	Std Plus	50	6.05	270.94	400	32	Acquisition for portal venous phase of the liver

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
26.11	Tri Phase Liver		Helical	LargeBody	1.375:1 55.0	0.6	5	40	120	150	50	Std Plus	50	5.19	92.16	130	32	Non contrast evaluation of upper abdominal structures including liver
		MIROI	Axial	LargeBody	1i	1	5 [0]	5	120	80	0	Std	0	112.67	56.34	0	32	Timing bolus for Multi-Group acquisition for early arterial, late arterial and portal venous phase of the liver
	Group1		Helical	LargeBody	1.375:1 55.0	0.5	2.5	40	120	275	50	Std Plus	50	7.92	140.87	130	32	Acquisition for early arterial phase of the liver
	Group2		Helical	LargeBody	1.375:1 55.0	0.5	5	40	120	225	50	Std Plus	50	6.48	290.29	400	32	Acquisition for late arterial phase of the liver
	Group3		Helical	LargeBody	1.375:1 55.0	0.5	5	40	120	225	50	Std Plus	50	6.48	290.29	400	32	Acquisition for portal venous phase of the liver
26.12	Dual Phase Pancreas SmartmA/ SmartPrep		Helical	LargeBody	0.984:1 39.375	0.5	5	40	120	SmartmA 80-650mA NI=12 Avg mA 200	50	Std Plus	50	8.05	82.74	70	32	Non contrast evaluation of upper abdominal structures including pancreas
	Group1		Helical	LargeBody	0.984:1 39.375	0.5	2.5	40	120	SmartmA 50-700mA NI=15 Avg mA 260	50	Std Plus	50	10.46	191.28	150	32	Acquisition with SmartmA and SmartPrep for arterial phase of the pancreas
	Group2		Helical	LargeBody	0.984:1 39.375	0.5	5	40	120	SmartmA 50-700mA NI=15 Avg mA 225	50	Std Plus	50	9.06	319.47	320	32	Acquisition for venous phase of the pancreas
26.13	Dual Phase Liver SmartmA/ SmartPrep		Helical	LargeBody	1.375:1 55.0	0.5	5	40	120	SmartmA 80-500mA NI=12 Avg mA 160	50	Std Plus	50	4.61	81.96	130	32	Non contrast evaluation of upper abdominal structures including liver
	Group1		Helical	LargeBody	1.375:1 55.0	0.5	2.5	40	120	SmartmA 50-700mA NI=15 Avg mA 275	50	Std Plus	50	7.92	140.87	130	32	Acquisition with SmartmA and SmartPrep for early arterial phase of the liver
	Group2		Helical	LargeBody	0.984:1 39.375	0.5	5	40	120	SmartmA 50-700mA NI=11 Avg mA 225	50	Std Plus	50	9.06	391.91	400	32	Acquisition with SmartmA and SmartPrep for portal venous phase of the liver
26.14	Tri Phase Liver SmartmA/ SmartPrep		Helical	LargeBody	0.984:1 39.375	0.5	5	40	120	SmartmA 80-500mA NI=12 Avg mA 200	50	Std Plus	50	8.05	131.04	130	32	Non contrast evaluation of upper abdominal structures including liver
	Group1		Helical	LargeBody	1.375:1 55.0	0.5	2.5	40	120	SmartmA 50-700mA NI=15 Avg mA 275	50	Std Plus	50	7.92	140.87	130	32	Acquisition with SmartmA and SmartPrep for early arterial phase of the liver
	Group2		Helical	LargeBody	1.375:1 55.0	0.5	5	40	120	SmartmA 50-700mA NI=11 Avg mA 225	50	Std Plus	50	6.48	290.29	400	32	Acquisition with SmartmA and SmartPrep for late arterial phase of the liver
	Group3		Helical	LargeBody	1.375:1 55.0	0.5	5	40	120	SmartmA 50-700mA NI=11 Avg mA 225	50	Std Plus	50	6.48	290.29	400	32	Acquisition with SmartmA and SmartPrep for portal venous phase of the liver
26.15	VHS Liver Perfusion		Helical	LargeBody	0.984:1 39.375	0.6	5	40	120	85	50	Std	50	4.11	66.80	130	32	Non contrast evaluation of upper abdominal structures including liver

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy-cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
		Perfusion	Helical-S/P	LargeBody	1.375:1 55.0 21 pass	0.4	5	40	120	150	50	Std Plus	50	109.99	1704.89	140	32	Volume Helical scan mode for evaluation of liver tumor perfusion over time
26.16	VHS 4D CTA		Helical	LargeBody	0.984:1 39.375	0.6	5	40	120	170	50	Std	50	8.21	133.61	130	32	Localizer scan for Volume Helical Shuttle scan mode for evaluation of abdominal vasculature
		MIROI	Axial	LargeBody	1i	1	5 [0]	5	120	80	0	Std	0	112.67	56.34	0	32	Timing bolus scan for Volume Helical Shuttle scan mode for evaluation of abdominal vasculature
			Helical-S	LargeBody	1.375:1 55.0 18 pass	0.5	5 [10]	40	140	SmartmA 150-515mA NI=12 Avg mA 175	50	Std Plus	50	162.84	3175.35	180	32	Volume Helical Shuttle scan mode for time resolved CT angiography of abdominal vasculature
26.17	Runoff 0.625mm	MIROI	Axial	LargeBody	1i	1	5 [0]	5	120	80	0	Std	0	112.67	56.34	0	32	Timing bolus scan for evaluation of vascular structure of abdomen, femurs and lower extremities (Runoff)
			Helical	LargeBody	1.375:1 55.0	0.5	0.625	40	140	310	40	Std Plus	40	12.65	1246.09	936.875	32	Helical scanmode 0.625mm, Evaluation of vascular structure of abdomen, femurs and lower extremities (Runoff)
26.18	Runoff 1.25mm	MIROI	Axial	LargeBody	1i	1	5 [0]	5	120	80	0	Std	0	112.67	56.34	0	32	Timing bolus scan for evaluation of vascular structure of abdomen, femurs and lower extremities (Runoff)
			Helical	LargeBody	1.375:1 55.0	0.5	1.25 [0.7]	40	140	240	40	Std Plus	40	9.80	1025.48	998.9	32	Helical scanmode 1.25mm, Evaluation of vascular structure of abdomen, femurs and lower extremities (Runoff)
26.19	Runoff 0.625mm SmartmA/ SmartPrep		Helical	LargeBody	1.375:1 55.0	0.5	0.625	40	140	SmartmA 100-515mA NI=19 Avg mA 310	40	Std Plus	40	12.65	1246.09	936.875	32	Helical scanmode 0.625mm with SmartmA and SmartPrep, Evaluation of vascular structure of abdomen, femurs and lower extremities (Runoff)
26.20	Runoff 1.25mm SmartmA/ SmartPrep		Helical	LargeBody	1.375:1 55.0	0.5	1.25	40	140	SmartmA 100-515mA NI=20 Avg mA 240	40	Std Plus	40	9.80	1025.48	998.9	32	Helical scanmode 1.25mm with SmartmA and SmartPrep, Evaluation of vascular structure of abdomen, femurs and lower extremities (Runoff)
26.22	CT Colonography 1.25mm	Advantage CTC Pro	Helical	LargeBody	1.375:1 55.0	0.5	1.25	40	120	100	50	Std Plus	50	2.88	129.02	400	32	Helical scan mode supine acquisition for using Advantage CTC Pro for evaluation of the colon
		Advantage CTC Pro	Helical	LargeBody	1.375:1 55.0	0.5	1.25	40	120	100	50	Std Plus	50	2.88	129.02	400	32	Helical scan mode prone acquisition for using Advantage CTC Pro for evaluation of the colon

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy-cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
26.41	Body Interventional 5mm/3i		Helical	LargeBody	1.375:1 27.5	0.5	5	20	120	SmartmA 50-650mA NI=11.57 Avg mA 360	0	Std	0	11.41	251.58	195	32	Localization scan for CT interventional scan
			SmartView SEG	LargeBody	3i	0.8	5	10	120	80	0	Std Segment	0	684.58	684.58	0	32	CT Fluoro scan for interventional scan
26.42	Body Interventional 5mm/1i		Helical	LargeBody	1.375:1 27.50	0.5	5	20	120	SmartmA 50-650mA NI=11.57 Avg mA 360	0	Std	0	11.41	251.58	195	32	Localization scan for CT interventional scan
			SmartView SEG	LargeBody	1i	0.8	5	5	120	80	0	Std Segment	0	845.04	422.52	0	32	CT Fluoro scan for interventional scan

L-Spine

Table 14-7

[Interval]* - For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
27.1	L-Spine 3 Level Axial Group1		Axial	LargeBody	8i	1	2.5	20	120	225	30	Std	40	19.08	38.15	17.5	32	Axial scan mode, Evaluation of soft tissue and boney structures lumbar spine
	Group2		Axial	LargeBody	8i	1	2.5	20	120	225	30	Std	40	19.08	38.15	17.5	32	Axial scan mode, Evaluation of soft tissue and boney structures lumbar spine
	Group3		Axial	LargeBody	8i	1	2.5	20	140	250	30	Std	40	30.03	60.06	17.5	32	Axial scan mode, Evaluation of soft tissue and boney structures lumbar spine
27.3	L-Spine Survey Helical 2.5mm 0.6s. SmartmA		Helical	LargeBody	0.516:1 20.625	0.6	2.5	40	140	SmartmA 100-600mA NI=14 Avg mA 380	30	Std Plus	40	49.64	1040.25	175	32	Helical scan mode 2.5mm (0.6sec), Limited evaluation of lumbar spine
27.4	L-Spine Survey Helical 2.5mm 0.8s. SmartmA		Helical	LargeBody	0.516:1 20.625	0.8	2.5	40	140	SmartmA 100-600mA NI=14 Avg mA 365	30	Std Plus	40	63.57	1332.02	175	32	Helical scan mode 2.5mm (0.8sec), Limited evaluation of lumbar spine
27.6	L-Spine Helical 0.625mm + DMPR/ SmartmA		Helical	LargeBody	0.516:1 20.625	0.6	0.625	40	140	SmartmA 100-600mA NI=20 Avg mA 425	30	Std Plus IQE	40	55.52	1163.43	175	32	Helical scan mode 0.625mm with DMPR and SmartmA, Evaluation of soft tissue and boney structures lumbar spine
27.7	L-Spine Helical 1.25mm + DMPR/ SmartmA - Large Patient	DMPR	Helical	LargeBody	0.516:1 20.625	1	1.25	40	140	SmartmA 100-600mA NI=18 Avg mA 420	30	Std Plus	40	91.44	1915.61	175	32	Helical scan mode 1.25mm with DMPR and SmartmA, Evaluation of soft tissue and boney structures lumbar spine

Pelvis

Table 14-8
[Interval]* - For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.
For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
28.1	Pelvis for Fracture + DMPR/ SmartmA	DMPR	Helical	LargeBody	0.984:1 39.375	0.8	1.25 [0.625]	40	120	SmartmA 100-700mA NI=20 Avg mA 300	40	Bone Plus IQE	40	19.32	258.70	101.25	32	Helical scan with DMPR and SmartmA, Evaluation of pelvis bony structures for fracture

Lower Extremity

Table 14-9

[Interval]* - For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
29.1	Lower Extremity Survey 2.5mm		Helical	LargeBody	0.516:1 20.625	0.5	2.5	40	120	175	30	Std Plus	30	13.45	126.94	60	32	Helical scan mode 2.5mm, Limited evaluation of lower extremity for abnormalities
29.2	Ankle 0.625 mm Single Joint		Helical	MediumBody	0.531:1 10.625	0.5	0.625	20	120	175	30	Std Plus IQE	40	14.50	75.64	35	32	Helical scan mode 0.625mm, Evaluation of soft tissue and bony anatomy of the ankle area
29.3	Ankle 0.625 mm Both Joints		Helical	LargeBody	0.531:1 10.625	0.5	0.625	20	120	175	30	Std Plus IQE	30	13.96	72.82	35	32	Helical scan mode 0.625mm, Evaluation of soft tissue and bony anatomy of the ankle area
29.4	Knee 0.625 mm Both Joints		Helical	LargeBody	0.531:1 10.625	0.6	0.625	20	120	Auto mA 100-335mA NI=20 Avg mA 190	30	Std Plus IQE	30	18.19	94.84	35	32	Helical scan mode 0.625mm, Evaluation of soft tissue and bony anatomy of both knees

Reference Pediatric Protocols - Installed Options : ASiR, VCT85kW and VCT Hi Power

Head

Table 15-1

[Interval]* - For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.
For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy-cm) ASiR	Scan Length (mm)	Phantom (cm)	Description
31.1	PED HEAD TO 18 Months		Axial	Ped Head	4i	0.5	5	20	120	170	30	Std	30	14.47	173.68	115	16	Routine head (0.5s) for infant (up to 18 months) 5mm
31.2	PED HEAD 18 MOS TO 5YRS		Axial	SmallHead	4i	0.5	5	20	120	235	30	Std	30	20.01	240.08	115	16	Routine head (0.5s) for children (18 months to 5 years) 5mm
31.3	PED HEAD TO 5YRS TO 18YRS Group1		Axial	SmallHead	8i	1	2.5	20	120	170	30	Std	30	28.95	115.79	37.5	16	Routine head (1.0s) for children (5 years to 18 years) 2.5mm
	Group2		Axial	SmallHead	4i	1	5	20	120	125	30	Std	30	21.28	170.27	75	16	Routine head (1.0s) for children (5 years to 18 years) 5mm

Orbit

Table 15-2

[Interval]* - For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy-cm) ASiR	Scan Length (mm)	Phantom (cm)	Description
32.1	Sinus Supine Helical + DMPR	DMPR	Helical	SmallHead	0.531:1 10.625	0.4	0.625	20	120	65	20	Bone Plus IQE	30	8.33	76.29	74.375	16	Routine Helical scan mode with DMPR, Evaluation of sinus area for abnormality in pediatric patients
32.3	Orbits Helical	DMPR	Helical	SmallHead	0.531:1 10.625	0.4	0.625	20	120	145	20	Std Plus IQE	30	18.59	146.96	61.875	16	Helical scan mode, Evaluation of orbital and face structures for abnormality in pediatric patients
32.6	Helical IAC 0.625mm Axial mode - Use Reformat for Coronal		Helical	SmallHead	0.531:1 10.625	0.4	0.625	20	120	130	20	Std Plus IQE	30	16.67	93.21	38.75	16	Helical scan mode, Evaluation of inner ear structures for abnormality in pediatric patients

Chest

Table 15-3

[Interval]* - For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy-cm) ASiR	Scan Length (mm)	Phantom (cm)	Description
35.1.1	RC 6.0-7.5 kg (13.2-16.5 lbs)		Helical	PedBody	1.375:1 27.5	0.4	3.75	20	100	55	30	Std Plus	30	0.76	7.62	75	32	Weight and height based routine chest protocol
35.1.3	SnapShot Segment 30-74 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	85	30	Std Plus	30	0.59	5.80	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	Std	0	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
			Cardiac SEG	CardiacSmall	0.180:1 7.2	0.4	0.625	40	80	ECG Mod 60-195mA Phase 40-80 AvgmA195	30	Std SS Seg	40	8.88	97.72	75	32	Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
35.1.4	SnapShot Burst 75-104 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	85	30	Std Plus	30	0.59	5.80	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	Std	0	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
			Cardiac SSB	CardiacSmall	0.180:1 7.2	0.4	0.625	40	80	ECG Mod 60-195mA Phase 40-80 AvgmA195	30	Std SSB	40	8.88	97.72	75	32	Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
35.1.5	SnapShot Burst+ >105 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	85	30	Std Plus	30	0.59	5.80	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	Std	0	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
			Cardiac SSBP	CardiacSmall	0.180:1 7.2	0.4	0.625	40	80	ECG Mod 60-195mA Phase 35-85 AvgmA195	30	Std SSB	40	9.50	104.47	75	32	Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
35.1.7	Non ECG Gated		Helical	PedBody	0.984:1 39.375	0.4	0.625	40	80	SmartmA 50-200mA NI=18 AvgmA140	30	Std Plus	30	1.36	14.70	75	32	Weight and height based non ECG gated protocol for cardiac with SmartmA
35.2.1	RC 7.5-9.5 kg (16.5-20.9lbs)		Helical	PedBody	1.375:1 27.5	0.4	3.75	20	100	85	30	Std Plus	30	1.18	11.77	75	32	Weight and height based routine chest protocol
35.2.3	SnapShot Segment 30-74 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	85	30	Std Plus	30	0.59	5.80	50	32	Weight and height based protocol for localization

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy-cm) ASiR	Scan Length (mm)	Phantom (cm)	Description
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	Std	0	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
			Cardiac SEG	CardiacSmall	0.180:1 7.2	0.4	0.625	40	80	ECG Mod 60-210mA Phase 40-80 AvgmA210	30	Std SS Seg	40	9.54	104.90	75	32	Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
35.2.4	SnapShot Burst 75-104 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	85	30	Std Plus	30	0.59	5.80	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	Std	0	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
			Cardiac SSB	CardiacSmall	0.180:1 7.2	0.4	0.625	40	80	ECG Mod 60-210mA Phase 40-80 AvgmA210	30	Std SSB	40	9.54	104.90	75	32	Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
35.2.5	SnapShot Burst+ >105 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	85	30	Std Plus	30	0.59	5.80	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	Std	0	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
			Cardiac SSBP	CardiacSmall	0.180:1 7.2	0.4	0.625	40	80	ECG Mod 60-210mA Phase 35-85 AvgmA210	30	Std SSB	40	10.22	112.39	75	32	Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
35.2.7	Non ECG Gated		Helical	PedBody	0.984:1 39.375	0.4	0.625	40	80	SmartmA 50-220mA NI=18 AvgmA155	30	Std Plus	30	1.51	16.28	75	32	Weight and height based non ECG gated protocol for cardiac with SmartmA
35.3.1	RC 9.5-11.5 kg (20.9-25.4lbs)		Helical	SmallBody	1.375:1 27.5	0.4	3.75	20	120	70	30	Std Plus	30	1.52	15.77	78.75	32	Weight and height based routine chest protocol
35.3.3	SnapShot Segment 30-74 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	85	30	Std Plus	30	0.59	5.80	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	Std	0	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
			Cardiac SEG	CardiacSmall	0.180:1 7.2	0.4	0.625	40	80	ECG Mod 70-225mA Phase 40-80 AvgmA225	30	Std SS Seg	40	10.26	112.87	75	32	Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
35.3.4	SnapShot Burst 75-104 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	85	30	Std Plus	30	0.59	5.80	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	Std	0	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy-cm) ASiR	Scan Length (mm)	Phantom (cm)	Description
			Cardiac SSB	CardiacSmall	0.180:1 7.2	0.4	0.625	40	80	ECG Mod 70-225mA Phase 40-80 AvgmA225	30	Std SSB	40	10.26	112.87	75	32	Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
35.3.5	SnapShot Burst+ >105 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	85	30	Std Plus	30	0.59	5.80	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	Std	0	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
			Cardiac SSBP	CardiacSmall	0.180:1 7.2	0.4	0.625	40	80	ECG Mod 70-225mA Phase 35-85 AvgmA225	30	Std SSB	40	10.96	120.61	75	32	Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
35.3.7	Non ECG Gated		Helical	PedBody	0.984:1 39.375	0.4	0.625	40	80	SmartmA 50-240mA NI=18 AvgmA170	30	Std Plus	30	1.65	17.85	75	32	Weight and height based non ECG gated protocol for cardiac with SmartmA
35.4.1	RC 11.5-14.5kg (25.4-32.0lbs)		Helical	SmallBody	1.375:1 27.5	0.4	5	20	120	65	30	Std Plus	30	1.41	16.23	90	32	Weight and height based routine chest protocol
35.4.3	SnapShot Segment 30-74 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	85	30	Std Plus	30	0.59	5.80	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	Std	0	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
			Cardiac SEG	CardiacSmall	0.180:1 7.2	0.4	0.625	40	80	ECG Mod 70-240mA Phase 40-80 AvgmA240	30	Std SS Seg	40	10.91	120.04	75	32	Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
35.4.4	SnapShot Burst 75-104 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	85	30	Std Plus	30	0.59	5.80	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	Std	0	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
			Cardiac SSB	CardiacSmall	0.180:1 7.2	0.4	0.625	40	80	ECG Mod 70-240mA Phase 40-80 AvgmA240	30	Std SSB	40	10.91	120.04	75	32	Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
35.4.5	SnapShot Burst+ >105 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	85	30	Std Plus	30	0.59	5.80	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	Std	0	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
			Cardiac SSBP	CardiacSmall	0.180:1 7.2	0.4	0.625	40	80	ECG Mod 70-240mA Phase 35-85 AvgmA240	30	Std SSB	40	11.68	128.53	75	32	Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy-cm) ASiR	Scan Length (mm)	Phantom (cm)	Description
35.4.7	Non ECG Gated		Helical	PedBody	0.984:1 39.375	0.4	0.625	40	80	SmartmA 50-280mA NI=18 AvgmA195	30	Std Plus	30	1.90	20.48	75	32	Weight and height based non ECG gated protocol for cardiac with SmartmA
35.5.1	RC 14.5-18.5kg (32.0-40.8lbs)		Helical	SmallBody	1.375:1 27.5	0.4	5	20	120	70	30	Std Plus	30	1.52	17.48	90	32	Weight and height based routine chest protocol
35.5.3	SnapShot Segment 30-74 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	85	30	Std Plus	30	0.59	5.80	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	Std	0	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
			Cardiac SEG	CardiacSmall	0.180:1 7.2	0.4	0.625	40	80	ECG Mod 80-265mA Phase 40-80 AvgmA265	30	Std SS Seg	40	12.07	132.78	75	32	Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
35.5.4	SnapShot Burst 75-104 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	85	30	Std Plus	30	0.59	5.80	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	Std	0	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
			Cardiac SSB	CardiacSmall	0.180:1 7.2	0.4	0.625	40	80	ECG Mod 80-265mA Phase 40-80 AvgmA265	30	Std SSB	40	12.07	132.78	75	32	Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
35.5.5	SnapShot Burst+ >105 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	85	30	Std Plus	30	0.59	5.80	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	Std	0	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
			Cardiac SSBP	CardiacSmall	0.180:1 7.2	0.4	0.625	40	80	ECG Mod 80-265mA Phase 35-85 AvgmA265	30	Std SSB	40	12.91	142.02	75	32	Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
35.5.7	Non ECG Gated		Helical	PedBody	0.984:1 39.375	0.4	0.625	40	80	SmartmA 50-300mA NI=18 AvgmA210	30	Std Plus	30	2.04	22.05	75	32	Weight and height based non ECG gated protocol for cardiac with SmartmA
35.6.1	RC 18.5-22.5kg (40.8-49.6lbs)		Helical	SmallBody	1.375:1 27.5	0.4	5	20	120	75	30	Std Plus	30	1.63	18.73	90	32	Weight and height based routine chest protocol
35.6.3	SnapShot Segment 30-74 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	85	30	Std Plus	30	0.59	5.80	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	Std	0	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy-cm) ASiR	Scan Length (mm)	Phantom (cm)	Description
			Cardiac SEG	CardiacSmall	0.180:1 7.2	0.4	0.625	40	80	ECG Mod 80-280mA Phase 40-80 AvgmA280	30	Std SS Seg	40	12.72	139.92	75	32	Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
35.6.4	SnapShot Burst 75-104 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	85	30	Std Plus	30	0.59	5.80	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	Std	0	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
			Cardiac SSB	CardiacSmall	0.180:1 7.2	0.4	0.625	40	80	ECG Mod 80-280mA Phase 40-80 AvgmA280	30	Std SSB	40	12.72	139.92	75	32	Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
35.6.5	SnapShot Burst+ >105 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	85	30	Std Plus	30	0.59	5.80	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	Std	0	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
			Cardiac SSBP	CardiacSmall	0.180:1 7.2	0.4	0.625	40	80	ECG Mod 80-280mA Phase 35-85 AvgmA280	30	Std SSB	40	13.63	149.91	75	32	Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
35.6.7	Non ECG Gated		Helical	PedBody	0.984:1 39.375	0.4	0.625	40	80	SmartmA 100-320mA NI=18 AvgmA225	30	Std Plus	30	2.19	23.63	75	32	Weight and height based non ECG gated protocol for cardiac with SmartmA
35.7.1	RC 22.5-31.5kg (49.6-69.5lbs)		Helical	LargeBody	1.375:1 55.0	0.4	5	40	120	75	30	Std Plus	30	1.73	22.98	85	32	Weight and height based routine chest protocol
35.7.3	SnapShot Segment 30-74 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	85	30	Std Plus	30	0.59	5.80	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	Std	0	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
			Cardiac SEG	CardiacSmall	0.180:1 7.2	0.4	0.625	40	80	ECG Mod 85-295mA Phase 40-80 AvgmA295	30	Std SS Seg	40	13.41	147.51	75	32	Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
35.7.4	SnapShot Burst 75-104 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	85	30	Std Plus	30	0.59	5.80	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	Std	0	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
			Cardiac SSB	CardiacSmall	0.180:1 7.2	0.4	0.625	40	80	ECG Mod 85-295mA Phase 40-80 AvgmA295	30	Std SSB	40	13.41	147.51	75	32	Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy-cm) ASiR	DLP (mGy-cm) ASiR	Scan Length (mm)	Phantom (cm)	Description
35.7.5	SnapShot Burst+ >105 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	85	30	Std Plus	30	0.59	5.80	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	Std	0	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
			Cardiac SSBP	CardiacSmall	0.180:1 7.2	0.4	0.625	40	80	ECG Mod 85-295mA Phase 35-85 AvgmA295	30	Std SSB	40	14.36	158.00	75	32	Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
35.7.7	Non ECG Gated		Helical	PedBody	0.984:1 39.375	0.4	0.625	40	80	SmartmA 100-360mA NI=18 AvgmA250	30	Std Plus	30	2.43	26.25	75	32	Weight and height based non ECG gated protocol for cardiac with SmartmA
35.8.1	RC 31.5-40.5kg (69.5-89.3lbs)		Helical	LargeBody	1.375:1 55.0	0.4	5	40	120	85	30	Std Plus	30	1.96	32.90	120	32	Weight and height based routine chest protocol
35.8.3	SnapShot Segment 30-74 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	100	85	30	Std Plus	30	1.10	10.76	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	100	50	0	Std	0	39.50	19.75	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
			Cardiac SEG	CardiacSmall	0.180:1 7.2	0.4	0.625	40	100	ECG Mod 80-265mA Phase 40-80 AvgmA265	30	Std SS Seg	30	22.40	246.43	75	32	Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
35.8.4	SnapShot Burst 75-104 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	100	85	30	Std Plus	30	1.10	10.76	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	100	50	0	Std	0	39.50	19.75	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
			Cardiac SSB	CardiacSmall	0.180:1 7.2	0.4	0.625	40	100	ECG Mod 80-265mA Phase 40-80 AvgmA265	30	Std SSB	30	22.40	246.43	75	32	Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
35.8.5	SnapShot Burst+ >105 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	100	85	30	Std Plus	30	1.10	10.76	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	100	50	0	Std	0	39.50	19.75	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
			Cardiac SSBP	CardiacSmall	0.180:1 7.2	0.4	0.625	40	100	ECG Mod 80-265mA Phase 35-85 AvgmA265	30	Std SSB	40	23.96	263.58	75	32	Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
35.8.7	Non ECG Gated		Helical	SmallBody	0.984:1 39.375	0.4	0.625	40	100	SmartmA 100-400mA NI=16 AvgmA280	30	Std Plus	30	5.06	54.57	75	32	Weight and height based non ECG gated protocol for cardiac with SmartmA

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy-cm) ASiR	Scan Length (mm)	Phantom (cm)	Description
35.9.1	RC 40.5-55.0kg(89.3-121.3lbs)		Helical	LargeBody	1.375:1 55.0	0.4	5	40	120	90	30	Std Plus	30	2.07	41.06	150	32	Weight and height based routine chest protocol
35.9.3	SnapShot Segment 30-74 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	100	85	30	Std Plus	30	1.10	10.76	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	100	50	0	Std	0	39.50	19.75	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
			Cardiac SEG	CardiacSmall	0.180:1 7.2	0.4	0.625	40	100	ECG Mod 85-295mA Phase 40-80 AvgmA295	30	Std SS Seg	30	24.89	273.78	75	32	Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
35.9.4	SnapShot Burst 75-104 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	100	85	30	Std Plus	30	1.10	10.76	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	100	50	0	Std	0	39.50	19.75	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
			Cardiac SSB	CardiacSmall	0.180:1 7.2	0.4	0.625	40	100	ECG Mod 85-295mA Phase 40-80 AvgmA295	30	Std SSB	30	24.89	273.78	75	32	Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
35.9.5	SnapShot Burst+ >105 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	100	85	30	Std Plus	30	1.10	10.76	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	100	50	0	Std	0	39.50	19.75	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
			Cardiac SSBP	CardiacSmall	0.180:1 7.2	0.4	0.625	40	100	ECG Mod 85-295mA Phase 35-85 AvgmA295	30	Std SSB	40	26.66	293.24	75	32	Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
35.9.7	Non ECG Gated		Helical	SmallBody	0.984:1 39.375	0.4	0.625	40	100	SmartmA 100-440mA NI=16 AvgmA310	30	Std Plus	30	5.60	60.42	75	32	Weight and height based non ECG gated protocol for cardiac with SmartmA

Abdomen

Table 15-4

[Interval]* - For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy-cm) ASiR	Scan Length (mm)	Phantom (cm)	Description
36.1.1	Abd 6.0-7.5kg (13.2-16.5lbs)		Helical	PedBody	1.375:1 27.5	0.4	3.75	20	120	70	30	Std Plus	30	1.52	15.77	78.75	32	Weight and height based routine abdomen protocol
36.2.1	Abd 7.5-9.5kg (16.5-20.9lbs)		Helical	PedBody	1.375:1 27.5	0.4	3.75	20	120	110	30	Std Plus	30	2.39	27.47	90	32	Weight and height based routine abdomen protocol
36.3.1	Abd 9.5-11.5kg (20.9-25.4lbs)		Helical	PedBody	1.375:1 27.5	0.4	3.75	20	120	125	30	Std Plus	30	2.71	34.26	101.25	32	Weight and height based routine abdomen protocol
36.4.1	Abd 11.5-14.5kg (25.4-32.0lbs)		Helical	SmallBody	1.375:1 27.5	0.4	5	20	120	85	30	Std Plus	30	1.84	24.91	110	32	Weight and height based routine abdomen protocol
36.5.1	Abd 14.5-18.5kg (32.0-40.8lbs)		Helical	SmallBody	1.375:1 27.5	0.4	5	20	120	95	30	Std Plus	30	2.06	27.84	110	32	Weight and height based routine abdomen protocol
36.6.1	Abd 18.5-22.5kg (40.8-49.6lbs)		Helical	Medium Body	1.375:1 27.5	0.4	5	20	120	85	30	Std Plus	30	2.18	33.77	130	32	Weight and height based routine abdomen protocol
36.7.1	Abd 22.5-31.5kg (49.6-69.5lbs)		Helical	LargeBody	1.375:1 55.0	0.4	5	40	120	105	30	Std Plus	30	2.42	47.91	150	32	Weight and height based routine abdomen protocol
36.8.1	Abd 31.5-40.5kg (69.5-89.3lbs)		Helical	LargeBody	1.375:1 55.0	0.4	5	40	120	120	30	Std Plus	30	2.77	54.75	150	32	Weight and height based routine abdomen protocol
36.9.1	Abd 40.5-55.0kg (89.3-121.3lbs)		Helical	LargeBody	1.375:1 55.0	0.4	5	40	120	140	30	Std Plus	30	3.23	80.01	200	32	Weight and height based routine abdomen protocol

Miscellaneous

Table 15-5

[Interval]* - For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy-cm) ASiR	Scan Length (mm)	Phantom (cm)	Description
40.1	Pink Zone Routine 0-20 lbs/0-9 kg		Helical	PedBody	1.375:1 55.0	0.5	5	40	80	SmartmA 65-130mA NI=5 Avg mA 70	0	Std	0	0.61	8.70	95	32	Pediatric and Small patient Routine procedure based protocols
40.2	Pink Zone Routine 21-60 lbs/9.1-27.2 kg		Helical	PedBody	1.375:1 55.0	0.5	5	40	100	SmartmA 80-160mA NI=7 Avg mA 70	0	Std	0	1.13	27.47	195	32	Pediatric and Small patient Routine procedure based protocols
40.3	Pink Zone Routine 61-100 lbs/27.3-45.4 kg		Helical	Medium Body	1.375:1 55.0	0.5	5	40	120	SmartmA 95-190mA NI=10 Avg mA 70	0	Std	0	2.09	71.81	295	32	Pediatric and Small patient Routine procedure based protocols
40.4	Pink Zone Routine 101-200 lbs/45.4-90.7 kg		Helical	LargeBody	1.375:1 55.0	0.5	5	40	120	SmartmA 110-220mA NI=12 Avg mA 70	0	Std	0	2.02	89.30	395	32	Pediatric and Small patient Routine procedure based protocols
40.5	Pink Zone Routine >200 lbs/>90.8 kg		Helical	LargeBody	1.375:1 55.0	0.5	5	40	120	SmartmA 125-300mA NI=15 Avg mA 70	0	Std	0	2.02	89.30	395	32	Pediatric and Small patient Routine procedure based protocols
40.7	Green Zone Lower Dose 0-20 lbs/0-9 kg		Helical	PedBody	1.375:1 55.0	0.5	5	40	80	SmartmA 50-100mA NI=7 Avg mA 70	0	Std	0	0.61	8.70	95	32	Pediatric and Small patient Lower Dose procedure based protocols
40.8	Green Zone Lower Dose 21-60 lbs/9.1-27.2 kg		Helical	PedBody	1.375:1 55.0	0.5	5	40	100	SmartmA 60-120mA NI=9 Avg mA 70	0	Std	0	1.13	27.47	195	32	Pediatric and Small patient Lower Dose procedure based protocols
40.9	Green Zone Lower Dose 61-100 lbs/27.3-45.4 kg		Helical	Medium Body	1.375:1 55.0	0.5	5	40	120	SmartmA 70-140mA NI=11 Avg mA 70	0	Std	0	2.09	71.81	295	32	Pediatric and Small patient Lower Dose procedure based protocols
40.10	Green Zone Lower Dose 101-200 lbs/45.4-90.7 kg		Helical	LargeBody	1.375:1 55.0	0.5	5	40	120	SmartmA 80-160mA NI=13 Avg mA 70	0	Std	0	2.02	89.30	395	32	Pediatric and Small patient Lower Dose procedure based protocols
40.11	Green Zone Lower Dose >200 lbs/>90.8 kg		Helical	LargeBody	1.375:1 55.0	0.5	5	40	120	SmartmA 90-240mA NI=16 Avg mA 70	0	Std	0	2.02	89.30	395	32	Pediatric and Small patient Lower Dose procedure based protocols
40.13	Grey Zone CT Angio 0-20 lbs/0-9 kg		Helical	PedBody	1.375:1 55.0	0.5	2.5	40	80	SmartmA 100-200mA NI=5 Avg mA 140	0	Std	0	1.22	17.71	97.5	32	Pediatric and Small patient Angio procedure based protocols

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy-cm) ASiR	Scan Length (mm)	Phantom (cm)	Description
40.14	Grey Zone CT Angio 21-60 lbs/9.1-27.2 kg		Helical	SmallBody	1.375:1 55.0	0.5	2.5	40	100	SmartmA 120-240mA NI=9.9 Avg mA 140	0	Std	0	2.26	55.50	197.5	32	Pediatric and Small patient Angio procedure based protocols
40.15	Grey Zone CT Angio 61-100 lbs/ 27.3-45.4 kg		Helical	LargeBody	1.375:1 55.0	0.5	2.5	40	120	SmartmA 120-240mA NI=10 Avg mA 140	0	Std	0	4.03	139.28	297.5	32	Pediatric and Small patient Angio procedure based protocols
40.16	Grey Zone CT Angio 101-200 lbs/ 45.4-90.7 kg		Helical	LargeBody	1.375:1 55.0	0.5	2.5	40	120	SmartmA 120-240mA NI=12 Avg mA 140	0	Std	0	4.03	179.62	397.5	32	Pediatric and Small patient Angio procedure based protocols
40.17	Grey Zone CT Angio >200 lbs/ >90.8 kg		Helical	LargeBody	1.375:1 55.0	0.5	2.5	40	120	SmartmA 120-300mA NI=15 Avg mA 140	0	Std	0	4.03	179.62	397.5	32	Pediatric and Small patient Angio procedure based protocols

Reference Adult Protocols - Installed

Options : ASiR

Head

Table 16-1

[Interval]* - For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (s)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (mm)	Phantom (cm)	Description
21.1	Routine Head 1s Group1		Axial	Head	4i	1	5	20	140	120	40	Std	40	32.00	192.01	55	16	Routine head protocol for evaluation of the brain for abnormalities.
	Group2		Axial	Head	4i	1	5	20	120	190	40	Std	40	36.47	291.78	75	16	
21.2	Routine Head 2s Group1		Axial	Head	4i	2	5	20	140	60	40	Std	40	32.00	192.01	55	16	Routine head protocol for evaluation of the brain for abnormalities.
	Group2		Axial	Head	4i	2	5	20	120	95	40	Std	40	36.47	291.78	75	16	
21.3	Trauma Head 2.5mm		Axial	Head	8i	0.5	2.5	20	140	310	40	Std	40	41.34	578.70	137.5	16	Emergency head protocol for evaluation of the brain and cranium for abnormalities.
21.4	Routine Head 0.5s Group1		Axial	Head	4i	0.5	5	20	140	240	40	Std	40	32.00	192.01	55	16	Routine head protocol for evaluation of the brain for abnormalities.
	Group2		Axial	Head	4i	0.5	5	20	120	360	40	Std	40	35.52	284.17	75	16	
21.5	Circle of Willis 0.4s	MIROI	Axial	Head	1i	1	5 [0]	5	120	50	0	Std	0	125.58	62.79	0	16	Test bolus for CT angiography of the head for evaluation of carotid and cerebral vasculature.
			Helical	Head	0.531:1 10.625	0.4	0.625	20	120	180	40	Std Plus IQE	50	26.02	252.82	80	16	CT angiography of the head for evaluation of carotid and cerebral vasculature.
21.6	CoW/ Carotid 0.625mm 0.4s.Smart mA	MIROI	Axial	Head	1i	1	5 [0]	5	120	80	0	Std	0	180.84	90.42	0	16	Test bolus for CT angiography of the head for evaluation of carotid and cerebral vasculature.
			Helical	Head	0.531:1 10.625	0.4	0.625	20	120	SmartmA 100-600mA NI=8 Avg mA 415	30	Std Plus IQE	40	61.66	1155.86	170	16	CT angiography of the head for evaluation of carotid and cerebral vasculature.
21.8	Helical Head	DMPR	Helical	Head	0.531:1 10.625	0.5	0.625	20	120	180	40	Std Plus IQE	40	32.52	527.31	145	16	Helical scan mode for evaluation of the brain for cerebral abnormality.
21.10	VHS Head 4D CTA	MIROI	Axial	Head	1i	1	5 [0]	5	120	50	0	Std	0	125.58	62.79	0	16	Test bolus for CT angiography of the head for evaluation of carotid and cerebral vasculature.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (s)	Slice Thickness (mm) [Interval] * (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (mm)	Phantom (cm)	Description
			Helical-S	Head	0.984:1 39.375 18 pass	0.4	5 [10]	40	120	150	50	Std Plus	50	248.10	3101.19	110	16	CT angiography of the head using Volume Helical scan mode for evaluation of cerebral vasculature.
21.11	VHS Perfusion 350-370 Strength Contrast		Axial	Head	4i	1	5	20	120	200	40	Std	40	38.39	537.48	135	16	Non-Enhance Brain, evaluation for hemorrhage or infarction.
		Perfusion	Helical-S/P	Head	0.984:1 39.375 28 pass	0.4	5	40	80	200	0	Std	50	186.02	2325.25	110	16	CT Perfusion using Volume Helical scan mode for evaluation of cerebral perfusion over time.
			Axial	Head	4i	1	5	20	120	200	40	Std	40	38.39	537.48	135	16	Post Contrast Brain, Evaluation for residual tissue enhancement.
21.12	CT Perfusion 300-320 Strength Contrast		Axial	Head	4i	1	5	20	120	200	40	Std	40	38.39	537.48	135	16	Non-Enhance Brain, evaluation for hemorrhage or infarction.
		Perfusion	Cine	Head	8i	1	5 [0]	40	80	200	0	Std	0	652.82	2611.27	35	16	CT Perfusion using Cine scan mode for evaluation of cerebral perfusion over time.
			Axial	Head	4i	1	5	20	120	200	40	Std	40	38.39	537.48	135	16	Non-Enhance Brain, evaluation for hemorrhage or infarction.
21.13	CT Perfusion 350-370 Strength Contrast		Axial	Head	4i	1	5	20	120	200	40	Std	40	38.39	537.48	135	16	Non-Enhance Brain, evaluation for hemorrhage or infarction.
		Perfusion	Cine	Head	8i	1	5 [0]	40	80	200	0	Std	0	587.55	2350.19	35	16	CT Perfusion using Cine scan mode for evaluation of cerebral perfusion over time.
			Axial	Head	4i	1	5	20	120	200	40	Std	40	38.39	537.48	135	16	Post Contrast Brain, Evaluation for residual tissue enhancement.
21.14	CT Perfusion Brain Tumor - Dual Phase		Axial	Head	4i	1	5	20	120	200	40	Std	40	38.39	537.48	135	16	Non-Enhance Brain, evaluation for hemorrhage or infarction.
	Group 1	Perfusion	Cine	Head	8i	1	5 [0]	40	80	200	0	Std	0	587.55	2350.19	35	16	CT Perfusion using Cine scan mode for evaluation of cerebral perfusion over time.
	Group 2	Perfusion	Axial	Head	8i	1	5 [0]	40	80	200	0	Std	0	130.54	522.15	35	16	Delayed phase CT Perfusion using Axial scan mode for evaluation of Brain tumor permeability.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (s)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (mm)	Phantom (cm)	Description
			Axial	Head	4i	1	5	20	120	200	40	Std	40	38.39	537.48	135	16	Post Contrast Brain, Evaluation for residual tissue enhancement.
21.15	CT Perfusion 350-370 Strength Contrast (Axial)		Axial	Head	4i	1	5	20	120	200	40	Std	40	38.39	537.48	135	16	Non-Enhance Brain, evaluation for hemorrhage or infarction.
		Perfusion	Axial	Head	8i	1	5 [0]	40	80	200	0	Std	0	287.18	1148.73	35	16	CT Perfusion using Axial scan mode for evaluation of cerebral perfusion over time.
			Axial	Head	4i	1	5	20	120	200	40	Std	40	38.39	537.48	135	16	Post Contrast Brain, Evaluation for residual tissue enhancement.
21.16	CT Perfusion 350-370 Strength Contrast (Axial Shuttle Mode)		Axial	Head	4i	1	5	20	120	200	40	Std	40	38.39	537.48	135	16	Non-Enhance Brain, evaluation for hemorrhage or infarction.
		Perfusion	Axial-S	Head	8i 17 pass	0.4	5	40	80	500	0	Std	0	221.91	1775.31	75	16	CT Perfusion using Volume Shuttle (Axial) scan mode for evaluation of cerebral perfusion over time.
			Axial	Head	4i	1	5	20	120	200	40	Std	40	38.39	537.48	135	16	Post Contrast Brain, Evaluation for residual tissue enhancement.

Orbit

Table 16-2

[Interval]* - For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy-cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
22.1	Sinus Supine Helical + DMPPR	DMPPR	Helical	Head	0.531:1 10.625	0.4	0.625	20	120	85	30	Bone Plus IQE	50	12.29	112.48	74.375	16	Routine Helical Sinus protocol, Evaluation for any abnormality
22.3	Orbits Helical + DMPPR	DMPPR	Helical	Head	0.531:1 10.625	0.5	0.625	20	120	155	30	Std Plus IQE	40	28.00	221.29	61.875	16	Routine Helical Orbit protocol, Evaluation for abnormality soft tissues and bone structures of the face.
22.6	Helical IAC 0.625mm Axial mode - Use Reformat for Coronal		Helical	Head	0.531:1 10.625	0.6	0.625	20	140	190	30	Std Plus IQE	40	57.23	319.79	38.75	16	Axial scan mode, Evaluation for inner ear abnormalities in Axial plane
22.8	Axial and Coronal IAC		Axial	Head	16i	1	1.25	20	140	195	30	Std	40	52.00	208.01	38.75	16	Helical scan mode, Evaluation for inner ear abnormalities in Axial plane
			Axial	Head	16i	1	1.25	20	140	235	30	Std	40	62.67	250.68	38.75	16	Helical scan mode, Evaluation for inner ear abnormalities in Coronal plane

Neck

Table 16-3

[Interval]* - For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
23.1	C-Spine C5-C7 Axial		Axial	SmallBody	8i	0.5	2.5	20	140	410	20	Std	40	22.29	312.12	137.5	32	C-Spine C5-C7 area Axial mode, Evaluation for soft tissue, disc and bone abnormalities.
23.2	C-Spine C1-C4 Axial		Axial	SmallBody	8i	1	2.5	20	120	190	20	Std	40	14.17	113.36	77.5	32	C-Spine C1-C4 area Axial mode, Evaluation for soft tissue, disc and bone abnormalities.
23.3	C-Spine Helical + DMPR	DMPR	Helical	SmallBody	0.516:1 20.625	0.8	0.625	40	120	270	20	Bone Plus IQE	40	29.20	421.46	110	32	C-Spine Helical mode with DMPR, Evaluation for soft tissue, disc and bone abnormalities.
23.4	C-Spine Helical + DMPR/ SmartmA	DMPR	Helical	SmallBody	0.516:1 20.625	0.8	0.625	40	120	SmartmA 100-600mA NI=11.34 Avg mA 250	20	Bone Plus IQE	40	27.04	390.24	110	32	C-Spine Helical mode with DMPR and SmartmA, Evaluation for soft tissue, disc and bone abnormalities.
23.8	Carotid 0.625mm 0.4s. 0.5:1 SmartmA/ SmartPrep		Helical	SmallBody	0.516:1 20.625	0.4	0.625	40	140	SmartmA 100-515mA NI=11 Avg mA 310	40	Std Plus IQE	50	23.78	486.15	170	32	Carotid CTA (pitch 0.5:1) with SmartmA and SmartPrep, Evaluation of carotid vasculature.
23.9	Carotid 0.625mm 0.4s. 0.9:1 SmartmA/ SmartPrep		Helical	SmallBody	0.984:1 39.375	0.4	0.625	40	120	SmartmA 80-600mA NI=11 Avg mA 360	40	Std Plus IQE	50	10.20	207.25	170	32	Carotid CTA (pitch 0.9:1) with SmartmA and SmartPrep, Evaluation of carotid vasculature.
23.10	Carotid/ CoW 0.625mm 0.4s. AutomA	MIROI	Axial	SmallBody	1i	1	5 [0]	5	120	80	0	Std	0	99.00	49.50	0	32	Test bolus for Carotid CTA (pitch 0.5:1) of the head for evaluation of carotid and cerebral vasculature.
			Helical	SmallBody	0.516:1 20.625	0.4	0.625	40	140	AutomA 80-515mA NI=11 Avg mA 310	40	Std Plus IQE	50	23.78	486.15	170	32	Carotid and Brain CTA (pitch 0.5:1) with AutomA, Evaluation of carotid and cerebral vasculature.
23.11	Carotid/ CoW 0.625mm 0.4s. Smart mA/ SmartPrep		Helical	SmallBody	0.969:1 19.375	0.4	0.625	20	140	SmartmA 100-515mA NI=11 Avg mA 310	40	Std Plus IQE	50	13.54	255.40	170	32	Carotid CTA (pitch 0.9:1) with SmartmA and SmartPrep, Evaluation of carotid and cerebral vasculature.
23.12	Soft Tissue Neck 2.5mm AutomA		Helical	SmallBody	0.984:1 39.375	0.5	2.5	40	120	AutomA 100-335mA NI=9.1 Avg mA 175	30	Std Plus	30	6.20	113.26	150	32	Neck with AutomA, Evaluation of soft tissue structures for abnormality

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
23.13	Soft Tissue Neck + DMPR/ SmartmA	DMPR	Helical	SmallBody	0.969:1 19.375	0.5	0.625	20	120	SmartmA 100-600mA NI=12.6 Avg mA 255	40	Std Plus IQE	50	9.82	165.46	150	32	Neck with DMPR and SmartmA, Evaluation of soft tissue structures for abnormality

Upper Extremity

Table 16-4

[Interval]* - For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
24.1	Shoulder 1.25mm + DMPR	DMPR	Helical	LargeBody	0.516:1 20.625	0.6	1.25 [0.625]	40	140	215	40	Std Plus IQE	40	28.09	377.41	100	32	Shoulder with DMPR, Evaluation of soft tissue and bone of shoulder area
24.2	Shoulder 1.25mm + DMPR/ SmartmA	DMPR	Helical	LargeBody	0.516:1 20.625	0.6	1.25 [0.625]	40	140	SmartmA 100-515mA NI=20 Avg mA 270	40	Std Plus IQE	40	35.27	473.96	100	32	Shoulder with DMPR and SmartmA, Evaluation of soft tissue and bone of shoulder area

Chest

Table 16-5

[Interval]* - For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
25.1	Routine Chest 0.4s. 5mm SmartmA		Helical	LargeBody	1.375:1 55.0	0.4	5	40	120	SmartmA 100-600mA NI=13 Avg mA 250	50	Std Plus	50	5.76	142.87	200	32	Chest with SmartmA, Evaluation of mediastinum and lungs fields
25.2	Routine Chest 0.4s.+ DMPR/ SmartmA	DMPR	Helical	LargeBody	0.984:1 39.375	0.4	0.625	40	120	SmartmA 100-600mA NI=22 Avg mA 300	50	Std Plus	50	9.66	224.97	200	32	Chest with SmartmA and DMPR, Evaluation of mediastinum and lungs fields
25.3	Chest with HiRes 0.4s. SmartmA		Helical	LargeBody	0.969:1 19.375	0.4	5	20	120	SmartmA 80-600mA NI=12 Avg mA 300	40	Chest Plus	40	10.50	229.59	200	32	Chest with SmartmA, Evaluation of mediastinum and lung fields with higher resolution images of the lungs
25.4	Hi Res Chest 0.5s. Axial		Axial	LargeBody	1i	0.5	1.25 [10]	1.25	120	190	50	Bone+	50	2.22	53.35	230	32	Axial scan mode, Evaluation of mediastinum and lung fields in high resolution mode
25.5	Chest Abd Pelvis 0.4/ 0.5s. SmartmA 5mm (2 groups) Group1		Helical	LargeBody	0.984:1 39.375	0.4	5	40	120	SmartmA 100-600mA NI=13 Avg mA 185	50	Std Plus	50	5.96	108.95	150	32	Multi group scan for Chest, Abdomen, Pelvis (0.4/0.5 sec) with SmartmA, Evaluation of chest and abdominal anatomy for abnormalities
	Group2		Helical	LargeBody	1.375:1 55.0	0.5	5	40	120	SmartmA 100-600mA NI=13 Avg mA 185	50	Std Plus	50	5.33	182.72	295	32	
25.6	Chest Abd Pelvis 0.5s + DMPR SmartPrep	DMPR	Helical	LargeBody	1.375:1 55.0	0.5	0.625	40	120	SmartmA 100-600mA NI=22 Avg mA 270	50	Std Plus	50	7.78	387.24	450	32	Helical scan mode (0.5sec) with SmartmA, Evaluation of chest and abdominal anatomy for abnormalities
25.7	Pulmonary Embolis 0.5s.	MIROI	Axial	LargeBody	1i	1	5 [0]	5	120	80	0	Std	0	93.89	46.95	0	32	Timing Bolus for Helical scan mode 1.25mm thickness and helical pitch of 0.9:1, evaluation of Chest for pulmonary embolism
			Helical	LargeBody	0.984:1 39.375	0.5	1.25	40	120	350	40	Std	40	14.09	300.25	180	32	Helical scan mode 1.25mm thickness and helical pitch of 0.9:1, Evaluation Chest for pulmonary embolism

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
25.8	Pulmonary Embolis 0.5s. SmartmA	MIROI	Axial	LargeBody	1i	1	5 [0]	5	120	80	0	Std	0	93.89	46.95	0	32	Timing Bolus for Helical scan mode 1.25mm thickness and helical pitch of 0.9:1 with SmartmA, evaluation of Chest for pulmonary embolism
			Helical	LargeBody	0.984:1 39.375	0.5	1.25	40	120	SmartmA 100-600mA NI=20 Avg mA 300	40	Std	40	12.07	256.93	180	32	Helical scan mode 1.25mm thickness and helical pitch of 0.9:1 with SmartmA, Evaluation Chest for pulmonary embolism
25.9	Aortic Dissection 0.625mm + DMPR/ SmartmA/ SmartPrep	DMPR	Helical	LargeBody	1.375:1 55.0	0.5	0.625	40	120	SmartmA 100-600mA NI=22 Avg mA 360	40	Std Plus	40	10.37	361.20	300	32	Helical scan mode 0.625mm thickness and helical pitch of 1.375:1 with SmartmA, SmartPrep and DMPR, Evaluation for aortic dissection
25.10	Aortic Dissection 1.25mm + DMPR/ SmartmA/ SmartPrep	DMPR	Helical	LargeBody	1.375:1 55.0	0.5	1.25 [0.8]	40	120	SmartmA 100-600mA NI=20 Avg mA 330	40	Std Plus	40	9.51	330.68	300	32	Helical scan mode 1.25mm thickness and helical pitch of 1.375:1 with SmartmA, SmartPrep and DMPR, Evaluation for aortic dissection
25.11	Lung Assessment	Advanced Lung Analysis	Helical	LargeBody	0.969:1 19.375	0.5	1.25	20	120	120	0	Bone	0	5.25	114.77	200	32	Helical scan mode for acquisition of data for Advanced Lung Analysis (ALA) software for lung nodule assessment
25.12	Gated Chest Cardiac Helical		Helical	LargeBody	1.375:1 55.0	0.4	5	40	120	60	0	Std Plus	0	1.38	13.55	50	32	Localizer scan for Helical Gated scan for evaluation of thoracic vasculature
		MIROI	Axial	LargeBody	1i	1	5 [0]	5	120	60	0	Std	0	98.59	49.29	0	32	Timing bolus scan for Helical Gated scan for evaluation of thoracic vasculature
			Cardiac SEG	CardiacLarge	0.180:1 7.2	0.4	0.625	40	120	ECG Mod 120-235mA Phase 70-80 Avg mA 235	40	Std SS Seg	40	30.28	1014.47	300	32	Helical Gated scan for evaluation of thoracic vasculature
25.13	SmartScore - Gated 0.4s	Smart Score	Cine	LargeBody	8i	0.4	2.5	20	120	375	0	Std Segment	0	9.15	91.51	97.5	32	Cine gated scan mode for acquisition of data for coronary artery calcification analysis
25.14	SnapShot Segment 30-74 BPM		Helical	LargeBody	1.375:1 55.0	0.4	5	40	120	60	0	Std Plus	0	1.38	13.55	50	32	Localizer scan for Retrospectively gated Cardiac Helical scan for evaluation of coronary arteries, intended for heart rates 30-74 bpm

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy-cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
		MIROI	Axial	LargeBody	1i	1	5 [0]	5	120	60	0	Std	0	98.59	49.29	0	32	Timing bolus scan for Retrospectively gated Cardiac Helical scan for evaluation of coronary arteries, intended for heart rates 30-74 bpm
			Cardiac SEG	CardiacLarge	0.180:1 7.2	0.4	0.625	40	120	ECG Mod 120-235mA Phase 70-80 Avg mA 235	40	Std SS Seg	40	31.29	422.36	100	32	Retrospectively gated Cardiac Helical scan for evaluation of coronary arteries, intended for heart rates 30-74 bpm
25.15	SnapShot Burst 75-113 BPM		Helical	LargeBody	1.375:1 55.0	0.4	5	40	120	60	0	Std Plus	0	1.38	13.55	50	32	Localizer scan for Retrospectively gated Cardiac Helical scan for evaluation of coronary arteries, intended for heart rates 75-113 bpm
		MIROI	Axial	LargeBody	1i	1	5 [0]	5	120	60	0	Std	0	98.59	49.29	0	32	Timing bolus scan for Retrospectively gated Cardiac Helical scan for evaluation of coronary arteries, intended for heart rates 75-113 bpm
			Cardiac SSB	CardiacLarge	0.180:1 7.2	0.4	0.625	40	120	ECG Mod 120-235mA Phase 70-80 Avg mA 235	40	Std SSB	40	31.29	422.36	100	32	Retrospectively gated Cardiac Helical scan for evaluation of coronary arteries, intended for heart rates 75-113 bpm
25.16	SnapShot Burst Plus 114+ BPM		Helical	LargeBody	1.375:1 55.0	0.4	5	40	120	60	0	Std Plus	0	1.38	13.55	50	32	Localizer scan for Retrospectively gated Cardiac Helical scan for evaluation of coronary arteries, intended for heart rates over 113 bpm
		MIROI	Axial	LargeBody	1i	1	5 [0]	5	120	60	0	Std	0	98.59	49.29	0	32	Timing bolus scan for Retrospectively gated Cardiac Helical scan for evaluation of coronary arteries, intended for heart rates over 113 bpm
			Cardiac SSBP	CardiacLarge	0.180:1 7.2	0.4	0.625	40	120	ECG Mod 120-235mA Phase 70-80 Avg mA 235	40	Std SSB	40	31.29	422.36	100	32	Retrospectively gated Cardiac Helical scan for evaluation of coronary arteries, intended for heart rates over 113 bpm
25.17	Advantage 4D		Helical	LargeBody	1.375:1 55.0	0.6	5	40	120	170	40	Std Plus	40	5.88	145.59	200	32	Localizer scan for Advantage 4D series
		Advantage 4D	Cine	LargeBody	8i	1	2.5	20	120	70	30	Std	30	35.67	570.67	157.5	32	Cine scan mode for acquisition data for Advantage 4D for retrospective gating
25.18	VHS Chest 4D CTA SmartPrep		Helical-S	LargeBody	1.375:1 55.0 11 pass	0.4	5 [10]	40	120	SmartmA 100-600mA NI=12 Avg mA 360	40	Std Plus	40	115.35	3633.55	300	32	Volume Helical scan mode with SmartPrep for time resolved CT angiography of chest vasculature

Abdomen

Table 16-6

[Interval]* - For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
26.1	Abdomen Pelvis 0.5s. 5mm SmartmA		Helical	LargeBody	1.375:1 55.0	0.5	5	40	120	SmartmA 50-600mA NI=11.57 Avg mA 250	50	Std Plus	50	7.20	322.54	400	32	Basic Abdomen Pelvis (0.5sec) with Smart mA, Evaluation for abnormalities of abdominal and pelvic structures
26.2	Abdomen Pelvis 0.5s. 5mm + DMPR SmartmA/ SmartPrep	DMPR	Helical	LargeBody	0.984:1 39.375	0.6	0.625	40	120	SmartmA 80-600mA NI=18 Avg mA 260	50	Std Plus	50	12.56	543.37	400	32	Basic Abdomen Pelvis (0.5sec) with Smart mA, SmartPrep and DMPR, Evaluation for abnormalities of abdominal and pelvic structures
26.4	Renal Stone 0.5s. 5mm		Helical	LargeBody	1.375:1 55.0	0.5	5	40	120	190	50	Std Plus	50	5.47	245.13	400	32	Helical scan, Evaluation of kidney, ureters and bladder for renal stones
26.5	Renal Stone 0.5s. 5mm SmartmA		Helical	LargeBody	1.375:1 55.0	0.5	5	40	120	SmartmA 50-600mA NI=12 Avg mA 190	50	Std Plus	50	5.47	245.13	400	32	Helical scan with SmartmA, Evaluation of kidney, ureters and bladder for renal stones
26.7	AAA 0.5s. + DMPR 0.625mm	MIROI	Axial	LargeBody	1i	1	5 [0]	5	120	80	0	Std	0	112.67	56.34	0	32	Timing bolus for Helical scan with DMPR, Evaluation for abdominal aortic aneurysm
		DMPR	Helical	LargeBody	1.375:1 55.0	0.5	0.625	40	140	255	50	Std Plus	50	10.41	371.80	309.375	32	Helical scan with DMPR, Evaluation for abdominal aortic aneurysm
26.8	AAA 0.5s. + DMPR 0.625mm SmartmA/ SmartPrep	DMPR	Helical	LargeBody	1.375:1 55.0	0.6	0.625	40	120	SmartmA 100-600mA NI=19 Avg mA 275	50	Std Plus	50	9.51	339.51	309.375	32	Helical scan with DMPR, Smart mA and SmartPrep, Evaluation for abdominal aortic aneurysm
26.10	Dual Phase Liver		Helical	LargeBody	1.375:1 55.0	0.6	5	40	120	150	50	Std Plus	50	5.19	92.16	130	32	Non contrast evaluation of upper abdominal structures including liver
		MIROI	Axial	LargeBody	1i	1	5 [0]	5	120	80	0	Std	0	112.67	56.34	0	32	Timing bolus for Multi-Group acquisition for early arterial and portal venous phase of the liver
	Group1		Helical	LargeBody	1.375:1 55.0	0.5	2.5	40	120	275	50	Std Plus	50	7.92	140.87	130	32	Acquisition for early arterial phase of the liver
	Group2		Helical	LargeBody	1.375:1 55.0	0.5	5	40	120	210	50	Std Plus	50	6.05	270.94	400	32	Acquisition for portal venous phase of the liver

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy-cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
26.11	Tri Phase Liver		Helical	LargeBody	1.375:1 55.0	0.6	5	40	120	150	50	Std Plus	50	5.19	92.16	130	32	Non contrast evaluation of upper abdominal structures including liver
		MIROI	Axial	LargeBody	1i	1	5 [0]	5	120	80	0	Std	0	112.67	56.34	0	32	Timing bolus for Multi-Group acquisition for early arterial, late arterial and portal venous phase of the liver
	Group1		Helical	LargeBody	1.375:1 55.0	0.5	2.5	40	120	275	50	Std Plus	50	7.92	140.87	130	32	Acquisition for early arterial phase of the liver
	Group2		Helical	LargeBody	1.375:1 55.0	0.5	5	40	120	225	50	Std Plus	50	6.48	290.29	400	32	Acquisition for late arterial phase of the liver
	Group3		Helical	LargeBody	1.375:1 55.0	0.5	5	40	120	225	50	Std Plus	50	6.48	290.29	400	32	Acquisition for portal venous phase of the liver
26.12	Dual Phase Pancreas SmartmA/ SmartPrep		Helical	LargeBody	0.984:1 39.375	0.5	5	40	120	SmartmA 80-600mA NI=12 Avg mA 200	50	Std Plus	50	8.05	82.74	70	32	Non contrast evaluation of upper abdominal structures including pancreas
	Group1		Helical	LargeBody	0.984:1 39.375	0.5	2.5	40	120	SmartmA 50-600mA NI=15 Avg mA 260	50	Std Plus	50	10.46	191.28	150	32	Acquisition with SmartmA and SmartPrep for arterial phase of the pancreas
	Group2		Helical	LargeBody	0.984:1 39.375	0.5	5	40	120	SmartmA 50-600mA NI=15 Avg mA 225	50	Std Plus	50	9.06	319.47	320	32	Acquisition for venous phase of the pancreas
26.13	Dual Phase Liver SmartmA/ SmartPrep		Helical	LargeBody	1.375:1 55.0	0.5	5	40	120	SmartmA 80-500mA NI=12 Avg mA 160	50	Std Plus	50	4.61	81.96	130	32	Non contrast evaluation of upper abdominal structures including liver
	Group1		Helical	LargeBody	1.375:1 55.0	0.5	2.5	40	120	SmartmA 50-600mA NI=15 Avg mA 275	50	Std Plus	50	7.92	140.87	130	32	Acquisition with SmartmA and SmartPrep for early arterial phase of the liver
	Group2		Helical	LargeBody	0.984:1 39.375	0.5	5	40	120	SmartmA 50-600mA NI=11 Avg mA 225	50	Std Plus	50	9.06	391.91	400	32	Acquisition with SmartmA and SmartPrep for portal venous phase of the liver
26.14	Tri Phase Liver SmartmA/ SmartPrep		Helical	LargeBody	0.984:1 39.375	0.5	5	40	120	SmartmA 80-500mA NI=12 Avg mA 200	50	Std Plus	50	8.05	131.04	130	32	Non contrast evaluation of upper abdominal structures including liver
	Group1		Helical	LargeBody	1.375:1 55.0	0.5	2.5	40	120	SmartmA 50-600mA NI=15 Avg mA 275	50	Std Plus	50	7.92	140.87	130	32	Acquisition with SmartmA and SmartPrep for early arterial phase of the liver
	Group2		Helical	LargeBody	1.375:1 55.0	0.5	5	40	120	SmartmA 50-600mA NI=11 Avg mA 225	50	Std Plus	50	6.48	290.29	400	32	Acquisition with SmartmA and SmartPrep for late arterial phase of the liver
	Group3		Helical	LargeBody	1.375:1 55.0	0.5	5	40	120	SmartmA 50-600mA NI=11 Avg mA 225	50	Std Plus	50	6.48	290.29	400	32	Acquisition with SmartmA and SmartPrep for portal venous phase of the liver
26.15	VHS Liver Perfusion		Helical	LargeBody	0.984:1 39.375	0.6	5	40	120	85	50	Std	50	4.11	66.80	130	32	Non contrast evaluation of upper abdominal structures including liver

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy-cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
		Perfusion	Helical-S/P	LargeBody	1.375:1 55.0 21 pass	0.4	5	40	120	150	50	Std Plus	50	109.99	1704.89	140	32	Volume Helical scan mode for evaluation of liver tumor perfusion over time
26.16	VHS 4D CTA		Helical	LargeBody	0.984:1 39.375	0.6	5	40	120	170	50	Std	50	8.21	133.61	130	32	Localizer scan for Volume Helical Shuttle scan mode for evaluation of abdominal vasculature
		MIROI	Axial	LargeBody	1i	1	5 [0]	5	120	80	0	Std	0	112.67	56.34	0	32	Timing bolus scan for Volume Helical Shuttle scan mode for evaluation of abdominal vasculature
			Helical-S	LargeBody	1.375:1 55.0 18 pass	0.5	5 [10]	40	140	SmartmA 150-515mA NI=12 Avg mA 175	50	Std Plus	50	162.84	3175.35	180	32	Volume Helical Shuttle scan mode for time resolved CT angiography of abdominal vasculature
26.17	Runoff 0.625mm	MIROI	Axial	LargeBody	1i	1	5 [0]	5	120	80	0	Std	0	112.67	56.34	0	32	Timing bolus scan for evaluation of vascular structure of abdomen, femurs and lower extremities (Runoff)
			Helical	LargeBody	1.375:1 55.0	0.5	0.625	40	140	310	40	Std Plus	40	12.65	1246.09	936.875	32	Helical scanmode 0.625mm, Evaluation of vascular structure of abdomen, femurs and lower extremities (Runoff)
26.18	Runoff 1.25mm	MIROI	Axial	LargeBody	1i	1	5 [0]	5	120	80	0	Std	0	112.67	56.34	0	32	Timing bolus scan for evaluation of vascular structure of abdomen, femurs and lower extremities (Runoff)
			Helical	LargeBody	1.375:1 55.0	0.5	1.25 [0.7]	40	140	240	40	Std Plus	40	9.80	1025.48	998.9	32	Helical scanmode 1.25mm, Evaluation of vascular structure of abdomen, femurs and lower extremities (Runoff)
26.19	Runoff 0.625mm SmartmA/ SmartPrep		Helical	LargeBody	1.375:1 55.0	0.5	0.625	40	140	SmartmA 100-515mA NI=19 Avg mA 310	40	Std Plus	40	12.65	1246.09	936.875	32	Helical scanmode 0.625mm with SmartmA and SmartPrep, Evaluation of vascular structure of abdomen, femurs and lower extremities (Runoff)
26.20	Runoff 1.25mm SmartmA/ SmartPrep		Helical	LargeBody	1.375:1 55.0	0.5	1.25 [0.7]	40	140	SmartmA 100-515mA NI=20 Avg mA 240	40	Std Plus	40	9.80	1025.48	998.9	32	Helical scanmode 1.25mm with SmartmA and SmartPrep, Evaluation of vascular structure of abdomen, femurs and lower extremities (Runoff)
26.22	CT Colonography 1.25mm	Advantage CTC Pro	Helical	LargeBody	1.375:1 55.0	0.5	1.25	40	120	100	50	Std Plus	50	2.88	129.02	400	32	Helical scan mode supine acquisition for using Advantage CTC Pro for evaluation of the colon
		Advantage CTC Pro	Helical	LargeBody	1.375:1 55.0	0.5	1.25	40	120	100	50	Std Plus	50	2.88	129.02	400	32	Helical scan mode prone acquisition for using Advantage CTC Pro for evaluation of the colon

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy-cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
26.41	Body Interventional 5mm/3i		Helical	LargeBody	1.375:1 27.5	0.5	5	20	120	SmartmA 50-600mA NI=11.57 Avg mA 360	0	Std	0	11.41	251.58	195	32	Localization scan for CT interventional scan
			SmartView SEG	LargeBody	3i	0.8	5	10	120	80	0	Std Segment	0	684.58	684.58	0	32	CT Fluoro scan for interventional scan
26.42	Body Interventional 5mm/1i		Helical	LargeBody	1.375:1 27.50	0.5	5	20	120	SmartmA 50-600mA NI=11.57 Avg mA 360	0	Std	0	11.41	251.58	195	32	Localization scan for CT interventional scan
			SmartView SEG	LargeBody	1i	0.8	5	5	120	80	0	Std Segment	0	845.04	422.52	0	32	CT Fluoro scan for interventional scan

L-Spine

Table 16-7

[Interval]* - For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
27.1	L-Spine 3 Level Axial Group1		Axial	LargeBody	8i	1	2.5	20	120	225	30	Std	40	19.08	38.15	17.5	32	Axial scan mode, Evaluation of soft tissue and boney structures lumbar spine
	Group2		Axial	LargeBody	8i	1	2.5	20	120	225	30	Std	40	19.08	38.15	17.5	32	Axial scan mode, Evaluation of soft tissue and boney structures lumbar spine
	Group3		Axial	LargeBody	8i	1	2.5	20	140	250	30	Std	40	30.03	60.06	17.5	32	Axial scan mode, Evaluation of soft tissue and boney structures lumbar spine
27.3	L-Spine Survey Helical 2.5mm 0.6s. SmartmA		Helical	LargeBody	0.516:1 20.625	0.6	2.5	40	140	SmartmA 100-515mA NI=14 Avg mA 360	30	Std Plus	40	47.03	985.50	75	32	Helical scan mode 2.5mm (0.6sec), Limited evaluation of lumbar spine
27.4	L-Spine Survey Helical 2.5mm 0.8s. SmartmA		Helical	LargeBody	0.516:1 20.625	0.8	2.5	40	140	SmartmA 100-515mA NI=14 Avg mA 360	30	Std Plus	40	62.70	1313.78	75	32	Helical scan mode 2.5mm (0.8sec), Limited evaluation of lumbar spine
27.6	L-Spine Helical 0.625mm + DMPR/ SmartmA	DMPR	Helical	LargeBody	0.516:1 20.625	0.6	0.625	40	140	SmartmA 100-515mA NI=20 Avg mA 360	30	Std Plus IQE	40	47.03	985.50	75	32	Helical scan mode 0.625mm with DMPR and SmartmA, Evaluation of soft tissue and boney structures lumbar spine
27.7	L-Spine Helical 1.25mm + DMPR/ SmartmA - Large Patient	DMPR	Helical	LargeBody	0.516:1 20.625	1	1.25	40	140	SmartmA 100-515mA NI=18 Avg mA 360	30	Std Plus	40	78.38	1641.95	75	32	Helical scan mode 1.25mm with DMPR and SmartmA, Evaluation of soft tissue and boney structures lumbar spine

Pelvis

Table 16-8

[Interval]* - For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.
For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
28.1	Pelvis for Fracture + DMPR/ SmartmA	DMPR	Helical	LargeBody	0.984:1 39.375	0.8	1.25 [0.625]	40	120	SmartmA 100-600mA NI=20 Avg mA 300	40	Bone Plus IQE	40	19.32	258.70	101.25	32	Helical scan with DMPR and SmartmA, Evaluation of pelvis bony structures for fracture

Lower Extremity

Table 16-9

[Interval]* - For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
29.1	Lower Extremity Survey 2.5mm		Helical	LargeBody	0.516:1 20.625	0.5	2.5	40	120	175	30	Std Plus	30	13.45	126.94	60	32	Helical scan mode 2.5mm, Limited evaluation of lower extremity for abnormalities
29.2	Ankle 0.625 mm Single Joint		Helical	MediumBody	0.531:1 10.625	0.5	0.625	20	120	175	30	Std Plus IQE	40	14.50	75.64	35	32	Helical scan mode 0.625mm, Evaluation of soft tissue and bony anatomy of the ankle area
29.3	Ankle 0.625 mm Both Joints		Helical	LargeBody	0.531:1 10.625	0.5	0.625	20	120	175	30	Std Plus IQE	30	13.96	72.82	35	32	Helical scan mode 0.625mm, Evaluation of soft tissue and bony anatomy of the ankle area
29.4	Knee 0.625 mm Both Joints		Helical	LargeBody	0.531:1 10.625	0.6	0.625	20	120	Auto mA 100-335mA NI=20 Avg mA 190	30	Std Plus IQE	30	18.19	94.84	35	32	Helical scan mode 0.625mm, Evaluation of soft tissue and bony anatomy of both knees

Reference Pediatric Protocols - Installed Options : ASiR

Head

Table 17-1

[Interval]* - For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy-cm) ASiR	Scan Length (mm)	Phantom (cm)	Description
31.1	PED HEAD TO 18 Months		Axial	Ped Head	4i	0.5	5	20	120	170	30	Std	30	14.47	173.68	115	16	Routine head (0.5s) for infant (up to 18 months) 5mm
31.2	PED HEAD 18 MOS TO 5YRS		Axial	SmallHead	4i	0.5	5	20	120	235	30	Std	30	20.01	240.08	115	16	Routine head (0.5s) for children (18 months to 5 years) 5mm
31.3	PED HEAD TO 5YRS TO 18YRS Group1		Axial	SmallHead	8i	1	2.5	20	120	170	30	Std	30	28.95	115.79	37.5	16	Routine head (1.0s) for children (5 years to 18 years) 2.5mm
	Group2		Axial	SmallHead	4i	1	5	20	120	125	30	Std	30	21.28	170.27	75	16	Routine head (1.0s) for children (5 years to 18 years) 5mm

Orbit

Table 17-2

[Interval]* - For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy-cm) ASiR	Scan Length (mm)	Phantom (cm)	Description
32.1	Sinus Supine Helical + DMPR	DMPR	Helical	SmallHead	0.531:1 10.625	0.4	0.625	20	120	65	20	Bone Plus IQE	30	8.33	76.29	74.375	16	Routine Helical scan mode with DMPR, Evaluation of sinus area for abnormality in pediatric patients
32.3	Orbits Helical	DMPR	Helical	SmallHead	0.531:1 10.625	0.4	0.625	20	120	145	20	Std Plus IQE	30	18.59	146.96	61.875	16	Helical scan mode, Evaluation of orbital and face structures for abnormality in pediatric patients
32.6	Helical IAC 0.625mm Axial mode - Use Reformat for Coronal		Helical	SmallHead	0.531:1 10.625	0.4	0.625	20	120	130	20	Std Plus IQE	30	16.67	93.21	38.75	16	Helical scan mode, Evaluation of inner ear structures for abnormality in pediatric patients

Chest

Table 17-3

[Interval]* - For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy-cm) ASiR	Scan Length (mm)	Phantom (cm)	Description
35.1.1	RC 6.0-7.5 kg (13.2-16.5 lbs)		Helical	PedBody	1.375:1 27.5	0.4	3.75	20	100	55	30	Std Plus	30	0.76	7.62	75	32	Weight and height based routine chest protocol
35.1.3	SnapShot Segment 30-74 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	85	30	Std Plus	30	0.59	5.80	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	Std	0	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
			Cardiac SEG	CardiacSmall	0.180:1 7.2	0.4	0.625	40	80	ECG Mod 60-195mA Phase 40-80 AvgmA195	30	Std SS Seg	40	8.88	97.72	75	32	Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
35.1.4	SnapShot Burst 75-104 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	85	30	Std Plus	30	0.59	5.80	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	Std	0	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
			Cardiac SSB	CardiacSmall	0.180:1 7.2	0.4	0.625	40	80	ECG Mod 60-195mA Phase 40-80 AvgmA195	30	Std SSB	40	8.88	97.72	75	32	Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
35.1.5	SnapShot Burst+ >105 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	85	30	Std Plus	30	0.59	5.80	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	Std	0	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
			Cardiac SSBP	CardiacSmall	0.180:1 7.2	0.4	0.625	40	80	ECG Mod 60-195mA Phase 35-85 AvgmA195	30	Std SSB	40	9.50	104.47	75	32	Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
35.1.7	Non ECG Gated		Helical	PedBody	0.984:1 39.375	0.4	0.625	40	80	SmartmA 50-200mA NI=18 AvgmA140	30	Std Plus	30	1.36	14.70	75	32	Weight and height based non ECG gated protocol for cardiac with SmartmA
35.2.1	RC 7.5-9.5 kg (16.5-20.9 lbs)		Helical	PedBody	1.375:1 27.5	0.4	3.75	20	100	85	30	Std Plus	30	1.18	11.77	75	32	Weight and height based routine chest protocol
35.2.3	SnapShot Segment 30-74 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	85	30	Std Plus	30	0.59	5.80	50	32	Weight and height based protocol for localization

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy-cm) ASiR	Scan Length (mm)	Phantom (cm)	Description
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	Std	0	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
			Cardiac SEG	CardiacSmall	0.180:1 7.2	0.4	0.625	40	80	ECG Mod 60-210mA Phase 40-80 AvgmA210	30	Std SS Seg	40	9.54	104.90	75	32	Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
35.2.4	SnapShot Burst 75-104 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	85	30	Std Plus	30	0.59	5.80	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	Std	0	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
			Cardiac SSB	CardiacSmall	0.180:1 7.2	0.4	0.625	40	80	ECG Mod 60-210mA Phase 40-80 AvgmA210	30	Std SSB	40	9.54	104.90	75	32	Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
35.2.5	SnapShot Burst+ >105 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	85	30	Std Plus	30	0.59	5.80	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	Std	0	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
			Cardiac SSBP	CardiacSmall	0.180:1 7.2	0.4	0.625	40	80	ECG Mod 60-210mA Phase 35-85 AvgmA210	30	Std SSB	40	10.22	112.39	75	32	Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
35.2.7	Non ECG Gated		Helical	PedBody	0.984:1 39.375	0.4	0.625	40	80	SmartmA 50-220mA NI=18 AvgmA155	30	Std Plus	30	1.51	16.28	75	32	Weight and height based non ECG gated protocol for cardiac with SmartmA
35.3.1	RC 9.5-11.5 kg (20.9-25.4lbs)		Helical	SmallBody	1.375:1 27.5	0.4	3.75	20	120	70	30	Std Plus	30	1.52	15.77	78.75	32	Weight and height based routine chest protocol
35.3.3	SnapShot Segment 30-74 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	85	30	Std Plus	30	0.59	5.80	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	Std	0	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
			Cardiac SEG	CardiacSmall	0.180:1 7.2	0.4	0.625	40	80	ECG Mod 70-225mA Phase 40-80 AvgmA225	30	Std SS Seg	40	10.26	112.87	75	32	Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
35.3.4	SnapShot Burst 75-104 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	85	30	Std Plus	30	0.59	5.80	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	Std	0	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy-cm) ASiR	Scan Length (mm)	Phantom (cm)	Description
			Cardiac SSB	CardiacSmall	0.180:1 7.2	0.4	0.625	40	80	ECG Mod 70-225mA Phase 40-80 AvgmA225	30	Std SSB	40	10.26	112.87	75	32	Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
35.3.5	SnapShot Burst+ >105 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	85	30	Std Plus	30	0.59	5.80	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	Std	0	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
			Cardiac SSBP	CardiacSmall	0.180:1 7.2	0.4	0.625	40	80	ECG Mod 70-225mA Phase 35-85 AvgmA225	30	Std SSB	40	10.96	120.61	75	32	Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
35.3.7	Non ECG Gated		Helical	PedBody	0.984:1 39.375	0.4	0.625	40	80	SmartmA 50-240mA NI=18 AvgmA170	30	Std Plus	30	1.65	17.85	75	32	Weight and height based non ECG gated protocol for cardiac with SmartmA
35.4.1	RC 11.5-14.5kg (25.4-32.0lbs)		Helical	SmallBody	1.375:1 27.5	0.4	5	20	120	65	30	Std Plus	30	1.41	16.23	90	32	Weight and height based routine chest protocol
35.4.3	SnapShot Segment 30-74 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	85	30	Std Plus	30	0.59	5.80	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	Std	0	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
			Cardiac SEG	CardiacSmall	0.180:1 7.2	0.4	0.625	40	80	ECG Mod 70-240mA Phase 40-80 AvgmA240	30	Std SS Seg	40	10.91	120.04	75	32	Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
35.4.4	SnapShot Burst 75-104 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	85	30	Std Plus	30	0.59	5.80	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	Std	0	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
			Cardiac SSB	CardiacSmall	0.180:1 7.2	0.4	0.625	40	80	ECG Mod 70-240mA Phase 40-80 AvgmA240	30	Std SSB	40	10.91	120.04	75	32	Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
35.4.5	SnapShot Burst+ >105 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	85	30	Std Plus	30	0.59	5.80	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	Std	0	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
			Cardiac SSBP	CardiacSmall	0.180:1 7.2	0.4	0.625	40	80	ECG Mod 70-240mA Phase 35-85 AvgmA240	30	Std SSB	40	11.68	128.53	75	32	Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy-cm) ASiR	Scan Length (mm)	Phantom (cm)	Description
35.4.7	Non ECG Gated		Helical	PedBody	0.984:1 39.375	0.4	0.625	40	80	SmartmA 50-280mA NI=18 AvgmA195	30	Std Plus	30	1.90	20.48	75	32	Weight and height based non ECG gated protocol for cardiac with SmartmA
35.5.1	RC 14.5-18.5kg (32.0-40.8lbs)		Helical	SmallBody	1.375:1 27.5	0.4	5	20	120	70	30	Std Plus	30	1.52	17.48	90	32	Weight and height based routine chest protocol
35.5.3	SnapShot Segment 30-74 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	85	30	Std Plus	30	0.59	5.80	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	Std	0	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
			Cardiac SEG	CardiacSmall	0.180:1 7.2	0.4	0.625	40	80	ECG Mod 80-265mA Phase 40-80 AvgmA265	30	Std SS Seg	40	12.07	132.78	75	32	Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
35.5.4	SnapShot Burst 75-104 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	85	30	Std Plus	30	0.59	5.80	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	Std	0	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
			Cardiac SSB	CardiacSmall	0.180:1 7.2	0.4	0.625	40	80	ECG Mod 80-265mA Phase 40-80 AvgmA265	30	Std SSB	40	12.07	132.78	75	32	Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
35.5.5	SnapShot Burst+ >105 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	85	30	Std Plus	30	0.59	5.80	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	Std	0	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
			Cardiac SSBP	CardiacSmall	0.180:1 7.2	0.4	0.625	40	80	ECG Mod 80-265mA Phase 35-85 AvgmA265	30	Std SSB	40	12.91	142.02	75	32	Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
35.5.7	Non ECG Gated		Helical	PedBody	0.984:1 39.375	0.4	0.625	40	80	SmartmA 50-300mA NI=18 AvgmA210	30	Std Plus	30	2.04	22.05	75	32	Weight and height based non ECG gated protocol for cardiac with SmartmA
35.6.1	RC 18.5-22.5kg (40.8-49.6lbs)		Helical	SmallBody	1.375:1 27.5	0.4	5	20	120	75	30	Std Plus	30	1.63	18.73	90	32	Weight and height based routine chest protocol
35.6.3	SnapShot Segment 30-74 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	85	30	Std Plus	30	0.59	5.80	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	Std	0	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy-cm) ASiR	Scan Length (mm)	Phantom (cm)	Description
			Cardiac SEG	CardiacSmall	0.180:1 7.2	0.4	0.625	40	80	ECG Mod 80-280mA Phase 40-80 AvgmA280	30	Std SS Seg	40	12.72	139.92	75	32	Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
35.6.4	SnapShot Burst 75-104 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	85	30	Std Plus	30	0.59	5.80	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	Std	0	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
			Cardiac SSB	CardiacSmall	0.180:1 7.2	0.4	0.625	40	80	ECG Mod 80-280mA Phase 40-80 AvgmA280	30	Std SSB	40	12.72	139.92	75	32	Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
35.6.5	SnapShot Burst+ >105 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	85	30	Std Plus	30	0.59	5.80	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	Std	0	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
			Cardiac SSBP	CardiacSmall	0.180:1 7.2	0.4	0.625	40	80	ECG Mod 80-280mA Phase 35-85 AvgmA280	30	Std SSB	40	13.63	149.91	75	32	Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
35.6.7	Non ECG Gated		Helical	PedBody	0.984:1 39.375	0.4	0.625	40	80	SmartmA 100-320mA NI=18 AvgmA225	30	Std Plus	30	2.19	23.63	75	32	Weight and height based non ECG gated protocol for cardiac with SmartmA
35.7.1	RC 22.5-31.5kg (49.6-69.5lbs)		Helical	LargeBody	1.375:1 55.0	0.4	5	40	120	75	30	Std Plus	30	1.73	22.98	85	32	Weight and height based routine chest protocol
35.7.3	SnapShot Segment 30-74 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	85	30	Std Plus	30	0.59	5.80	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	Std	0	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
			Cardiac SEG	CardiacSmall	0.180:1 7.2	0.4	0.625	40	80	ECG Mod 85-295mA Phase 40-80 AvgmA295	30	Std SS Seg	40	13.41	147.51	75	32	Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
35.7.4	SnapShot Burst 75-104 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	85	30	Std Plus	30	0.59	5.80	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	Std	0	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
			Cardiac SSB	CardiacSmall	0.180:1 7.2	0.4	0.625	40	80	ECG Mod 85-295mA Phase 40-80 AvgmA295	30	Std SSB	40	13.41	147.51	75	32	Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy-c m) ASiR	DLP (mGy-c m) ASiR	Scan Length (mm)	Phantom (cm)	Description
35.7.5	SnapShot Burst+ >105 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	85	30	Std Plus	30	0.59	5.80	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	Std	0	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
			Cardiac SSBP	CardiacSmall	0.180:1 7.2	0.4	0.625	40	80	ECG Mod 85-295mA Phase 35-85 AvgmA295	30	Std SSB	40	14.36	158.00	75	32	Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
35.7.7	Non ECG Gated		Helical	PedBody	0.984:1 39.375	0.4	0.625	40	80	SmartmA 100-360mA NI=18 AvgmA250	30	Std Plus	30	2.43	26.25	75	32	Weight and height based non ECG gated protocol for cardiac with SmartmA
35.8.1	RC 31.5-40.5kg (69.5-89.3lbs)		Helical	LargeBody	1.375:1 55.0	0.4	5	40	120	85	30	Std Plus	30	1.96	32.90	120	32	Weight and height based routine chest protocol
35.8.3	SnapShot Segment 30-74 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	100	85	30	Std Plus	30	1.10	10.76	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	100	50	0	Std	0	39.50	19.75	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
			Cardiac SEG	CardiacSmall	0.180:1 7.2	0.4	0.625	40	100	ECG Mod 80-265mA Phase 40-80 AvgmA265	30	Std SS Seg	30	22.40	246.43	75	32	Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
35.8.4	SnapShot Burst 75-104 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	100	85	30	Std Plus	30	1.10	10.76	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	100	50	0	Std	0	39.50	19.75	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
			Cardiac SSB	CardiacSmall	0.180:1 7.2	0.4	0.625	40	100	ECG Mod 80-265mA Phase 40-80 AvgmA265	30	Std SSB	30	22.40	246.43	75	32	Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
35.8.5	SnapShot Burst+ >105 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	100	85	30	Std Plus	30	1.10	10.76	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	100	50	0	Std	0	39.50	19.75	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
			Cardiac SSBP	CardiacSmall	0.180:1 7.2	0.4	0.625	40	100	ECG Mod 80-265mA Phase 35-85 AvgmA265	30	Std SSB	40	23.96	263.58	75	32	Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
35.8.7	Non ECG Gated		Helical	SmallBody	0.984:1 39.375	0.4	0.625	40	100	SmartmA 100-400mA NI=16 AvgmA280	30	Std Plus	30	5.06	54.57	75	32	Weight and height based non ECG gated protocol for cardiac with SmartmA

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy-cm) ASiR	Scan Length (mm)	Phantom (cm)	Description
35.9.1	RC 40.5-55.0kg(89.3-121.3lbs)		Helical	LargeBody	1.375:1 55.0	0.4	5	40	120	90	30	Std Plus	30	2.07	41.06	150	32	Weight and height based routine chest protocol
35.9.3	SnapShot Segment 30-74 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	100	85	30	Std Plus	30	1.10	10.76	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	100	50	0	Std	0	39.50	19.75	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
			Cardiac SEG	CardiacSmall	0.180:1 7.2	0.4	0.625	40	100	ECG Mod 85-295mA Phase 40-80 AvgmA295	30	Std SS Seg	30	24.89	273.78	75	32	Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
35.9.4	SnapShot Burst 75-104 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	100	85	30	Std Plus	30	1.10	10.76	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	100	50	0	Std	0	39.50	19.75	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
			Cardiac SSB	CardiacSmall	0.180:1 7.2	0.4	0.625	40	100	ECG Mod 85-295mA Phase 40-80 AvgmA295	30	Std SSB	30	24.89	273.78	75	32	Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
35.9.5	SnapShot Burst+ >105 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	100	85	30	Std Plus	30	1.10	10.76	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	100	50	0	Std	0	39.50	19.75	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
			Cardiac SSBP	CardiacSmall	0.180:1 7.2	0.4	0.625	40	100	ECG Mod 85-295mA Phase 35-85 AvgmA295	30	Std SSB	40	26.66	293.24	75	32	Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
35.9.7	Non ECG Gated		Helical	SmallBody	0.984:1 39.375	0.4	0.625	40	100	SmartmA 100-440mA NI=16 AvgmA310	30	Std Plus	30	5.60	60.42	75	32	Weight and height based non ECG gated protocol for cardiac with SmartmA

Abdomen

Table 17-4

[Interval]* - For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy-cm) ASiR	Scan Length (mm)	Phantom (cm)	Description
36.1.1	Abd 6.0-7.5kg (13.2-16.5lbs)		Helical	PedBody	1.375:1 27.5	0.4	3.75	20	120	70	30	Std Plus	30	1.52	15.77	78.75	32	Weight and height based routine abdomen protocol
36.2.1	Abd 7.5-9.5kg (16.5-20.9lbs)		Helical	PedBody	1.375:1 27.5	0.4	3.75	20	120	110	30	Std Plus	30	2.39	27.47	90	32	Weight and height based routine abdomen protocol
36.3.1	Abd 9.5-11.5kg (20.9-25.4lbs)		Helical	PedBody	1.375:1 27.5	0.4	3.75	20	120	125	30	Std Plus	30	2.71	34.26	101.25	32	Weight and height based routine abdomen protocol
36.4.1	Abd 11.5-14.5kg (25.4-32.0lbs)		Helical	SmallBody	1.375:1 27.5	0.4	5	20	120	85	30	Std Plus	30	1.84	24.91	110	32	Weight and height based routine abdomen protocol
36.5.1	Abd 14.5-18.5kg (32.0-40.8lbs)		Helical	SmallBody	1.375:1 27.5	0.4	5	20	120	95	30	Std Plus	30	2.06	27.84	110	32	Weight and height based routine abdomen protocol
36.6.1	Abd 18.5-22.5kg (40.8-49.6lbs)		Helical	Medium Body	1.375:1 27.5	0.4	5	20	120	85	30	Std Plus	30	2.18	33.77	130	32	Weight and height based routine abdomen protocol
36.7.1	Abd 22.5-31.5kg (49.6-69.5lbs)		Helical	LargeBody	1.375:1 55.0	0.4	5	40	120	105	30	Std Plus	30	2.42	47.91	150	32	Weight and height based routine abdomen protocol
36.8.1	Abd 31.5-40.5kg (69.5-89.3lbs)		Helical	LargeBody	1.375:1 55.0	0.4	5	40	120	120	30	Std Plus	30	2.77	54.75	150	32	Weight and height based routine abdomen protocol
36.9.1	Abd 40.5-55.0kg (89.3-121.3lbs)		Helical	LargeBody	1.375:1 55.0	0.4	5	40	120	140	30	Std Plus	30	3.23	80.01	200	32	Weight and height based routine abdomen protocol

Miscellaneous

Table 17-5

[Interval]* - For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy-cm) ASiR	Scan Length (mm)	Phantom (cm)	Description
40.1	Pink Zone Routine 0-20 lbs/0-9 kg		Helical	PedBody	1.375:1 55.0	0.5	5	40	80	SmartmA 65-130mA NI=5 Avg mA 70	0	Std	0	0.61	8.70	95	32	Pediatric and Small patient Routine procedure based protocols
40.2	Pink Zone Routine 21-60 lbs/9.1-27.2 kg		Helical	PedBody	1.375:1 55.0	0.5	5	40	100	SmartmA 80-160mA NI=7 Avg mA 70	0	Std	0	1.13	27.47	195	32	Pediatric and Small patient Routine procedure based protocols
40.3	Pink Zone Routine 61-100 lbs/27.3-45.4 kg		Helical	Medium Body	1.375:1 55.0	0.5	5	40	120	SmartmA 95-190mA NI=10 Avg mA 70	0	Std	0	2.09	71.81	295	32	Pediatric and Small patient Routine procedure based protocols
40.4	Pink Zone Routine 101-200 lbs/45.4-90.7 kg		Helical	LargeBody	1.375:1 55.0	0.5	5	40	120	SmartmA 110-220mA NI=12 Avg mA 70	0	Std	0	2.02	89.30	395	32	Pediatric and Small patient Routine procedure based protocols
40.5	Pink Zone Routine >200 lbs/>90.8 kg		Helical	LargeBody	1.375:1 55.0	0.5	5	40	120	SmartmA 125-300mA NI=15 Avg mA 70	0	Std	0	2.02	89.30	395	32	Pediatric and Small patient Routine procedure based protocols
40.7	Green Zone Lower Dose 0-20 lbs/0-9 kg		Helical	PedBody	1.375:1 55.0	0.5	5	40	80	SmartmA 50-100mA NI=7 Avg mA 70	0	Std	0	0.61	8.70	95	32	Pediatric and Small patient Lower Dose procedure based protocols
40.8	Green Zone Lower Dose 21-60 lbs/9.1-27.2 kg		Helical	PedBody	1.375:1 55.0	0.5	5	40	100	SmartmA 60-120mA NI=9 Avg mA 70	0	Std	0	1.13	27.47	195	32	Pediatric and Small patient Lower Dose procedure based protocols
40.9	Green Zone Lower Dose 61-100 lbs/27.3-45.4 kg		Helical	Medium Body	1.375:1 55.0	0.5	5	40	120	SmartmA 70-140mA NI=11 Avg mA 70	0	Std	0	2.09	71.81	295	32	Pediatric and Small patient Lower Dose procedure based protocols
40.10	Green Zone Lower Dose 101-200 lbs/45.4-90.7 kg		Helical	LargeBody	1.375:1 55.0	0.5	5	40	120	SmartmA 80-160mA NI=13 Avg mA 70	0	Std	0	2.02	89.30	395	32	Pediatric and Small patient Lower Dose procedure based protocols
40.11	Green Zone Lower Dose >200 lbs/>90.8 kg		Helical	LargeBody	1.375:1 55.0	0.5	5	40	120	SmartmA 90-240mA NI=16 Avg mA 70	0	Std	0	2.02	89.30	395	32	Pediatric and Small patient Lower Dose procedure based protocols
40.13	Grey Zone CT Angio 0-20 lbs/0-9 kg		Helical	PedBody	1.375:1 55.0	0.5	2.5	40	80	SmartmA 100-200mA NI=5 Avg mA 140	0	Std	0	1.22	17.71	97.5	32	Pediatric and Small patient Angio procedure based protocols

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy-cm) ASiR	Scan Length (mm)	Phantom (cm)	Description
40.14	Grey Zone CT Angio 21-60 lbs/9.1-27.2 kg		Helical	SmallBody	1.375:1 55.0	0.5	2.5	40	100	SmartmA 120-240mA NI=9.9 Avg mA 140	0	Std	0	2.26	55.50	197.5	32	Pediatric and Small patient Angio procedure based protocols
40.15	Grey Zone CT Angio 61-100 lbs/ 27.3-45.4 kg		Helical	LargeBody	1.375:1 55.0	0.5	2.5	40	120	SmartmA 120-240mA NI=10 Avg mA 140	0	Std	0	4.03	139.28	297.5	32	Pediatric and Small patient Angio procedure based protocols
40.16	Grey Zone CT Angio 101-200 lbs/ 45.4-90.7 kg		Helical	LargeBody	1.375:1 55.0	0.5	2.5	40	120	SmartmA 120-240mA NI=12 Avg mA 140	0	Std	0	4.03	179.62	397.5	32	Pediatric and Small patient Angio procedure based protocols
40.17	Grey Zone CT Angio >200 lbs/ >90.8 kg		Helical	LargeBody	1.375:1 55.0	0.5	2.5	40	120	SmartmA 120-300mA NI=15 Avg mA 140	0	Std	0	4.03	179.62	397.5	32	Pediatric and Small patient Angio procedure based protocols

Reference Adult Protocols - Installed Options : VCT85kW and/or VCT Hi Power

Head

Table 18-1

[Interval]* - For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (s)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (mm)	Phantom (cm)	Description
21.1	Routine Head 1s Group1		Axial	Head	4i	1	5	20	140	200	Std	53.34	320.02	55	16	Routine head protocol for evaluation of the brain for abnormalities.
	Group2		Axial	Head	4i	1	5	20	120	320	Std	61.43	491.41	75	16	
21.2	Routine Head 2s Group1		Axial	Head	4i	2	5	20	140	100	Std	53.34	320.02	55	16	Routine head protocol for evaluation of the brain for abnormalities.
	Group2		Axial	Head	4i	2	5	20	120	160	Std	61.43	491.41	75	16	
21.3	Trauma Head 2.5mm		Axial	Head	8i	0.5	2.5	20	140	560	Std	76.76	1074.71	137.5	16	Emergency head protocol for evaluation of the brain and cranium for abnormalities.
21.4	Routine Head 0.5s Group1		Axial	Head	4i	0.5	5	20	140	400	Std	54.83	328.99	55	16	Routine head protocol for evaluation of the brain for abnormalities.
	Group2		Axial	Head	4i	0.5	5	20	120	600	Std	59.20	473.62	75	16	
21.5	Circle of Willis 0.4s	MIROI	Axial	Head	1i	1	5 [0]	5	120	50	Std	125.58	62.79	0	16	Test bolus for CT angiography of the head for evaluation of carotid and cerebral vasculature.
			Helical	Head	0.531:1 10.625	0.4	0.625	20	120	300	Std Plus IQE	43.36	421.37	80	16	CT angiography of the head for evaluation of carotid and cerebral vasculature.
21.6	CoW/ Carotid 0.625mm 0.4s.Smart mA	MIROI	Axial	Head	1i	1	5 [0]	5	120	80	Std	180.84	90.42	0	16	Test bolus for CT angiography of the head for evaluation of carotid and cerebral vasculature.
			Helical	Head	0.531:1 10.625	0.4	0.625	20	120	SmartmA 100-600mA NI=8 Avg mA 595	Std Plus IQE	88.41	1657.19	170	16	CT angiography of the head for evaluation of carotid and cerebral vasculature.
21.8	Helical Head	DMPR	Helical	Head	0.531:1 10.625	0.5	0.625	20	120	300	Std Plus IQE	54.20	878.85	145	16	Helical scan mode for evaluation of the brain for cerebral abnormality.
21.10	VHS Head 4D CTA	MIROI	Axial	Head	1i	1	5 [0]	5	120	50	Std	125.58	62.79	0	16	Test bolus for CT angiography of the head for evaluation of carotid and cerebral vasculature.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (s)	Slice Thickness (mm) [Interval] J* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (mm)	Phantom (cm)	Description
			Helical-S	Head	0.984:1 39.375 18 pass	0.4	5 [10]	40	120	300	Std Plus	496.19	6202.39	110	16	CT angiography of the head using Volume Helical scan mode for evaluation of cerebral vasculature.
21.11	VHS Perfusion 350-370 Strength Contrast		Axial	Head	4i	1	5	20	120	335	Std	64.31	900.28	135	16	Non-Enhance Brain, evaluation for hemorrhage or infarction.
		Perfusion	Helical-S/P	Head	0.984:1 39.375 28 pass	0.4	5	40	80	200	Std	186.02	2325.25	110	16	CT Perfusion using Volume Helical scan mode for evaluation of cerebral perfusion over time.
			Axial	Head	4i	1	5	20	120	335	Std	64.31	900.28	135	16	Post Contrast Brain, Evaluation for residual tissue enhancement.
21.12	CT Perfusion 300-320 Strength Contrast		Axial	Head	4i	1	5	20	120	335	Std	64.31	900.28	135	16	Non-Enhance Brain, evaluation for hemorrhage or infarction.
		Perfusion	Cine	Head	8i	1	5 [0]	40	80	200	Std	652.82	2611.27	35	16	CT Perfusion using Cine scan mode for evaluation of cerebral perfusion over time.
			Axial	Head	4i	1	5	20	120	335	Std	64.31	900.28	135	16	Non-Enhance Brain, evaluation for hemorrhage or infarction.
21.13	CT Perfusion 350-370 Strength Contrast		Axial	Head	4i	1	5	20	120	335	Std	64.31	900.28	135	16	Non-Enhance Brain, evaluation for hemorrhage or infarction.
		Perfusion	Cine	Head	8i	1	5 [0]	40	80	200	Std	587.55	2350.19	35	16	CT Perfusion using Cine scan mode for evaluation of cerebral perfusion over time.
			Axial	Head	4i	1	5	20	120	335	Std	64.31	900.28	135	16	Post Contrast Brain, Evaluation for residual tissue enhancement.
21.14	CT Perfusion Brain Tumor - Dual Phase		Axial	Head	4i	1	5	20	120	335	Std	64.31	900.28	135	16	Non-Enhance Brain, evaluation for hemorrhage or infarction.
	Group 1	Perfusion	Cine	Head	8i	1	5 [0]	40	80	200	Std	587.55	2350.19	35	16	CT Perfusion using Cine scan mode for evaluation of cerebral perfusion over time.
	Group 2	Perfusion	Axial	Head	8i	1	5 [0]	40	80	200	Std	130.54	522.15	35	16	Delayed phase CT Perfusion using Axial scan mode for evaluation of Brain tumor permeability.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (s)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (mm)	Phantom (cm)	Description
			Axial	Head	4i	1	5	20	120	335	Std	64.31	900.28	135	16	Post Contrast Brain, Evaluation for residual tissue enhancement.
21.15	CT Perfusion 350-370 Strength Contrast (Axial)		Axial	Head	4i	1	5	20	120	335	Std	64.31	900.28	135	16	Non-Enhance Brain, evaluation for hemorrhage or infarction.
		Perfusion	Axial	Head	8i	1	5 [0]	40	80	200	Std	287.18	1148.73	35	16	CT Perfusion using Axial scan mode for evaluation of cerebral perfusion over time.
			Axial	Head	4i	1	5	20	120	335	Std	64.31	900.28	135	16	Post Contrast Brain, Evaluation for residual tissue enhancement.
21.16	CT Perfusion 350-370 Strength Contrast (Axial Shuttle Mode)		Axial	Head	4i	1	5	20	120	335	Std	64.31	900.28	135	16	Non-Enhance Brain, evaluation for hemorrhage or infarction.
		Perfusion	Axial-S	Head	8i 17 pass	0.4	5	40	80	500	Std	221.91	1775.31	75	16	CT Perfusion using Volume Shuttle (Axial) scan mode for evaluation of cerebral perfusion over time.
			Axial	Head	4i	1	5	20	120	335	Std	64.31	900.28	135	16	Post Contrast Brain, Evaluation for residual tissue enhancement.

Orbit

Table 18-2

[Interval]* - For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy-cm)	Scan Length (cm)	Phantom (cm)	Description
22.1	Sinus Supine Helical + DMPR	DMPR	Helical	Head	0.531:1 10.625	0.4	0.625	20	120	120	Bone Plus IQE	17.34	158.79	74.375	16	Routine Helical Sinus protocol, Evaluation for any abnormality
22.3	Orbits Helical + DMPR	DMPR	Helical	Head	0.531:1 10.625	0.5	0.625	20	120	220	Std Plus IQE	39.75	314.09	61.875	16	Routine Helical Orbit protocol, Evaluation for abnormality soft tissues and bone structures of the face.
22.6	Helical IAC 0.625mm Axial mode - Use Reformat for Coronal		Helical	Head	0.531:1 10.625	0.6	0.625	20	140	270	Std Plus IQE	81.32	454.44	38.75	16	Axial scan mode, Evaluation for inner ear abnormalities in Axial plane
22.8	Axial and Coronal IAC		Axial	Head	16i	1	1.25	20	140	280	Std	74.67	298.68	38.75	16	Helical scan mode, Evaluation for inner ear abnormalities in Axial plane
			Axial	Head	16i	1	1.25	20	140	335	Std	89.34	357.35	38.75	16	Helical scan mode, Evaluation for inner ear abnormalities in Coronal plane

Neck

Table 18-3

[Interval]* - For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (cm)	Phantom (cm)	Description
23.1	C-Spine C5-C7 Axial		Axial	SmallBody	8i	0.5	2.5	20	140	610	Std	33.17	464.38	137.5	32	C-Spine C5-C7 area Axial mode, Evaluation for soft tissue, disc and bone abnormalities.
23.2	C-Spine C1-C4 Axial		Axial	SmallBody	8i	1	2.5	20	120	240	Std	17.90	143.19	77.5	32	C-Spine C1-C4 area Axial mode, Evaluation for soft tissue, disc and bone abnormalities.
23.3	C-Spine Helical + DMPR	DMPR	Helical	SmallBody	0.516:1 20.625	0.8	0.625	40	120	335	Bone Plus IQE	36.23	522.92	110	32	C-Spine Helical mode with DMPR, Evaluation for soft tissue, disc and bone abnormalities.
23.4	C-Spine Helical + DMPR/ SmartmA	DMPR	Helical	SmallBody	0.516:1 20.625	0.8	0.625	40	120	SmartmA 100-600mA NI=11.34 Avg mA 310	Bone Plus IQE	33.52	483.89	110	32	C-Spine Helical mode with DMPR and SmartmA, Evaluation for soft tissue, disc and bone abnormalities.
23.8	Carotid 0.625mm 0.4s. 0.5:1 SmartmA/ SmartPrep		Helical	SmallBody	0.516:1 20.625	0.4	0.625	40	140	SmartmA 100-610mA NI=11 Avg mA 575	Std Plus IQE	44.10	902.52	170	32	Carotid CTA (pitch 0.5:1) with SmartmA and SmartPrep, Evaluation of carotid vasculature.
23.9	Carotid 0.625mm 0.4s. 0.9:1 SmartmA/ SmartPrep		Helical	SmallBody	0.984:1 39.375	0.4	0.625	40	120	SmartmA 80-610mA NI=11 Avg mA 625	Std Plus IQE	17.70	359.81	170	32	Carotid CTA (pitch 0.9:1) with SmartmA and SmartPrep, Evaluation of carotid vasculature.
23.10	Carotid/ CoW 0.625mm 0.4s. AutomA	MIROI	Axial	SmallBody	1i	1	5 [0]	5	120	80	Std	99.00	49.50	0	32	Test bolus for Carotid CTA (pitch 0.5:1) of the head for evaluation of carotid and cerebral vasculature.
			Helical	SmallBody	0.516:1 20.625	0.4	0.625	40	140	AutomA 80-515mA NI=11 Avg mA 515	Std Plus IQE	39.50	808.35	170	32	Carotid and Brain CTA (pitch 0.5:1) with AutomA, Evaluation of carotid and cerebral vasculature.
23.11	Carotid/ CoW 0.625mm 0.4s. SmartmA/ SmartPrep		Helical	SmallBody	0.969:1 19.375	0.4	0.625	20	140	SmartmA 100-610mA NI=11 Avg mA 600	Std Plus IQE	26.94	508.94	170	32	Carotid CTA (pitch 0.9:1) with SmartmA and SmartPrep, Evaluation of carotid and cerebral vasculature.
23.12	Soft Tissue Neck 2.5mm AutomA		Helical	SmallBody	0.984:1 39.375	0.5	2.5	40	120	AutomA 100-335mA NI=9.1 Avg mA 250	Std Plus	8.85	161.79	150	32	Neck with AutomA, Evaluation of soft tissue structures for abnormality

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy-cm)	Scan Length (cm)	Phantom (cm)	Description
23.13	Soft Tissue Neck + DMPR/ SmartmA	DMPR	Helical	SmallBody	0.969:1 19.375	0.5	0.625	20	120	SmartmA 100-600mA NI=12.6 Avg mA 425	Std Plus IQE	16.82	283.95	150	32	Neck with DMPR and SmartmA, Evaluation of soft tissue structures for abnormality

Upper Extremity

Table 18-4

[Interval]* - For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy-cm)	Scan Length (cm)	Phantom (cm)	Description
24.1	Shoulder 1.25mm + DMPR	DMPR	Helical	LargeBody	0.516:1 20.625	0.6	1.25 [0.625]	40	140	360	Std Plus IQE	47.03	632.79	100	32	Shoulder with DMPR, Evaluation of soft tissue and bone of shoulder area
24.2	Shoulder 1.25mm + DMPR/ SmartmA	DMPR	Helical	LargeBody	0.516:1 20.625	0.6	1.25 [0.625]	40	140	SmartmA 100-610mA NI=20 Avg mA 450	Std Plus IQE	58.78	790.99	100	32	Shoulder with DMPR and SmartmA, Evaluation of soft tissue and bone of shoulder area

Chest

Table 18-5

[Interval]* - For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (second s)	Slice Thickne ss (mm) [Interval]* (mm)	Beam Collimat ion (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy·c m)	Scan Length (cm)	Phantom (cm)	Description
25.1	Routine Chest 0.4s. 5mm SmartmA		Helical	LargeBody	1.375:1 55.0	0.4	5	40	120	SmartmA 100-650mA NI=13 Avg mA 500	Std Plus	11.52	286.26	200	32	Chest with SmartmA, Evaluation of mediastinum and lungs fields
25.2	Routine Chest 0.4s.+ DMPR/ SmartmA	DMPR	Helical	LargeBody	0.984:1 39.375	0.4	0.625	40	120	SmartmA 100-700mA NI=22 Avg mA 620	Std Plus	19.96	465.64	200	32	Chest with SmartmA and DMPR, Evaluation of mediastinum and lungs fields
25.3	Chest with HiRes 0.4s. SmartmA		Helical	LargeBody	0.969:1 19.375	0.4	5	20	120	SmartmA 80-650mA NI=12 Avg mA 500	Chest Plus	17.99	393.87	200	32	Chest with SmartmA, Evaluation of mediastinum and lung fields with higher resolution images of the lungs
25.4	Hi Res Chest 0.5s. Axial		Axial	LargeBody	1i	0.5	1.25 [10]	1.25	120	380	Bone+	4.48	107.55	230	32	Axial scan mode, Evaluation of mediastinum and lung fields in high resolution mode
25.5	Chest Abd Pelvis 0.4/ 0.5s. SmartmA 5mm (2 groups) Group1		Helical	LargeBody	0.984:1 39.375	0.4	5	40	120	SmartmA 100-650mA NI=13 Avg mA 370	Std Plus	11.91	218.32	150	32	Multi group scan for Chest, Abdomen, Pelvis (0.4/0.5 sec) with SmartmA, Evaluation of chest and abdominal anatomy for abnormalities
	Group2		Helical	LargeBody	1.375:1 55.0	0.5	5	40	120	SmartmA 100-650mA NI=13 Avg mA 370	Std Plus	10.66	365.90	295	32	
25.6	Chest Abd Pelvis 0.5s + DMPR SmartPrep	DMPR	Helical	LargeBody	1.375:1 55.0	0.5	0.625	40	120	SmartmA 100-700mA NI=22 Avg mA 540	Std Plus	15.56	775.18	450	32	Helical scan mode (0.5sec) with SmartmA, Evaluation of chest and abdominal anatomy for abnormalities
25.7	Pulmonary Embolis 0.5s.	MIROI	Axial	LargeBody	1i	1	5 [0]	5	120	80	Std	93.89	46.95	0	32	Timing Bolus for Helical scan mode 1.25mm thickness and helical pitch of 0.9:1, evaluation of Chest for pulmonary embolism
			Helical	LargeBody	0.984:1 39.375	0.5	1.25	40	120	580	Std	23.34	497.55	180	32	Helical scan mode 1.25mm thickness and helical pitch of 0.9:1, Evaluation Chest for pulmonary embolism

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy-cm)	Scan Length (cm)	Phantom (cm)	Description
25.8	Pulmonary Embolism 0.5s. SmartmA	MIROI	Axial	LargeBody	1i	1	5 [0]	5	120	80	Std	93.89	46.95	0	32	Timing Bolus for Helical scan mode 1.25mm thickness and helical pitch of 0.9:1 with SmartmA, evaluation of Chest for pulmonary embolism
			Helical	LargeBody	0.984:1 39.375	0.5	1.25	40	120	SmartmA 100-600mA NI=20 Avg mA 500	Std	20.12	428.92	180	32	Helical scan mode 1.25mm thickness and helical pitch of 0.9:1 with SmartmA, Evaluation Chest for pulmonary embolism
25.9	Aortic Dissection 0.625mm + DMPR/ SmartmA/ SmartPrep	DMPR	Helical	LargeBody	1.375:1 55.0	0.5	0.625	40	120	SmartmA 100-700mA NI=22 Avg mA 600	Std Plus	17.29	602.00	300	32	Helical scan mode 0.625mm thickness and helical pitch of 1.375:1 with SmartmA, SmartPrep and DMPR, Evaluation for aortic dissection
25.10	Aortic Dissection 1.25mm + DMPR/ SmartmA/ SmartPrep	DMPR	Helical	LargeBody	1.375:1 55.0	0.5	1.25 [0.8]	40	120	SmartmA 100-700mA NI=20 Avg mA 550	Std Plus	15.85	551.83	300	32	Helical scan mode 1.25mm thickness and helical pitch of 1.375:1 with SmartmA, SmartPrep and DMPR, Evaluation for aortic dissection
25.11	Lung Assessment	Advanced Lung Analysis	Helical	LargeBody	0.969:1 19.375	0.5	1.25	20	120	120	Bone	5.25	114.77	200	32	Helical scan mode for acquisition of data for Advanced Lung Analysis (ALA) software for lung nodule assessment
25.12	Gated Chest Cardiac Helical		Helical	LargeBody	1.375:1 55.0	0.4	5	40	120	60	Std Plus	1.38	13.55	50	32	Localizer scan for Helical Gated scan for evaluation of thoracic vasculature
		MIROI	Axial	LargeBody	1i	1	5 [0]	5	120	60	Std	98.59	49.29	0	32	Timing bolus scan for Helical Gated scan for evaluation of thoracic vasculature
			Cardiac SEG	CardiacLarge	0.180:1 7.2	0.4	0.625	40	120	ECG Mod 200-395mA Phase 70-80 Avg mA 395	Std SS Seg	50.74	1699.82	300	32	Helical Gated scan for evaluation of thoracic vasculature
25.13	SmartScore - Gated 0.4s	Smart Score	Cine	LargeBody	8i	0.4	2.5	20	120	375	Std Segment	9.15	91.51	97.5	32	Cine gated scan mode for acquisition of data for coronary artery calcification analysis
25.14	SnapShot Segment 30-74 BPM		Helical	LargeBody	1.375:1 55.0	0.4	5	40	120	60	Std Plus	1.38	13.55	50	32	Localizer scan for Retrospectively gated Cardiac Helical scan for evaluation of coronary arteries, intended for heart rates 30-74 bpm

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy-cm)	Scan Length (cm)	Phantom (cm)	Description
		MIROI	Axial	LargeBody	1i	1	5 [0]	5	120	60	Std	98.59	49.29	0	32	Timing bolus scan for Retrospectively gated Cardiac Helical scan for evaluation of coronary arteries, intended for heart rates 30-74 bpm
			Cardiac SEG	CardiacLarge	0.180:1 7.2	0.4	0.625	40	120	ECG Mod 200-395mA Phase 70-80 Avg mA 395	Std SS Seg	52.44	707.94	100	32	Retrospectively gated Cardiac Helical scan for evaluation of coronary arteries, intended for heart rates 30-74 bpm
25.15	SnapShot Burst 75-113 BPM		Helical	LargeBody	1.375:1 55.0	0.4	5	40	120	60	Std Plus	1.38	13.55	50	32	Localizer scan for Retrospectively gated Cardiac Helical scan for evaluation of coronary arteries, intended for heart rates 75-113 bpm
		MIROI	Axial	LargeBody	1i	1	5 [0]	5	120	60	Std	98.59	49.29	0	32	Timing bolus scan for Retrospectively gated Cardiac Helical scan for evaluation of coronary arteries, intended for heart rates 75-113 bpm
			Cardiac SSB	CardiacLarge	0.180:1 7.2	0.4	0.625	40	120	ECG Mod 200-395mA Phase 70-80 Avg mA 395	Std SSB	52.44	707.94	100	32	Retrospectively gated Cardiac Helical scan for evaluation of coronary arteries, intended for heart rates 75-113 bpm
25.16	SnapShot Burst Plus 114+ BPM		Helical	LargeBody	1.375:1 55.0	0.4	5	40	120	60	Std Plus	1.38	13.55	50	32	Localizer scan for Retrospectively gated Cardiac Helical scan for evaluation of coronary arteries, intended for heart rates over 113 bpm
		MIROI	Axial	LargeBody	1i	1	5 [0]	5	120	60	Std	98.59	49.29	0	32	Timing bolus scan for Retrospectively gated Cardiac Helical scan for evaluation of coronary arteries, intended for heart rates over 113 bpm
			Cardiac SSBP	CardiacLarge	0.180:1 7.2	0.4	0.625	40	120	ECG Mod 200-395mA Phase 70-80 Avg mA 395	Std SSB	52.44	707.94	100	32	Retrospectively gated Cardiac Helical scan for evaluation of coronary arteries, intended for heart rates over 113 bpm
25.17	Advantage 4D		Helical	LargeBody	1.375:1 55.0	0.6	5	40	120	285	Std Plus	9.85	244.08	200	32	Localizer scan for Advantage 4D series
		Advantage 4D	Cine	LargeBody	8i	1	2.5	20	120	100	Std	50.95	815.24	157.5	32	Cine scan mode for acquisition data for Advantage 4D for retrospective gating
25.18	VHS Chest 4D CTA SmartPrep		Helical-S	LargeBody	1.375:1 55.0 11 pass	0.4	5 [10]	40	120	SmartmA 100-600mA NI=12 Avg mA 600	Std Plus	192.25	6055.91	300	32	Volume Helical scan mode with SmartPrep for timr resolved CT angiography of chest vasculature

Abdomen

Table 18-6

[Interval]* - For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy-cm)	Scan Length (cm)	Phantom (cm)	Description
26.1	Abdomen Pelvis 0.5s. 5mm SmartmA		Helical	LargeBody	1.375:1 55.0	0.5	5	40	120	SmartmA 50-650mA NI=11.57 Avg mA 500	Std Plus	14.41	645.73	400	32	Basic Abdomen Pelvis (0.5sec) with Smart mA, Evaluation for abnormalities of abdominal and pelvic structures
26.2	Abdomen Pelvis 0.5s. 5mm + DMPR SmartmA/ SmartPrep	DMPR	Helical	LargeBody	0.984:1 39.375	0.6	0.625	40	120	SmartmA 80-700mA NI=18 Avg mA 520	Std Plus	25.11	1087.62	400	32	Basic Abdomen Pelvis (0.5sec) with Smart mA, SmartPrep and DMPR, Evaluation for abnormalities of abdominal and pelvic structures
26.4	Renal Stone 0.5s. 5mm		Helical	LargeBody	1.375:1 55.0	0.5	5	40	120	380	Std Plus	10.95	490.75	400	32	Helical scan, Evaluation of kidney, ureters and bladder for renal stones
26.5	Renal Stone 0.5s. 5mm SmartmA		Helical	LargeBody	1.375:1 55.0	0.5	5	40	120	SmartmA 50-650mA NI=12 Avg mA 380	Std Plus	10.95	490.75	400	32	Helical scan with SmartmA, Evaluation of kidney, ureters and bladder for renal stones
26.7	AAA 0.5s. + DMPR 0.625mm	MIROI	Axial	LargeBody	1i	1	5 [0]	5	120	80	Std	112.67	56.34	0	32	Timing bolus for Helical scan with DMPR, Evaluation for abdominal aortic aneurysm
		DMPR	Helical	LargeBody	1.375:1 55.0	0.5	0.625	40	140	515	Std Plus	21.02	751.82	309.375	32	Helical scan with DMPR, Evaluation for abdominal aortic aneurysm
26.8	AAA 0.6s. + DMPR 0.625mm SmartmA/ SmartPrep	DMPR	Helical	LargeBody	1.375:1 55.0	0.6	0.625	40	120	SmartmA 100-700mA NI=19 Avg mA 550	Std Plus	19.02	679.85	309.375	32	Helical scan with DMPR, Smart mA and SmartPrep, Evaluation for abdominal aortic aneurysm
26.10	Dual Phase Liver		Helical	LargeBody	1.375:1 55.0	0.6	5	40	120	300	Std Plus	10.37	184.32	130	32	Non contrast evaluation of upper abdominal structures including liver
		MIROI	Axial	LargeBody	1i	1	5 [0]	5	120	80	Std	112.67	56.34	0	32	Timing bolus for Multi-Group acquisition for early arterial and portal venous phase of the liver
	Group1		Helical	LargeBody	1.375:1 55.0	0.5	2.5	40	120	550	Std Plus	15.85	282.44	130	32	Acquisition for early arterial phase of the liver
	Group2		Helical	LargeBody	1.375:1 55.0	0.5	5	40	120	420	Std Plus	12.10	542.41	400	32	Acquisition for portal venous phase of the liver

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy-cm)	Scan Length (cm)	Phantom (cm)	Description
26.11	Tri Phase Liver		Helical	LargeBody	1.375:1 55.0	0.6	5	40	120	300	Std Plus	10.37	184.32	130	32	Non contrast evaluation of upper abdominal structures including liver
		MIROI	Axial	LargeBody	1i	1	5 [0]	5	120	80	Std	112.67	56.34	0	32	Timing bolus for Multi-Group acquisition for early arterial, late arterial and portal venous phase of the liver
	Group1		Helical	LargeBody	1.375:1 55.0	0.5	2.5	40	120	550	Std Plus	15.85	282.44	130	32	Acquisition for early arterial phase of the liver
	Group2		Helical	LargeBody	1.375:1 55.0	0.5	5	40	120	450	Std Plus	12.97	581.15	400	32	Acquisition for late arterial phase of the liver
	Group3		Helical	LargeBody	1.375:1 55.0	0.5	5	40	120	450	Std Plus	12.97	581.15	400	32	Acquisition for portal venous phase of the liver
26.12	Dual Phase Pancreas SmartmA/ SmartPrep		Helical	LargeBody	0.984:1 39.375	0.5	5	40	120	SmartmA 80-650mA NI=12 Avg mA 400	Std Plus	16.10	166.06	70	32	Non contrast evaluation of upper abdominal structures including pancreas
	Group1		Helical	LargeBody	0.984:1 39.375	0.5	2.5	40	120	SmartmA 50-700mA NI=15 Avg mA 520	Std Plus	20.93	383.30	150	32	Acquisition with SmartmA and SmartPrep for arterial phase of the pancreas
	Group2		Helical	LargeBody	0.984:1 39.375	0.5	5	40	120	SmartmA 50-700mA NI=15 Avg mA 450	Std Plus	18.11	639.58	320	32	Acquisition for venous phase of the pancreas
26.13	Dual Phase Liver SmartmA/ SmartPrep		Helical	LargeBody	1.375:1 55.0	0.5	5	40	120	SmartmA 80-500mA NI=12 Avg mA 320	Std Plus	9.22	163.92	130	32	Non contrast evaluation of upper abdominal structures including liver
	Group1		Helical	LargeBody	1.375:1 55.0	0.5	2.5	40	120	SmartmA 50-700mA NI=15 Avg mA 550	Std Plus	15.85	282.44	130	32	Acquisition with SmartmA and SmartPrep for early arterial phase of the liver
	Group2		Helical	LargeBody	0.984:1 39.375	0.5	5	40	120	SmartmA 50-700mA NI=11 Avg mA 450	Std Plus	18.11	784.46	400	32	Acquisition with SmartmA and SmartPrep for portal venous phase of the liver
26.14	Tri Phase Liver SmartmA/ SmartPrep		Helical	LargeBody	0.984:1 39.375	0.5	5	40	120	SmartmA 80-500mA NI=12 Avg mA 400	Std Plus	16.10	262.65	130	32	Non contrast evaluation of upper abdominal structures including liver
	Group1		Helical	LargeBody	1.375:1 55.0	0.5	2.5	40	120	SmartmA 50-700mA NI=15 Avg mA 550	Std Plus	15.85	282.44	130	32	Acquisition with SmartmA and SmartPrep for early arterial phase of the liver
	Group2		Helical	LargeBody	1.375:1 55.0	0.5	5	40	120	SmartmA 50-700mA NI=11 Avg mA 450	Std Plus	12.97	581.15	400	32	Acquisition with SmartmA and SmartPrep for late arterial phase of the liver
	Group3		Helical	LargeBody	1.375:1 55.0	0.5	5	40	120	SmartmA 50-700mA NI=11 Avg mA 450	Std Plus	12.97	581.15	400	32	Acquisition with SmartmA and SmartPrep for portal venous phase of the liver
26.15	VHS Liver Perfusion		Helical	LargeBody	0.984:1 39.375	0.6	5	40	120	170	Std	8.21	133.61	130	32	Non contrast evaluation of upper abdominal structures including liver

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy-cm)	Scan Length (cm)	Phantom (cm)	Description
		Perfusion	Helical-S/P	LargeBody	1.375:1 55.0 21 pass	0.4	5	40	120	300	Std Plus	219.99	3409.79	140	32	Volume Helical scan mode for evaluation of liver tumor perfusion over time
26.16	VHS 4D CTA		Helical	LargeBody	0.984:1 39.375	0.6	5	40	120	340	Std	16.42	267.79	130	32	Localizer scan for Volume Helical Shuttle scan mode for evaluation of abdominal vasculature
		MIROI	Axial	LargeBody	1i	1	5 [0]	5	120	80	Std	112.67	56.34	0	32	Timing bolus scan for Volume Helical Shuttle scan mode for evaluation of abdominal vasculature
			Helical-S	LargeBody	1.375:1 55.0 18 pass	0.5	5 [10]	40	140	SmartmA 150-515mA NI=12 Avg mA 350	Std Plus	325.68	6350.70	180	32	Volume Helical Shuttle scan mode for time resolved CT angiography of abdominal vasculature
26.17	Runoff 0.625mm	MIROI	Axial	LargeBody	1i	1	5 [0]	5	120	80	Std	112.67	56.34	0	32	Timing bolus scan for evaluation of vascular structure of abdomen, femurs and lower extremities (Runoff)
			Helical	LargeBody	1.375:1 55.0	0.5	0.625	40	140	515	Std Plus	21.02	2071.05	936.875	32	Helical scanmode 0.625mm, Evaluation of vascular structure of abdomen, femurs and lower extremities (Runoff)
26.18	Runoff 1.25mm	MIROI	Axial	LargeBody	1i	1	5 [0]	5	120	80	Std	112.67	56.34	0	32	Timing bolus scan for evaluation of vascular structure of abdomen, femurs and lower extremities (Runoff)
			Helical	LargeBody	1.375:1 55.0	0.5	1.25 [0.7]	40	140	400	Std Plus	16.33	1709.86	998.9	32	Helical scanmode 1.25mm, Evaluation of vascular structure of abdomen, femurs and lower extremities (Runoff)
26.19	Runoff 0.625mm SmartmA/ SmartPrep		Helical	LargeBody	1.375:1 55.0	0.5	0.625	40	140	SmartmA 100-515mA NI=19 Avg mA 515	Std Plus	21.02	2071.05	936.875	32	Helical scanmode 0.625mm with SmartmA and SmartPrep, Evaluation of vascular structure of abdomen, femurs and lower extremities (Runoff)
26.20	Runoff 1.25mm SmartmA/ SmartPrep		Helical	LargeBody	1.375:1 55.0	0.5	1.25 [0.7]	40	140	SmartmA 100-515mA NI=20 Avg mA 400	Std Plus	16.33	1709.86	998.9	32	Helical scanmode 1.25mm with SmartmA and SmartPrep, Evaluation of vascular structure of abdomen, femurs and lower extremities (Runoff)
26.22	CT Colonography 1.25mm	Advantage CTC Pro	Helical	LargeBody	1.375:1 55.0	0.5	1.25	40	120	200	Std Plus	5.76	258.04	400	32	Helical scan mode supine acquisition for using Advantage CTC Pro for evaluation of the colon
		Advantage CTC Pro	Helical	LargeBody	1.375:1 55.0	0.5	1.25	40	120	200	Std Plus	5.76	258.04	400	32	Helical scan mode prone acquisition for using Advantage CTC Pro for evaluation of the colon

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy-cm)	Scan Length (cm)	Phantom (cm)	Description
26.41	Body Interventional 5mm/3i		Helical	LargeBody	1.375:1 27.5	0.5	5	20	120	SmartmA 50-650mA NI=11.57 Avg mA 360	Std	11.41	251.58	195	32	Localization scan for CT interventional scan
			SmartView SEG	LargeBody	3i	0.8	5	10	120	80	Std Segment	684.58	684.58	0	32	CT Fluoro scan for interventional scan
26.42	Body Interventional 5mm/1i		Helical	LargeBody	1.375:1 27.50	0.5	5	20	120	SmartmA 50-650mA NI=11.57 Avg mA 360	Std	11.41	251.58	195	32	Localization scan for CT interventional scan
			SmartView SEG	LargeBody	1i	0.8	5	5	120	80	Std Segment	845.04	422.52	0	32	CT Fluoro scan for interventional scan

L-Spine

Table 18-7

[Interval]* - For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy-cm)	Scan Length (cm)	Phantom (cm)	Description
27.1	L-Spine 3 Level Axial Group1		Axial	LargeBody	8i	1	2.5	20	120	320	Std	27.13	54.26	17.5	32	Axial scan mode, Evaluation of soft tissue and bony structures lumbar spine
	Group2		Axial	LargeBody	8i	1	2.5	20	120	320	Std	27.13	54.26	17.5	32	Axial scan mode, Evaluation of soft tissue and bony structures lumbar spine
	Group3		Axial	LargeBody	8i	1	2.5	20	140	360	Std	44.46	88.91	17.5	32	Axial scan mode, Evaluation of soft tissue and bony structures lumbar spine
27.3	L-Spine Survey Helical 2.5mm 0.6s. SmartmA		Helical	LargeBody	0.516:1 20.625	0.6	2.5	40	140	SmartmA 100-600mA NI=14 Avg mA 540	Std Plus	70.54	1478.24	75	32	Helical scan mode 2.5mm (0.6sec), Limited evaluation of lumbar spine
27.4	L-Spine Survey Helical 2.5mm 0.8s. SmartmA		Helical	LargeBody	0.516:1 20.625	0.8	2.5	40	140	SmartmA 100-600mA NI=14 Avg mA 525	Std Plus	91.44	1915.92	75	32	Helical scan mode 2.5mm (0.8sec), Limited evaluation of lumbar spine
27.6	L-Spine Helical 0.625mm + DMPR/ SmartmA	DMPR	Helical	LargeBody	0.516:1 20.625	0.6	0.625	40	140	SmartmA 100-600mA NI=20 Avg mA 610	Std Plus IQE	79.69	1669.87	75	32	Helical scan mode 0.625mm with DMPR and SmartmA, Evaluation of soft tissue and bony structures lumbar spine
27.7	L-Spine Helical 1.25mm + DMPR/ SmartmA - Large Patient	DMPR	Helical	LargeBody	0.516:1 20.625	1	1.25	40	140	SmartmA 100-600mA NI=18 Avg mA 600	Std Plus	130.63	2736.58	75	32	Helical scan mode 1.25mm with DMPR and SmartmA, Evaluation of soft tissue and bony structures lumbar spine

Pelvis

Table 18-8

[Interval]* - For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (cm)	Phantom (cm)	Description
28.1	Pelvis for Fracture + DMPR/ SmartmA	DMPR	Helical	LargeBody	0.984:1 39.375	0.8	1.25 [0.625]	40	120	SmartmA 100-700mA NI=20 Avg mA 500	Bone Plus IQE	32.20	432.30	101.25	32	Helical scan with DMPR and SmartmA, Evaluation of pelvis bony structures for fracture

Lower Extremity

Table 18-9

[Interval]* - For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (cm)	Phantom (cm)	Description
29.1	Lower Extremity Survey 2.5mm		Helical	LargeBody	0.516:1 20.625	0.5	2.5	40	120	250	Std Plus	19.21	181.35	60	32	Helical scan mode 2.5mm, Limited evaluation of lower extremity for abnormalities
29.2	Ankle 0.625 mm Single Joint		Helical	MediumBody	0.531:1 10.625	0.5	0.625	20	120	250	Std Plus IQE	20.72	108.05	35	32	Helical scan mode 0.625mm, Evaluation of soft tissue and bony anatomy of the ankle area
29.3	Ankle 0.625 mm Both Joints		Helical	LargeBody	0.531:1 10.625	0.5	0.625	20	120	250	Std Plus IQE	19.95	104.03	35	32	Helical scan mode 0.625mm, Evaluation of soft tissue and bony anatomy of the ankle area
29.4	Knee 0.625 mm Both Joints		Helical	LargeBody	0.531:1 10.625	0.6	0.625	20	120	Auto mA 100-335mA NI=20 Avg mA 275	Std Plus IQE	26.33	137.27	35	32	Helical scan mode 0.625mm, Evaluation of soft tissue and bony anatomy of both knees

Reference Pediatric Protocols - Installed Options : VCT85kW and/or VCT Hi Power

Head

Table 19-1

[Interval]* - For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (mm)	Phantom (cm)	Description
31.1	PED HEAD TO 18 Months		Axial	Ped Head	4i	0.5	5	20	120	240	Std	20.43	245.19	115	16	Routine head (0.5s) for infant (up to 18 months) 5mm
31.2	PED HEAD 18 MOS TO 5YRS		Axial	SmallHead	4i	0.5	5	20	120	335	Std	28.52	342.25	115	16	Routine head (0.5s) for children (18 months to 5 years) 5mm
31.3	PED HEAD TO 5YRS TO 18YRS Group1		Axial	SmallHead	8i	1	2.5	20	120	240	Std	40.87	163.46	37.5	16	Routine head (1.0s) for children (5 years to 18 years) 2.5mm
	Group2		Axial	SmallHead	4i	1	5	20	120	180	Std	30.65	245.19	75	16	Routine head (1.0s) for children (5 years to 18 years) 5mm

Orbit

Table 19-2

[Interval]* - For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (mm)	Phantom (cm)	Description
32.1	Sinus Supine Helical + DMPR	DMPR	Helical	SmallHead	0.531:1 10.625	0.4	0.625	20	120	80	Bone Plus IQE	10.26	93.90	74.375	16	Routine Helical scan mode with DMPR, Evaluation of sinus area for abnormality in pediatric patients
32.3	Orbits Helical	DMPR	Helical	SmallHead	0.531:1 10.625	0.4	0.625	20	120	180	Std Plus IQE	23.08	182.43	61.875	16	Helical scan mode, Evaluation of orbital and face structures for abnormality in pediatric patients
32.6	Helical IAC 0.625mm Axial mode - Use Reformat for Coronal		Helical	SmallHead	0.531:1 10.625	0.4	0.625	20	120	160	Std Plus IQE	20.51	114.73	38.75	16	Helical scan mode, Evaluation of inner ear structures for abnormality in pediatric patients

Chest

Table 19-3

[Interval]* - For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (mm)	Phantom (cm)	Description
35.1.1	RC 6.0-7.5 kg (13.2-16.5 lbs)		Helical	PedBody	1.375:1 27.5	0.4	3.75	20	100	80	Std Plus	1.11	11.08	75	32	Weight and height based routine chest protocol
35.1.3	SnapShot Segment 30-74 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	120	Std Plus	0.84	8.19	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	Std	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
			Cardiac SEG	CardiacSmall	0.180:1 7.2	0.4	0.625	40	80	ECG Mod 85-280mA Phase 40-80 AvgmA280	Std SS Seg	12.76	140.31	75	32	Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
35.1.4	SnapShot Burst 75-104 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	120	Std Plus	0.84	8.19	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	Std	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
			Cardiac SSB	CardiacSmall	0.180:1 7.2	0.4	0.625	40	80	ECG Mod 85-280mA Phase 40-80 AvgmA280	Std SSB	12.76	140.31	75	32	Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
35.1.5	SnapShot Burst+ >105 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	120	Std Plus	0.84	8.19	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	Std	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
			Cardiac SSBP	CardiacSmall	0.180:1 7.2	0.4	0.625	40	80	ECG Mod 85-280mA Phase 35-85 AvgmA280	Std SSB	13.64	150.05	75	32	Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
35.1.7	Non ECG Gated		Helical	PedBody	0.984:1 39.375	0.4	0.625	40	80	SmartmA 50-200mA NI=18 AvgmA200	Std Plus	1.95	21.00	75	32	Weight and height based non ECG gated protocol for cardiac with SmartmA
35.2.1	RC 7.5-9.5 kg (16.5-20.9lbs)		Helical	PedBody	1.375:1 27.5	0.4	3.75	20	100	120	Std Plus	1.66	16.62	75	32	Weight and height based routine chest protocol
35.2.3	SnapShot Segment 30-74 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	120	Std Plus	0.84	8.19	50	32	Weight and height based protocol for localization

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (mm)	Phantom (cm)	Description
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	Std	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
			Cardiac SEG	CardiacSmall	0.180:1 7.2	0.4	0.625	40	80	ECG Mod 85-300mA Phase 40-80 AvgmA300	Std SS Seg	13.63	149.92	75	32	Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
35.2.4	SnapShot Burst 75-104 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	120	Std Plus	0.84	8.19	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	Std	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
			Cardiac SSB	CardiacSmall	0.180:1 7.2	0.4	0.625	40	80	ECG Mod 85-300mA Phase 40-80 AvgmA300	Std SSB	13.63	149.92	75	32	Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
35.2.5	SnapShot Burst+ >105 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	120	Std Plus	0.84	8.19	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	Std	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
			Cardiac SSBP	CardiacSmall	0.180:1 7.2	0.4	0.625	40	80	ECG Mod 85-300mA Phase 35-85 AvgmA300	Std SSB	14.61	160.66	75	32	Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
35.2.7	Non ECG Gated		Helical	PedBody	0.984:1 39.375	0.4	0.625	40	80	SmartmA 50-220mA NI=18 AvgmA220	Std Plus	2.14	23.10	75	32	Weight and height based non ECG gated protocol for cardiac with SmartmA
35.3.1	RC 9.5-11.5 kg (20.9-25.4lbs)		Helical	SmallBody	1.375:1 27.5	0.4	3.75	20	120	100	Std Plus	2.17	22.53	78.75	32	Weight and height based routine chest protocol
35.3.3	SnapShot Segment 30-74 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	120	Std Plus	0.84	8.19	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	Std	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
			Cardiac SEG	CardiacSmall	0.180:1 7.2	0.4	0.625	40	80	ECG Mod 100-320mA Phase 40-80 AvgmA320	Std SS Seg	14.59	160.53	75	32	Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
35.3.4	SnapShot Burst 75-104 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	120	Std Plus	0.84	8.19	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	Std	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (mm)	Phantom (cm)	Description
			Cardiac SSB	CardiacSmall	0.180:1 7.2	0.4	0.625	40	80	ECG Mod 100-320mA Phase 40-80 AvgmA320	Std SSB	14.59	160.53	75	32	Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
35.3.5	SnapShot Burst+ >105 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	120	Std Plus	0.84	8.19	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	Std	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
			Cardiac SSBP	CardiacSmall	0.180:1 7.2	0.4	0.625	40	80	ECG Mod 100-320mA Phase 35-85 AvgmA320	Std SSB	15.59	171.51	75	32	Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
35.3.7	Non ECG Gated		Helical	PedBody	0.984:1 39.375	0.4	0.625	40	80	SmartmA 50-240mA NI=18 AvgmA240	Std Plus	2.34	25.20	75	32	Weight and height based non ECG gated protocol for cardiac with SmartmA
35.4.1	RC 11.5-14.5kg (25.4-32.0lbs)		Helical	SmallBody	1.375:1 27.5	0.4	5	20	120	90	Std Plus	1.95	22.47	90	32	Weight and height based routine chest protocol
35.4.3	SnapShot Segment 30-74 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	120	Std Plus	0.84	8.19	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	Std	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
			Cardiac SEG	CardiacSmall	0.180:1 7.2	0.4	0.625	40	80	ECG Mod 100-340mA Phase 40-80 AvgmA340	Std SS Seg	15.45	169.99	75	32	Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
35.4.4	SnapShot Burst 75-104 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	120	Std Plus	0.84	8.19	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	Std	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
			Cardiac SSB	CardiacSmall	0.180:1 7.2	0.4	0.625	40	80	ECG Mod 100-340mA Phase 40-80 AvgmA340	Std SSB	15.45	169.99	75	32	Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
35.4.5	SnapShot Burst+ >105 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	120	Std Plus	0.84	8.19	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	Std	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
			Cardiac SSBP	CardiacSmall	0.180:1 7.2	0.4	0.625	40	80	ECG Mod 100-340mA Phase 35-85 AvgmA340	Std SSB	16.54	181.98	75	32	Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval] (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (mm)	Phantom (cm)	Description
35.4.7	Non ECG Gated		Helical	PedBody	0.984:1 39.375	0.4	0.625	40	80	SmartmA 50-280mA NI=18 AvgmA280	Std Plus	2.72	29.40	75	32	Weight and height based non ECG gated protocol for cardiac with SmartmA
35.5.1	RC 14.5-18.5kg (32.0-40.8lbs)		Helical	SmallBody	1.375:1 27.5	0.4	5	20	120	100	Std Plus	2.17	24.97	90	32	Weight and height based routine chest protocol
35.5.3	SnapShot Segment 30-74 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	120	Std Plus	0.84	8.19	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	Std	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
			Cardiac SEG	CardiacSmall	0.180:1 7.2	0.4	0.625	40	80	ECG Mod 115-380mA Phase 40-80 AvgmA380	Std SS Seg	17.30	190.30	75	32	Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
35.5.4	SnapShot Burst 75-104 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	120	Std Plus	0.84	8.19	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	Std	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
			Cardiac SSB	CardiacSmall	0.180:1 7.2	0.4	0.625	40	80	ECG Mod 115-380mA Phase 40-80 AvgmA380	Std SSB	17.30	190.30	75	32	Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
35.5.5	SnapShot Burst+ >105 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	120	Std Plus	0.84	8.19	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	Std	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
			Cardiac SSBP	CardiacSmall	0.180:1 7.2	0.4	0.625	40	80	ECG Mod 115-380mA Phase 35-85 AvgmA380	Std SSB	18.50	203.54	75	32	Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
35.5.7	Non ECG Gated		Helical	PedBody	0.984:1 39.375	0.4	0.625	40	80	SmartmA 50-300mA NI=18 AvgmA300	Std Plus	2.92	31.50	75	32	Weight and height based non ECG gated protocol for cardiac with SmartmA
35.6.1	RC 18.5-22.5kg (40.8-49.6lbs)		Helical	SmallBody	1.375:1 27.5	0.4	5	20	120	110	Std Plus	2.39	27.47	90	32	Weight and height based routine chest protocol
35.6.3	SnapShot Segment 30-74 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	120	Std Plus	0.84	8.19	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	Std	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval] (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (mm)	Phantom (cm)	Description
			Cardiac SEG	CardiacSmall	0.180:17.2	0.4	0.625	40	80	ECG Mod 115-400mA Phase 40-80 AvgmA400	Std SS Seg	18.17	199.84	75	32	Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
35.6.4	SnapShot Burst 75-104 BPM		Helical	SmallBody	1.375:155.0	0.4	2.5	40	80	120	Std Plus	0.84	8.19	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	Std	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
			Cardiac SSB	CardiacSmall	0.180:17.2	0.4	0.625	40	80	ECG Mod 115-400mA Phase 40-80 AvgmA400	Std SSB	18.17	199.84	75	32	Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
35.6.5	SnapShot Burst+ >105 BPM		Helical	SmallBody	1.375:155.0	0.4	2.5	40	80	120	Std Plus	0.84	8.19	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	Std	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
			Cardiac SSBP	CardiacSmall	0.180:17.2	0.4	0.625	40	80	ECG Mod 115-400mA Phase 35-85 AvgmA400	Std SSB	19.46	214.07	75	32	Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
35.6.7	Non ECG Gated		Helical	PedBody	0.984:139.375	0.4	0.625	40	80	SmartmA 100-320mA NI=18 AvgmA320	Std Plus	3.11	33.71	75	32	Weight and height based non ECG gated protocol for cardiac with SmartmA
35.7.1	RC 22.5-31.5kg (49.6-69.5lbs)		Helical	LargeBody	1.375:155.0	0.4	5	40	120	110	Std Plus	2.54	33.71	85	32	Weight and height based routine chest protocol
35.7.3	SnapShot Segment 30-74 BPM		Helical	SmallBody	1.375:155.0	0.4	2.5	40	80	120	Std Plus	0.84	8.19	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	Std	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
			Cardiac SEG	CardiacSmall	0.180:17.2	0.4	0.625	40	80	ECG Mod 120-420mA Phase 40-80 AvgmA420	Std SS Seg	19.07	209.80	75	32	Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
35.7.4	SnapShot Burst 75-104 BPM		Helical	SmallBody	1.375:155.0	0.4	2.5	40	80	120	Std Plus	0.84	8.19	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	Std	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
			Cardiac SSB	CardiacSmall	0.180:17.2	0.4	0.625	40	80	ECG Mod 120-420mA Phase 40-80 AvgmA420	Std SSB	19.07	209.80	75	32	Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval] (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (mm)	Phantom (cm)	Description
35.7.5	SnapShot Burst+ >105 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	120	Std Plus	0.84	8.19	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	Std	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
			Cardiac SSBP	CardiacSmall	0.180:1 7.2	0.4	0.625	40	80	ECG Mod 120-420mA Phase 35-85 AvgmA420	Std SSB	20.44	224.79	75	32	Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
35.7.7	Non ECG Gated		Helical	PedBody	0.984:1 39.375	0.4	0.625	40	80	SmartmA 100-360mA NI=18 AvgmA360	Std Plus	3.50	37.93	75	32	Weight and height based non ECG gated protocol for cardiac with SmartmA
35.8.1	RC 31.5-40.5kg (69.5-89.3lbs)		Helical	LargeBody	1.375:1 55.0	0.4	5	40	120	120	Std Plus	2.77	46.45	120	32	Weight and height based routine chest protocol
35.8.3	SnapShot Segment 30-74 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	100	120	Std Plus	1.55	15.20	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	100	50	Std	39.50	19.75	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
			Cardiac SEG	CardiacSmall	0.180:1 7.2	0.4	0.625	40	100	ECG Mod 115-380mA Phase 40-80 AvgmA380	Std SS Seg	32.11	353.20	75	32	Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
35.8.4	SnapShot Burst 75-104 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	100	120	Std Plus	1.55	15.20	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	100	50	Std	39.50	19.75	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
			Cardiac SSB	CardiacSmall	0.180:1 7.2	0.4	0.625	40	100	ECG Mod 115-380mA Phase 40-80 AvgmA380	Std SSB	32.11	353.20	75	32	Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
35.8.5	SnapShot Burst+ >105 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	100	120	Std Plus	1.55	15.20	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	100	50	Std	39.50	19.75	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
			Cardiac SSBP	CardiacSmall	0.180:1 7.2	0.4	0.625	40	100	ECG Mod 115-380mA Phase 35-85 AvgmA380	Std SSB	34.34	377.76	75	32	Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
35.8.7	Non ECG Gated		Helical	SmallBody	0.984:1 39.375	0.4	0.625	40	100	SmartmA 100-400mA NI=16 AvgmA400	Std Plus	7.22	78.21	75	32	Weight and height based non ECG gated protocol for cardiac with SmartmA

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval] (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (mm)	Phantom (cm)	Description
35.9.1	RC 40.5-55.0kg(89.3-121.3lbs)		Helical	LargeBody	1.375:1 55.0	0.4	5	40	120	130	Std Plus	3.00	59.31	150	32	Weight and height based routine chest protocol
35.9.3	SnapShot Segment 30-74 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	100	120	Std Plus	1.55	15.20	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	100	50	Std	39.50	19.75	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
			Cardiac SEG	CardiacSmall	0.180:1 7.2	0.4	0.625	40	100	ECG Mod 120-420mA Phase 40-80 AvgmA420	Std SS Seg	35.40	389.39	75	32	Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
35.9.4	SnapShot Burst 75-104 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	100	120	Std Plus	1.55	15.20	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	100	50	Std	39.50	19.75	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
			Cardiac SSB	CardiacSmall	0.180:1 7.2	0.4	0.625	40	100	ECG Mod 120-420mA Phase 40-80 AvgmA420	Std SSB	35.40	389.39	75	32	Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
35.9.5	SnapShot Burst+ >105 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	100	120	Std Plus	1.55	15.20	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	100	50	Std	39.50	19.75	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
			Cardiac SSBP	CardiacSmall	0.180:1 7.2	0.4	0.625	40	100	ECG Mod 120-420mA Phase 35-85 AvgmA420	Std SSB	37.93	417.20	75	32	Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
35.9.7	Non ECG Gated		Helical	SmallBody	0.984:1 39.375	0.4	0.625	40	100	SmartmA 100-440mA NI=16 AvgmA440	Std Plus	7.95	86.03	75	32	Weight and height based non ECG gated protocol for cardiac with SmartmA

Abdomen

Table 19-4

[Interval]* - For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickne ss (mm) [Interval]* (mm)	Beam Collimat ion (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy·c m)	Scan Length (mm)	Phantom (cm)	Description
36.1.1	Abd 6.0-7.5kg (13.2-16.5lbs)		Helical	PedBody	1.375:1 27.5	0.4	3.75	20	120	100	Std Plus	2.17	22.53	78.75	32	Weight and height based routine abdomen protocol
36.2.1	Abd 7.5-9.5kg (16.5-20.9lbs)		Helical	PedBody	1.375:1 27.5	0.4	3.75	20	120	160	Std Plus	3.47	39.95	90	32	Weight and height based routine abdomen protocol
36.3.1	Abd 9.5-11.5kg (20.9-25.4lbs)		Helical	PedBody	1.375:1 27.5	0.4	3.75	20	120	180	Std Plus	3.91	49.34	101.25	32	Weight and height based routine abdomen protocol
36.4.1	Abd 11.5-14.5kg (25.4-32.0lbs)		Helical	SmallBo dy	1.375:1 27.5	0.4	5	20	120	125	Std Plus	2.71	36.64	110	32	Weight and height based routine abdomen protocol
36.5.1	Abd 14.5-18.5kg (32.0-40.8lbs)		Helical	SmallBo dy	1.375:1 27.5	0.4	5	20	120	135	Std Plus	2.93	39.57	110	32	Weight and height based routine abdomen protocol
36.6.1	Abd 18.5-22.5kg (40.8-49.6lbs)		Helical	Medium Body	1.375:1 27.5	0.4	5	20	120	120	Std Plus	3.07	47.68	130	32	Weight and height based routine abdomen protocol
36.7.1	Abd 22.5- 31.5kg(49.6- 69.5lbs)		Helical	LargeBo dy	1.375:1 55.0	0.4	5	40	120	150	Std Plus	3.46	68.44	150	32	Weight and height based routine abdomen protocol
36.8.1	Abd 31.5- 40.5kg(69.5- 89.3lbs)		Helical	LargeBo dy	1.375:1 55.0	0.4	5	40	120	170	Std Plus	3.92	77.56	150	32	Weight and height based routine abdomen protocol
36.9.1	Abd 40.5- 55.0kg(89.3- 121.3lbs)		Helical	LargeBo dy	1.375:1 55.0	0.4	5	40	120	200	Std Plus	4.61	114.30	200	32	Weight and height based routine abdomen protocol

Miscellaneous

Table 19-5

[Interval]* - For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (mm)	Phantom (cm)	Description
40.1	Pink Zone Routine 0-20 lbs/0-9 kg		Helical	PedBody	1.375:1 55.0	0.5	5	40	80	SmartmA 65-130mA NI=5 Avg mA 70	Std	0.61	8.70	95	32	Pediatric and Small patient Routine procedure based protocols
40.2	Pink Zone Routine 21-60 lbs/9.1-27.2 kg		Helical	PedBody	1.375:1 55.0	0.5	5	40	100	SmartmA 80-160mA NI=7 Avg mA 70	Std	1.13	27.47	195	32	Pediatric and Small patient Routine procedure based protocols
40.3	Pink Zone Routine 61-100 lbs/27.3-45.4 kg		Helical	Medium Body	1.375:1 55.0	0.5	5	40	120	SmartmA 95-190mA NI=10 Avg mA 70	Std	2.09	71.81	295	32	Pediatric and Small patient Routine procedure based protocols
40.4	Pink Zone Routine 101-200 lbs/45.4-90.7 kg		Helical	LargeBody	1.375:1 55.0	0.5	5	40	120	SmartmA 110-220mA NI=12 Avg mA 70	Std	2.02	89.30	395	32	Pediatric and Small patient Routine procedure based protocols
40.5	Pink Zone Routine >200 lbs/>90.8 kg		Helical	LargeBody	1.375:1 55.0	0.5	5	40	120	SmartmA 125-300mA NI=15 Avg mA 70	Std	2.02	89.30	395	32	Pediatric and Small patient Routine procedure based protocols
40.7	Green Zone Lower Dose 0-20 lbs/0-9 kg		Helical	PedBody	1.375:1 55.0	0.5	5	40	80	SmartmA 50-100mA NI=7 Avg mA 70	Std	0.61	8.70	95	32	Pediatric and Small patient Lower Dose procedure based protocols
40.8	Green Zone Lower Dose 21-60 lbs/9.1-27.2 kg		Helical	PedBody	1.375:1 55.0	0.5	5	40	100	SmartmA 60-120mA NI=9 Avg mA 70	Std	1.13	27.47	195	32	Pediatric and Small patient Lower Dose procedure based protocols
40.9	Green Zone Lower Dose 61-100 lbs/27.3-45.4 kg		Helical	Medium Body	1.375:1 55.0	0.5	5	40	120	SmartmA 70-140mA NI=11 Avg mA 70	Std	2.09	71.81	295	32	Pediatric and Small patient Lower Dose procedure based protocols
40.10	Green Zone Lower Dose 101-200 lbs/45.4-90.7 kg		Helical	LargeBody	1.375:1 55.0	0.5	5	40	120	SmartmA 80-160mA NI=13 Avg mA 70	Std	2.02	89.30	395	32	Pediatric and Small patient Lower Dose procedure based protocols
40.11	Green Zone Lower Dose >200 lbs/>90.8 kg		Helical	LargeBody	1.375:1 55.0	0.5	5	40	120	SmartmA 90-240mA NI=16 Avg mA 70	Std	2.02	89.30	395	32	Pediatric and Small patient Lower Dose procedure based protocols
40.13	Grey Zone CT Angio 0-20 lbs/0-9 kg		Helical	PedBody	1.375:1 55.0	0.5	2.5	40	80	SmartmA 100-200mA NI=5 Avg mA 140	Std	1.22	17.71	97.5	32	Pediatric and Small patient Angio procedure based protocols

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval J* (mm)]	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (mm)	Phantom (cm)	Description
40.14	Grey Zone CT Angio 21-60 lbs/9.1-27.2 kg		Helical	SmallBody	1.375:1 55.0	0.5	2.5	40	100	SmartmA 120-240mA NI=9.9 Avg mA 140	Std	2.26	55.50	197.5	32	Pediatric and Small patient Angio procedure based protocols
40.15	Grey Zone CT Angio 61-100 lbs/ 27.3-45.4 kg		Helical	LargeBody	1.375:1 55.0	0.5	2.5	40	120	SmartmA 120-240mA NI=10 Avg mA 140	Std	4.03	139.28	297.5	32	Pediatric and Small patient Angio procedure based protocols
40.16	Grey Zone CT Angio 101-200 lbs/ 45.4-90.7 kg		Helical	LargeBody	1.375:1 55.0	0.5	2.5	40	120	SmartmA 120-240mA NI=12 Avg mA 140	Std	4.03	179.62	397.5	32	Pediatric and Small patient Angio procedure based protocols
40.17	Grey Zone CT Angio >200 lbs/ >90.8 kg		Helical	LargeBody	1.375:1 55.0	0.5	2.5	40	120	SmartmA 120-300mA NI=15 Avg mA 140	Std	4.03	179.62	397.5	32	Pediatric and Small patient Angio procedure based protocols

Reference Adult Protocols - Installed

Options : 72kW

Head

Table 20-1

[Interval]* - For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (s)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (mm)	Phantom (cm)	Description
21.1	Routine Head 1s Group1		Axial	Head	4i	1	5	20	140	200	Std	53.34	320.02	55	16	Routine head protocol for evaluation of the brain for abnormalities.
	Group2		Axial	Head	4i	1	5	20	120	320	Std	61.43	491.41	75	16	
21.2	Routine Head 2s Group1		Axial	Head	4i	2	5	20	140	100	Std	53.34	320.02	55	16	Routine head protocol for evaluation of the brain for abnormalities.
	Group2		Axial	Head	4i	2	5	20	120	160	Std	61.43	491.41	75	16	
21.3	Trauma Head 2.5mm		Axial	Head	8i	0.5	2.5	20	140	515	Std	70.60	988.35	137.5	16	Emergency head protocol for evaluation of the brain and cranium for abnormalities.
21.4	Routine Head 0.5s Group1		Axial	Head	4i	0.5	5	20	140	400	Std	54.83	328.99	55	16	Routine head protocol for evaluation of the brain for abnormalities.
	Group2		Axial	Head	4i	0.5	5	20	120	600	Std	59.20	473.62	75	16	
21.5	Circle of Willis 0.4s	MIROI	Axial	Head	1i	1	5 [0]	5	120	50	Std	125.58	62.79	0	16	Test bolus for CT angiography of the head for evaluation of carotid and cerebral vasculature.
			Helical	Head	0.531:1 10.625	0.4	0.625	20	120	300	Std Plus IQE	43.36	421.37	80	16	CT angiography of the head for evaluation of carotid and cerebral vasculature.
21.6	CoW/ Carotid 0.625mm 0.4s.Smart mA	MIROI	Axial	Head	1i	1	5 [0]	5	120	80	Std	180.84	90.42	0	16	Test bolus for CT angiography of the head for evaluation of carotid and cerebral vasculature.
			Helical	Head	0.531:1 10.625	0.4	0.625	20	120	SmartmA 100-600mA NI=8 Avg mA 595	Std Plus IQE	88.41	1657.19	170	16	CT angiography of the head for evaluation of carotid and cerebral vasculature.
21.8	Helical Head	DMPR	Helical	Head	0.531:1 10.625	0.5	0.625	20	120	300	Std Plus IQE	54.20	878.85	145	16	Helical scan mode for evaluation of the brain for cerebral abnormality.
21.10	VHS Head 4D CTA	MIROI	Axial	Head	1i	1	5 [0]	5	120	50	Std	125.58	62.79	0	16	Test bolus for CT angiography of the head for evaluation of carotid and cerebral vasculature.
			Helical-S	Head	0.984:1 39.375 18 pass	0.4	5 [10]	40	120	300	Std Plus	496.19	6202.39	110	16	CT angiography of the head using Volume Helical scan mode for evaluation of cerebral vasculature.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (s)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (mm)	Phantom (cm)	Description
21.11	VHS Perfusion 350-370 Strength Contrast		Axial	Head	4i	1	5	20	120	335	Std	64.31	900.28	135	16	Non-Enhance Brain, evaluation for hemorrhage or infarction.
		Perfusion	Helical-S/P	Head	0.984:1 39.375 28 pass	0.4	5	40	80	200	Std	186.02	2325.25	110	16	CT Perfusion using Volume Helical scan mode for evaluation of cerebral perfusion over time.
			Axial	Head	4i	1	5	20	120	335	Std	64.31	900.28	135	16	Post Contrast Brain, Evaluation for residual tissue enhancement.
21.12	CT Perfusion 300-320 Strength Contrast		Axial	Head	4i	1	5	20	120	335	Std	64.31	900.28	135	16	Non-Enhance Brain, evaluation for hemorrhage or infarction.
		Perfusion	Cine	Head	8i	1	5 [0]	40	80	200	Std	652.82	2611.27	35	16	CT Perfusion using Cine scan mode for evaluation of cerebral perfusion over time.
			Axial	Head	4i	1	5	20	120	335	Std	64.31	900.28	135	16	Non-Enhance Brain, evaluation for hemorrhage or infarction.
21.13	CT Perfusion 350-370 Strength Contrast		Axial	Head	4i	1	5	20	120	335	Std	64.31	900.28	135	16	Non-Enhance Brain, evaluation for hemorrhage or infarction.
		Perfusion	Cine	Head	8i	1	5 [0]	40	80	200	Std	587.55	2350.19	35	16	CT Perfusion using Cine scan mode for evaluation of cerebral perfusion over time.
			Axial	Head	4i	1	5	20	120	335	Std	64.31	900.28	135	16	Post Contrast Brain, Evaluation for residual tissue enhancement.
21.14	CT Perfusion Brain Tumor - Dual Phase		Axial	Head	4i	1	5	20	120	335	Std	64.31	900.28	135	16	Non-Enhance Brain, evaluation for hemorrhage or infarction.
	Group1	Perfusion	Cine	Head	8i	1	5 [0]	40	80	200	Std	587.55	2350.19	35	16	CT Perfusion using Cine scan mode for evaluation of cerebral perfusion over time.
	Group2	Perfusion	Axial	Head	8i	1	5 [0]	40	80	200	Std	130.54	522.15	35	16	Delayed phase CT Perfusion using Axial scan mode for evaluation of Brain tumor permeability.
			Axial	Head	4i	1	5	20	120	335	Std	64.31	900.28	135	16	Post Contrast Brain, Evaluation for residual tissue enhancement.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (s)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (mm)	Phantom (cm)	Description
21.15	CT Perfusion 350-370 Strength Contrast (Axial)		Axial	Head	4i	1	5	20	120	335	Std	64.31	900.28	135	16	Non-Enhance Brain, evaluation for hemorrhage or infarction.
		Perfusion	Axial	Head	8i	1	5 [0]	40	80	200	Std	287.18	1148.73	35	16	CT Perfusion using Axial scan mode for evaluation of cerebral perfusion over time.
			Axial	Head	4i	1	5	20	120	335	Std	64.31	900.28	135	16	Post Contrast Brain, Evaluation for residual tissue enhancement.
21.16	CT Perfusion 350-370 Strength Contrast (Axial Shuttle Mode)		Axial	Head	4i	1	5	20	120	335	Std	64.31	900.28	135	16	Non-Enhance Brain, evaluation for hemorrhage or infarction.
		Perfusion	Axial-S	Head	8i 17 pass	0.4	5	40	80	500	Std	221.91	1775.31	75	16	CT Perfusion using Volume Shuttle (Axial) scan mode for evaluation of cerebral perfusion over time.
			Axial	Head	4i	1	5	20	120	335	Std	64.31	900.28	135	16	Post Contrast Brain, Evaluation for residual tissue enhancement.

Orbit

Table 20-2

[Interval]* - For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy-cm)	Scan Length (cm)	Phantom (cm)	Description
22.1	Sinus Supine Helical + DMPR	DMPR	Helical	Head	0.531:1 10.625	0.4	0.625	20	120	120	Bone Plus IQE	17.34	158.79	74.375	16	Routine Helical Sinus protocol, Evaluation for any abnormality
22.3	Orbits Helical + DMPR	DMPR	Helical	Head	0.531:1 10.625	0.5	0.625	20	120	220	Std Plus IQE	39.75	314.09	61.875	16	Routine Helical Orbit protocol, Evaluation for abnormality soft tissues and bone structures of the face.
22.6	Helical IAC 0.625mm Axial mode - Use Reformat for Coronal		Helical	Head	0.531:1 10.625	0.6	0.625	20	140	270	Std Plus IQE	81.32	454.44	38.75	16	Axial scan mode, Evaluation for inner ear abnormalities in Axial plane
22.8	Axial and Coronal IAC		Axial	Head	16i	1	1.25	20	140	280	Std	74.67	298.68	38.75	16	Helical scan mode, Evaluation for inner ear abnormalities in Axial plane
			Axial	Head	16i	1	1.25	20	140	335	Std	89.34	357.35	38.75	16	Helical scan mode, Evaluation for inner ear abnormalities in Coronal plane

Neck

Table 20-3

[Interval]* - For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (second s)	Slice Thickne ss (mm) [Interval]* (mm)	Beam Collimat ion (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy·c m)	Scan Length (cm)	Phantom (cm)	Description
23.1	C-Spine C5-C7 Axial		Axial	SmallBody	8i	0.5	2.5	20	140	515	Std	28.00	392.06	137.5	32	C-Spine C5-C7 area Axial mode, Evaluation for soft tissue, disc and bone abnormalities.
23.2	C-Spine C1-C4 Axial		Axial	SmallBody	8i	1	2.5	20	120	240	Std	17.90	143.19	77.5	32	C-Spine C1-C4 area Axial mode, Evaluation for soft tissue, disc and bone abnormalities.
23.3	C-Spine Helical + DMPR	DMPR	Helical	SmallBody	0.516:1 20.625	0.8	0.625	40	120	335	Bone Plus IQE	36.23	522.92	110	32	C-Spine Helical mode with DMPR, Evaluation for soft tissue, disc and bone abnormalities.
23.4	C-Spine Helical + DMPR/ SmartmA	DMPR	Helical	SmallBody	0.516:1 20.625	0.8	0.625	40	120	SmartmA 100-600mA NI=11.34 Avg mA 310	Bone Plus IQE	33.52	483.89	110	32	C-Spine Helical mode with DMPR and SmartmA, Evaluation for soft tissue, disc and bone abnormalities.
23.8	Carotid 0.625mm 0.4s. 0.5:1 SmartmA/ SmartPrep		Helical	SmallBody	0.516:1 20.625	0.4	0.625	40	140	SmartmA 100-515mA NI=11 Avg mA 515	Std Plus IQE	39.50	808.35	170	32	Carotid CTA (pitch 0.5:1) with SmartmA and SmartPrep, Evaluation of carotid vasculature.
23.9	Carotid 0.625mm 0.4s. 0.9:1 SmartmA/ SmartPrep		Helical	SmallBody	0.984:1 39.375	0.4	0.625	40	120	SmartmA 80-600mA NI=11 Avg mA 600	Std Plus IQE	16.99	345.42	170	32	Carotid CTA (pitch 0.9:1) with SmartmA and SmartPrep, Evaluation of carotid vasculature.
23.10	Carotid/ CoW 0.625mm 0.4s. AutomA	MIROI	Axial	SmallBody	1i	1	5 [0]	5	120	80	Std	99.00	49.50	0	32	Test bolus for Carotid CTA (pitch 0.5:1) of the head for evaluation of carotid and cerebral vasculature.
			Helical	SmallBody	0.516:1 20.625	0.4	0.625	40	140	AutomA 80-515mA NI=11 Avg mA 515	Std Plus IQE	39.50	808.35	170	32	Carotid and Brain CTA (pitch 0.5:1) with AutomA, Evaluation of carotid and cerebral vasculature.
23.11	Carotid/ CoW 0.625mm 0.4s. SmartmA/ SmartPrep		Helical	SmallBody	0.969:1 19.375	0.4	0.625	20	140	SmartmA 100-515mA NI=11 Avg mA 515	Std Plus IQE	23.13	436.84	170	32	Carotid CTA (pitch 0.9:1) with SmartmA and SmartPrep, Evaluation of carotid and cerebral vasculature.
23.12	Soft Tissue Neck 2.5mm AutomA		Helical	SmallBody	0.984:1 39.375	0.5	2.5	40	120	AutomA 100-335mA NI=9.1 Avg mA 250	Std Plus	8.85	161.79	150	32	Neck with AutomA, Evaluation of soft tissue structures for abnormality

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy-cm)	Scan Length (cm)	Phantom (cm)	Description
23.13	Soft Tissue Neck + DMPR/ SmartmA	DMPR	Helical	SmallBody	0.969:1 19.375	0.5	0.625	20	120	SmartmA 100-600mA NI=12.6 Avg mA 425	Std Plus IQE	16.82	283.95	150	32	Neck with DMPR and SmartmA, Evaluation of soft tissue structures for abnormality

Upper Extremity

Table 20-4

[Interval]* - For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.
For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy-cm)	Scan Length (cm)	Phantom (cm)	Description
24.1	Shoulder 1.25mm + DMPR	DMPR	Helical	LargeBody	0.516:1 20.625	0.6	1.25 [0.625]	40	140	360	Std Plus IQE	47.03	632.79	100	32	Shoulder with DMPR, Evaluation of soft tissue and bone of shoulder area
24.2	Shoulder 1.25mm + DMPR/ SmartmA	DMPR	Helical	LargeBody	0.516:1 20.625	0.6	1.25 [0.625]	40	140	SmartmA 100-515mA NI=20 Avg mA 450	Std Plus IQE	58.78	790.99	100	32	Shoulder with DMPR and SmartmA, Evaluation of soft tissue and bone of shoulder area

Chest

Table 20-5

[Interval]* - For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (second s)	Slice Thickne ss (mm) [Interval]* (mm)	Beam Collimat ion (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy·c m)	Scan Length (cm)	Phantom (cm)	Description
25.1	Routine Chest 0.4s. 5mm SmartmA		Helical	LargeBody	1.375:1 55.0	0.4	5	40	120	SmartmA 100-600mA NI=13 Avg mA 500	Std Plus	11.52	286.26	200	32	Chest with SmartmA, Evaluation of mediastinum and lungs fields
25.2	Routine Chest 0.4s.+ DMPR/ SmartmA	DMPR	Helical	LargeBody	0.984:1 39.375	0.4	0.625	40	120	SmartmA 100-600mA NI=22 Avg mA 600	Std Plus	19.32	450.62	200	32	Chest with SmartmA and DMPR, Evaluation of mediastinum and lungs fields
25.3	Chest with HiRes 0.4s. SmartmA		Helical	LargeBody	0.969:1 19.375	0.4	5	20	120	SmartmA 80-600mA NI=12 Avg mA 500	Chest Plus	17.99	393.87	200	32	Chest with SmartmA, Evaluation of mediastinum and lung fields with higher resolution images of the lungs
25.4	Hi Res Chest 0.5s. Axial		Axial	LargeBody	1i	0.5	1.25 [10]	1.25	120	380	Bone+	4.48	107.55	230	32	Axial scan mode, Evaluation of mediastinum and lung fields in high resolution mode
25.5	Chest Abd Pelvis 0.4/ 0.5s. SmartmA 5mm (2 groups) Group1		Helical	LargeBody	0.984:1 39.375	0.4	5	40	120	SmartmA 100-600mA NI=13 Avg mA 370	Std Plus	11.91	218.32	150	32	Multi group scan for Chest, Abdomen, Pelvis (0.4/0.5sec) with SmartmA, Evaluation of chest and abdominal anatomy for abnormalities
	Group2		Helical	LargeBody	1.375:1 55.0	0.5	5	40	120	SmartmA 100-600mA NI=13 Avg mA 370	Std Plus	10.66	365.90	295	32	
25.6	Chest Abd Pelvis 0.5s + DMPR SmartPrep	DMPR	Helical	LargeBody	1.375:1 55.0	0.5	0.625	40	120	SmartmA 100-600mA NI=22 Avg mA 540	Std Plus	15.56	775.18	450	32	Helical scan mode (0.5sec) with SmartmA, Evaluation of chest and abdominal anatomy for abnormalities
25.7	Pulmonary Embolis 0.5s.	MIROI	Axial	LargeBody	1i	1	5 [0]	5	120	80	Std	93.89	46.95	0	32	Timing Bolus for Helical scan mode 1.25mm thickness and helical pitch of 0.9:1, evaluation of Chest for pulmonary embolism
			Helical	LargeBody	0.984:1 39.375	0.5	1.25	40	120	580	Std	23.34	497.55	180	32	Helical scan mode 1.25mm thickness and helical pitch of 0.9:1, Evaluation Chest for pulmonary embolism

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy-cm)	Scan Length (cm)	Phantom (cm)	Description
25.8	Pulmonary Embolis 0.5s. SmartmA	MIROI	Axial	LargeBody	1i	1	5 [0]	5	120	80	Std	93.89	46.95	0	32	Timing Bolus for Helical scan mode 1.25mm thickness and helical pitch of 0.9:1 with SmartmA, evaluation of Chest for pulmonary embolism
			Helical	LargeBody	0.984:1 39.375	0.5	1.25	40	120	SmartmA 100-600mA NI=20 Avg mA 500	Std	20.12	428.92	180	32	Helical scan mode 1.25mm thickness and helical pitch of 0.9:1 with SmartmA, Evaluation Chest for pulmonary embolism
25.9	Aortic Dissection 0.625mm + DMPR/ SmartmA/ SmartPrep	DMPR	Helical	LargeBody	1.375:1 55.0	0.5	0.625	40	120	SmartmA 100-600mA NI=22 Avg mA 600	Std Plus	17.29	602.00	300	32	Helical scan mode 0.625mm thickness and helical pitch of 1.375:1 with SmartmA, SmartPrep and DMPR, Evaluation for aortic dissection
25.10	Aortic Dissection 1.25mm + DMPR/ SmartmA/ SmartPrep	DMPR	Helical	LargeBody	1.375:1 55.0	0.5	1.25 [0.8]	40	120	SmartmA 100-600mA NI=20 Avg mA 550	Std Plus	15.85	551.83	300	32	Helical scan mode 1.25mm thickness and helical pitch of 1.375:1 with SmartmA, SmartPrep and DMPR, Evaluation for aortic dissection
25.11	Lung Assessment	Advanced Lung Analysis	Helical	LargeBody	0.969:1 19.375	0.5	1.25	20	120	120	Bone	5.25	114.77	200	32	Helical scan mode for acquisition of data for Advanced Lung Analysis (ALA) software for lung nodule assessment
25.12	Gated Chest Cardiac Helical		Helical	LargeBody	1.375:1 55.0	0.4	5	40	120	60	Std Plus	1.38	13.55	50	32	Localizer scan for Helical Gated scan for evaluation of thoracic vasculature
		MIROI	Axial	LargeBody	1i	1	5 [0]	5	120	60	Std	98.59	49.29	0	32	Timing bolus scan for Helical Gated scan for evaluation of thoracic vasculature
			Cardiac SEG	CardiacLarge	0.180:1 7.2	0.4	0.625	40	120	ECG Mod 200-395mA Phase 70-80 Avg mA 395	Std SS Seg	50.74	1699.82	300	32	Helical Gated scan for evaluation of thoracic vasculature
25.13	SmartScore - Gated 0.4s	Smart Score	Cine	LargeBody	8i	0.4	2.5	20	120	375	Std Segment	9.15	91.51	97.5	32	Cine gated scan mode for acquisition of data for coronary artery calcification analysis
25.14	SnapShot Segment 30-74 BPM		Helical	LargeBody	1.375:1 55.0	0.4	5	40	120	60	Std Plus	1.38	13.55	50	32	Localizer scan for Retrospectively gated Cardiac Helical scan for evaluation of coronary arteries, intended for heart rates 30-74 bpm

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy-cm)	Scan Length (cm)	Phantom (cm)	Description
		MIROI	Axial	LargeBody	1i	1	5 [0]	5	120	60	Std	98.59	49.29	0	32	Timing bolus scan for Retrospectively gated Cardiac Helical scan for evaluation of coronary arteries, intended for heart rates 30-74 bpm
			Cardiac SEG	CardiacLarge	0.180:1 7.2	0.4	0.625	40	120	ECG Mod 200-395mA Phase 70-80 Avg mA 395	Std SS Seg	52.44	707.94	100	32	Retrospectively gated Cardiac Helical scan for evaluation of coronary arteries, intended for heart rates 30-74 bpm
25.15	SnapShot Burst 75-113 BPM		Helical	LargeBody	1.375:1 55.0	0.4	5	40	120	60	Std Plus	1.38	13.55	50	32	Localizer scan for Retrospectively gated Cardiac Helical scan for evaluation of coronary arteries, intended for heart rates 75-113 bpm
		MIROI	Axial	LargeBody	1i	1	5 [0]	5	120	60	Std	98.59	49.29	0	32	Timing bolus scan for Retrospectively gated Cardiac Helical scan for evaluation of coronary arteries, intended for heart rates 75-113 bpm
			Cardiac SSB	CardiacLarge	0.180:1 7.2	0.4	0.625	40	120	ECG Mod 200-395mA Phase 70-80 Avg mA 395	Std SSB	52.44	707.94	100	32	Retrospectively gated Cardiac Helical scan for evaluation of coronary arteries, intended for heart rates 75-113 bpm
25.16	SnapShot Burst Plus 114+ BPM		Helical	LargeBody	1.375:1 55.0	0.4	5	40	120	60	Std Plus	1.38	13.55	50	32	Localizer scan for Retrospectively gated Cardiac Helical scan for evaluation of coronary arteries, intended for heart rates over 113 bpm
		MIROI	Axial	LargeBody	1i	1	5 [0]	5	120	60	Std	98.59	49.29	0	32	Timing bolus scan for Retrospectively gated Cardiac Helical scan for evaluation of coronary arteries, intended for heart rates over 113 bpm
			Cardiac SSBP	CardiacLarge	0.180:1 7.2	0.4	0.625	40	120	ECG Mod 200-395mA Phase 70-80 Avg mA 395	Std SSB	52.44	707.94	100	32	Retrospectively gated Cardiac Helical scan for evaluation of coronary arteries, intended for heart rates over 113 bpm
25.17	Advantage 4D		Helical	LargeBody	1.375:1 55.0	0.6	5	40	120	285	Std Plus	9.85	244.08	200	32	Localizer scan for Advantage 4D series
		Advantage 4D	Cine	LargeBody	8i	1	2.5	20	120	100	Std	50.95	815.24	157.5	32	Cine scan mode for acquisition data for Advantage 4D for retrospective gating
25.18	VHS Chest 4D CTA SmartPrep		Helical-S	LargeBody	1.375:1 55.0 11 pass	0.4	5 [10]	40	120	SmartmA 100-600mA NI=12 Avg mA 600	Std Plus	192.25	6055.91	300	32	Volume Helical scan mode with SmartPrep for timr resolved CT angiography of chest vasculature

Abdomen

Table 20-6

[Interval]* - For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (cm)	Phantom (cm)	Description
26.1	Abdomen Pelvis 0.5s. 5mm SmartmA		Helical	LargeBody	1.375:1 55.0	0.5	5	40	120	SmartmA 50-600mA NI=11.57 Avg mA 500	Std Plus	14.41	645.73	400	32	Basic Abdomen Pelvis (0.5sec) with Smart mA, Evaluation for abnormalities of abdominal and pelvic structures
26.2	Abdomen Pelvis 0.5s. 5mm + DMPR SmartmA/ SmartPrep	DMPR	Helical	LargeBody	0.984:1 39.375	0.6	0.625	40	120	SmartmA 80-600mA NI=18 Avg mA 520	Std Plus	25.11	1087.62	400	32	Basic Abdomen Pelvis (0.5sec) with Smart mA, SmartPrep and DMPR, Evaluation for abnormalities of abdominal and pelvic structures
26.4	Renal Stone 0.5s. 5mm		Helical	LargeBody	1.375:1 55.0	0.5	5	40	120	380	Std Plus	10.95	490.75	400	32	Helical scan, Evaluation of kidney, ureters and bladder for renal stones
26.5	Renal Stone 0.5s. 5mm SmartmA		Helical	LargeBody	1.375:1 55.0	0.5	5	40	120	SmartmA 50-600mA NI=12 Avg mA 380	Std Plus	10.95	490.75	400	32	Helical scan with SmartmA, Evaluation of kidney, ureters and bladder for renal stones
26.7	AAA 0.5s. + DMPR 0.625mm	MIROI	Axial	LargeBody	1i	1	5 [0]	5	120	80	Std	112.67	56.34	0	32	Timing bolus for Helical scan with DMPR, Evaluation for abdominal aortic aneurysm
		DMPR	Helical	LargeBody	1.375:1 55.0	0.5	0.625	40	140	515	Std Plus	21.02	751.82	309.375	32	Helical scan with DMPR, Evaluation for abdominal aortic aneurysm
26.8	AAA 0.5s. + DMPR 0.625mm SmartmA/ SmartPrep	DMPR	Helical	LargeBody	1.375:1 55.0	0.6	0.625	40	120	SmartmA 100-600mA NI=19 Avg mA 550	Std Plus	19.02	679.85	309.375	32	Helical scan with DMPR, Smart mA and SmartPrep, Evaluation for abdominal aortic aneurysm
26.10	Dual Phase Liver		Helical	LargeBody	1.375:1 55.0	0.6	5	40	120	300	Std Plus	10.37	184.32	130	32	Non contrast evaluation of upper abdominal structures including liver
		MIROI	Axial	LargeBody	1i	1	5 [0]	5	120	80	Std	112.67	56.34	0	32	Timing bolus for Multi-Group acquisition for early arterial and portal venous phase of the liver
	Group1		Helical	LargeBody	1.375:1 55.0	0.5	2.5	40	120	550	Std Plus	15.85	282.44	130	32	Acquisition for early arterial phase of the liver
	Group2		Helical	LargeBody	1.375:1 55.0	0.5	5	40	120	420	Std Plus	12.10	542.41	400	32	Acquisition for portal venous phase of the liver

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy-cm)	Scan Length (cm)	Phantom (cm)	Description
26.11	Tri Phase Liver		Helical	LargeBody	1.375:1 55.0	0.6	5	40	120	300	Std Plus	10.37	184.32	130	32	Non contrast evaluation of upper abdominal structures including liver
		MIROI	Axial	LargeBody	1i	1	5 [0]	5	120	80	Std	112.67	56.34	0	32	Timing bolus for Multi-Group acquisition for early arterial, late arterial and portal venous phase of the liver
	Group1		Helical	LargeBody	1.375:1 55.0	0.5	2.5	40	120	550	Std Plus	15.85	282.44	130	32	Acquisition for early arterial phase of the liver
	Group2		Helical	LargeBody	1.375:1 55.0	0.5	5	40	120	450	Std Plus	12.97	581.15	400	32	Acquisition for late arterial phase of the liver
	Group3		Helical	LargeBody	1.375:1 55.0	0.5	5	40	120	450	Std Plus	12.97	581.15	400	32	Acquisition for portal venous phase of the liver
26.12	Dual Phase Pancreas SmartmA/ SmartPrep		Helical	LargeBody	0.984:1 39.375	0.5	5	40	120	SmartmA 80-600mA NI=12 Avg mA 400	Std Plus	16.10	166.06	70	32	Non contrast evaluation of upper abdominal structures including pancreas
	Group1		Helical	LargeBody	0.984:1 39.375	0.5	2.5	40	120	SmartmA 50-600mA NI=15 Avg mA 520	Std Plus	20.93	383.30	150	32	Acquisition with SmartmA and SmartPrep for arterial phase of the pancreas
	Group2		Helical	LargeBody	0.984:1 39.375	0.5	5	40	120	SmartmA 50-600mA NI=15 Avg mA 450	Std Plus	18.11	639.58	320	32	Acquisition for venous phase of the pancreas
26.13	Dual Phase Liver SmartmA/ SmartPrep		Helical	LargeBody	1.375:1 55.0	0.5	5	40	120	SmartmA 80-500mA NI=12 Avg mA 320	Std Plus	9.22	163.92	130	32	Non contrast evaluation of upper abdominal structures including liver
	Group1		Helical	LargeBody	1.375:1 55.0	0.5	2.5	40	120	SmartmA 50-600mA NI=15 Avg mA 550	Std Plus	15.85	282.44	130	32	Acquisition with SmartmA and SmartPrep for early arterial phase of the liver
	Group2		Helical	LargeBody	0.984:1 39.375	0.5	5	40	120	SmartmA 50-600mA NI=11 Avg mA 450	Std Plus	18.11	784.46	400	32	Acquisition with SmartmA and SmartPrep for portal venous phase of the liver
26.14	Tri Phase Liver SmartmA/ SmartPrep		Helical	LargeBody	0.984:1 39.375	0.5	5	40	120	SmartmA 80-500mA NI=12 Avg mA 400	Std Plus	16.10	262.65	130	32	Non contrast evaluation of upper abdominal structures including liver
	Group1		Helical	LargeBody	1.375:1 55.0	0.5	2.5	40	120	SmartmA 50-600mA NI=15 Avg mA 550	Std Plus	15.85	282.44	130	32	Acquisition with SmartmA and SmartPrep for early arterial phase of the liver
	Group2		Helical	LargeBody	1.375:1 55.0	0.5	5	40	120	SmartmA 50-600mA NI=11 Avg mA 450	Std Plus	12.97	581.15	400	32	Acquisition with SmartmA and SmartPrep for late arterial phase of the liver
	Group3		Helical	LargeBody	1.375:1 55.0	0.5	5	40	120	SmartmA 50-600mA NI=11 Avg mA 450	Std Plus	12.97	581.15	400	32	Acquisition with SmartmA and SmartPrep for portal venous phase of the liver
26.15	VHS Liver Perfusion		Helical	LargeBody	0.984:1 39.375	0.6	5	40	120	170	Std	8.21	133.61	130	32	Non contrast evaluation of upper abdominal structures including liver

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy-cm)	Scan Length (cm)	Phantom (cm)	Description
		Perfusion	Helical-S/P	LargeBody	1.375:1 55.0 21 pass	0.4	5	40	120	300	Std Plus	219.99	3409.79	140	32	Volume Helical scan mode for evaluation of liver tumor perfusion over time
26.16	VHS 4D CTA		Helical	LargeBody	0.984:1 39.375	0.6	5	40	120	340	Std	16.42	267.79	130	32	Localizer scan for Volume Helical Shuttle scan mode for evaluation of abdominal vasculature
		MIROI	Axial	LargeBody	1i	1	5 [0]	5	120	80	Std	112.67	56.34	0	32	Timing bolus scan for Volume Helical Shuttle scan mode for evaluation of abdominal vasculature
			Helical-S	LargeBody	1.375:1 55.0 18 pass	0.5	5 [10]	40	140	SmartmA 150-515mA NI=12 Avg mA 350	Std Plus	325.68	6350.70	180	32	Volume Helical Shuttle scan mode for time resolved CT angiography of abdominal vasculature
26.17	Runoff 0.625mm	MIROI	Axial	LargeBody	1i	1	5 [0]	5	120	80	Std	112.67	56.34	0	32	Timing bolus scan for evaluation of vascular structure of abdomen, femurs and lower extremities (Runoff)
			Helical	LargeBody	1.375:1 55.0	0.5	0.625	40	140	515	Std Plus	21.02	2071.05	936.875	32	Helical scanmode 0.625mm, Evaluation of vascular structure of abdomen, femurs and lower extremities (Runoff)
26.18	Runoff 1.25mm	MIROI	Axial	LargeBody	1i	1	5 [0]	5	120	80	Std	112.67	56.34	0	32	Timing bolus scan for evaluation of vascular structure of abdomen, femurs and lower extremities (Runoff)
			Helical	LargeBody	1.375:1 55.0	0.5	1.25 [0.7]	40	140	400	Std Plus	16.33	1709.86	998.9	32	Helical scanmode 1.25mm, Evaluation of vascular structure of abdomen, femurs and lower extremities (Runoff)
26.19	Runoff 0.625mm SmartmA/ SmartPrep		Helical	LargeBody	1.375:1 55.0	0.5	0.625	40	140	SmartmA 100-515mA NI=19 Avg mA 515	Std Plus	21.02	2071.05	936.875	32	Helical scanmode 0.625mm with SmartmA and SmartPrep, Evaluation of vascular structure of abdomen, femurs and lower extremities (Runoff)
26.20	Runoff 1.25mm SmartmA/ SmartPrep		Helical	LargeBody	1.375:1 55.0	0.5	1.25 [0.7]	40	140	SmartmA 100-515mA NI=20 Avg mA 400	Std Plus	16.33	1709.86	998.9	32	Helical scanmode 1.25mm with SmartmA and SmartPrep, Evaluation of vascular structure of abdomen, femurs and lower extremities (Runoff)
26.22	CT Colonography 1.25mm	Advantage CTC Pro	Helical	LargeBody	1.375:1 55.0	0.5	1.25	40	120	200	Std Plus	5.76	258.04	400	32	Helical scan mode supine acquisition for using Advantage CTC Pro for evaluation of the colon
		Advantage CTC Pro	Helical	LargeBody	1.375:1 55.0	0.5	1.25	40	120	200	Std Plus	5.76	258.04	400	32	Helical scan mode prone acquisition for using Advantage CTC Pro for evaluation of the colon

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy-cm)	Scan Length (cm)	Phantom (cm)	Description
26.41	Body Interventional 5mm/3i		Helical	LargeBody	1.375:1 27.5	0.5	5	20	120	SmartmA 50-600mA NI=11.57 Avg mA 360	Std	11.41	251.58	195	32	Localization scan for CT interventional scan
			SmartView SEG	LargeBody	3i	0.8	5	10	120	80	Std Segment	684.58	684.58	0	32	CT Fluoro scan for interventional scan
26.42	Body Interventional 5mm/1i		Helical	LargeBody	1.375:1 27.50	0.5	5	20	120	SmartmA 50-600mA NI=11.57 Avg mA 360	Std	11.41	251.58	195	32	Localization scan for CT interventional scan
			SmartView SEG	LargeBody	1i	0.8	5	5	120	80	Std Segment	845.04	422.52	0	32	CT Fluoro scan for interventional scan

L-Spine

Table 20-7

[Interval]* - For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy-cm)	Scan Length (cm)	Phantom (cm)	Description
27.1	L-Spine 3 Level Axial Group1		Axial	LargeBody	8i	1	2.5	20	120	320	Std	27.13	54.26	17.5	32	Axial scan mode, Evaluation of soft tissue and bony structures lumbar spine
	Group2		Axial	LargeBody	8i	1	2.5	20	120	320	Std	27.13	54.26	17.5	32	Axial scan mode, Evaluation of soft tissue and bony structures lumbar spine
	Group3		Axial	LargeBody	8i	1	2.5	20	140	360	Std	44.46	88.91	17.5	32	Axial scan mode, Evaluation of soft tissue and bony structures lumbar spine
27.3	L-Spine Survey Helical 2.5mm 0.6s. SmartmA		Helical	LargeBody	0.516:1 20.625	0.6	2.5	40	140	SmartmA 100-515mA NI=14 Avg mA 515	Std Plus	67.28	1409.81	75	32	Helical scan mode 2.5mm (0.6sec), Limited evaluation of lumbar spine
27.4	L-Spine Survey Helical 2.5mm 0.8s. SmartmA		Helical	LargeBody	0.516:1 20.625	0.8	2.5	40	140	SmartmA 100-515mA NI=14 Avg mA 515	Std Plus	89.70	1879.43	75	32	Helical scan mode 2.5mm (0.8sec), Limited evaluation of lumbar spine
27.6	L-Spine Helical 0.625mm + DMPR/ SmartmA	DMPR	Helical	LargeBody	0.516:1 20.625	0.6	0.625	40	140	SmartmA 100-515mA NI=20 Avg mA 515	Std Plus IQE	67.28	1409.81	75	32	Helical scan mode 0.625mm with DMPR and SmartmA, Evaluation of soft tissue and bony structures lumbar spine
27.7	L-Spine Helical 1.25mm + DMPR/ SmartmA - Large Patient	DMPR	Helical	LargeBody	0.516:1 20.625	1	1.25	40	140	SmartmA 100-515mA NI=18 Avg mA 515	Std Plus	112.13	2348.90	75	32	Helical scan mode 1.25mm with DMPR and SmartmA, Evaluation of soft tissue and bony structures lumbar spine

Pelvis

Table 20-8
[Interval]* - For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.
For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (second s)	Slice Thickne ss (mm) [Interval]* (mm)	Beam Collimat ion (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy·c m)	Scan Length (cm)	Phantom (cm)	Description
28.1	Pelvis for Fracture + DMPR/ SmartmA	DMPR	Helical	LargeBody	0.984:1 39.375	0.8	1.25 [0.625]	40	120	SmartmA 100-600mA NI=20 Avg mA 500	Bone Plus IQE	32.20	432.30	101.25	32	Helical scan with DMPR and SmartmA, Evaluation of pelvis bony structures for fracture

Lower Extremity

Table 20-9

[Interval]* - For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (cm)	Phantom (cm)	Description
29.1	Lower Extremity Survey 2.5mm		Helical	LargeBody	0.516:1 20.625	0.5	2.5	40	120	250	Std Plus	19.21	181.35	60	32	Helical scan mode 2.5mm, Limited evaluation of lower extremity for abnormalities
29.2	Ankle 0.625 mm Single Joint		Helical	MediumBody	0.531:1 10.625	0.5	0.625	20	120	250	Std Plus IQE	20.72	108.05	35	32	Helical scan mode 0.625mm, Evaluation of soft tissue and bony anatomy of the ankle area
29.3	Ankle 0.625 mm Both Joints		Helical	LargeBody	0.531:1 10.625	0.5	0.625	20	120	250	Std Plus IQE	19.95	104.03	35	32	Helical scan mode 0.625mm, Evaluation of soft tissue and bony anatomy of the ankle area
29.4	Knee 0.625 mm Both Joints		Helical	LargeBody	0.531:1 10.625	0.6	0.625	20	120	Auto mA 100-335mA NI=20 Avg mA 275	Std Plus IQE	26.33	137.27	35	32	Helical scan mode 0.625mm, Evaluation of soft tissue and bony anatomy of both knees

Reference Pediatric Protocols - Installed Options : 72kW

Head

Table 21-1

[Interval]* - For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.
For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (mm)	Phantom (cm)	Description
31.1	PED HEAD TO 18 Months		Axial	Ped Head	4i	0.5	5	20	120	240	Std	20.43	245.19	115	16	Routine head (0.5s) for infant (up to 18 months) 5mm
31.2	PED HEAD 18 MOS TO 5YRS		Axial	SmallHead	4i	0.5	5	20	120	335	Std	28.52	342.25	115	16	Routine head (0.5s) for children (18 months to 5 years) 5mm
31.3	PED HEAD TO 5YRS TO 18YRS Group1		Axial	SmallHead	8i	1	2.5	20	120	240	Std	40.87	163.46	37.5	16	Routine head (1.0s) for children (5 years to 18 years) 2.5mm
	Group2		Axial	SmallHead	4i	1	5	20	120	180	Std	30.65	245.19	75	16	Routine head (1.0s) for children (5 years to 18 years) 5mm

Orbit

Table 21-2

[Interval]* - For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (mm)	Phantom (cm)	Description
32.1	Sinus Supine Helical + DMPR	DMPR	Helical	SmallHead	0.531:1 10.625	0.4	0.625	20	120	80	Bone Plus IQE	10.26	93.90	74.375	16	Routine Helical scan mode with DMPR, Evaluation of sinus area for abnormality in pediatric patients
32.3	Orbits Helical	DMPR	Helical	SmallHead	0.531:1 10.625	0.4	0.625	20	120	180	Std Plus IQE	23.08	182.43	61.875	16	Helical scan mode, Evaluation of orbital and face structures for abnormality in pediatric patients
32.6	Helical IAC 0.625mm Axial mode - Use Reformat for Coronal		Helical	SmallHead	0.531:1 10.625	0.4	0.625	20	120	160	Std Plus IQE	20.51	114.73	38.75	16	Helical scan mode, Evaluation of inner ear structures for abnormality in pediatric patients

Chest

Table 21-3

[Interval]* - For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (mm)	Phantom (cm)	Description
35.1.1	RC 6.0-7.5 kg (13.2-16.5 lbs)		Helical	PedBody	1.375:1 27.5	0.4	3.75	20	100	80	Std Plus	1.11	11.08	75	32	Weight and height based routine chest protocol
35.1.3	SnapShot Segment 30-74 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	120	Std Plus	0.84	8.19	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	Std	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
			Cardiac SEG	CardiacSmall	0.180:1 7.2	0.4	0.625	40	80	ECG Mod 85-280mA Phase 40-80 AvgmA280	Std SS Seg	12.76	140.31	75	32	Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
35.1.4	SnapShot Burst 75-104 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	120	Std Plus	0.84	8.19	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	Std	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
			Cardiac SSB	CardiacSmall	0.180:1 7.2	0.4	0.625	40	80	ECG Mod 85-280mA Phase 40-80 AvgmA280	Std SSB	12.76	140.31	75	32	Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
35.1.5	SnapShot Burst+ >105 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	120	Std Plus	0.84	8.19	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	Std	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
			Cardiac SSBP	CardiacSmall	0.180:1 7.2	0.4	0.625	40	80	ECG Mod 85-280mA Phase 35-85 AvgmA280	Std SSB	13.64	150.05	75	32	Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
35.1.7	Non ECG Gated		Helical	PedBody	0.984:1 39.375	0.4	0.625	40	80	SmartmA 50-200mA NI=18 AvgmA200	Std Plus	1.95	21.00	75	32	Weight and height based non ECG gated protocol for cardiac with SmartmA
35.2.1	RC 7.5-9.5 kg (16.5-20.9lbs)		Helical	PedBody	1.375:1 27.5	0.4	3.75	20	100	120	Std Plus	1.66	16.62	75	32	Weight and height based routine chest protocol
35.2.3	SnapShot Segment 30-74 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	120	Std Plus	0.84	8.19	50	32	Weight and height based protocol for localization

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (mm)	Phantom (cm)	Description
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	Std	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
			Cardiac SEG	CardiacSmall	0.180:1 7.2	0.4	0.625	40	80	ECG Mod 85-300mA Phase 40-80 AvgmA300	Std SS Seg	13.63	149.92	75	32	Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
35.2.4	SnapShot Burst 75-104 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	120	Std Plus	0.84	8.19	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	Std	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
			Cardiac SSB	CardiacSmall	0.180:1 7.2	0.4	0.625	40	80	ECG Mod 85-300mA Phase 40-80 AvgmA300	Std SSB	13.63	149.92	75	32	Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
35.2.5	SnapShot Burst+ >105 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	120	Std Plus	0.84	8.19	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	Std	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
			Cardiac SSBP	CardiacSmall	0.180:1 7.2	0.4	0.625	40	80	ECG Mod 85-300mA Phase 35-85 AvgmA300	Std SSB	14.61	160.66	75	32	Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
35.2.7	Non ECG Gated		Helical	PedBody	0.984:1 39.375	0.4	0.625	40	80	SmartmA 50-220mA NI=18 AvgmA220	Std Plus	2.14	23.10	75	32	Weight and height based non ECG gated protocol for cardiac with SmartmA
35.3.1	RC 9.5-11.5 kg (20.9-25.4lbs)		Helical	SmallBody	1.375:1 27.5	0.4	3.75	20	120	100	Std Plus	2.17	22.53	78.75	32	Weight and height based routine chest protocol
35.3.3	SnapShot Segment 30-74 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	120	Std Plus	0.84	8.19	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	Std	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
			Cardiac SEG	CardiacSmall	0.180:1 7.2	0.4	0.625	40	80	ECG Mod 100-320mA Phase 40-80 AvgmA320	Std SS Seg	14.59	160.53	75	32	Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
35.3.4	SnapShot Burst 75-104 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	120	Std Plus	0.84	8.19	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	Std	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (mm)	Phantom (cm)	Description
			Cardiac SSB	CardiacSmall	0.180:17.2	0.4	0.625	40	80	ECG Mod 100-320mA Phase 40-80 AvgmA320	Std SSB	14.59	160.53	75	32	Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
35.3.5	SnapShot Burst+ >105 BPM		Helical	SmallBody	1.375:155.0	0.4	2.5	40	80	120	Std Plus	0.84	8.19	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	Std	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
			Cardiac SSBP	CardiacSmall	0.180:17.2	0.4	0.625	40	80	ECG Mod 100-320mA Phase 35-85 AvgmA320	Std SSB	15.59	171.51	75	32	Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
35.3.7	Non ECG Gated		Helical	PedBody	0.984:139.375	0.4	0.625	40	80	SmartmA 50-240mA NI=18 AvgmA240	Std Plus	2.34	25.20	75	32	Weight and height based non ECG gated protocol for cardiac with SmartmA
35.4.1	RC 11.5-14.5kg (25.4-32.0lbs)		Helical	SmallBody	1.375:127.5	0.4	5	20	120	90	Std Plus	1.95	22.47	90	32	Weight and height based routine chest protocol
35.4.3	SnapShot Segment 30-74 BPM		Helical	SmallBody	1.375:155.0	0.4	2.5	40	80	120	Std Plus	0.84	8.19	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	Std	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
			Cardiac SEG	CardiacSmall	0.180:17.2	0.4	0.625	40	80	ECG Mod 100-340mA Phase 40-80 AvgmA340	Std SS Seg	15.45	169.99	75	32	Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
35.4.4	SnapShot Burst 75-104 BPM		Helical	SmallBody	1.375:155.0	0.4	2.5	40	80	120	Std Plus	0.84	8.19	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	Std	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
			Cardiac SSB	CardiacSmall	0.180:17.2	0.4	0.625	40	80	ECG Mod 100-340mA Phase 40-80 AvgmA340	Std SSB	15.45	169.99	75	32	Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
35.4.5	SnapShot Burst+ >105 BPM		Helical	SmallBody	1.375:155.0	0.4	2.5	40	80	120	Std Plus	0.84	8.19	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	Std	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
			Cardiac SSBP	CardiacSmall	0.180:17.2	0.4	0.625	40	80	ECG Mod 100-340mA Phase 35-85 AvgmA340	Std SSB	16.54	181.98	75	32	Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval] (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (mm)	Phantom (cm)	Description
35.4.7	Non ECG Gated		Helical	PedBody	0.984:1 39.375	0.4	0.625	40	80	SmartmA 50-280mA NI=18 AvgmA280	Std Plus	2.72	29.40	75	32	Weight and height based non ECG gated protocol for cardiac with SmartmA
35.5.1	RC 14.5-18.5kg (32.0-40.8lbs)		Helical	SmallBody	1.375:1 27.5	0.4	5	20	120	100	Std Plus	2.17	24.97	90	32	Weight and height based routine chest protocol
35.5.3	SnapShot Segment 30-74 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	120	Std Plus	0.84	8.19	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	Std	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
			Cardiac SEG	CardiacSmall	0.180:1 7.2	0.4	0.625	40	80	ECG Mod 115-380mA Phase 40-80 AvgmA380	Std SS Seg	17.30	190.30	75	32	Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
35.5.4	SnapShot Burst 75-104 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	120	Std Plus	0.84	8.19	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	Std	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
			Cardiac SSB	CardiacSmall	0.180:1 7.2	0.4	0.625	40	80	ECG Mod 115-380mA Phase 40-80 AvgmA380	Std SSB	17.30	190.30	75	32	Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
35.5.5	SnapShot Burst+ >105 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	120	Std Plus	0.84	8.19	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	Std	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
			Cardiac SSBP	CardiacSmall	0.180:1 7.2	0.4	0.625	40	80	ECG Mod 115-380mA Phase 35-85 AvgmA380	Std SSB	18.50	203.54	75	32	Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
35.5.7	Non ECG Gated		Helical	PedBody	0.984:1 39.375	0.4	0.625	40	80	SmartmA 50-300mA NI=18 AvgmA300	Std Plus	2.92	31.50	75	32	Weight and height based non ECG gated protocol for cardiac with SmartmA
35.6.1	RC 18.5-22.5kg (40.8-49.6lbs)		Helical	SmallBody	1.375:1 27.5	0.4	5	20	120	110	Std Plus	2.39	27.47	90	32	Weight and height based routine chest protocol
35.6.3	SnapShot Segment 30-74 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	120	Std Plus	0.84	8.19	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	Std	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval] (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (mm)	Phantom (cm)	Description
			Cardiac SEG	CardiacSmall	0.180:17.2	0.4	0.625	40	80	ECG Mod 115-400mA Phase 40-80 AvgmA400	Std SS Seg	18.17	199.84	75	32	Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
35.6.4	SnapShot Burst 75-104 BPM		Helical	SmallBody	1.375:155.0	0.4	2.5	40	80	120	Std Plus	0.84	8.19	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	Std	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
			Cardiac SSB	CardiacSmall	0.180:17.2	0.4	0.625	40	80	ECG Mod 115-400mA Phase 40-80 AvgmA400	Std SSB	18.17	199.84	75	32	Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
35.6.5	SnapShot Burst+ >105 BPM		Helical	SmallBody	1.375:155.0	0.4	2.5	40	80	120	Std Plus	0.84	8.19	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	Std	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
			Cardiac SSBP	CardiacSmall	0.180:17.2	0.4	0.625	40	80	ECG Mod 115-400mA Phase 35-85 AvgmA400	Std SSB	19.46	214.07	75	32	Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
35.6.7	Non ECG Gated		Helical	PedBody	0.984:139.375	0.4	0.625	40	80	SmartmA 100-320mA NI=18 AvgmA320	Std Plus	3.11	33.71	75	32	Weight and height based non ECG gated protocol for cardiac with SmartmA
35.7.1	RC 22.5-31.5kg (49.6-69.5lbs)		Helical	LargeBody	1.375:155.0	0.4	5	40	120	110	Std Plus	2.54	33.71	85	32	Weight and height based routine chest protocol
35.7.3	SnapShot Segment 30-74 BPM		Helical	SmallBody	1.375:155.0	0.4	2.5	40	80	120	Std Plus	0.84	8.19	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	Std	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
			Cardiac SEG	CardiacSmall	0.180:17.2	0.4	0.625	40	80	ECG Mod 120-420mA Phase 40-80 AvgmA420	Std SS Seg	19.07	209.80	75	32	Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
35.7.4	SnapShot Burst 75-104 BPM		Helical	SmallBody	1.375:155.0	0.4	2.5	40	80	120	Std Plus	0.84	8.19	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	Std	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
			Cardiac SSB	CardiacSmall	0.180:17.2	0.4	0.625	40	80	ECG Mod 120-420mA Phase 40-80 AvgmA420	Std SSB	19.07	209.80	75	32	Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval] (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (mm)	Phantom (cm)	Description
35.7.5	SnapShot Burst+ >105 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	80	120	Std Plus	0.84	8.19	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	Std	21.31	10.66	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
			Cardiac SSBP	CardiacSmall	0.180:1 7.2	0.4	0.625	40	80	ECG Mod 120-420mA Phase 35-85 AvgmA420	Std SSB	20.44	224.79	75	32	Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
35.7.7	Non ECG Gated		Helical	PedBody	0.984:1 39.375	0.4	0.625	40	80	SmartmA 100-360mA NI=18 AvgmA360	Std Plus	3.50	37.93	75	32	Weight and height based non ECG gated protocol for cardiac with SmartmA
35.8.1	RC 31.5-40.5kg (69.5-89.3lbs)		Helical	LargeBody	1.375:1 55.0	0.4	5	40	120	120	Std Plus	2.77	46.45	120	32	Weight and height based routine chest protocol
35.8.3	SnapShot Segment 30-74 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	100	120	Std Plus	1.55	15.20	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	100	50	Std	39.50	19.75	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
			Cardiac SEG	CardiacSmall	0.180:1 7.2	0.4	0.625	40	100	ECG Mod 115-380mA Phase 40-80 AvgmA380	Std SS Seg	32.11	353.20	75	32	Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
35.8.4	SnapShot Burst 75-104 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	100	120	Std Plus	1.55	15.20	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	100	50	Std	39.50	19.75	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
			Cardiac SSB	CardiacSmall	0.180:1 7.2	0.4	0.625	40	100	ECG Mod 115-380mA Phase 40-80 AvgmA380	Std SSB	32.11	353.20	75	32	Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
35.8.5	SnapShot Burst+ >105 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	100	120	Std Plus	1.55	15.20	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	100	50	Std	39.50	19.75	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
			Cardiac SSBP	CardiacSmall	0.180:1 7.2	0.4	0.625	40	100	ECG Mod 115-380mA Phase 35-85 AvgmA380	Std SSB	34.34	377.76	75	32	Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
35.8.7	Non ECG Gated		Helical	SmallBody	0.984:1 39.375	0.4	0.625	40	100	SmartmA 100-400mA NI=16 AvgmA400	Std Plus	7.22	78.21	75	32	Weight and height based non ECG gated protocol for cardiac with SmartmA

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval] (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (mm)	Phantom (cm)	Description
35.9.1	RC 40.5-55.0kg(89.3-121.3lbs)		Helical	LargeBody	1.375:1 55.0	0.4	5	40	120	130	Std Plus	3.00	59.31	150	32	Weight and height based routine chest protocol
35.9.3	SnapShot Segment 30-74 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	100	120	Std Plus	1.55	15.20	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	100	50	Std	39.50	19.75	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
			Cardiac SEG	CardiacSmall	0.180:1 7.2	0.4	0.625	40	100	ECG Mod 120-420mA Phase 40-80 AvgmA420	Std SS Seg	35.40	389.39	75	32	Weight and height based cardiac protocol with ECG modulated mA (30-74 bpm)
35.9.4	SnapShot Burst 75-104 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	100	120	Std Plus	1.55	15.20	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	100	50	Std	39.50	19.75	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
			Cardiac SSB	CardiacSmall	0.180:1 7.2	0.4	0.625	40	100	ECG Mod 120-420mA Phase 40-80 AvgmA420	Std SSB	35.40	389.39	75	32	Weight and height based cardiac protocol with ECG modulated mA (75-104 bpm)
35.9.5	SnapShot Burst+ >105 BPM		Helical	SmallBody	1.375:1 55.0	0.4	2.5	40	100	120	Std Plus	1.55	15.20	50	32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	100	50	Std	39.50	19.75	0	32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
			Cardiac SSBP	CardiacSmall	0.180:1 7.2	0.4	0.625	40	100	ECG Mod 120-420mA Phase 35-85 AvgmA420	Std SSB	37.93	417.20	75	32	Weight and height based cardiac protocol with ECG modulated mA (>105 bpm)
35.9.7	Non ECG Gated		Helical	SmallBody	0.984:1 39.375	0.4	0.625	40	100	SmartmA 100-440mA NI=16 AvgmA440	Std Plus	7.95	86.03	75	32	Weight and height based non ECG gated protocol for cardiac with SmartmA

Abdomen

Table 21-4

[Interval]* - For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (second s)	Slice Thickne ss (mm) [Interval]* (mm)	Beam Collimat ion (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy·c m)	Scan Length (mm)	Phantom (cm)	Description
36.1.1	Abd 6.0-7.5kg (13.2-16.5lbs)		Helical	PedBody	1.375:1 27.5	0.4	3.75	20	120	100	Std Plus	2.17	22.53	78.75	32	Weight and height based routine abdomen protocol
36.2.1	Abd 7.5-9.5kg (16.5-20.9lbs)		Helical	PedBody	1.375:1 27.5	0.4	3.75	20	120	160	Std Plus	3.47	39.95	90	32	Weight and height based routine abdomen protocol
36.3.1	Abd 9.5-11.5kg (20.9-25.4lbs)		Helical	PedBody	1.375:1 27.5	0.4	3.75	20	120	180	Std Plus	3.91	49.34	101.25	32	Weight and height based routine abdomen protocol
36.4.1	Abd 11.5-14.5kg (25.4-32.0lbs)		Helical	SmallBo dy	1.375:1 27.5	0.4	5	20	120	125	Std Plus	2.71	36.64	110	32	Weight and height based routine abdomen protocol
36.5.1	Abd 14.5-18.5kg (32.0-40.8lbs)		Helical	SmallBo dy	1.375:1 27.5	0.4	5	20	120	135	Std Plus	2.93	39.57	110	32	Weight and height based routine abdomen protocol
36.6.1	Abd 18.5-22.5kg (40.8-49.6lbs)		Helical	Medium Body	1.375:1 27.5	0.4	5	20	120	120	Std Plus	3.07	47.68	130	32	Weight and height based routine abdomen protocol
36.7.1	Abd 22.5- 31.5kg(49.6- 69.5lbs)		Helical	LargeBo dy	1.375:1 55.0	0.4	5	40	120	150	Std Plus	3.46	68.44	150	32	Weight and height based routine abdomen protocol
36.8.1	Abd 31.5- 40.5kg(69.5- 89.3lbs)		Helical	LargeBo dy	1.375:1 55.0	0.4	5	40	120	170	Std Plus	3.92	77.56	150	32	Weight and height based routine abdomen protocol
36.9.1	Abd 40.5- 55.0kg(89.3- 121.3lbs)		Helical	LargeBo dy	1.375:1 55.0	0.4	5	40	120	200	Std Plus	4.61	114.30	200	32	Weight and height based routine abdomen protocol

Miscellaneous

Table 21-5

[Interval]* - For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (mm)	Phantom (cm)	Description
40.1	Pink Zone Routine 0-20 lbs/0-9 kg		Helical	PedBody	1.375:1 55.0	0.5	5	40	80	SmartmA 65-130mA NI=5 Avg mA 70	Std	0.61	8.70	95	32	Pediatric and Small patient Routine procedure based protocols
40.2	Pink Zone Routine 21-60 lbs/9.1-27.2 kg		Helical	PedBody	1.375:1 55.0	0.5	5	40	100	SmartmA 80-160mA NI=7 Avg mA 70	Std	1.13	27.47	195	32	Pediatric and Small patient Routine procedure based protocols
40.3	Pink Zone Routine 61-100 lbs/27.3-45.4 kg		Helical	Medium Body	1.375:1 55.0	0.5	5	40	120	SmartmA 95-190mA NI=10 Avg mA 70	Std	2.09	71.81	295	32	Pediatric and Small patient Routine procedure based protocols
40.4	Pink Zone Routine 101-200 lbs/45.4-90.7 kg		Helical	LargeBody	1.375:1 55.0	0.5	5	40	120	SmartmA 110-220mA NI=12 Avg mA 70	Std	2.02	89.30	395	32	Pediatric and Small patient Routine procedure based protocols
40.5	Pink Zone Routine >200 lbs/>90.8 kg		Helical	LargeBody	1.375:1 55.0	0.5	5	40	120	SmartmA 125-300mA NI=15 Avg mA 70	Std	2.02	89.30	395	32	Pediatric and Small patient Routine procedure based protocols
40.7	Green Zone Lower Dose 0-20 lbs/0-9 kg		Helical	PedBody	1.375:1 55.0	0.5	5	40	80	SmartmA 50-100mA NI=7 Avg mA 70	Std	0.61	8.70	95	32	Pediatric and Small patient Lower Dose procedure based protocols
40.8	Green Zone Lower Dose 21-60 lbs/9.1-27.2 kg		Helical	PedBody	1.375:1 55.0	0.5	5	40	100	SmartmA 60-120mA NI=9 Avg mA 70	Std	1.13	27.47	195	32	Pediatric and Small patient Lower Dose procedure based protocols
40.9	Green Zone Lower Dose 61-100 lbs/ 27.3-45.4 kg		Helical	Medium Body	1.375:1 55.0	0.5	5	40	120	SmartmA 70-140mA NI=11 Avg mA 70	Std	2.09	71.81	295	32	Pediatric and Small patient Lower Dose procedure based protocols
40.10	Green Zone Lower Dose 101-200 lbs/ 45.4-90.7 kg		Helical	LargeBody	1.375:1 55.0	0.5	5	40	120	SmartmA 80-160mA NI=13 Avg mA 70	Std	2.02	89.30	395	32	Pediatric and Small patient Lower Dose procedure based protocols
40.11	Green Zone Lower Dose >200 lbs/ >90.8 kg		Helical	LargeBody	1.375:1 55.0	0.5	5	40	120	SmartmA 90-240mA NI=16 Avg mA 70	Std	2.02	89.30	395	32	Pediatric and Small patient Lower Dose procedure based protocols
40.13	Grey Zone CT Angio 0-20 lbs/0-9 kg		Helical	PedBody	1.375:1 55.0	0.5	2.5	40	80	SmartmA 100-200mA NI=5 Avg mA 140	Std	1.22	17.71	97.5	32	Pediatric and Small patient Angio procedure based protocols

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval J* (mm)]	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Recon Type	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (mm)	Phantom (cm)	Description
40.14	Grey Zone CT Angio 21-60 lbs/9.1-27.2 kg		Helical	SmallBody	1.375:1 55.0	0.5	2.5	40	100	SmartmA 120-240mA NI=9.9 Avg mA 140	Std	2.26	55.50	197.5	32	Pediatric and Small patient Angio procedure based protocols
40.15	Grey Zone CT Angio 61-100 lbs/ 27.3-45.4 kg		Helical	LargeBody	1.375:1 55.0	0.5	2.5	40	120	SmartmA 120-240mA NI=10 Avg mA 140	Std	4.03	139.28	297.5	32	Pediatric and Small patient Angio procedure based protocols
40.16	Grey Zone CT Angio 101-200 lbs/ 45.4-90.7 kg		Helical	LargeBody	1.375:1 55.0	0.5	2.5	40	120	SmartmA 120-240mA NI=12 Avg mA 140	Std	4.03	179.62	397.5	32	Pediatric and Small patient Angio procedure based protocols
40.17	Grey Zone CT Angio >200 lbs/ >90.8 kg		Helical	LargeBody	1.375:1 55.0	0.5	2.5	40	120	SmartmA 120-300mA NI=15 Avg mA 140	Std	4.03	179.62	397.5	32	Pediatric and Small patient Angio procedure based protocols

Lexicon

Lexicon

For the CT technologist who operates multiple scanner models, perhaps from multiple manufacturers, the variability in names for important scan acquisition and reconstruction parameters can lead to confusion, reduced comfort and an increased potential for error. The intent of this CT terminology lexicon is to allow users to translate important CT acquisition and reconstruction terms between different manufacturers' systems.

The following tables provide the CT terminology for GE and is a result of the AAPM CT Terminology working group. It identifies relevant terms from established lexicon (e.g., Radlex and DICOM) and other relevant literature published.

The AAPM website will provide updates when changes in standardization of terminology occur. Visit www.aapm.org, search CT Protocols, and select Lexicon for full listing of terminology across various manufactures of CT equipment.

The generic descriptions or terms in the first column are intended to orient the user to the relevant concepts; they are not consensus "preferred terms." The generic descriptions are not based on any single existing or pending terminology standard; however the references cited below were consulted in developing the generic descriptions.

A number of individuals and groups have advocated for terminology standardization in CT, including at a March 30-31, 2010 FDA public meeting entitled "Device Improvements to Reduce Unnecessary Radiation Exposure from Medical Imaging" (transcripts available at the FDA website: www.fda.gov in Medical Devices). Participants proposed a cooperative effort among professional organizations (AAPM, ASRT, ACR, etc.), industry, and the FDA.

Scan Acquisition and User Interface Basics

Table 22-1 Scan acquisition and user interface terms

Generic Description	GE Name
The portion of the user interface where scans are prescribed	Exam Rx
Other portions of the user interface , such as where reconstructed images are viewed	Desktop
CT localizer radiograph (i.e. the scanned projection radiograph, often acquired by the CT system to allow the user to prescribe the start and end locations of the scan range)	Scout
Axial scan mode: Data acquisition while the patient table remains stationary; the table position may be incremented between X-ray exposures to collect data over a longer z axis range.	Axial
Helical or Spiral scan mode: Data acquisition while the patient table is continuously moving along the z axis.	Helical
Dynamic scan mode - single detector width: Data acquisition at multiple time points over the same anatomic location(s) while the patient table remains stationary; X-ray exposure can be continuous or intermittent	Cine or zero interval Axial
Dynamic scan mode - multiple detector widths: Data acquisition at multiple time points over the same anatomic location(s) while the patient table cycles back and forth between designated start and end locations in order image a region wider than the detector	Shuttle
Interventional CT - Intermittent X-ray exposures	SmartStep
Interventional CT - Continuous X-ray exposures	SmartView

Generic Description	GE Name
Table increment (mm) per 360 degree rotation of the X-ray tube (axial scan mode)	Interval
Table feed per 360 degree rotation of the X-ray tube (helical scan mode)	Speed (mm/rot)
Field of measurement: Diameter of the circular region within the scan plane over which projection data are collected. Nominally equal to the diameter of the primary beam at isocenter in the axial plane.	Scan Field of View (SFOV, cm)
Tube current: Number of electrons accelerated across an X-ray tube per unit time, expressed in units of milliamperage (mA)	mA
Tube current-time product: The product of tube current and exposure time per rotation, expressed in units of milliamperage • seconds (mAs). In axial scan mode, this is equal to tube current × (scan angle ÷ 360) × rotation time. In helical scan mode, this is equal to tube current × rotation time.	Not used on this system
Effective tube current-time product: In helical scan mode, this is the product of tube current and rotation time (expressed in units of milliamperage • seconds (mAs) ÷ pitch)	Not used on this system
Tube potential: The electric potential applied across an X-ray tube to accelerate electrons towards a target material, expressed in units of kilovolts (kV)	kV
Pitch: Unitless parameter used to describe the table travel during helical CT; equal to table travel (mm) per gantry rotation ÷ total nominal beam width (mm)	Pitch
Automated patient instructions	AutoVoice

Dose Modulation and Reduction Tools

Table 22-2 Dose modulation and reduction tool terms

Generic Description	GE Name
Automatic exposure control (AEC): A scanner feature that automatically adapts the X-ray tube current to the overall patient size to achieve a specified level of image quality	Available in AutomA and SmartmA
Angular tube current modulation	SmartScan (CT/i only)
Longitudinal tube current modulation	AutomA
Angular and longitudinal tube current modulation	SmartmA (x, y, z)
ECG-based tube current modulation	ECG Modulated mA
Image quality reference parameter for AEC	Noise Index

Multi-slice Detector Geometry

Table 22-3 Multi-slice detector geometry terms

Generic Description	GE Name
Multi-slice detector array design	Fixed
Detector configuration	Detector Configuration

Figure 22-1 Detector Coverage and Slice Thickness - Detector rows (N) and slice thickness (T) selection console screen image

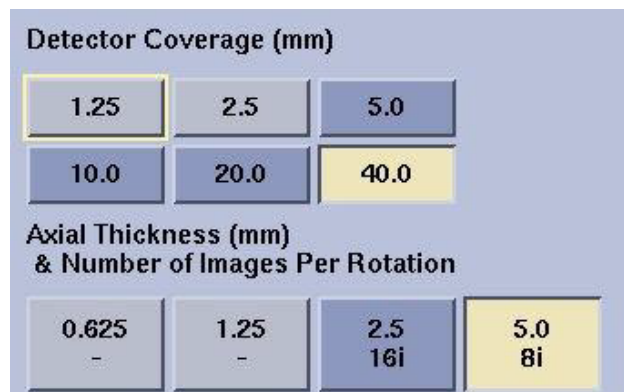


Image Reconstruction and Display

Table 22-4 Image reconstruction and display terms

Generic Description	GE Name
Window width: Range of CT numbers (maximum - minimum) that are distributed over the viewable grey scale of the display device or film	Window Width
Window center: The CT number in the center of the viewable grey scale	Window Level
Reconstruction field of view: Width of the square region mapped to the reconstructed image matrix	Display Field of View (DFOV) (cm)
Prescribing the reconstruction parameters prior to scan acquisition	Prospective recon
Prescribing the reconstruction parameters after scan acquisition	Retrospective recon
Reconstruction property that determines sharpness or smoothness of image in the axial plane	Algorithm
Helical interpolation options to achieve a wider or narrower section sensitivity profile	Full (narrower) or Plus (wider) mode
Nominal width of reconstructed image along the z axis	Thickness (mm)
Distance between two consecutive reconstructed images	Interval
Fast but lower-quality reconstructed images for rapid review of entire exam	Image Check
Off-center reconstruction coordinates are called	RL Center AP Center
Flip or rotate the image orientation is called	Flip/rotate
Image modifications to alter sharpness or smoothness (done in image space without reconstructing images)	Image Filters

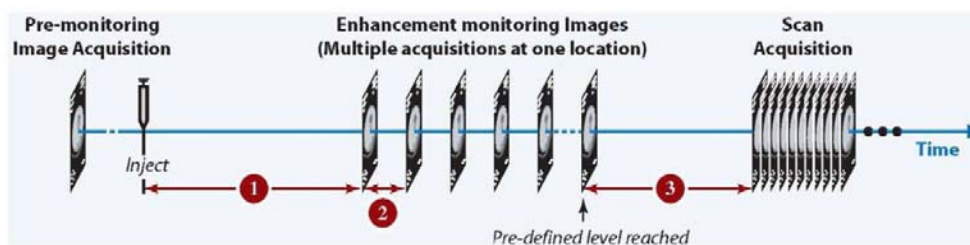
Contrast Media Tools

Table 22-5 Contrast media tool terms

Generic Description	GE Name
Bolus tracking: Scanner feature to automatically initiate a prescribed axial, helical or dynamic scan when a threshold level of contrast enhancement is reached at a specified region of interest	Smart Prep
Test Bolus: Scan mode used to measure the contrast transit time using a small injection of contrast media	Take axial scans at zero table feed and process with MIROI

Generic Description	GE Name
Time-attenuation curve (TAC): Graph of the contrast enhancement versus time	Smart Prep graph or MIROI graph
Threshold: CT number (HU) where bolus tracking tool will trigger the system to begin the scan	Transition ROI Threshold
Scanner feature used to quantitatively evaluate the TAC	MIROI (multiple image region of interest)
Monitoring delay: Time from injection to the start of monitoring scans (Time 1 in figure below)	Monitoring Delay
Monitoring interval: Time between consecutive monitoring scans to (Time 2 in figure below)	Monitor ISD (InterScan Delay)
Scan delay: Time from when threshold is reached and prescribed axial, helical or dynamic scan begins (Time 3 in figure below)	Diagnostic delay

Figure 22-2



Multi-planar Formats and 3D Processing

Table 22-6 Multi-planar formats and 3D processing terms

Generic Description	GE Name
Reformatted image at an oblique plane (not an axial, coronal, or sagittal)	Oblique reformat
Saving images at various viewing angles about a volume or surface rendered object	Batch Loop
Saving images at various planes through a volume	Batch Reformat
Surface-rendered object	3D
Volume-rendered object	Volume Rendered image (VR)

Service and Application Tools

Table 22-7 Service and application tool terms

Generic Description	GE Name
X-ray tube warm up	Tube Warm-up (tube warm up)
Daily calibrations	Fast Cals (done in daily prep)
Application information	Learning Solutions or User Manual
Application support assistance	Insite or Ilinq

Workflow

Table 22-8 Workflow terms

Generic Description	GE Name
Scheduled (but not yet scanned) patient list is called	Patient Schedule
Already scanned patient list is called	List/Select
User comments or text added to an image is called	User annotation
Filming tools are called	Auto/manual film composer
Data page summarizing scan parameters, CTDIvol and DLP	Exam Text Page or Series Text Page
Sorting patient list	Sort

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